Basic Ecological Concepts and Data in R

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Preface

Students (that's you) and other interested parties should read this preface. Consider it the real chapter 1.

This book provides an activity-based approach introduction to the science and practice of ecology. It is designed to form the basis of an undergraduate general ecology course. It includes real-world assessment activities, and so answer keys are not generally available, even if they might be appropriate in some situations.

Each chapter provides,

- sources in the form of video lecture, references to peer-reviewed publications, and other sources were relevant,
- data exploration and modeling, and
- one or more writing assignments.

Each chapter mixes (i) ecological content and (ii) coding practices. You will get the most out of the content if you understand and accept this from the start. Avoiding one or the other reduces your ability to understand, appreciate, and use both.

Guidance

Use these tips to help reduce frustration and make things work better:

- When the text provides an enumerated list or *numbered* bullet items (e.g., 1, 2, 3 or a, b, c), it is usually very important or even essential to do these steps in sequence. There will be times that it is impossible to succeed unless you do the steps in *sequence*.
- When the text provides a simple, unnumbered list, it is usually not important to consider these items as representing a sequence.
- I included a few sections entitled *Overthinking*, and I got this idea from Richard McElreath (McElreath 2020). These are sections that you can skip on the first read, and are not essentially for getting things to *work*. They provide deeper, more detailed explanations than is typically required.

McElreath, Richard. 2020. *Statistical Rethinking: A Bayesian Course with Examples in R and Stan*. Second Edition. Texts in Statistical Science. CRC Press.

Skills for a résumé

You may never apply for a job that requires coding but putting it on your résumé or answering questions about it in an interview will show eployers that you have qualities that they want. Demonstrating that you have done coding shows that you can,

- manage details,
- troubleshoot problems,
- learn to new skills to complete tasks.

You will show that you can understand tasks you've taken on and can communicate them to others. In the event you work with coders or do some coding yourself, you will be better prepared to work with others to accomplish meaningful work.

Two R skills recur across nearly every chapter: *wrangling* and *displaying* data. "Wrangling" data refers to the majority of data management and data set rearrangement that we inevitably need to do in any serious work. "Displaying" data in ways that communicate effectively is not simple, and doing it badly hampers communication and reflects poorly on whoever made it. It must communicate an important idea, and not hide any relevant information.

In this book, you will wrangle data with **tidyverse**-style functions (things like `read_csv`, `subset`, `mutate`), and display or visualize it with `ggplot2`. This book will introduce these functions and styles organically, as they are mixed throughout each chapter. You will practice these so often that, by the end of the book, they should feel like second nature.

If you ever want help translating that practice into language for a résumé, cover letter, or job application, see the chapter-by-chapter breakdown of how these skills translate into résumé language, see the Appendix A, From This Book to Your Résumé: A Skills Crosswalk.

Coding for Science in a Changing Workplace Landscape

AI is doing more and more for us, and here I describe some issues that I think are critical to keep in mind throughout your schooling and career.

AI can now write passable R code, summarize articles, and define terms almost instantly. That doesn't reduce how much you need to learn — it shifts what you need to learn. Because AI is fluent in the jargon of ecology, it becomes more important, not less, that you develop real understanding of ecological mechanisms, model assumptions, and data interpretation: only a trained ecologist can recognize when a fluent AI answer is subtly (or not so subtly) wrong.

Durable skills vs. AI-augmentable tasks

Durable human skills — ecological intuition, mechanistic and mathematical reasoning, model-assumption literacy, data literacy, multi-scale synthesis — require judgment AI doesn't reliably have. AI-augmentable tasks (summarizing papers, boilerplate code, definitions, outlines) are things AI already does well. But AI-augmentable doesn't mean unimportant: you still need to understand a task well enough to judge whether the AI did it correctly.

Why AI makes ecological understanding more important

Offloading a task to AI without the underlying knowledge means you can't tell whether its output — code, an explanation, a derivation — is right. This is the "plausibility trap": AI-generated ecological content is trained to sound like an expert wrote it, not to be correct about the natural world. The better AI sounds, the more ecological training you need to catch what it gets wrong.

Six categories of skills to develop

Mechanistic reasoning explains why a pattern exists, not just that it exists. Mathematical reasoning is itself a form of ecological understanding — deriving a result is how you learn what controls a system, not a hurdle before the real learning. Model assumption literacy means knowing what a model assumes and when that breaks. Predicting an outcome before running the code (and being wrong) is where real learning happens. Data literacy means judging bias, sample size, and what data can't tell you. Multi-scale synthesis connects individual-level processes to population, community, and ecosystem patterns.

Understanding vs. doing

Code that displays data is checked by its output — does it look right? Code that builds a mechanistic or statistical model has to be checked by examining the assumptions built into it. When a model's results seem off, the cause is usually one of three things: a plain coding error, a mismatch between the math and the intended ecology, or ecological assumptions with unexpected (but real) consequences. Mathematical reasoning is the tool for telling these apart, and for verifying that an AI's code actually implements the biology you intended.

Using AI responsibly in this classroom setting using this book

AI is useful the way a calculator is useful — it saves time on mechanical tasks but can't replace the thinking that makes them meaningful. Good uses: debugging code, syntax refreshers, alternative explanations of material

you've already studied. Risky uses: first drafts, learning a completely new concept from AI, running AI code you don't understand, or letting AI interpret your results. Rules of thumb: always ask AI to explain what it does; if you can't explain every line of AI-written code, you haven't learned what you needed to.

The skills that will define your career

Future ecologists will work alongside AI tools that can process imagery, classify species, and generate statistical code. Human value won't lie in doing these tasks manually — it will lie in asking the right ecological questions, designing studies that can answer them, recognizing when a tool is wrong, connecting math to biological reality, and communicating findings to non-specialists. The tools will keep changing; the need for ecologists who understand the living world will not.

A note for instructors

Design assignments that require judgment, not just output that AI could produce alone. Make AI use transparent rather than banned. Foreground model assumptions explicitly as content, not fine print. Build in a predict-before-you-compute step. Use AI's mistakes as teaching moments for critical evaluation.

app-ai-skills

For a more extensive discussion of these topics, see (**app-ai-skills?**).

Other books

My other books include *A Primer of Ecology using R* and *A Primer of Ecosystem Modeling*. The former (Primer of Ecology) is designed primarily for beginning graduate students as a primer to theory in population and community ecology. It was first published with Springer in 2009, and the current GitHub version is quite updated. I use the latter (ecosystem modeling) when I co-instruct a graduate course led by colleagues who are real ecosystem ecologists (Drs. Melany Fisk, Mike Vanni, and Lesley Knoll).

Questions for you

Career skills

1. Have you ever written computer code before this class? Have you ever typed a function into a spreadsheet like Microsoft Excel? Explain briefly the level of your experience with coding.
2. What do you anticipate the coding in this book will be like? What is your *a priori* expectation?

3. Have you ever put computer use on a résumé? If so, how did you list or describe your experience, knowledge, or skill?
4. What kind of job do you have now, and what do you hope to do some day? At this moment, can you imagine how writing and using code to do scientific research in ecology might help you–directly or indirectly–in your future career?

Generative AI

1. How do you use generative AI in your current life?
2. Describe environmental impacts of AI, and identify (as much as you can), the sources of that information. Do you hear or read something in a reputable news story? Did you learn about it in a class (and if so, can you remember the source)? Did you hear about it from a friend? Was the information quantitative?
3. List potential risks and benefits of AI generated content (not data center risks). Rank them, based on your intuition, knowledge, or evidence at hand, from most beneficial to most harmful. Consider both the likelihood of something happening and also the size or magnitude of the benefit/risk.
4. How do you feel about AI?

Mazel tov!

1. *Thinking about thinking about concepts*

"...in which we build our metacognition."

Before you start

- **Assumptions about background and guidance:** no background beyond curiosity and high-school algebra. The figures were made in R, but the code is hidden, and you do not need R to follow this chapter. If an idea about probability (such as an average count, or a distribution of counts) is new to you, read the figure captions slowly and bring your questions to class.

1.1. Goals

In this chapter, you will,

- Understand and explain six concepts and practices that are fundamental to science.
- Apply each concept to your daily life.

1.2. Introduction

How do we think?

Thinking clearly is the most critical skill in all of Science. While we understand elements of how to do this, we are a long way off from really understanding how the brain works. In the meantime, let's focus on tools that humanity has found to be very useful in thinking clearly.

1. Observation
2. Learning (Bayesian updating)
3. Logic
4. Mathematics
5. Causal inference

These ideas and tools all help us think more clearly about whatever we're interested in.

In addition to the metacognitive activities and tools, we'll also describe some cross-cutting concepts that apply to all fields of study. By considering these concepts in what we do, in are field of research, we can view our own research more broadly and poossibly connect it more easily to other research areas. There concepts are,

1. Scale
2. Hierarchy
3. Emergence
4. Complexity

These help us place our own relatively narrow research focus into a broader realm of ideas, by seeing how it is constrained in space and time (scale), what its is made up of and what small role it plays in a bigger entity (hierarchy), how it is more than the sum of its parts (emergence and complexity).

Let us begin by considering the following tools of the human mind.

1.3. Observation

Observing the world around us is the most fundamental process of science. Period.

The goal of science is to describe and explain the natural or material world. The true test of whether we can do that is to compare our *understanding* of the world to the world itself. Science makes great strides when we invent technology to observe the world in new ways: microscopes and teloscopes, sequencers and environmenal DNA, binoculars and satellites, particle accelerators and LIGO.

Science also makes great strides when our theory helps us see new things that have always been right in front of us, in new ways.

1.3.0.1. For you to do

Of the thousands of small observations you made today, describe one of them here, in your own handwriting:

1.4. Learning, or Bayesian Updating

All organisms have systems for learning.

Learning consists of i. having a pre-existing set of behaviors or thoughts, ii. encountering new situations or information, and iii. updating our sets of behaviors or thoughts.

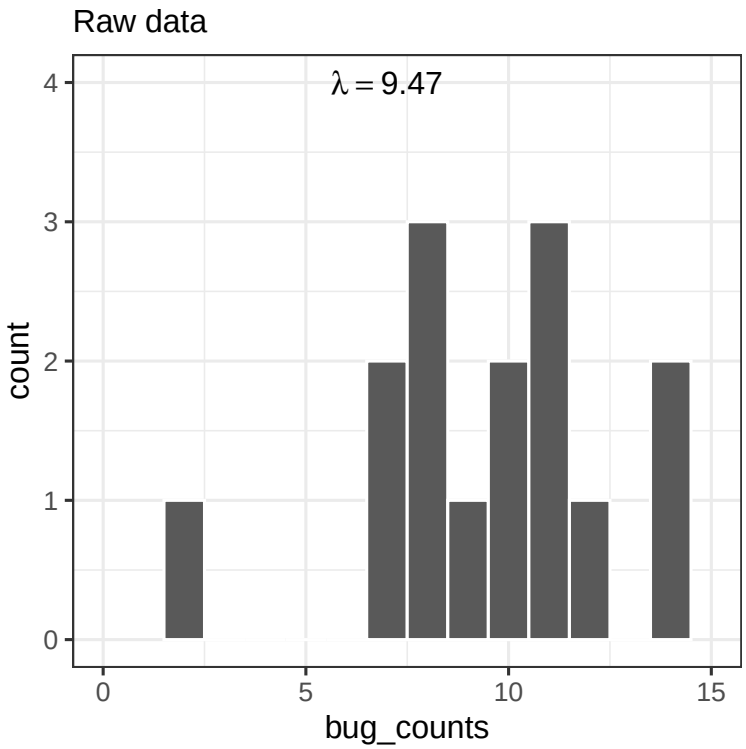


Figure 1.1.: I counted the number of arthropods on ten plants. The average of those counts (lambda) was 9.47.

This sequence of prior knowledge -> new observation -> updated knowledge appears to be fundamental to learning. Mathematicians have formalized it to use in all kinds of ways, perhaps most notably as Bayesian epistemology and statistics.

1.4.1. Four Examples of Bayesian Reasoning

Consider a toddler learning to walk, or a child learning to ride a bicycle. Our nervous systems and muscles have a prior knowledge of how to keep us upright. We encounter or observe or sense a new situation (being vertical, on our feet or on skinny bicycle tires), and our nervous system and muscles update their behaviors to keep us upright.

When I was very young, my parents led me to think there was a person called Santa Claus who delivered presents all around the world every December. This constituted my prior knowledge when I was young. As I grew older, I made increasingly sophisticated observations around gift giving, and updated my knowledge about Santa Claus such that I now believe that it is very, very unlikely that Santa Claus exists.

Every October at the beginning of the NHL hockey season, I believe that my favorite NHL hockey team (the Pittsburgh Penguins) has a good chance to get to the Stanley Cup final (the professional league championship). My prior belief is that the team has a particular level of talent. As the season progresses and I watch the team play against other teams, I gather more knowledge about the actual level of talent. I update my belief each night when I see them and other teams play. Following each game, I have a

new, updated, belief regarding the team's probability of making it to the championship.

Consider climate change. In the early 1800s, Alexander von Humboldt argued that human development was leading to changes in climate (Wulf 2015, Humboldt's legacy. *Nat Ecol Evol* 3, 1265–1266 (2019). https://doi.org/10.1038/s41559-019-0980-5). In 1896, chemist Svante Arrhenius showed that CO2 in the atmosphere probably contributed to global cycles of glaciation and that fossil fuel combustion by humans would likely cause global warming. In 1980s, scientists at Exxon showed with very high confidence, in secret Exxon documents, that burning fossil fuels would increase average global temperatures. In 1988, NASA scientist Dr. James Hansen testified before the U.S. Congress regarding the high probability of severe climate change due to human activity. Since then, thousands of research studies have documented details supporting predictions of a warming climate. Scientists have updated their prior knowledge in the face of new observations.

To conceptualize these examples, we will state:

prior understanding + new observations => updated understanding.

Write down a brief example where you learn something. Identify your prior knowledge, the new information or data, and your updated understanding or ability.

In all of these situations, we use new observations to update our prior knowledge to create new understanding. This is how we learn.

1.4.1.1. Formal Bayesian reasoning

We use Bayesian statistics to do this formally. We,

1. express our prior beliefs in the form of a mathematical probability distribution; we refer to this our the prior probabilities.
2. make new observations and express these using a probability distribution, and
3. update our knowledge by combining these probability distributions, and obtain a new probability distribution that we refer to as our posterior probability distribution.

It boils down to this:

Statistics

Prior distribution x new data => posterior distribution

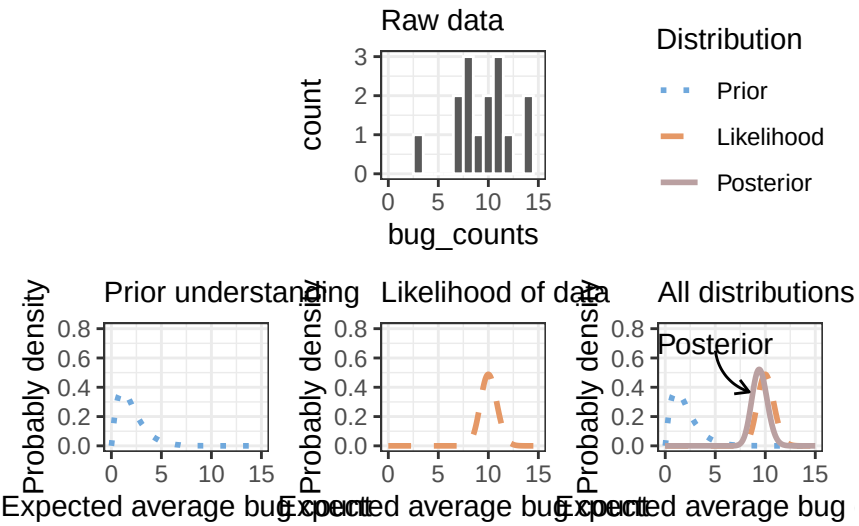


Figure 1.2.: Before I ever counted arthropods on plants, I believed there would be about 2 bugs on each plant (on average), and probably not more than 5 or 6. The *prior distribution* describes that initial guess and uncertainty. The normalized *likelihood* shown here is the probability of observing those counts, assuming a Poisson distribution and that the true mean is the observed mean. The *posterior* distribution shows us the combination (literally the product) of the prior and normalized likelihood. The posterior is our updated understanding of what the mean count number is for the entire population of plants I sampled from.

1.4.2. Three important notes

First, uncertainty is part of learning. Sometimes more data causes us to realize we know less than we thought; sometimes it allows us to be more certain. When I first believed in Santa Claus, I was completely certain in my belief that there was a Santa Claus. As I began to gather more data, I became less certain, but I still thought Santa Claus probably existed. Now, I have accumulated sufficient evidence that I am quite certain — I have reduced my certainty — regarding the existence of Santa Claus.

Second, we refer to strength of evidence or our degree of rational belief as credence. Credence our degree of belief about something, as in "don't give any credence to that gossip." In this book, and elsewhere, we use credence in both an informal sense, and a formal quantitative sense.

Third, Bayesian epistemology views certainty, probability, and credence as subjective probabilities. Each of us comes to new observations with different prior knowledge, and so new data has a different effect on each of our beliefs. For most intents and purposes, we can all agree that a flipped coin lands heads-up about 50% of the time, not merely because it is an objective fact, but rather because we all have extensive observations with coins and with the logic of a hypothetical fair coin. We have updated any prior beliefs with so much data that we have come to a similar belief about flipped coins.

1.4.2.1. For you to do

Write down one example of Bayesian learning that you did. Identify your prior knowledge, the new observations you made, and how it altered your expectation or your level of credence or certainty regarding that expectation.

1.5. Logic

We use logic all the time, and sometimes we forego it. In science, we often use intuition to get started, but must be logical in the end.

Here we'll just mention three types of logic we encounter and use routinely in science.

1.5.1. Deduction

Deductive arguments can be valid or invalid, and have conclusions that are true or false. Here we describe the most common form, a categorical syllogism. A categorical syllogism has a major proposition, a minor proposition, and a conclusion:

Major proposition 1: If P, then Q Minor proposition 2: P Conclusion: Therefore, Q

1.5.1.1. Three contrasting examples

Valid argument and true conclusion. The conclusion is true because the argument is valid and the propositions are true.

1. All women are mortal (True; If a person is a woman, then she is mortal)
2. E. Lucy Braun was a woman (True)
3. Therefore, E. Lucy Braun was mortal (True)

Valid argument, false conclusion. The conclusion is false because the argument is valid, but one of more of the propositions is false.

All women are immortal (False) E. Lucy Braun was a woman (True) Therefore, E. Lucy Braun was immortal (False)

Invalid argument. Note that all that statements are true, but the conclusion is not supported by the argument.

All women are mortal (True) My cat's name is Dahlia (True) Therefore, Dahlia is mortal (True but not supported by these premises)

1.5.2. Induction

Inductive arguments are arguments of extrapolation from examples. They are not technically valid, but as you will see, they are useful if used carefully.

All the swans that I have ever seen are white. Therefore, all swans are white.

The sun has always risen in the east every morning. Therefore, the sun always will rise in the east each morning.

In my survey of University students, 86% said they liked ice cream. Therefore, 86% of all University students like ice cream.

All of the above express their conclusions with complete certainty which is unwarranted, given the preceding observations.

In science, we rely on induction all the time to help guide us in the right direction, but unlike the examples above, we introduce uncertainty.

Many plants produce nectar on its leaves to attract ants and the ants protect the plants from herbivores. Therefore, this partridge pea that produces nectar on its leaves seems likely to be defended by ants.

Past El Niño events have tended to precede spikes in global average air temperature. Therefore, the 2026-2027 El Niño seems more likely than not to cause temperature to spike in 2027.

1.5.3. Abduction

Abduction is less frequently mentioned in introductory logic, but it is how we do science. It is the process of seeking the most likely and simplest explanation, given a set of observations. Abduction is quite flexible, and so formal abstractions are more complicated and numerous.

In abduction, we often argue backwards, in an invalid argument, referred to as affirming the consequent:

This morning my driveway is wet. I know that when it rains at night, my driveway is wet in the morning. Therefore, perhaps it rained last night.

1. If rain then wet
2. Wet
3. Therefore, rain

Note that other events could have made my driveway wet - perhaps I forgot the sprinkler was on, or my son empty his ice chest in my driveway. We have to be very cautious if we choose to try to make conclusions that way.

In useful abduction, we use multiple lines of evidence. Did I leave the sprinkler on last night? What does my son say? Were other driveways or the street wet?

Here is another example, about the effect of El Niño on global average surface temperature.

Pattern: The El Niño–Southern Oscillation (ENSO) is a gigantic atmospheric and southern Pacific Ocean circulation pattern that swings back and forth across the southern Pacific a little like a bathtub, resulting from and causing changes in the distribution of heat in the ocean and atmospher. The El Niño phase occurs on a somewhat unpredictable cycle of 4-10 years. It causes big weather changes across the global, resulting droughts in some places and torrential rains and floods in other places around the globe.

Mechanism: El Niño events prevent cold water from welling up to the surface along the coast of South America in the eastern Pacific, and keep warm

water on the surface in the western Pacific. This seems likely to warm surface air more than non-El Niño conditions. Observations: Past El Niño events have tended to precede spikes in global average air temperature. Tentative conclusion: Given the observations and a plausible mechanism, it seems very plausible that the 2026-2027 El Niño will cause temperature to spike in 2026-2027.

In this section, we've introduced these forms of logic (deduction, induction, abduction) to help you improve your metacognition, to become more aware of your thinking. Your goal should be to become more expert at using evidence to help you make sense of the world.

1.5.3.1. For you to do

Write out an informal argument that you have made today. Maybe it concerned where you should have breakfast, or maybe to help you make a prediction in your own research. Regardless, write your argument below, and identify its type and structure.

1.6. Mathematics

Mathematics is an unreasonably useful tool for describing and explaining nature. It requires us to be unambiguous and precise, but rewards us with indisputable logic. A famous statistician (G. E. P. Box) said that all models are wrong, but some are useful; it was his humble way of saying that models are gross abstraction that can nonetheless reveal partial truths that are otherwise hidden.

Throughout this book, you will use mathematical models and statistical models to help you understand the logic of your biological assumptions and to make predictions about the future.

1.7. Causal Inference

Causal inference is a specific area of formal logic and mathematics that helps us make sense of networks of relationships. It provides a set of tools that we should be aware of. It allows us to convert words into simple pictures (network graphs) that we can analyze formally. We can use it to help us figure out how to design experiments and analyze data in valid ways.

Consider my wet driveway:

Rain -> Driveway <- Sprinkler;

This graph tells me that rain can affect my driveway (get it wet) and an independent event, my sprinkler (which I use to water the adjacent garden) can also affect my driveway. However, in this graph, the rain and my sprinkler are independent - they don't affect each other. Note also the

direction of the arrows. The direction represents cause or a causal effect. Rain causes my driveway to get wet, but my driveway does not cause the sky to release rain.

Let's look at another example, in which ice cream sales are associated with drowning deaths. In this case, imagine a conspiracy theorist arguing that because people are buying (and eating) more ice cream, they are more at risk of drowning if they at e too much ice cream. But let's look at the causal graph.

Ice cream sales <- Heat -> Drowning deaths

It is merely that summer time heat is leading people to buy (and eat) more ice cream. It is also encouraging more people to spend more time around water, and this increased time around water leads to more drowning deaths by chance alone.

Let's look at one more example, this time a chain of causes. Rising CO_2 levels leads to more summer heat extremes and more summer heat extremes leads to higher mortality.

CO2 -> extreme atmospheric heat -> heat-related mortality

Note the CO_2 does not directly cause people to die. They are not suffocating. In this graph, CO2 has a direct causal effect on temperatures and temperature has a direct causal effect on people.

The above graphs are the foundation of causal inference. They all are made of variables and arrows, sometimes referred to as vertices and edges (think geometry). In addition they do not have any loops or cycles; if you follow the arrows along any path, it always comes to an end. These graphs are called directed acyclic graphs (DAGs). They have causal direction (arrows), and they contain no loops or cycles (acyclic).

They come in three flavors: Collider: two separate variables affect a third. Fork: one variable affects two others. Chain: a cascade where one variable affects a second, which affects a third.

A collider is a variable in the middle that has arrows only pointed into it from two other variables, and no arrows leaving. A fork has a variable in the middle with arrows leading only to two other variables. A chain is a linear path from one variable, into the central variable, and then from the central variable to the third variable.

Think about about these in an ecological context. With a collider, there are two potential causes for something to happen, so we have to consider multiple possibilities. With fork, two variables can appear correlated even when they don't actually affect each other - they simply share a common cause. With a chain, we have both direct and indirect causes for some phenomenon.

Now consider this graph:

Rain -> Driveway <- Sprinkler; Rain -> Sprinkler.

Once again, the rain and the sprinkler can make my driveway wet. However, now the rain affects the sprinkler. How? Because I am less likely to turn on the sprinkler if it rains. Now the rain and sprinkler are no longer independent; the behavior of the sprinkler depends on the rain.

Within this graph, we have one collider, one fork, and one chain. Identify them now.

Draw and explain a causal graph or DAG of something in your life. Remember, it needs to be directed (with causes) and acyclic (no loops or cycles).

1.8. Emergence

In ecology, everything we study is made up of smaller things; they emerge from the combination and interaction of smaller often faster things. Ecological communities are made of interacting populations, and of course those populations are made of interacting individuals. All things we study emerge from the interaction of simpler things.

The same is true for the rest of biology, chemistry, geology, and almost every other discipline. The only features of our world that are so fundamental that they are not made of anything else is the quantum wave function of the Universe. Everything else emerges at higher levels of organization.

Within biology, we are fond of describing levels of organization, from components to the whole, like this (cf. Scheiner 2010):

Biosphere <-> landscapes <-> ecosystems <-> metacommunities <-> communities populations interacting individuals, individuals, organ systems, organs, tissues, cells, organelles, molecules, atoms.

At each level, there are both physical and biological elements. For instance, blood is a tissue where the cells are floating in an aqueous medium. Take away the medium, and you no longer have blood. An ecosystem is made of organisms and the elements we track from pool to pool.

An entity at each level that has properties that we can study. I can study the dynamics of a population of coneflowers without studying the organs, tissues, and cells that make up an individual coneflower. I can study nutrient cycling from a watershed to a lake without studying individual fish or bacterial cells that live in that system. Each level of organization has properties that emerge — emergent properties — that show persistence and respond to causal forces.

1.8.1. Interactions of emergent properties

To understand why an entity behaves as it does, we often study its pieces at some level or levels below. To understanding the cause of a blood disease, we are likely to study sickle cell anemia, we might study just the red blood cells and the genetic basis form that sickle shape. To understand why one

lake is eutrophic and filled with algae while another is not, we are likely to study the one of two nutrients in that lake rather than simply study a group of lakes. To find out why, we typically examine subsystems to get to the cause.

Sickle cell anemia causation: Genes -> cell shape -> tissue -> individual

Lake algal blooms: phosphorus -> algae cells -> algal population -> lake condition

In contrast, we sometimes find a cause occurs at a hierarchical level above our entity of interest. Perhaps our skin problem is sunburn, caused by the behavior of the entire individual. Perhaps our desert shrubland is overtaken with Eurasian native cheatgrass (Bromis tectorum) because this species that evolved in a different biogeographic region in a different landscape was brought in by people, and it has left behind its usual controls.

Sunburn: Clear sky -> skin damage

Cheatgrass invasion: Biogeography + Human-mediated dispersal -> cheatgrass invasion

By now you may be thinking that emergent properties interact across many levels. Let's use a causal graph (DAG) to make those connections.

Lake algal blooms: Landscape/watershed surrounding the lake -> human behavior -> phosphorus on land -> phosphorus in water -> algal cells -> algae population -> lake condition

Cheatgrass invasion: Biogeography + Human-mediated dispersal -> cheatgrass invasion <- community dynamics <- cheatgrass population <- individual (cheatgrass traits); community dynamics <- populations of other species <- traits of those individuals

1.9. Review

As you work through the content in this book, keep in mind these fundamental processes we describe very briefly above: Observation, Learning (Bayesian updating), Logic, Mathematics, Causal inference, Emergence

Test yourself. What can you remember about each of these topics?

1.10. Cross-cutting concepts

1.10.1. Scale

1.10.2. Hierarchy

1.10.3. Emergence and complexity

Reductionism vs. compression.

2. Computer setup and first R script

Before you start

- **Assumptions about background and guidance:** no experience with R, RStudio, or programming — only a laptop or desktop computer on which you can install software. If you get stuck, see Getting help at the end of this chapter.

2.1. Goals

In this chapter, you will,

- Learn about your computer and how to organize its files.
- Install R, an open source programming language.
- Begin to use R effectively.

2.2. Deliverables

For this assignment, upload to the LMS three files that you created:

1. `myPlot.png`, from your `figs` folder,
2. `myDataframe.csv`, from your `output` folder,
3. the R script, from the top level of `Rwork`, into which you wrote all your code.
    a. This file must be written in R/RStudio, and the name must end with the file name extension, ".R", such as "Stevens_script_1.R",
    b. comments, including at least a title, the date, your name, and the path to your Rwork folder.

If you cannot find one of these files, that is useful information rather than a disaster: it means the file went somewhere other than where you thought, which is exactly the problem this folder structure exists to prevent. Check the top level of `Rwork` first, since that is where a file lands when you forget the `figs/` or `output/` prefix.

Congratulations. *C'est fini*!

2.3. Background

In this project, you will begin to use tools for data exploration and analysis. This tools and your computer are really important for helping you make sense of our really complicated world. Ecology is way more complicated than physics or engineering, and the data are noisy and often unpredictable. We need to use computers to make sense of it.

We will use open source and free software to help us understand ecological systems. You will download and install R, known formally as the R programming language and environment. You will also install RStudio, an integrated development environment (IDE), designed to be used with R. The R interface looks a little different on Windows, Macs, and Linux computers. RStudio will allow us all share an identical appearance and user experience. You will use R by opening RStudio.

In this class, you will use the English language to write about ecology, and the R language to explore and analyze ecological data.

2.3.1. Understanding you computer

First, we need to make sure we know some basics about computers.

2.3.2. Folders, directories, and directory structure

On your computer,

- a folder is a place to put files and other folders,
- "a directory is a file system cataloging structure that contains references to other computer files, and possibly other directories." (Wikipedia)
- A folder is a part of your computer's directory system.

If you have not already done so, **you need to organize your computer.**

Do these:

- Make different folders (or directories) for different classes that you are taking.
- Make different folders for different kinds of content (e.g., data, articles, your writing).

In addition,

- do *not* use the Downloads folder to store anything important, and
- do *not* mix personal content and course content.

Watch one of these to help you understand how your computer is organized. This will save you hours of time and frustration throughout your life.

- For Windows users (PCs)
- For Mac users

- If you are a Linux user, I am betting that you already know this stuff.

2.3.3. The location of a folder or file: The Path

A "path" is the complete location and name of a file, starting from the hard drive, down to the file or folder of interest. Here are a two examples:

- (Windows 11) `C:\Users\Stevenmh\Documents/BIO209W`
- (Mac) `/Users/stevenmh/courses/209W/`

Note they both start big and get more narrow. On Windows, the directory for the harddrive is typically "C:" and on a Mac it is just "/".

Here are ways to find those paths.

2.3.3.1. File paths in Windows 11 operating system

Open File Explorer, navigate to the file, and click the three-dot menu (...) in the top toolbar to select Copy path. Alternatively, right-click the file and select "Copy as path" (or "Show more options" then "Copy as path") to paste the full location.

Here are the fastest methods to find a file path on Windows 11:

- Copy as Path (Fastest): Right-click the file in File Explorer and select Copy as path. If you don't see it, select "Show more options" first.
- Top Menu Option: Click the file, then click the three dots (...) in the top menu and select Copy path.
- Address Bar Method: Navigate to the file in File Explorer, click the blank space in the address bar at the top, and it will highlight the full path, which you can copy with Ctrl+C.
- Properties Menu: Right-click the file, select Properties, and look at the "Location" field to see the folder path.

2.3.3.2. File paths on Mac Tahoe operating system

To find a file's path on a Mac, select the file in Finder and press Option-Command-C to copy the path directly. Alternatively, right-click (or Control-click) the file and select "Copy" while holding the Option key to change it to "Copy as Pathname".

Other Methods to Find File Paths:

- Path Bar: In Finder, select View > Show Path Bar to see the file location at the bottom of the window.
- Get Info: Select the file and press Command-I, then look under "Where".
- In the Terminal app: Drag and drop the file into the Terminal window to display its full path.

2.3.4. File types and file name extensions

Here we describe three file types (.R, .csv, and .txt) that you are probably not familiar with. First, we need some background.

Different **file types** allow you to do different sorts of things, and different types are labelled with different **file extensions**. File extensions are little tags at the end of file names that identify the *type* of file it is. A file extension is the little label at the end of a file name that comes after a dot.

Common file extensions include .jpg or .jpeg for many image files, .docx for Microsoft Word documents, and .pages for Apple Pages documents.

Think of a file extension as a label on a container that tells both you and your computer what's inside. Just like you'd label a jar in your fridge as "pickles" or "jam" so you know what to expect, file extensions (those letters after the dot in a filename) tell your computer what kind of information is in that file and which program should open it.

2.3.4.1. Three important file types

- A **.R** file contains your code—the commands you write to analyze data and make graphs. Save your work as "my_Chapter_1.R" so R knows it's code, not just notes.
- A **.csv** file (comma-separated values) stores your data in rows and columns. This is the standard format for datasets because almost every program—-Excel, Numbers, R, Google Sheets, even Word-—can read it. When you have data, such as measurements, it's probably in a .csv file.
- A **.txt** file is plain text with no formatting—useful for notes, project descriptions, or simple lists. Almost any application can open these types of files.

Why it matters: Without the right extension, your computer won't know what to do with a file. If you save code without .R, R and R RStudio won't recognize it properly. If data aren't marked as .csv, R might not read it correctly.

One essential tip: Make extensions visible on your computer. On Mac, go to Finder preferences and check "show all filename extensions." On Windows, uncheck "hide extensions for known file types" in folder options. You'll always know what you're working with.

You will need to be able to see file extensions on your computer.

2.3.4.2. Windows

On Windows 10, you need to make changes to see file extensions by revealing "hidden items".

Windows 10:

1. Open File Explorer; if you do not have an icon for this in the task bar; click Start, click Windows System, and then File Explorer.

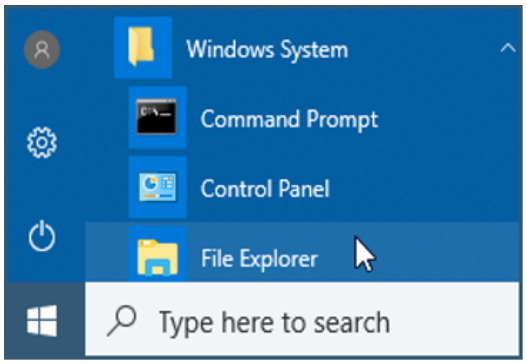


Figure 2.1.: Finding File Explorer on Windows.

2. Click the View tab in File Explorer

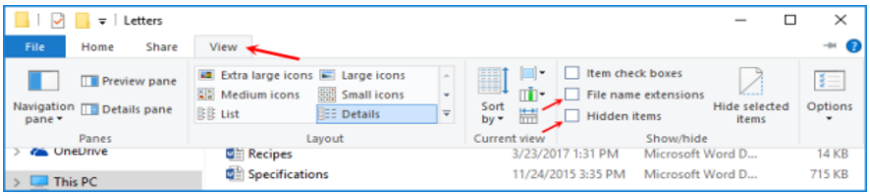


Figure 2.2.: View tab in File Explorer

3. Click the box next to File name extensions to see file extensions.
4. Click the box next to Hidden items to see hidden files.

You should now be able to see file name extensions (e.g., .docx, .R) and hidden files.

Windows 11 has a similar, and slightly less complicated layout. In File Explorer, select "Show" and then "File extensions".

2.3.4.3. *Macs*

To show all file extensions in macOS (including the latest Tahoe version), open Finder, select Finder > Settings (or Preferences, Command-,) in the top menu bar, click the Advanced tab, and check the box for "Show all filename extensions".

2.4. *Working with CSV files on Windows and Apple computers*

It is important that you keep CSV files unchanged. These are our data, and *data are sacrosanct*.

CSV files are easily opened in Microsoft Excel and Apple Numbers, but these applications also encourage you, or force you, to change their format. Below, we explain how to fix that.

Microsoft Excel usually recognizes your CSV file and opens it without any problem. At other times, it will ask you to confirm what kind of file it is, and you will need to specify that values are "delimited" or separated by commas.

When you close a .csv file in Excel, you will be prompted to save it in a different format, as a .xlsx file. Do not do that. Save it as a .csv file.

Apple Numbers will recognize and open your .csv file, similar to Excel. Unfortunately, it always automatically converts it to a different format, .numbers. That means that you have to always **export** to a .csv file type.

When you want to close your file in Numbers - pause! *You must export your file* from Numbers, not merely save it. Do not bother to save it. Just export it. Use the File menu, specifically, File > Export To > CSV, and use whatever name you need to.

2.5. Installing R

Armed with a little knowledge of your computer, you now install R.

2.5.1. What and Why

We will use the Open Source R programming language and environment for doing science, including,

- exploring data,
- testing ideas, and
- modeling dynamical systems like populations of rare and endangered species, disease dynamics, and ecosystems.

You need to download the application to get started.

- R is free.
- R runs on all operating systems.
- R is powerful.

2.5.2. How-to videos (optional)

The following videos might be useful, but they are not required! The next sections describe (in words) how to install R and RStudio.

Optional videos:

1. Installing R and RStudio
2. Set up your computer and start to use R via RStudio
3. Saving figures made with ggplot()
4. Introduction to ggplot2 (20 min)

2.5.3. How-to instructions

1. Find out what version of operating system (OS) your computer is using (e.g., Windows 10 or 11, or Linux Ubuntu, or Mac Sonoma, or other recent editions). This will help determine which version of R to use. If you are using a Mac, this also requires that you determine which version of processor or chip your computer uses (Intel processor vs. Apple M1 chip or higher). Check the processor or chip under "About this Mac" in the *Apple icon menu* in the upper left corner of your screen.
2. Google R (just the letter "R") to navigate to the *R Project for Statistical Computing*.
3. Find the link in the left Task Bar for *CRAN* (Comprehensive R Archive Network).
4. Find and navigate to a CRAN site near you (e.g., United States or Midwest US or Ohio). These are called *mirrors* because they are copies of each other.
5. Download and install the latest release of R for your operating system (Mac, Windows, or Linux).
6. As of January 2026, it was version 4.5.2. It is a single file to download. You should use whatever the latest version is.
7. In January 2026, it was worked like this:

    a. Mac - select R-4.2.2.arm64.pkg (for the Apple M1, M2, etc. chip) or R-4.5.2-x86_.pkg (for the older Intel chip)
    b. Windows - select the "base" package, then "Download R-4.5.2 for Windows".
    c. Linux - select your flavor, and follow instructions.

2.5.4. Where

R (and RStudio) will be installed along with all your other applications.

2.5.5. Installing RStudio

RStudio is an integrated development environment (IDE), designed to be used with R. Most importantly for us, it provides a graphical user interface (GUI) that appears identical on Windows, Mac, and Linux computers.

1. Navigate online to the RStudio website.
2. Under Products find RStudio IDE.
3. Select and download the Open Source Edition (RStudio Desktop). You want the desktop version (not the server version).
4. Follow instructions to install on your computer.

2.5.6. R ≠ RStudio

RStudio is an interface to R. When we use R in this class, we will use RStudio as our interface, but you need to realize that we are primarily *using R*. We don't need RStudio to do the work, but it makes my job as instructor easier, and makes it a little easier for you as well. So, *when you start to use R, just open RStudio, and it will load R for you*.

As we said at the beginning, you will use the English language to write about ecology, and the R language to explore and analyze ecological data.

2.6. Set up your folders for this class

1. Set up a folder for this class. Call it "BIO209W". Do not name it "BIO_209W" or "Bio209W" or "ecology". Just "BIO209W". It will help us. I don't really care where you put this folder, but you should have a place for all of your classes, and you could put it there. Don't put it in "Downloads".
2. Next, set up a **subfolder** inside BIO209W called "Rwork". Do not call it "RWORK" or "R_work" or anything else - just "Rwork". It will help us. A lot. `Rwork` will be your **working directory**, which is where R looks automatically for data and where it automatically puts output.
3. Finally, set up three subfolders *inside* Rwork, called "data", "figs", and "output". All lower case, no spaces, no capital letters. Your finished structure should look exactly like this:

```
BIO209W/
  Rwork/            <- your working directory and RStudio project
      Rwork.Rproj   <- this appears after you create the project,
    ↪  below
      data/         <- files you download or are given; you read
    ↪  these IN
      figs/         <- figures that R draws and saves
      output/       <- everything else R writes OUT (.csv files,
    ↪  mostly)
      my_script.R   <- your scripts live here, at the top of Rwork
```

The rule for deciding where a file goes is short:

- You didn't make it, you were given it, and you must not change it → `data/`
- R made it and it is a picture → `figs/`
- R made it and it is not a picture → `output/`

Yes, figures are a kind of output too. `figs` gets its own folder because you will make far more figures than anything else, and because a folder containing nothing but pictures is a genuinely useful thing to be able to scroll through.

2.6.1. Why we all use the same structure

You could organize `Rwork` any way you liked, and R would not care. There are three reasons we all use the same arrangement anyway.

Everyone's code becomes portable. If your data are in `data/` and mine are in `data/`, then a line of code I hand you, such as `read.csv("data/temps.csv")`, runs on your computer without any editing, and a line you send me runs on mine. That short, relative path is what makes code shareable. The alternative is a path like `/Users/stevenmh/courses/209W/Rwork/temps.csv`, which works on exactly one computer in the world - mine. **Never put a path like that in a script.**

It keeps data separate from output. Recall that *data are sacrosanct*. Files in `data/` were given to you and must not be modified. Files in `figs/` and `output/` are things R made, and R can always make them again by re-running your script. If you delete everything in `figs/` by accident, you have lost nothing but a few seconds. If you overwrite something in `data/`, you may have lost a great deal. Separate folders make that distinction visible every time you look at your project.

It scales. Right now you have one script and three files. By the end of the semester you will have a dozen scripts, a pile of .csv files, and several dozen .png figures. A single folder holding all sixty of them is a junk drawer, and you will spend real time hunting through it.

There are costs, and you should know them.

- R does **not** create folders for you. If you write `ggsave("figs/myPlot.png")` and there is no `figs` folder, you get an error rather than a folder. Make all three folders yourself, in Finder or File Explorer, right now, before you need them.
- A misspelled folder is a broken path. `Figs/`, `figures/`, and `figs /` are all different from `figs/`. Macs and Windows machines are often forgiving about capitalization; R, and every server you will ever use, is not.
- Every command that writes a file gets a little longer: `ggsave("figs/myPlot.png")` instead of just `ggsave("myPlot.png")`. That is a small price for always knowing where the file went.

2.6.2. Where do scripts go?

Notice that scripts are **not** in a subfolder. They sit at the top level of `Rwork`, right next to `Rwork.Rproj`. Many professional projects do keep their scripts in a `scripts/` folder, and you will meet that convention if you go on to a thesis or a job. We are not doing it here, on purpose, because of the way R actually behaves.

Your working directory is the project folder, not the folder your script is sitting in. When you open the `Rwork` project, R sets its working directory

to `Rwork` and leaves it there, no matter which script you happen to have open. If you moved a script down into `Rwork/scripts/`, R would *still* be looking in `Rwork`. The path `data/temps.csv` would still be correct, and the path `../data/temps.csv` - which looks as though it ought to work, since the data really are one level up from the script - would fail.

That is a sensible design, and it is also a reliable source of confusion. Keeping your scripts at the top of `Rwork` makes the simplest possible statement true: *the folder my script is in is the folder R is looking in.* Every path you write starts from that one place. Once paths are second nature, moving scripts into their own subfolder costs you nothing, and you should feel free to do it in your own later projects.

2.7. Working in R

To work in R in this class, we will open RStudio. It will open R for you.

2.7.1. Set your "working directory"

In this section, we learn what the "working directory" is and how to set it.

Your **working directory** is where R looks to find stuff by default. R can look anywhere you tell it to, but it looks automatically in your "working directory".

In this class, you should always use "Rwork" as your working directory.

I can find out where my working directory is by using `getwd()` to **get** my working directory:

```
getwd()
```

```
[1] "/Users/stevenmh/projects/intro_to_ecology"
```

This tells me what R is using as my working directory right now. My current working directory will differ from yours.

You should run this code, and see what you get. What do you get when you run this code (`getwd()`)?

Next, use the "Session" in pulldown menu in RStudio to `Set Working Directory` and then `Choose your working directory` and always choose `Rwork` (which should be inside BIO209W).

2.7.2. *Create a* project

RStudio allows you to organize your work in *projects*. Here we create a new project that we will call `Rwork`.

This requires that you already have a folder called "BIO209W" and a subfolder in that called "Rwork".

1. In RStudio, use the File menu to select "New Project", and select "Existing Directory".
2. In that dialogue, navigate to "Rwork", and select "Open."
3. Select "Create Project".

RStudio will now shift the working directory to your Rwork directory. You can see evidence of this: in the upper right pane which should have Rwork identified in the upper right corner, or in the lower right pane in the Files tab.

From now on in this course, make sure that you are using this project. You can open R by clicking the "Rwork.proj" file in your Rwork directory, or opening RStudio, and selecting the Rwork project in the upper right corner.

2.7.3. *Scripts: Start and save a script*

Goal: Write a script in R and learn several important programming operations.

After you have prepared your computer as above, you are ready to begin exploring and modeling data. We do that using *scripts*.

A script is a **plain text** file that contains the code and and your own comments that you use to address your questions.

The following steps will let you accomplish the goal of this section. The video above showed me opening R with RStudio, and starting a new script.

Writing comments Anything following a hashtag (#) is a comment and R ignores it. You and I will use lots of comments to ask questions of ourselves, and to describe what we are trying to do.

- Use the File menu in RStudio to start a New File, specifically, an R Script. This will open a file in the upper left of RStudio.
- write the following in the new script, and include the hashtags.

```
## [title] e.g., My first R script in BIO 209W
## [Date]
## [your name]
## [the full path to your Rwork directory]
```

Make sure to start each comment with at least one hashtag, #, and remove the square brackets seen above.

ALL OF THE WORK YOU DO MUST BE IN A SCRIPT.

2.7.4. Entering and running code

To enter code, place your cursor on any line in the script, and type, like this.

```
-1:5
```

To run code, simply place you cursor anywhere on a line of code and hit Control-Enter or Control-Return or Command-return (Mac).

Try it with this:

```
-1:5
```

```
[1] -1  0  1  2  3  4  5
```

This will pop up in the Console, which is the lower left pane in RStudio.

What did this create? Do you get what I got above?

2.7.5. Assignment operator

W?hen we create something, we typically want to *assign* it to a labeled object. In this class, we will use an arrow to indicate that we are putting a creation into an object. We create the arrow on our keyboard using shift-comma and a dash ('less than' and a dash)

```
a <- -1:5
```

This puts the series of integers from -1 to 5 into the object `a`.

2.7.6. Examining objects

If we have an object, like `a`, how can we figure out what it is? There are several ways of doing that. The simplest is to "print" it out. This doesn't mean that we print it on paper. Rather we are printing it to the console, like this:

```
# show or 'print' a
a
```

```
[1] -1  0  1  2  3  4  5
```

Another way is to look at its "structure":

```
str(a)
```

```
 int [1:7] -1 0 1 2 3 4 5
```

This tells us that this is a vector of integers that is 7 elements long.

Another way is to look in the RStudio *Environment pane* in the upper right hand side of RStudio.

2.7.7. Things you'll find out about R

If you are not used to writing code (and nobody is born that way), here are some thing to keep in mind about R and about coding in general.

1. Computers do **exactly** what you tell them. This is both frustrating and also satisfying.
2. R does not know anything or do anything until you **run** the code. Typing or pasting code into a script is just the first step; you also need to **run** the code, with Control-return (Windows or Mac) or Command-return (Mac).
3. A corollary of no. 5 is that you usually need to run lines of code in sequence. For instance, R won't be able to run `str(a)` unless you previously defined `a`, such as `a <- -1:5` (this is because R is an *interpreted* language - R interprets it one line at a time).
4. Case matters in R. R interprets `a` as something completely different than `A`.
5. Spaces *between* things usually don't matter in R. R interprets `2+3` the same as `2 + 3`.
6. Space *within* things usually breaks them, just the way "thin" "gs" is not the same as "things". For example, the pseudo-arrow we use in `f <- 9` is a thing. The compound symbol `<-` means something different than `< -`.
    i. `<-` is the *assignment operator* and assigns values to objects.
    ii. `< -` means "less than, minus", where `2 < -3` tests whether 2 is less than minus 3 (it isn't).

Try it.

```
f <- 9
f
```

```
[1] 9
```

Now `f` has the value of 9.

Try this:

```
2 < -3
```

```
[1] FALSE
```

That tested whether 2 is less than -3.

Moving on....

Now let's create a set of 7 uniformly distributed random numbers that are between 0 and 1.

```
b <- runif(7, min=0, max=1)
b
```

```
[1] 0.1124833 0.7158261 0.7586930 0.1674902 0.7079866 0.7887181
↪  0.1317637
```

These will differ every time you do this.

Now we multiply each element in `a` times the corresponding elements in `b`.

```
# multiply a and b
ab <- a * b
## Show a, b, and ab
a
```

```
[1] -1  0  1  2  3  4  5
```

```
b
```

```
[1] 0.1124833 0.7158261 0.7586930 0.1674902 0.7079866 0.7887181
↪  0.1317637
```

```
ab
```

```
[1] -0.1124833  0.0000000  0.7586930  0.3349804  2.1239598
↪  3.1548726  0.6588186
```

You can confirm at a glance that the first and second elements in `a` times the first and second elements in `b` equal the first and second elements in `ab`.

2.7.8. Combining vectors into a data frame

We will use lots of **data frames** in this course. Think of a data frame as a kind of spreadsheet of data, in which every column has the same number of entries and every row is an observation comprising all of the variables across all the columns.

Here is one way to create a data frame. We assign variables using "=".

```
d <- data.frame(a=a, b=b, a_times_b = ab)
d
```

```
   a         b  a_times_b
1 -1 0.1124833 -0.1124833
2  0 0.7158261  0.0000000
3  1 0.7586930  0.7586930
4  2 0.1674902  0.3349804
5  3 0.7079866  2.1239598
6  4 0.7887181  3.1548726
7  5 0.1317637  0.6588186
```

You could examine the structure of this data frame with `str()`.

```
str(d)
```

```
'data.frame':   7 obs. of  3 variables:
 $ a        : int  -1 0 1 2 3 4 5
 $ b        : num  0.112 0.716 0.759 0.167 0.708 ...
 $ a_times_b: num  -0.112 0 0.759 0.335 2.124 ...
```

You will also notice now that you can see all of our objects in the Environment pane, in the upper right of RStudio.

You can export data frames from R as well, and programmers call that "writing" a file. The command below will save our data frame as a .CSV file that you could open in a spreadsheet application. Note the `output/` at the front of the file name: that is us telling R which folder to put it in.

```
# export or write a file
write.csv(x=d, file="output/myDataframe.csv", row.names=FALSE)
# including row.names=FALSE prevents R from adding row names.
```

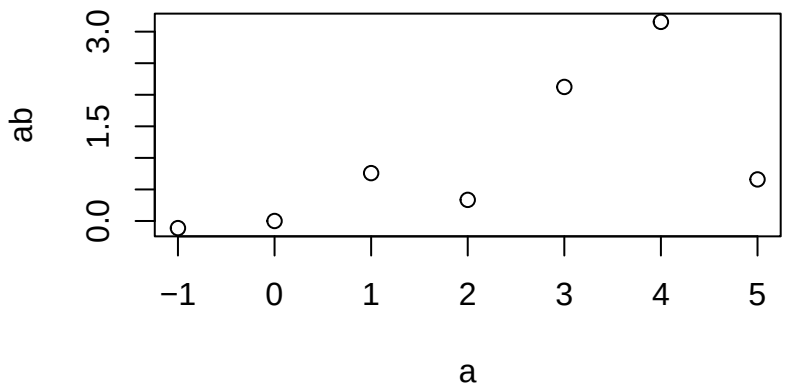
This should cause the data frame `d` to get saved into the `output` folder inside your working directory. Go look in that folder. Did it work for you?

If instead you got an error that says something like "cannot open file 'output/myDataframe.csv': No such file or directory", that almost always means the `output` folder does not exist yet, or is spelled differently than you think. Go make it, or fix the spelling, and run the line again.

2.7.9. Plotting data

Now let's plot some data, including both points and lines connecting them.

```
# type='p' is for points only
plot(a, ab, type='p')
```



R is well known for its graphics capabilities, and we will only just scratch the surface in this course.

2.8. Install an R package

R is open source, and its strength derives in part from the thousands of scientists creating extensions or add-on packages to do particular tasks. By itself, R can do amazing things. With thousands of *contributed packages* (as they are known), R's capabilities have exploded.

You can add a package in a few ways, including.

- Use the Tools menu, select "install packages" and type the name of the package you want.
- use a function, "install.packages("[packagename]"), with quotes but without brackets.

In this class, you will need to install several packages. Let's try installing "ggplot2".

```
install.packages("ggplot2")
```

This will install ggplot2, and it will also install anything upon which ggplot2 depends. It will install all of its dependencies.

Once you have installed a package, it is on your computer for good. You do not have to install it again, although you may want to update it once in a while to stay current.

Now let's use the plotting functions in the `ggplot2` package.

2.9. Load and use an R package

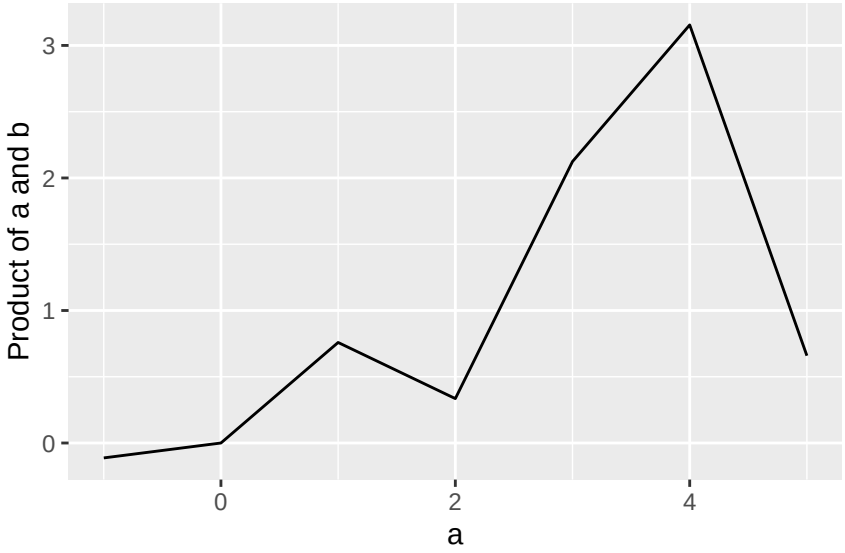
To use a contributed package, we need to *load* it. This simply means waking it up, and putting its code in R's working memeory. We do that with `library()`.

```
library(ggplot2)
```

Here we use `ggplot()` and `geom_smooth()` to plot our data, and add a fitted curve.

When we use ggplot(), we tell it the name of our data frame and the variables we want to use go inside `aes()`, which stands for "aesthetics". We also begin to add bells and whistles.

```
## plot data and fit a curve
ggplot(data=d, aes(x=a, y=a_times_b)) +
  geom_line() +
  labs(x="a", y="Product of a and b")
```



Note that with ggplot, we use the plus sign (+) to add elements to our graph.

2.9.1. Saving your graphs

We can save this figure as well, in a variety of formats and sizes. If we gave `ggsave()` nothing but a file name, R would drop the figure straight into our working directory. Instead we put `figs/` at the front of each name, which sends every figure into the `figs` folder where it belongs.

```
ggsave("figs/myPlot.png")
```

```
Saving 4.5 x 3 in image
```

```
ggsave("figs/myPlot.jpg", width=7, height = 3)
ggsave("figs/myPlot.pdf", width=3, height = 5)
```

Look in your `figs` folder and confirm that all three files are there. From here on, **every figure you save in this course goes in `figs/`, and every data file you write goes in `output/`.** I will do the same in every chapter that follows, so the code you read is the code you run.

2.10. Getting help

R has a bit of a learning curve. What should you do if you get stuck? It all depends what sort of problem it is.

1. First, ask a classmate.
2. Use the Help tab in the lower right pane in RStudio. (Similarly, type `?mean` in the Console to get help on the function `mean()`.)
3. Ask another classmate.

4. Ask me.
5. Search online.
6. Did I mention you could ask me?

3. *Environmental Justice: How income relates to environment.*

Before you start

- **Assumptions about background and guidance:** that you can open your `Rwork` project in RStudio and run a short script; if not, start with Computer setup. You will also need access to the class Google Sheet, which your instructor will share. No statistics are assumed beyond reading a scatterplot. If the R code gives you trouble, see Getting help.

3.1. Goals

In this chapter, you will,

- Test Wilson's *biophilia* hypothesis — that humans prefer greater connection to nature — using real socioeconomic and tree-cover data.
- Practice web-based, crowdsourced data collection: contribute your own city or location to a shared class Google Sheet using the Tree Equity Score and i-Tree/OurTrees platforms.
- Graph and interpret the relationship between wealth (median income, percent poverty) and tree cover.
- Consider what else, beyond wealth, might explain–or obscure–a relationship between people and tree cover.

3.2. Deliverables

Results and discussion

- In one document, put two figures side-by-side:
    i. your bargraph of % tree cover vs. % people in poverty (copied from the Tree Equity site) side-by-side, and
    ii. your scatterplot of tree cover vs. median income that you made using everyone's data, based on the iTree/OurTrees website that you shared in the Google spreadsheet.
    iii. include a figure legend that states what the graphs are in a way that is intelligible to a naive reader.

3. Environmental Justice: How income relates to environment.

- Provide a brief explanation of your findings, including
  i. the results described in a factual manner, without interpretation,
  ii. an explanation of whether the results are consistent with the biophilia hypothesis, and
  iii. a new question that emerges due to the inconsistencies between the data and the hypothesis; explain why this question is a logical extension of your current results and interpration.

3.3. Background

The types of places that organisms live–their habitats–depends on a variety factors. *Habitat selection*, the selection of particular habitats, results from real and perceived tradeoffs between the benefits from foraging and reproducing versus perceived risks of mortality (Stephens and Krebs 1986; Morin 1999).

Stephens, D W, and J R Krebs. 1986. *Foraging Theory*. Edited by J. R. Krebs and T. H. Clutton-Brock. Monographs in Behavior and Ecology. Princeton University Press.

Morin, P J. 1999. *Community Ecology*. Blackwell Science, Inc.

In addition to selecting their habitats, organisms also construct elements of their own habitats, including nests and burrows, as well as broader habitat alteration. This is known as *niche construction* (Odling-Smee and Erwin 2013). Beavers build dens and dams, birds and wasps build nests, and bacteria, ants, and humans all construct elaborate multi-layered cities of thousands or millions of individuals. At the same time, organisms often damage the very habitats they create, through over-consumption of resources and over-production of waste products.

Odling-Smee, J, and DH Erwin. 2013. "Niche Construction Theory: A Practical Guide for Ecologists." *The Quarterly Review of Biology* 88 (1): 3–28. https://www.jstor.org/stable/10.1086/669266.

We know a great deal about how our own species (*H. sapiens*) construct elements of their own habitats. We also know a lot about how we migrate and disperse from place to place. Nonetheless, we have a lot to learn about our own species, and, in particular, about which factors determine the our preferred habitats and determine the changes we make to those habitats.

Economic wealth seems to influence many of choices by *H sapiens*, including our preferred habitats. Ed Wilson (1984), a famous evolutionary biologist, have proposed the *Biophilia hypothesis*, that humans love nature. Biomedical researchers have shown that we are happier and heal from injury and illness faster when we have access to plants and to the outdoors (Franklin 2012). If wealth generally gives us greater access to what we want, will wealthier people more often put themselves in closer connection to nature?

Wilson, Edward O. 1984. *Biophiia*. : Harvard University Press.

Franklin, Deborah. 2012. "Nature That Nutures." *Scientific American* 306 (3): 24–25.

Specific socioeconomic data about *H. sapiens* populations may be particularly useful understanding both the causes and consequences of relations between wealth and nature. Wealth can be measured in a variety of ways. Median income, and percent of the population below the poverty line are common ways to measure the average wealth of individuals or small family groups.

Specific data about our landscapes maybe be particularly useful in generating ecological understanding of *H. sapiens*. Tree cover is the two-dimensional coverage of tree canopies that you would perceive in an aerial photo in seasons when trees have their leaves. Tree cover is relatively easy to measure

from satellite imagery. Temperature, or the heat experienced by residents, can be a serious problem in urban areas because they create heat "islands". Urban heat islands are areas which tend to absorb more solar radiation than would a natural landscape and then re-radiate this as heat experienced by its human and non-human residents. During heat waves, residents in urban areas who are especially vulnerable can suffer severe heat exhaustion or death. Some urban areas tend to get much hotter than others, and it is often related to tree cover and green space generally.

Trees provide enormous ecosystem service benefits, including to help mitigate pollution via uptake of carbon dioxide and carbon monoxide (CO_2 and CO) and other greehouse gases. Trees can also help reduce pollutants that cause direct negative effects on human health, such as fine and ultrafine particulate matter ($PM_{2.5}$, $PM_{0.1}$).

In this research project, we will use freely available online data to address whether there is an association between wealth and the wild. Specifically, we will address whether, if we compare different geographic locations, will we see a relationship between the median income of an area, and the percent cover of trees? The data we collect will allow us to explore a variety of related ideas, but we will start with testing Wilson's *biophilia* hypothesis, which predicts that humans love nature. We will test my corollary of this, that humans who have the greatest degree of freedom of choice (the wealthy) will tend to live in areas with the greatest connection to Nature (a lot of trees).

View your mission here: https://youtu.be/BABM3EUo990

or here:

3.4. Methods

Work in pairs. Share introductions with someone and get to work.

We will collect data on tree cover (%), heat, and wealth using two different platforms:

1. Tree Equity Score website
2. OurTrees application in *i-Tree*

The first of these two platforms, Tree Equity Score, gives fairly detailed information on cities over 50,000 residents in size. It provides information related to environmental justice, including levels of poverty, tree cover, and heat. Our second data source is from *i-Tree.org* and it provides fairly detailed information about tree cover and the ecosystem services they provide. (Ecosystem services are features of ecosystems that benefit humans, such as storm water protection, and shade for buildings). i-Tree also provides median household income.

3. *Environmental Justice: How income relates to environment.*

3.4.1. Instructions

Tree Equity Score data

1. Go to our class Google Sheet for *Tree Equity Score cities*, and enter the name of a city (pop. > 50,000) that you will investigate. It must be different than what anyone else has selected. Add your name to your selected city.
2. Go to the Tree Equity Score website.
3. Search for your city.
4. Find the data for the city: click on the map *inside the area of your city*. Note that a map will typically display more than just your city. It will also display adjacent regions. For instance, the map for "Cincinnati, OH" displays Hamilton County and Covington, KY. Selecting an area will cause a data summary window to pop up on the left side. **Confirm** that when you click on an area of the map, it is **inside the city you intended**.
5. Selecting an area link to a **Municipal Report**. This report provides information for the city as a whole. **Confirm** that when you click on an area of the map, it is **inside the city you intended**. Select the link, Municipal Report.
6. Pick the data most relevant to our question: adjust the setting for the right-hand column graph to show *Tree Cover* (%) *vs. People in Poverty* %. You will use this column graph in addressing our overall question.
7. The Municipal Report provides a Tree Equity Score, urbanized area population estimate, percent seniors, and a percentage of people in poverty. Enter these into the Google Sheet. Add one Miami uniqueID per group to each observation (row).
8. Save the web page as a PDF, and name it with the city name and "TES" (e.g., "Cincinnati_TES.pdf"). If you would like assistance saving the PDF, read the instructions under the button "Share report ->". Make sure to close the instructions before you save the page. This file will be a record of your data.
9. Make sure that you have entered the Tree Equity score and poverty percentage in the Google Sheet, and that you saved to your own computer your column graph of tree cover vs. percent of people in poverty.

OurTrees app, in *i-Tree*

1. **OurTrees application** in *i-Tree*. Select a location in which you are interested. Start with any location name, and select "Search". The app will show you a location. If you would like to, you can change the location by simply clicking somewhere else on the map. You can choose either Street view or Satellite view (the data will be the same).
2. Enter the name of your selected location into our class a new spreadsheet, Google Sheet for **OurTrees**.
3. In the OurTrees app, select "Get Results! ->".
4. From the "Benefits" tab, collect data on tree canopy cover (%), annual uptake of CO_2 equivalents and 2.5 micron particulate matter ($PM_{2.5}$).

Enter these into the OurTrees Google Sheet.

5. From the "Community" tab, collect data on total population size, number of seniors (>64), percent minority, median annual income, and percent impoverished. Enter these into the OurTrees Google Sheet.
6. Repeat the above steps for *nine more* locations anywhere in the world. Your locations must be different than what anyone else has selected in the OurTrees spreadsheet. Add one Miami uniqueID per group to each observation (row).

3.4.2. Graphing data to assess our predictions

I argued that the Biophilia Hypothesis leads to the prediction that tree cover will correlate positively with wealth. To evaluate this prediction, we should make a graph in which wealth is on the x-axis (horizontal axis) and tree cover is y-axis (vertical axis).

A couple of minor points: (1) Our prediction also implies that we should see a negative relation between poverty and tree cover. (2) We typically assume that the variable on the y-axis is a response to the variable on the x-axis, that is, that the x variable is causing variation in the y variable.

Given our data are *continuous variables* rather than categories, we should start by making a *scatterplot* of median income (x) vs. tree cover (y).

As the semester progresses, you will find R is very useful. Below, I provide you with code for an R script to easily make this graph.

```
# if needed, install these packages
# install.packages("googlesheets4")
# install.packages("ggplot2")

# load these packages
library(googlesheets4)
library(ggplot2)

# Import data.
# This code will require that you have permission to access our
 ↪  G-Sheet.
# You may be asked to select various options and/or sign in to your
# Miami Google account.

# The function read_sheet() requires a Google Sheet URL in quotes,
 ↪  and
# allows you to skip lines that are not data,
# such as a table description.

d <-
 ↪  read_sheet("https://docs.google.com/spreadsheets/d/1uYCuTziJiLNvVOQ9Q65BxdWcNgDF6EnF8L1T3O_yATw/edit#gid=0",
               skip=2)
```

```
v Reading from "iTree_OurTrees".
```

3. *Environmental Justice: How income relates to environment.*

```
v Range '3:10000000'.
```

```
# Uncomment to show us the names of the variables
# names(d)

# make a scatterplot
plot1 <- ggplot(data=d, aes(x=Median_Income,
                    y=Tree_canopy_cover,
                    label=`City (only)`)) + geom_text() +
  labs(y="Tree cover (%)", x="Median income", title="No relation
   ↪  twixt tree cover and income?")

ggsave("figs/myCoverIncome_text.png", plot=plot1, width=10, height =
 ↪  10)

plot2 <- ggplot(data=d, aes(x=Median_Income,
                    y=Tree_canopy_cover,
                    label=`City (only)`)) +
  geom_point() +
  labs(y="Tree cover (%)", x="Median income", title="No relation
   ↪  twixt tree cover and income?")

# save the figure (dimensions in inches)
ggsave("figs/myCoverIncome_points.png", plot=plot2, width=7, height
 ↪  = 7)
```

Here is a subset of data I collected on the places where members of my family lives.

```
d10 <- d[1:10,]
myPlot <- ggplot(data=d10, aes(x=Median_Income,
                    y=Tree_canopy_cover,
                    label=`City (only)`)) + geom_text() +
  labs(y="Tree cover (%)", x="Median income", title="No relation
   ↪  twixt tree cover and income?")

myPlot
```

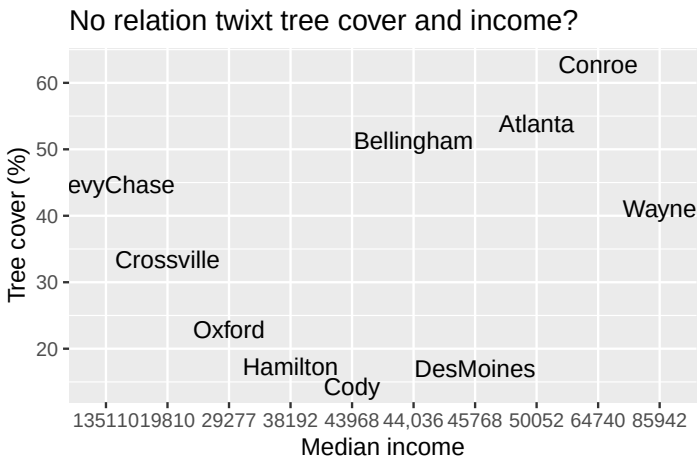


Figure 3.1.: The noisy but perhaps positive relation between percent of an location covered in trees and the median income of that location.

Refer to the scatterplot of my data (Figure 3.1). What do we find? Discuss the following questions with your colleagues, and propose and explain some possible answers.

1. Is there a relationship that is consistent with our hypothesis? Does tree cover increase with income?
2. Notice the cities that I selected. What else do you know about these locations, aside from income, that might influence tree cover?
3. What else might be going on that helps create this relation or might obscure a stronger relation?
4. After looking at these data, would you want to select locations in a different way?

4. *Physical Environment: A Fate of Earth*

...in which we reveal part of our ongoing climate reality.

Before you start

- **Assumptions about background and guidance:** your `Rwork` project set up as in Computer setup, and comfort running a script line by line. No statistics course is assumed: this chapter explains what a confidence interval is and walks through each piece of the forecasting (GAMM) code. If `ggplot()` is new to you, Displaying data with ggplot2 is a short reference.

4.1. Goals

In this chapter, you will

- Learn about the history of our perspective on global warming.
- Use R to display estimates of global temperature anomalies.
- Describe the current temperature trajectory.
- Compare and contrast changes in temperature between humans and Earth.
- **Transferable skill:** beyond the data-wrangling and `ggplot2` visualization skills you're building throughout this book (see the Preface), this chapter gives you hands-on exposure to time-series forecasting with a generalized additive mixed model, or GAMM (the `mgcv` package)–the same general approach used well beyond climate science, in any field that forecasts a trend over time (sales, web traffic, disease spread, sensor data). Running this model and correctly reading its forecast is real, useful experience; it is not the same as understanding the statistical theory behind it, and you shouldn't claim more than that–but you should understand what each piece of the code is doing, and this chapter now walks you through that. See the *Skills Crosswalk* appendix for how to describe this honestly on a résumé.
- Background reading: Hansen et al. (2012). Perception of climate change. *Proceedings of the National Academy of Sciences*, U.S.A.

4.2. Deliverables

Turn these into the relevant Canvas assignment before the due date.

1. First, put all your answers, including your graph with the confidence interval, into a .doc or .docx file and make fit on a single page. Turn in your answers and your graph to the relevant Canvas assignment Make sure that your name is on the file in the header!
2. Second, turn in your R script, named `global_temps_YourUniqueID.R` as a second file. This must NOT be a Word document. It must be a .R file.

4.3. Background

The year that you were born was much warmer than usual–and it won't ever be that *cool* again in your lifetime.

Earth is getting warmer due to human activities, it will push it well beyond what earth has experienced in hundreds of thousands of years. How and why Earth is warming has been known for many decades. In the early 1980's, even Exxon's scientists accurately predicted the global warming that has occurred because we failed to act to change our behavior.[1]

[1] Exxon suppressed these internal reports, and they became available only due to a lawsuit.

In this exercise, you'll graph data from the National Centers for Environmental Information, part of NOAA (US Dept of Commerce). These temperatures are anomalies from a long term mean. The reference they use is the period from 1951-1980.

Using the R script, you will construct a 95% confidence interval on expected mean surface anomalies. A "95% confidence interval of a mean" is an interval, if so constructed, that would tend to include the true mean about 95% of the time, given many, many repeated experiments conducted under identical conditions. It is based on frequentist statistical methods. It is actually pretty difficult to interpret, but it is a standard measure of uncertainty. You will construct such an interval for past and future global surface temperature anomalies.

Overthinking: *Frequentist statistical methods (those that you will all attempt to learn in intro stats class) give you a P-value, which is the probability of your data or more extreme data, given that your null hypothesis is true (i.e., your model is correct). That's weird. The data are real - they don't need a probability. A 95% confidence interval is an interval that we construct in a way where we would capture a "true" effect 95% of the time we repeat the same experiment under the same conditions. In contrast, Bayesian statistics would give us what we actually want: the probability that our model or hypothesis is correct, given our data. That is called conditional probability and is what a Bayesian analysis would give you. We might do that later in the course.*

4.4. Activity

In this activity, you will download and import data, and then make a forecast about future temperatures. That forecast will include an *expected* value and a measure of uncertainty. The *expected* value is the most likely value, but we also acknowledge that any time we estimate a most-likely value that there is always uncertainty.

Below, we set up or work environment, load data, work with data, and then save deliverables. This work flow will be natural after a while.

4.4.1. Set up

First, if you have not already done so, create a script file that you will call `global_anomalies_YourUniqueID.R`.

After starting a script, we always select the appropriate project and *working directory*. I do this by...

...glancing at the upper right corner:

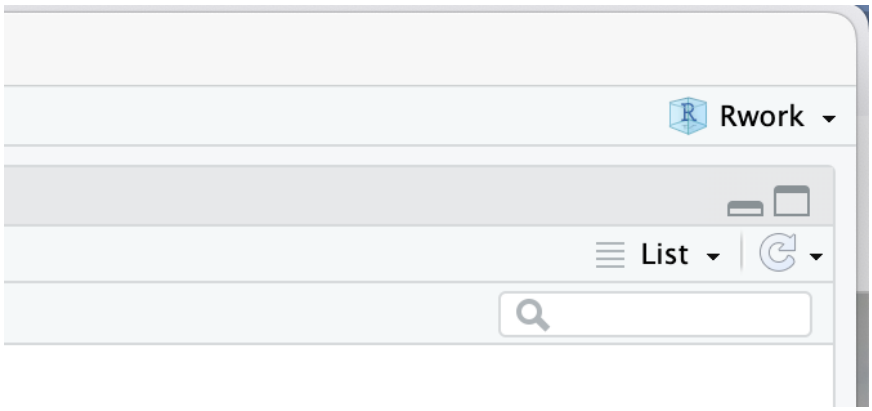


Figure 4.1.: RStudio shows which project you are using in the upper-right corner

...and by looking in the File tab in the lower right pane - that should be your working directory, which should be Rwork.

The next thing we do is **install** and then **load** packages that we know we will need. Remember, you only need to *install* a package onto your computer *once*. We **load** contributed packages in each session that we need them. In this course, you should always load `tidyverse`, at a minimum.

First we install.

```
## install tidyverse
## tidyverse is a"universe" or collection of related R packages
## that have a particular coding style that helps a lot.
install.packages("tidyverse", repos = "https://cloud.r-project.org")
```

Remember, we only have to do that *once*, and then it is sitting on our computer.

However, anytime we start a new session in R, we have to *load* these packages in order to use them. We do that here.

```
## load R packages
# this makes these packages available to R in this session.
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------
 ↪  tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------
 ↪  tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to
 ↪  force all conflicts to become errors
```

4.4.2. Import data

1. Go to our class data repository, and download `global_surface_anomalies_2025.csv`.
2. Make sure the file remains in the same simple format, "comma separated values" (.csv).
3. Put `global_surface_anomalies_2025.csv` in your data directory, `Rwork/data`.
4. Open `global_surface_anomalies_2025.csv` in a spreadsheet application. Don't save the file in the spreadsheet application. Do **not** save it in the spreadsheet application because it may try to change the format.
5. With the file open, learn what is on
    a. the first three lines
    b. the fourth line
    c. the remaining lines.
6. Do **not** save the file in the spreadsheet application. Just close and don't save it - it will still be there, unchanged.
7. Continue to make sure the file remains in the same simple format, "comma separated values" (.csv).

We import or "read in" the data so that R can work on it. Here we put it in an object called `temps`. We skip lines in the file

```
# read in data
temps <- read.csv(file = "data/global_surface_anomalies_2025.csv", #
 ↪  full file name
```

```
                skip=3, # skip these lines because they contain
                 ↪  metadata.
                header = TRUE # Tell R that the first row is
                 ↪  column namaes
                ) # last parenthesis
```

4.4.3. Finding out information about the temperature anomalies

Next we take a peek at the data with a summary.

```
summary(temps)
```

```
      year          deg_C
 Min.   :1880   Min.   :-0.49000
 1st Qu.:1916   1st Qu.:-0.20000
 Median :1952   Median :-0.03000
 Mean   :1952   Mean   : 0.08158
 3rd Qu.:1989   3rd Qu.: 0.31750
 Max.   :2025   Max.   : 1.28000
```

This shows us the minimum, maximum and mean of each numeric variable.

It also shows us the 1st, 2nd, and 3rd quartiles. The median is the 2nd quartile. Quartiles are three divisions in ranked data, such they divide the data into four equal parts, or quarters. The 1st quartile is the value that divides the lowest 25% of the values from the larger values. The 2nd quartile (i.e.) is the value that divides the lower 50% from the upper 50% of the observations. The 3rd quartile is the value that divides the highest 25% of the observations from the lower values.

4.4.4. Minimum and maximum

I'd like to know *when* the minimum and the maximum occurred. To find this out, I'll use a two step process: I will find the min and the max, and then I will use those to take a *subset* of the data frame where the anomaly is equal to either the minimum or equal to the maximum.

First, I find the minimum and the maximum.

```
## get the min and max
min.anomaly <- min(temps$deg_C)
max.anomaly <- max(temps$deg_C)
```

Now, I will use these to take a *subset* of the data frame where the anomaly is equal to either the minimum or equal to the maximum. The "OR" operator is the vertical line that programmers call a *pipe*. On your keyboard, it is above the return or enter key.

```
## use the min and the max
# note the two equal signs, a pipe, and then two more equals signs.
subset(temps, deg_C==min.anomaly | deg_C==max.anomaly )
```

```
    year deg_C
30  1909 -0.49
145 2024  1.28
```

These show us all of the entries in our data that the anomaly is equal to the minimum OR equal to the maximum.

4.4.5. Using a year

Here I find out the average temperature anomaly in the year I was born. I want YOU to find out the anomaly in the year *you were born.*

```
## Find the anomaly in the year you were born
subset(temps, year==1960) # use your year with two equal signs
```

```
   year deg_C
81 1960 -0.03
```

In my birth year, it was just a touch below the 1951-1980 average.

4.4.6. Plotting information

Here we are going to start visually examining the anomalies.

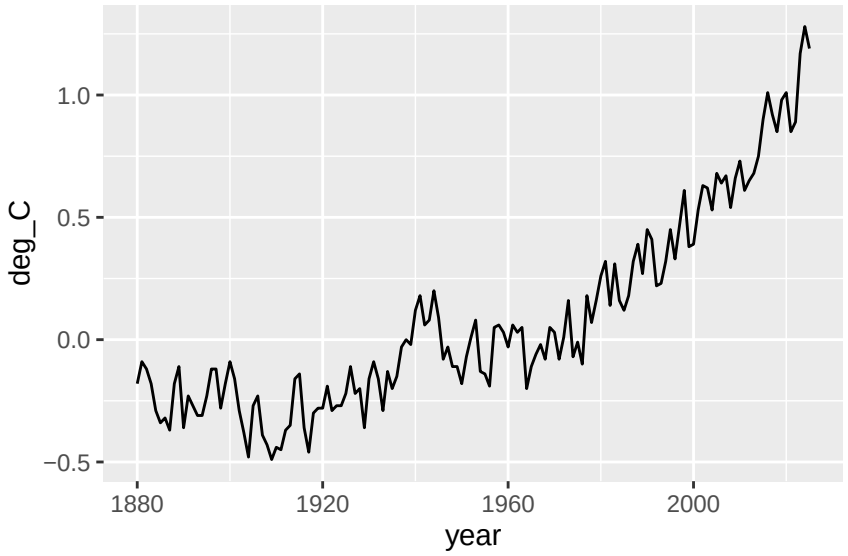
We will create a linegraph using `ggplot()`. Our first step will create the graph, but it won't display it right away. I include comments on each line to remind you what everything does.

```
## we will create a plot object and stick it in "p"
p <- ggplot(data=temps, # the data frame to use
       ## aes() stand for "aesthetics"
       aes(x=year, y=deg_C), # define x and y variables to plot
       ) + # closing parenthesis and add a plus sign to add another
 ↪  graphical element
  geom_line() # the type of graph to plot using x and y
```

If you do this correctly it will seem as though nothing happened. If you did something wrong, you might get an error message in the console. If R can't find the function "ggplot", that is because you didn't load `tidyverse` (or ggplot2 which is a subset of tidyverse). Another common error is an incomplete statement that might come from a missing parensthesis or something. In that case, R might think that you are not finished making your statement and it will keep adding plus signs so that you can finish your coding statement.

Now we display the graph.

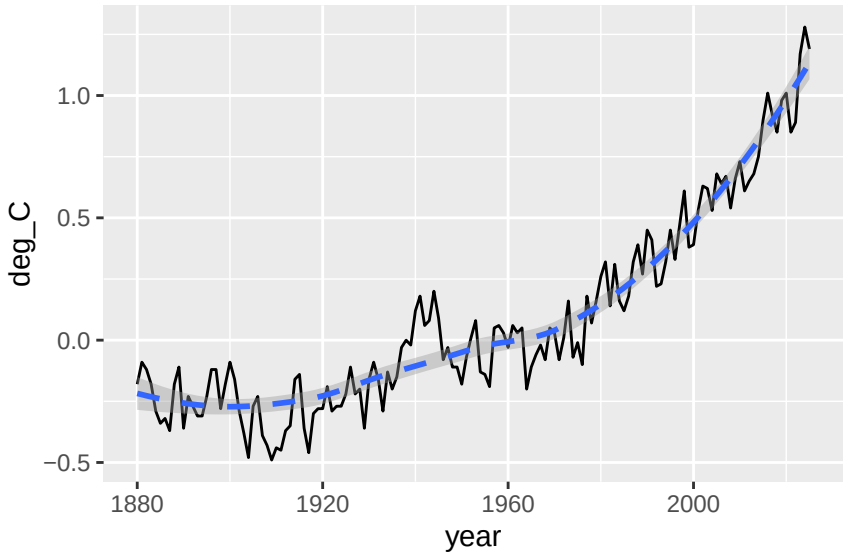
```
p
```



Here we add a trend line using a fancy statistical smoothing technique. By default, it will include a grey uncertainty region.

```
## add a trend line
p + geom_smooth(linetype="dashed")
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



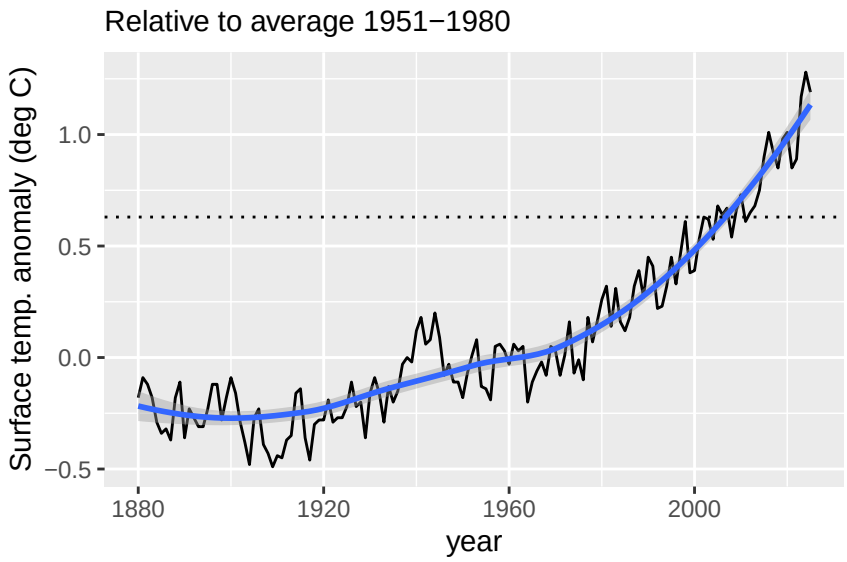
Here we add prettier, more descriptive labels. Add a line for the anomaly in 2002.

```
# first find the anomaly in 2002
anomaly.2002 <- temps$deg_C[temps$year==2002]
anomaly.2002
```

```
[1] 0.63
```

```
p + geom_smooth() +
  labs(
    y="Surface temp. anomaly (deg C)",
    subtitle="Relative to average 1951-1980"
    ) +
  geom_hline(yintercept = anomaly.2002, linetype=3)
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



4.5. Predicting future anomalies

Major goals of science are understanding and prediction. We could spend a long time discussing the connection between these two, but form now, we just want to forecast want our future might look like.

The statistical details of this are way beyond this course, and they are not even the way you would do it if you were serious about. Nonetheless, it won't be too far off, and the intuitive part applies generally to making any set of predictions. We will merely ask the computer to do some math to accomplish the following:

1. create an educated guess for the temperature trend. This will include,
    a. uncertainty that comes from how much the anomalies bounce up and down from year to year, and
    b. a trend that comes from averaging nearby values.
    c. information that nearby years may be more similar to each other than years that are really far apart.
2. given the observed uncertainty and the trend, make an educated guess for future obervations, taking into account that *each future year adds uncertainty that we can only guess at*.

3. Make a figure showing the future forecast.

So, lets start with step one. and create a statistical model of the observations. We start by loading another package.

```
## Load a necessary R package
library(mgcv) # stands for "Mixed Generalized Additive Model
 ↪  Computation Vehicle"
```

```
Loading required package: nlme
```

```
Attaching package: 'nlme'
```

```
The following object is masked from 'package:dplyr':

    collapse
```

```
This is mgcv 1.9-4. For overview type '?mgcv'.
```

Skill spotlight: *You've already been wrangling data and building `ggplot2` figures throughout this book–skills that recur in almost every chapter (see the Preface). What's new here is the model you're about to run: a Generalized Additive Mixed Model, or GAMM, fit with the `mgcv` package. Running one and correctly reading its forecast and uncertainty is real, useful experience–the same general approach shows up in forecasting sales, web traffic, disease spread, or sensor readings, well outside climate science. It is not the same as understanding the statistical theory behind a GAMM (that's graduate-level material)–but unlike a total black box, you can understand what each piece of the code below is asking R to do, even without the underlying math. See the section right after the code for that breakdown, and see the* Skills Crosswalk *appendix for a worked example of describing this experience accurately on a résumé.*

Now we ask R to fit the model.

```
## Create the model
mod <- gamm(deg_C ~ s(year),  # "deg_C as a function of year
            data=temps, # data frame
           correlation = corCAR1(form=~year), # specify
            ↪  autocorrelation among samples
           method="REML" # fit using restricted maximum likelihood
           )
```

Decoding the model: *You don't need to know the underlying math to understand what you just asked R to do. Here's what each piece of that line actually means, in plain language.*

`gamm(deg_C ~ s(year), ...)`: We want to predict `deg_C` from `year`–but instead of forcing a single straight line through the data (which would assume Earth has been warming at the same steady rate since 1880), the `s()` wrapped around `year` tells R: "fit a smooth, curving trend, and figure out the shape of the curve from the data itself." Look at the figure once you make it–the temperature trend

visibly bends upward in recent decades. Warming hasn't just been happening; it's been accelerating. A straight line can't show that. A smooth curve can.

`correlation = corCAR1(form=~year)`: Ordinary regression usually assumes that once you've accounted for the trend, whatever is "left over" in one year tells you nothing about what's left over in the next year. That's a bad assumption for climate data–a warm year tends to be followed by another warm year more often than by a cold one, because year-to-year climate has momentum, not independence. `corCAR1` ("continuous-time autoregressive order 1") tells the model to expect nearby years to be more similar to each other than distant years are, instead of treating every year as if it were an unrelated coin flip. This is also why the function is called `gamm()` ("mixed" model) rather than `gam()`–handling correlated errors like this requires the mixed-model machinery underneath. Skipping this step wouldn't just be a technicality: it would make the model overconfident, producing a narrower (falsely precise) confidence band than the data actually support.

`method="REML"` ("restricted maximum likelihood"): this controls how* R decides how wiggly the smooth curve in `s(year)` should be–wiggly enough to capture a real, meaningful trend, but not so wiggly that it's just chasing random year-to-year noise. REML is the standard, more reliable choice for this kind of model in `mgcv`, which is why it's used here instead of plain maximum likelihood.*

Put together: this one line of code asks R to find the best-supported curving trend in the temperature data, while being honest both about the fact that nearby years are correlated and about how much of the up-and-down wiggle is real signal versus noise. That honesty is exactly what shows up later as the width of the grey confidence band in your figure–and weighing "how much should I trust this curve?" is exactly the kind of judgment call that separates a defensible analysis from an overconfident one.

Next we have to decide what years we want predictions for. Let's go for 50 more years, to 2072

```
## years for fit and prediction
years = data.frame(year=1880:2072)
```

Now we apply the model to past and future years and make a data frame and keep the uncertainty estimates it generates. We also combine our specified years with the model.

```
## include uncertainty with se.fit = TRUE
preds <- data.frame(
  predict(mod$gam, newdata=years, type="response", se.fit=TRUE)
  )

# combine columns of data frames
my.predictions <- cbind(years, preds)
```

Let's look at the first three rows of what we've made.

```
# first three lines , and all columns
my.predictions[1:3, ]
```

```
  year        fit     se.fit
1 1880 -0.1958016 0.06214976
2 1881 -0.2004046 0.05778451
3 1882 -0.2050110 0.05365336
```

This shows that the first column is the year, the second (`fit`) is the expected local estimate, and the third (`se.fit`) is the estimate of the uncertainty, which is the standard error of the expected value.

Now we'll create 95% confidence intervals for each year. Confidence intervals are intervals which would contain the "true" value (if there is one) a specified proportion of times, if you repeated the experiment a really, really large number of times. Ninety-five percent confidence intervals exclude the lowest 2.5% and the upper 97.5% regions.

```
### Create 95% confidence intervals
crit.t.low <- qt(0.025, df = df.residual(mod$gam))
crit.t.high <- qt(0.975, df = df.residual(mod$gam))

### this requires that you loaded the tidyverse package
my.predictions <- my.predictions %>% # my data
  ## transform the data set to include two new columns
  transform(
    upper.conf.lim=fit + (crit.t.high * se.fit),
    lower.conf.lim=fit + (crit.t.low * se.fit))
###
## view the first three rows
my.predictions[1:3, ]
```

```
  year        fit     se.fit upper.conf.lim lower.conf.lim
1 1880 -0.1958016 0.06214976    -0.07292127     -0.3186820
2 1881 -0.2004046 0.05778451    -0.08615504     -0.3146541
3 1882 -0.2050110 0.05365336    -0.09892948     -0.3110926
```

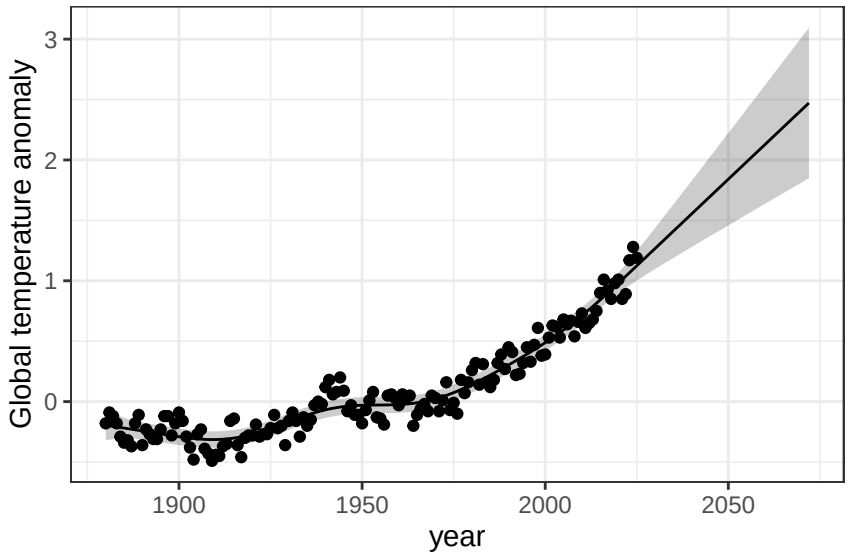
Now we see we've added upper and lower confidence intervals. Next we'll create a graph of our predictions, using several particular graphical features.

```
## basic data and x, y values
g <- ggplot(data = my.predictions, aes(x = year, y = fit)) +
  ## adding a wide region or band (a ribbon)
  geom_ribbon(aes(ymin = lower.conf.lim, ymax = upper.conf.lim,
                  x = year),
              alpha = 0.2, fill = "black") +
  ## add the expected local trend
  geom_line() +
  ## adding the points for each year
  geom_point(data = temps, mapping = aes(x = year, y = deg_C),
             inherit.aes = FALSE) +
  ## add a better label for the y axis
```

```
  labs(y = "Global temp. anomaly")
  ## simplify the image
```

Now we show the graph we created.

```
g
```



The grey area represents the 95% confidence interval for the trend observed years, and also for the trend in the future years. It gets wider because each year's uncertainty compunds with the next year's uncertainty. [Your figure will look a little different the this one.]

Finally, we save your figure. This is the graph you should use for your assignment.

```
## Save a PNG file into your figs folder
## THIS IS THE GRAPH TO INSERT INTO YOUR ASSIGNMENT
ggsave("figs/Global_temps.png", # name for your figure
       plot=g, ## identify which graph you want to save,
       height=5, width=5)
```

4.6. Does Earth have a fever?

In this last section, we ask whether Earth's temperature may be a little too high by comparing it to a human who has a fever.

First we calculate the percent increase in temperature for Earth over the past 70 years and a human experiencing a serious fever.

Average temperature for humans is 37 C, and a serious fever is considered 39 C, so an *increase* of 2 C.

Earth's 20th century average temperature is 13.9 C. Let's calculate the increase since 1950.

```
## First we take a subset of our data set.
a <- subset(temps, year==1950)
b <- subset(temps, year==2022)
increase <- b$deg_C - a$deg_C
increase
```

```
[1] 1.07
```

Next we calculate these increases (fever and global warming) as *percentage change*.

```
## percent increase for Earth
perc.Earth <- increase / 13.9 * 100

## percent increase for humans
perc.fever <- 2/37 * 100

perc.Earth; perc.fever
```

```
[1] 7.697842
```

```
[1] 5.405405
```

So, what do you think? Does Earth have a fever? In what ways do you think comparing temperature variation of Earth to that of a human is or is not a useful thing to do.

4.7. Questions to answer

1. What is a temperature anomaly?
2. In the file "global_surface_anomalies.csv", what is on
    a. the first three lines?
    b. the fourth line?
    c. the rest of the data?
3. What are the minimum, median, and maximum anomalies?
4. When (in what year) were the minimum and maximum anomalies?
5. What was the global average surface temperature anomaly in the year you were born? Provide the year and the anomaly.
6. What does the grey band in the figure represent, and why does it get wider in future years?
7. Explain whether you think Earth currently does or does not have a fever, and whether comparing temperature variation of Earth to that of a human is or is not a useful thing to do.

5. *Traits, Physiology, and the Global Spectrum of Plant Form and Function*

Before you start

- **Assumptions about background and guidance:** most of this chapter is reading, and the code that draws the null-model figure is hidden, so you can follow it without R. The data exercise at the end assumes your `Rwork` project from Computer setup and some familiarity with `filter()`, `mutate()`, and `ggplot()`; Extra-pair copulation and Displaying data with ggplot2 are good places to pick those up. For more on downloading from databases like BIEN, see Accessing public databases.

5.1. Goals

By the end of this chapter, you should be able to:

- Identify the six traits that explain most of the variation in plant form across species.
- Explain why organism design is limited by physical trade-offs in resource allocation and energy.
- Explain how the four null models differ from each other, and how they differ from observed (real) data.
- Acquire data from a curated public database that specifies selected constraints.
- Use public data to illustrate the extent of trade-offs in plant traits across a selected set of species.

Deliverables

For your assignment, write up three short paragraphs, and add your scatterplot with a descriptive caption. Your paragraphs will address:

1. which species subset and two traits you chose, and why;

2. what the scatterplot and correlation coefficient show — are the traits trading off (negative correlation), coordinating (positive correlation), or largely unrelated (correlation near zero) across your species; and
3. how that result connects to the physiological trade-off framework from this chapter — what would have to be true, mechanistically, for your two traits to relate the way they do in your data?

5.2. Starting with the individual

We visit the subject of traits of the individual organism, because population growth rates, community assembly, and ecosystem processes are all, at bottom, aggregates of what individual organisms are physiologically capable of doing. Before we can ask how large a population can grow, or why one species persists where another doesn't, we need a vocabulary for describing what an individual organism actually is — physiologically and morphologically — and why it is built the way it is.

This chapter asks a deceptively simple question: why is one plant species tall and another short, why one leaf is cheap and short-lived and another expensive and long-lived, or why one seed tiny and abundant and another large and well-provisioned? The answer to these anatomical differences lies in physiology — the internal machinery of growth, water transport, carbon gain, and reproduction — and in the traits that machinery produces. Traits are where physiology becomes visible, measurable, and comparable across species. The framework built here — traits as compact, physiologically-grounded summaries of an individual's strategy — is the foundation that later chapters on populations, communities, and ecosystems build on.

This chapter is built around a single influential peer-reviewed article: Díaz, S. et al. (2016), "The global spectrum of plant form and function," published in Nature. It is one of the most ambitious attempts to address how, across all vascular plants on Earth, individual-level traits vary along a relatively small number of general axes — or whether the possibilities are essentially limitless.

5.3. What is a functional trait?

A functional trait is a measurable feature of an organism — morphological, physiological, or phenological — that influences its growth, survival, and reproduction. Traits sit at the interface between an organism's physiology and the ecological filters (light, water, nutrients, herbivores, competitors) that act on it. Leaf nitrogen content, for example, is a trait, but it is really a proxy for something physiological: how much of a leaf's mass is invested in the photosynthetic enzyme machinery. Stem density is a trait, but it stands in for the anatomy of the wood — how much of the stem cross-section is thick-walled fiber versus water-conducting vessels.

Because traits are measurable on individuals and comparable across species, they let ecologists ask general questions — is there such a thing as a plant "strategy space," and if so, how is it organized — without needing a physiological model for every one of the roughly 400,000 species of land plants.

5.4. The search for universal axes of strategy

Díaz et al. (2016) did not start from nothing. Plant ecologists had long suspected that whole-plant strategy could be summarized along a few axes. Grime's competitor–stress-tolerator–ruderal (CSR) triangle proposed three basic strategies shaped by disturbance and resource availability. Westoby's "LHS" (leaf-height-seed) scheme proposed three simple, widely measurable traits as a first-pass strategy scheme. Most influentially for this chapter, Wright et al. (2004) assembled a worldwide leaf economics spectrum (LES): across biomes and growth forms, leaf traits fall along a single axis running from cheaply-built, short-lived, high-nutrient "acquisitive" leaves to expensive, long-lived, low-nutrient "conservative" leaves.

These efforts were each restricted to a single organ (mostly leaves) or a small trait set. The open question was whether a similarly compact structure would hold once traits from every major plant organ — stems, leaves, seeds — and the whole plant (height) were considered together, across the full range of vascular plant diversity.

5.5. Díaz et al. (2016): scale and design

Díaz and 31 co-authors compiled trait records for 46,085 vascular plant species from 423 growth forms and geographic regions — by far the largest and most taxonomically and geographically broad plant trait dataset assembled at the time, built on the TRY plant trait database plus additional unpublished contributions. They focused on six traits chosen because each indexes a fundamentally different aspect of plant form and function, each is measurable across nearly all vascular plants, and together they span growth, survival, and reproduction — the three components of Darwinian fitness.

Table 5.1.: The six core traits identified by Díaz et al. (2016)

Trait	Units	What it measures	Physiological / ecological significance
Adult plant height (H)	m	Typical height of the main photosynthetic tissue at maturity	Pre-emption of light; trades off against construction/maintenance cost and breakage risk with height.
Stem specific density (SSD)	mg/mm^3	Dry mass per unit fresh stem volume	Index of mechanical strength and hydraulic safety; low-SSD stems grow fast but are more vulnerable to damage and cavitation.
Leaf area (LA)	mm^2	One-sided surface area of an individual leaf	Shapes light interception, boundary-layer thickness, and leaf energy/water balance.
Leaf mass per area (LMA)	g/m^2	Leaf dry mass per unit fresh leaf surface area	Core axis of the leaf economics spectrum: construction cost traded off against leaf lifespan and light-capture efficiency.
Leaf nitrogen content (Nmass)	mg/g	N content per unit leaf dry mass	Tracks investment in photosynthetic machinery; higher Nmass raises photosynthetic potential but also palatability to herbivores.

Trait	Units	What it measures	Physiological / ecological significance
Diaspore mass (SM)	mg	Mass of an individual seed or spore	Trades off dispersal ability and colonization potential against per-propagule provisioning and seedling survival.

5.6. Two dimensions capture most of the variation

With six traits, the space of possible trait combinations is six-dimensional. If plants filled that space freely, there would be no simple picture to draw. Instead, Díaz et al. found that just two principal component axes — a plane — captured 74% of the variation among species. Two dimensions stood out within that plane:

- **The size dimension** (height and diaspore mass, H & SM): runs from short species with small seeds to tall species with large seeds. Physiologically, this dimension reflects the size of the whole plant and its parts — how much structural investment (biomechanical support, hydraulic transport distance) and reproductive investment a plant makes.
- **The leaf economics dimension** (nitrogen content - leaf mass per area, N_{mass} - LMA): runs from species with cheaply-constructed, nitrogen-rich, "acquisitive" leaves to species with expensive, nitrogen-poor, "conservative" leaves that persist longer and tolerate abiotic and biotic hazards. This is, essentially, the leaf economics spectrum (sensu Wright et al. 2004) re-discovered inside a much larger, whole-plant analysis.

Stem specific density (SSD) and leaf area (LA) load onto both dimensions rather than standing apart: SSD tracks with the height–seed mass axis, while leaf area shows independent variation tied to both. The headline result is not just that two axes explain most of the variance, but that they correspond to two distinct and physiologically intuitive trade-offs: how big to grow, and how to run your leaf economy.

5.7. A constrained, "lumpy" trait space

A second, arguably more surprising finding concerns how much of that six-dimensional space is actually occupied. Díaz et al. compared the observed trait hypervolume (the volume of trait space vascular plants actually

occupy) against four null models of increasing realism — from traits varying completely independently and uniformly, to traits varying as observed but uncorrelated. Even against the most conservative null model, the observed hypervolume was about 20% smaller, and compared to the simplest null models it was 72–98% smaller.

In other words, of all the trait combinations that seem geometrically possible, only a narrow, non-random subset are evolutionarily viable. The occupied space is also "lumpy": species cluster into two particularly dense functional hotspots — one dominated by small-seeded, small-leaved herbaceous and graminoid plants, the other by moderately tall, large-leaved, large-seeded woody species — rather than spreading evenly across the plane. Many distantly related lineages land in the same hotspot, which the authors read as evidence that a limited number of trait combinations have been discovered repeatedly and independently over the evolutionary history of land plants — convergence on a small set of workable solutions to the shared physiological problems of growing, surviving, and reproducing.

A toy example, with two made-up variables

The real analysis used six traits at once, which is hard to draw. Here is a much simpler, entirely hypothetical version with just two made-up variables — call them "Structural Investment" and "Leaf Economy" (arbitrary units, not measured data) — chosen only to echo the chapter's two real dimensions (size, and leaf economics). The goal is to see, on ordinary x–y axes, what each null model actually looks like, and how differently a real, evolved pattern can behave.

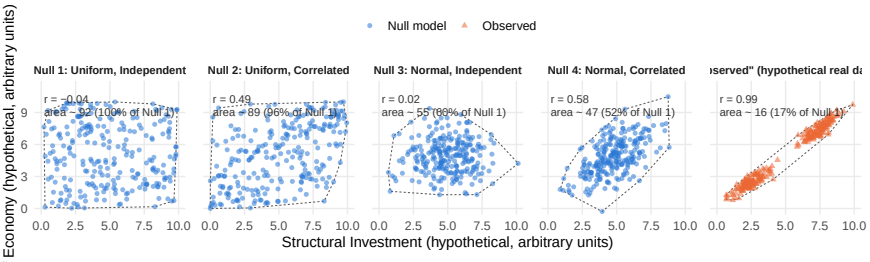


Figure 5.1.: Four null models (blue circles) built from every combination of uniform/normal margins and correlated/independent structure, compared to a hypothetical "observed" pattern (orange triangles) with two tight clusters that are positively correlated both within and between clusters. Dashed outlines are convex hulls (a 2-D stand-in for Díaz et al.'s six-dimensional hypervolume); r is the Pearson correlation for each panel.

Two things stand out, and both echo the real paper's findings. First, correlation: the observed data ($r \approx 0.99$) is far more tightly correlated than any null model managed, including the two null models that were deliberately built with a correlation baked in ($r \approx 0.5$–0.6). Second, occupied area: the observed cluster's convex hull covers only about a sixth of the space Null Model 1 covers, and roughly a third of even the most restrictive null model (Normal, Correlated) — despite being built from the same number of points. That combination — tighter correlation and a smaller, clumped footprint — is exactly what "lumpy" means in this context: not just fewer viable combinations, but combinations that clump into distinct, well-separated solutions (here, two clusters that both sit on the same trend line; in the real data, the herbaceous and woody hotspots) rather than filling one smooth, undifferentiated patch.

7. Traits as windows into physiology

Each of the six traits is a compact summary of an underlying physiological trade-off:

- **Height** sets how much light a plant can pre-empt from shorter neighbors, but taller stature requires longer hydraulic pathways (raising the risk of xylem cavitation and greater tension at the leaf) and more biomechanical support tissue, and it raises the cost and risk of reaching reproductive structures above the canopy.
- **Stem specific density** reflects a trade-off between mechanical strength and hydraulic safety on one hand, and growth rate on the other: low-density, loosely packed wood is cheap to build and supports fast volumetric growth, but is more vulnerable to mechanical failure, pathogens, and hydraulic dysfunction than dense wood.
- **Leaf area** governs the boundary layer around a leaf and therefore its energy balance and water loss: large leaves shed heat less efficiently and are more exposed to evaporative demand, while small leaves buffer temperature and water stress more effectively at some cost to light interception.
- **Leaf mass per area** is a direct proxy for construction cost per unit light-intercepting area; high-LMA leaves cost more carbon to build but tend to last longer and resist physical and herbivore damage, while low-LMA leaves are cheap and fast to replace but short-lived.
- **Leaf nitrogen content** per unit mass tracks investment in Rubisco and other photosynthetic proteins; high Nmass raises photosynthetic capacity and growth potential but also raises the nutritional value — and thus the attractiveness — of leaf tissue to herbivores and pathogens.
- **Diaspore mass** captures a trade-off between the number of seeds or spores a plant can produce and how far and reliably each can disperse, versus how well-provisioned and likely to survive each individual propagule is once it germinates.

None of these trade-offs act in isolation. A short-statured, fast-growing herb with cheap leaves and abundant tiny seeds is a physiologically coherent package — one that would be undermined by, say, also investing heavily in dense, long-lived wood. This is the deeper reason the six-dimensional space collapses onto a plane: physiology imposes internal consistency on which trait combinations can function together in a single organism.

8. From individual traits to communities and ecosystems

Because traits are measured on individuals but summarize physiological strategy, they scale upward in ways that population counts alone do not. Community ecologists use trait distributions — community-weighted means, functional diversity indices — to characterize how a community's collective physiology responds to disturbance, resource availability, or climate. Global vegetation models increasingly use continuous trait axes like

the ones Díaz et al. describe, rather than a small number of discrete plant functional types, to represent how vegetation might respond to environmental change. The two-dimensional global spectrum also maps loosely onto older, verbally-defined strategy schemes — the size dimension echoes the colonization (r) versus competition (K) continuum, and the leaf economics dimension echoes the adversity-selection (A) or leaf-economics continuum — giving those older frameworks a quantitative, globally-tested backbone.

You will meet several of these ideas again once this book turns to population growth models later on. Growth rates, mortality risk, and fecundity — the parameters those models take as inputs — are not arbitrary numbers; they are, in large part, downstream consequences of the trait strategy an organism has committed to. A species positioned at the fast, cheap-leaf, small-seeded end of the global spectrum is built for a different population dynamic than one at the slow, dense-wood, large-seeded end, before a single population equation is written down.

9. Why anchor a chapter on a single paper?

Centering this chapter on one globally general, quantitative study — rather than, say, the natural history of a single growth form — trades breadth of examples for a framework that still grounds every trait in individual physiology. We see a concrete, well-illustrated case (Figure 2 of the Diaz et al. , the two-axis trait plane with its herbaceous and woody hotspots) while building toward the more general skill of thinking about organisms as bundles of correlated, physiologically-constrained traits — a lens that applies just as well to a liana, a desert annual, or a temperate tree. A liana example fits naturally as a short illustrative case study here (e.g., where a woody vine that borrows structural support from other plants would plot on the H–SM and LMA–Nmass axes, compared to a free-standing tree) — see the reflection questions below.

10. Data exercise: exploring trait trade-offs with BIEN

The toy example in Section 6 used invented data to show what null models look like. Now you'll do the real version, in miniature: pull real trait measurements for a small set of plants, and see whether, and how strongly, two traits trade off (or coordinate) across those plants.

The BIEN database holds millions of records, so this exercise is *deliberately narrow*. You will make three choices, each from a short menu below, and you will not need to know anything about plant taxonomy or BIEN's trait vocabulary in advance:

1. **Two traits** (Table 5.2).
2. **One group of plants** (Table 5.3, Table 5.4, or Table 5.5).
3. **One route for getting the data** (A, B, or C below, matched to the group you chose).

Please do **not** download data for a whole country, state, or plant family. Those queries return hundreds of thousands of records, can take many minutes, and will probably freeze your R session. If a query is still running after two minutes, stop it (the red stop sign in RStudio) and pick a smaller group.

5.7.1. Choice 1: two traits

Each pair below comes from the six traits in Table 5.1, and each has been a standard comparison in the plant trait literature for decades. If you have no strong preference, use Pair 1.

Table 5.2.: Three pre-approved trait pairs. Names are BIEN's exact `trait_name` strings; Step 1 below has you confirm that they still exist.

Pair	Trait 1 (x-axis)	Trait 2 (y-axis)	What it tests
1	`leaf area per leaf dry mass`	`leaf nitrogen content per leaf dry mass`	The leaf economics spectrum: do cheap, thin leaves also carry more nitrogen per gram?
2	`whole plant height`	`seed mass`	Do taller plants make bigger seeds? (Westoby's leaf-height-seed scheme)
3	`whole plant height`	`leaf area`	Do taller plants carry larger leaves?

5.7.2. Choice 2: one group of plants

Choose **one** of the three groups below. Every entry has been picked because it is a common, well-studied plant, so BIEN is likely to have trait measurements for it. (Likely, not guaranteed: some trait pairs will have few measurements for some groups, and Step 6 shows you how many species you ended up with.)

Genera (use with Route A). Pick exactly one genus.

Table 5.3.: Genera that work well for Route A. A genus with fewer than about 10 species that have *both* of your traits will give a thin scatterplot; if that happens, pick a different genus.

Growth form	Genera
Temperate trees	*Quercus* (oaks), *Acer* (maples), *Pinus* (pines), *Betula* (birches), *Salix* (willows), *Populus* (poplars, aspens), *Fraxinus* (ashes), *Prunus* (cherries, plums)
Tropical trees and shrubs	*Ficus* (figs), *Inga*, *Piper*, *Miconia*
Grasses, sedges, and herbs	*Poa* (bluegrasses), *Carex* (sedges), *Solidago* (goldenrods), *Trifolium* (clovers), *Plantago* (plantains)

Ready-made species lists (use with Route B). Pick one list. Each is a community you could walk through, which makes this the best route for comparing a woody community against an herbaceous one (compare your result with the "two hotspots" in Díaz et al., 2016).

Table 5.4.: Ready-made species lists for Route B. BIEN matches the current accepted name, so if one species returns nothing, that is normal; just continue with the rest.

List	Species
Ohio hardwood forest trees (28 species)	*Acer saccharum*, *Acer rubrum*, *Acer negundo*, *Acer saccharinum*, *Aesculus glabra*, *Betula lenta*, *Betula nigra*, *Carya ovata*, *Carya cordiformis*, *Celtis occidentalis*, *Fagus grandifolia*, *Fraxinus americana*, *Fraxinus pennsylvanica*, *Juglans nigra*, *Liriodendron tulipifera*, *Liquidambar styraciflua*, *Platanus occidentalis*, *Populus deltoides*, *Prunus serotina*, *Quercus alba*, *Quercus rubra*, *Quercus velutina*, *Quercus macrocarpa*, *Quercus palustris*, *Sassafras albidum*, *Tilia americana*, *Ulmus americana*, *Ulmus rubra*

List	Species
Midwestern tallgrass prairie plants (24 species)	*Andropogon gerardii*, *Schizachyrium scoparium*, *Sorghastrum nutans*, *Panicum virgatum*, *Elymus canadensis*, *Bouteloua curtipendula*, *Solidago canadensis*, *Solidago rigida*, *Echinacea purpurea*, *Rudbeckia hirta*, *Ratibida pinnata*, *Silphium perfoliatum*, *Silphium laciniatum*, *Helianthus annuus*, *Monarda fistulosa*, *Asclepias syriaca*, *Asclepias tuberosa*, *Liatris spicata*, *Lupinus perennis*, *Baptisia australis*, *Symphyotrichum novae-angliae*, *Coreopsis lanceolata*, *Heliopsis helianthoides*, *Dalea purpurea*
Costa Rican lowland tropical trees (22 species)	*Cecropia peltata*, *Ochroma pyramidale*, *Ceiba pentandra*, *Swietenia macrophylla*, *Inga edulis*, *Guazuma ulmifolia*, *Hura crepitans*, *Spondias mombin*, *Tabebuia rosea*, *Gliricidia sepium*, *Cordia alliodora*, *Bursera simaruba*, *Brosimum alicastrum*, *Carapa guianensis*, *Virola surinamensis*, *Pentaclethra macroloba*, *Vochysia ferruginea*, *Dipteryx panamensis*, *Hyeronima alchorneoides*, *Schizolobium parahyba*, *Trema micrantha*, *Byrsonima crassifolia*

Countries (use with Route C, the most advanced). Pick one country *and* one of its suggested genera. The country query returns every species measured in that country, so you will trim to one genus immediately.

Table 5.5.: Countries for Route C. Ask your instructor before choosing a country that is not on this list.

Country	Habitat	Suggested genera to trim to
Costa Rica	tropical wet and dry forest	*Piper*, *Inga*, *Ficus*, *Miconia*
Panama	tropical forest	*Inga*, *Ficus*, *Piper*, *Psychotria*
South Africa	fynbos and savanna	*Protea*, *Leucadendron*, *Aloe*, *Searsia*
New Zealand	temperate rainforest	*Nothofagus*, *Podocarpus*, *Coprosma*, *Veronica*
Chile	temperate rainforest	*Nothofagus*, *Berberis*, *Fuchsia*, *Escallonia*

5.7.3. Choice 3: run your route

Run Steps 1 and 2 for everyone. Then run **only the one route** in Step 3 that matches your group (A for a genus, B for a species list, C for a country plus genus), and skip the others.

```
library(BIEN)
library(dplyr)
library(tidyr)
library(ggplot2)

# Step 1: confirm that your traits exist in BIEN ----
## BIEN_trait_list() returns every trait in the database (dozens of
 ↪  rows),
## so we filter it down to the handful we care about. Run this
 ↪  yourself and
## check that the names in the trait-pair table appear exactly as
## written. The set of available traits can change over time.
all_traits <- BIEN_trait_list()
all_traits |>
  filter(grepl("height|leaf area|nitrogen|seed mass", trait_name))
```

```
# Step 2: choose your two traits ----
## Copy one pair from the trait-pair table. Pair 1 shown here.
my_traits <- c("leaf area per leaf dry mass",
               "leaf nitrogen content per leaf dry mass")
```

```
# Step 3: get the data for your group (RUN ONLY ONE ROUTE) ----

## Route A: one genus from the genus table ----
## One call, no species list needed. Replace "Quercus" with your
 ↪  genus.
my_traits_data <- BIEN_trait_traitbygenus(genus = "Quercus", trait =
 ↪  my_traits)

## Route B: one species list from the species-list table ----
## Type (or copy) the species names from your chosen list inside
 ↪  c(...),
## with each name in quotes. The Ohio hardwood list is shown here,
 ↪  shortened
## to its first five species as a template; paste in the rest of
 ↪  yours.
# my_species <- c("Acer saccharum", "Acer rubrum", "Acer negundo",
#                 "Acer saccharinum", "Aesculus glabra")
# length(my_species)   # should be roughly 20-30, never hundreds
# my_traits_data <- BIEN_trait_traitbyspecies(species = my_species,
#                                             trait = my_traits)

## Route C: one country, trimmed to one genus from the country table
 ↪  ----
## Download ONE trait at a time for the whole country, then keep
 ↪  only your genus.
```

```
# country_data <- BIEN_trait_country(country = "Costa Rica", trait =
 ↪  my_traits[1])
# my_traits_data <- country_data |> filter(grepl("^Piper ",
 ↪  scrubbed_species_binomial))
## Then repeat for the second trait and combine with bind_rows():
# country_data2 <- BIEN_trait_country(country = "Costa Rica", trait
 ↪  = my_traits[2])
# my_traits_data <- bind_rows(my_traits_data,
#   country_data2 |> filter(grepl("^Piper ",
 ↪  scrubbed_species_binomial)))
```

```
# Step 4: look at what came back before assuming column names ----
## BIEN returns long-format data (one row per species-trait record).
nrow(my_traits_data)
names(my_traits_data)
table(my_traits_data$trait_name)    # how many records for each of
 ↪  your two traits?
```

```
# Step 5: reshape to one row per species, one column per trait ----
## Column names below (scrubbed_species_binomial, trait_name,
 ↪  trait_value)
## match current BIEN documentation -- confirm against Step 4's
 ↪  output.
traits_wide <- my_traits_data |>
  mutate(trait_value = as.numeric(trait_value)) |>
  group_by(scrubbed_species_binomial, trait_name) |>
  summarise(trait_value = mean(trait_value, na.rm = TRUE), .groups =
   ↪  "drop") |>
  pivot_wider(names_from = trait_name, values_from = trait_value) |>
  drop_na()

nrow(traits_wide)    # number of species with BOTH traits
## If this is fewer than about 10, choose a different genus, list,
 ↪  or trait pair.
```

```
# Step 6: plot your two traits against each other ----
ggplot(traits_wide, aes(x = .data[[my_traits[1]]], y =
 ↪  .data[[my_traits[2]]])) +
  geom_point(alpha = 0.7, colour = "#2a78d6", size = 2) +
  labs(x = my_traits[1], y = my_traits[2],
       title = paste("Trait relationship across", nrow(traits_wide),
        ↪  "species")) +
  theme_minimal()
ggsave("figs/my_trait_relationship.png", width = 5, height = 5)
```

```
# Step 7: quantify the relationship ----
cor(traits_wide[[my_traits[1]]], traits_wide[[my_traits[2]]], use =
 ↪  "complete.obs")
## Trait values often span orders of magnitude; if your plot looks
 ↪  like a
## "comet" crowded in one corner, try the correlation on
 ↪  log10-transformed values:
```

```
cor(log10(traits_wide[[my_traits[1]]]),
 ↪  log10(traits_wide[[my_traits[2]]]))
```

11. Reflection and discussion questions

1. Pick two traits from Table 5.1 that are not on the same principal-component axis (e.g., leaf area and diaspore mass). Can you think of a plant where these two traits would pull in "conflicting" physiological directions? What might resolve that conflict?
2. The observed trait hypervolume is smaller than all four null models, but least different from null model 4 (which already assumes traits are correlated as observed). What does that comparison tell you about how much of the "constraint" in plant trait space comes from correlations among traits, versus limits on any single trait's range?
3. The two functional hotspots in Figure 2 of Díaz et al. (2016) are described as occupied by many distantly related lineages. What does that convergence suggest about the role of natural selection versus shared ancestry in shaping plant form?
4. How would you expect a liana — a woody vine that borrows structural support from other plants rather than building its own — to plot on the height–diaspore mass axis, compared to a free-standing tree of similar leaf economics? What does that comparison reveal about the limits of using height alone as a proxy for "size" investment?
5. This chapter frames traits as compact summaries of physiological trade-offs. Choose one trait and describe, in your own words, the underlying physiological mechanism it stands in for.
6. In your BIEN data exercise, was the relationship between your two traits closer to the null-model pattern (weak or no correlation) or the toy "observed" pattern (strong correlation, clumped rather than evenly spread)? What might explain the difference, given how many species — and how narrow a slice of the plant world — you sampled compared to Díaz et al.'s 46,085 species?

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6. Behavior: Extra-pair copulation in a monogamous species

Figure 6.1.: Voles are small hamster-like rodents that live in grasslands and fields.

Before you start

- **Assumptions about background and guidance:** your `Rwork` project set up as in Computer setup, and that you can read a .csv file into R. We build graphs in `ggplot()` step by step here, so no prior plotting experience is assumed. The evolution and behavior background comes from this chapter and the introduction of Keane et al. (2024). For more plotting habits and recipes, see Displaying data with ggplot2.

6.1. Goals

In this chapter, you will

- Develop and use your understanding of evolution, behavior, and prairie voles to make predictions about determinants of extra-pair copulation in a monogamous species.
- Enhance your quantitative reasoning and data analytic skills.

6.2. Deliverables

Create one document (presentation slide or other document) that,

- places the hand-drawn graph of your predictions side by side with the graph of the observed data,
- includes a succinct *explanation* of
    i. basis of your original prediction; explain why voles would behave the way you predicted they would. Do not rely on your intuition. Instead, use your knowledge of evolutionary biology, behavior, and voles in your explanation, and
    ii. your observed result.
- includes a *headline*, that is, a title that provides your result or takeaway message, such as "Living in small groups enhances monogamy".

Turn this in to the assignment on Canvas. (*Previous versions asked to submit this to a Google Slide deck, but we will no longer do that.*)

6.3. Preparatory Steps

Download Keane et al. (2024), which is available in our PDF archive. Read the introduction to help you understand the context of this chapter.

6.4. Background: What causes extra pair copulation in prairie voles?

Monogamy, the mating system in which individuals pair-bond to raise offspring, is relatively rare. We don't know what causes this rarity, and nor do we know why it persists where it does occur. Prairie voles (*Microtus ochrogaster*), like humans, are famously monogamous, and behavioral ecologists have studied them for that reason. And, just like humans, voles also engage in extra-pair copulations that result in offspring, and scientists have been trying to understand the causes and consequences of those extra-pair copulations (Solomon and Keane 2018).

Monogamy is a mating system (*sexual monogamy*), but it is also one of many types of social system (*social monogamy*). Most species are entirely *solitary* (think grasshoppers, frogs, flies), but some species engage in persistant social behavior, or exhibit *sociality* (Nowak et al. 2010; Nordell and Valone 2014; Kappeler 2019). For example, all species of ants and many species of bees live in large colonies with several castes of individuals (*eusociality*). Many species live in family social units or have primitive forms of eusociality (Nordell and Valone 2014). Social monogamy is the particular form of sociality in which a female-male pair lives together, often resulting in more successful rearing of offspring.

Solomon, Nancy G., and Brian Keane. 2018. "Dispatches from the Field: Sociality and Reproductive Success in Prairie Voles." *Animal Behaviour* 143 (September): 193–203. https://doi.org/10.1016/j.anbehav.2018.07.001.

Nowak, Martin A, Corina E Tarnita, and Edward O Wilson. 2010. "The Evolution of Eusociality." *Nature* 466 (7310): 1057–62.

Nordell, Shawn E., and Thomas J. Valone. 2014. *Animal Behavior: Concepts, Methods, and Applications*. Oxford University Press.

Kappeler, Peter M. 2019. "A Framework for Studying Social Complexity." *Behavioral Ecology and Sociobiology* 73 (1): 13. https://doi.org/10.1007/s00265-018-2601-8.

Nordell, Shawn E., and Thomas J. Valone. 2014. *Animal Behavior: Concepts, Methods, and Applications*. Oxford University Press.

So why would socially monogamous individuals engage in extra-pair copulation?

One reason that individuals engage in extra-pair copulations is in order to avoid inbreeding depression. Breeding that results from matings between highly related individuals typically causes a decline in vigor and health of individual offspring and a reduction in fitness for those phenotypes. Inbreeding depression is the reduction in fitness due to breeding among relatives.

Fitness is the quantitative representation of natural and sexual selection within evolutionary biology. It can be defined either with respect to a genotype or to a phenotype in a given environment. In either case, it describes individual reproductive success and is equal to the average contribution to the gene pool of the next generation that is made by individuals of the specified genotype or phenotype. The fitness of a genotype is manifested through its phenotype, which is also affected by the developmental environment. The fitness of a given phenotype can also be different in different selective environments. (Quoted from Wikipedia)

6.4.1. Tinbergen's four questions

This section is from an online course at Oxford University

In 1963, Niko Tinbergen published a seminal paper entitled *On the aims and methods of ethology*, which laid the foundation for how to conduct research in the fledging field of animal behaviour. The lasting contribution of the paper is that in it Tinbergen formulated four different, albeit somewhat interlinked, approaches to the study of animal behaviour, or four different types of questions we can ask about an observed behaviour.

The four questions are:

1. Function (or adaption)

Why is the animal performing the behaviour? In which way does the behaviour increase the animal's fitness (i.e. its survival and reproduction)? Examples are plentiful and include, among many others, nurturing of young to increase their chance of survival, migration to warmer (and more food rich) habitats, escaping or avoiding attention from predators etc.

2. Evolution (or phylogeny)

How did the behaviour evolve? How has natural selection modified the behaviour over evolutionary time? This is typically addressed by the comparative approach, where the behaviour in question is compared among closely related species. Examples include how flight in birds may have evolved from gliding in dinosaurs or how the vertebrate and cephalopod eyes have evolved by convergent evolution, with the former having a blind spot, while the latter does not.

3. Causation (or mechanism)

What causes the behaviour to be performed? Which stimuli elicit or what physiological mechanisms cause the behaviour? Examples include the role of pheromones and hormones, such as increasing testosterone levels (caused by increasing day length) causing male display behaviour in many species of birds, moving shadows causing ragworms to withdraw into their burrows or contrast on beaks causing herring gull chicks to peck.

4. Development (or ontogeny)

How has the behaviour developed during the lifetime of the individual? In what way has it been influenced by experience and learning? Examples include how courtship behaviour improves with age in many birds and how predators learn to avoid toxic or dangerous prey with experience.

The four types of questions operate at two different levels. Questions 1 and 2 give ultimate or evolutionary explanations. These are answers that take a longer perspective and try to explain *why* the behaviour has evolved. On the other hand, questions 3 and 4 give proximate explanations. These are answers that look into the immediate mechanical reasons for *how* a behaviour is expressed. To obtain a full understanding of the costs, benefits and constraints that have shaped a given behaviour both ultimate and proximate answers must be obtained.

6.4.2. Determinants of extra-pair copulation in voles

So, what causes extra-pair copulation? What influences the probability that an individual engages in extra-pair copulation? We can think about these questions in any of the four contexts that Tinbergen's questions provide.

Here are some factors that may help determine to the probability of extra-pair copulation:

- the relatedness of the extra-pair partner, and avoidance of inbreeding.
- the relatedness of the nest- vs. extra-pair partner.
- sex (perhaps females are more or less likely than males to seek opportunities)
- number in nest, if crowding is a proximate cue to mate with a nest partner or extra-pair individual.

It is not always clear how animals and plants identify individuals that are closely related or not related, but under many circumstances, they somehow seem to mate with individuals that are less related than by chance alone.

6.5. Data from the Solomon/Keane lab

Dr. Nancy Solomon and Dr. Brain Keane have been studying prairie voles for decades, and here we investigate our questions using one of their data sets.

The data file `epr_209.csv` contains observations on voles related to those we read about in Keane et al. (2024). These data are from sampling of voles from the Indiana site, collected over three years (2006-2008). These data are available in our data repository

Each row in the data set is a record of a successful copulation that resulted in at least one offspring. Included in each row we have the ofllowing columns:

- id - identification number of the individual
- sex - biological sex of the individual
- year - calendar year of data collection
- nest - the identity of the home nest of the individual
- size - number of individuals in the nest
- r - relatedness to the partner.
- pairtype - nest partner or extra-pair copulation.

```
    id    sex year     nest size       r pairtype
1  101   male 2006 19/20/21    7  0.3776     nest
2  101   male 2006 19/20/21    7 -0.1830     nest
3 1021 female 2007       29    3  0.3208     nest
4 1021 female 2006       32    2 -0.1093    extra
```

```
Warning in rm(d209): object 'd209' not found
```

Relatedness to a partner varies between -1 and +1. A value near zero indicates no more related than expected by chance; 1 is self-identity (e.g., clone) and negative values indicate even less related than we expect due to chance.

6.6. Your objectives

Ask yourself, "what the heck are these voles doing?" What is might be a cause of extra-pair copulations?

You consider what you learned in your readings, and come up with a question or hypothesis about what conditions to might lead to more extra-pair copulations. In particular, you will

1. Hypothesis: Pose a hypothesis that you can address *using these data*. Your hypothesis should reflect a mechanism of what you think is going on. Rely on your understanding of evolution, behavior, and prairie voles to justify your the mechanism and hypothesis.
2. Prediction: Make a prediction that is based on your hypothesis. Draw *on paper* a mock-up of the graph that uses the sort of data we have access to, and is consistent with the mechanism you propose.
3. Data: Write R code to create a graph that mimicks your prediction, but uses the actual data you download.
4. Evaluation: Figure out what the graph tells you about the answer to your question. In what ways are your results consistent with your prediction or not? Rely on your understanding of evolution, behavior, and prairie voles to interpret your result.

Put all of the above (hypothesis, hand-drawn prediction, R figure of data, data-based evalution) in a document, and turn it in online. See **Deliverables**, near the start of this chapter.

6.6.1. Write your own script

To create the graph, you will write your own script. Start as usual, by finding out what your current working directory. Make sure it is `Rwork`. The read in your data and look at the variables. Then create the graph you want.

First we check the current working directory.

```
getwd()
```

This should show a path to `Rwork`. If not, use the "Session" menu to set your working directory. If you don't have an `Rwork` folder, see Chapter 1.

Next we load some packages that we will use.

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------
 ↪  tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------
 ↪  tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to
 ↪  force all conflicts to become errors
```

Next, we get the data.

1. Go to our class data repository, and download `epr.csv`.
2. Make sure the file remains in the same simple format, "comma separated values" (.csv).
3. Put `epr.csv` in your data directory, `Rwork/data`.
4. Continue to make sure the file remains in the same simple format, "comma separated values" (.csv).

Now import the data, and display its *str*ucture.

```
d209 <- read.csv("data/epr_209.csv")
str(d209)
```

```
'data.frame':   250 obs. of  7 variables:
 $ id      : chr  "101" "101" "1021" "1021" ...
 $ sex     : chr  "male" "male" "female" "female" ...
 $ year    : int  2006 2006 2007 2006 2006 2007 2007 2006 2006 2006
  ↪  ...
 $ nest    : chr  "19/20/21" "19/20/21" "29" "32" ...
 $ size    : int  7 7 3 2 2 2 2 7 7 2 ...
 $ r       : num  0.378 -0.183 0.321 -0.109 -0.032 ...
 $ pairtype: chr  "nest" "nest" "nest" "extra" ...
```

Use everything you have learned thus far in previous chapters, and the tutorial below to make the graph you want. After you make the graph you want, you can use `ggsave("figs/myVoles.png")` to save the graph into your `figs` folder.

6.6.2. Recoding "pairtype"

We can estimate the probability of something happening or not happening based on a series of successes and failures, like a coin coming up heads. A coin either comes up heads or it doesn't - success or failure. We can estimate the probability that it comes up heads ($P = 0.5$), even though we have only heads or tails, successes or failures.

We need to create a variable that represents either an extra-pair copulation or a nest-pair copulation. We will code extra-pair copulation as 1 (success) and nest-pair copulation (failure).

```
d209 <- d209 %>%
  mutate(
    epc = ifelse(pairtype=="extra", 1, 0)
  )
str(d209)
```

```
'data.frame':   250 obs. of  8 variables:
 $ id      : chr  "101" "101" "1021" "1021" ...
 $ sex     : chr  "male" "male" "female" "female" ...
 $ year    : int  2006 2006 2007 2006 2006 2007 2007 2006 2006 2006
  ↪  ...
 $ nest    : chr  "19/20/21" "19/20/21" "29" "32" ...
 $ size    : int  7 7 3 2 2 2 2 7 7 2 ...
 $ r       : num  0.378 -0.183 0.321 -0.109 -0.032 ...
 $ pairtype: chr  "nest" "nest" "nest" "extra" ...
 $ epc     : num  0 0 0 1 1 1 1 0 0 1 ...
```

Now we see that extra-pair copulation is coded as a number, just 0 or 1.

See below for an example of graphing zeros and ones on the y-axis, and a continuous variable on the x-axis, where we describe the probability that a car model has front wheel drive (y-axis), based its city gas mileage.

6.7. Graphing in `ggplot()`

Here is some general guidance for using `ggplot()`:

```
ggplot(data= myDataset, aes(x= X-variable, y= Y-variable)) +
geom_typeOfPlot()
```

- for a box-and-whisker plot: `geom_boxplot()` This is a more transparent and useful alternative to standard barplot or column graph.
- for a scatterplot `geom_point()`
- for a scatterplot with a fitted straight line `geom_point() + geom_smooth(method="lm")`

First, we load the packages of the tidyverse. If you already loaded this package as part of `tidyverse`, then you have already done this step.

```
library(tidyverse)
```

Here we use a data set (`mpg`) that comes with R to help demonstrate examples. This data set, `mpg` includes fuel economy data from 1999 to 2008 for 38 popular models of cars.

```
glimpse(mpg)
```

```
Rows: 234
Columns: 11
$ manufacturer <chr> "audi", "audi", "audi", "audi", "audi", "audi",
 ↪  "audi", "~
$ model        <chr> "a4", "a4", "a4", "a4", "a4", "a4", "a4", "a4
 ↪  quattro", "~
$ displ        <dbl> 1.8, 1.8, 2.0, 2.0, 2.8, 2.8, 3.1, 1.8, 1.8,
 ↪  2.0, 2.0, 2.~
$ year         <int> 1999, 1999, 2008, 2008, 1999, 1999, 2008, 1999,
 ↪  1999, 200~
$ cyl          <int> 4, 4, 4, 4, 6, 6, 6, 4, 4, 4, 4, 6, 6, 6, 6, 6,
 ↪  6, 8, 8, ~
$ trans        <chr> "auto(l5)", "manual(m5)", "manual(m6)",
 ↪  "auto(av)", "auto~
$ drv          <chr> "f", "f", "f", "f", "f", "f", "f", "4", "4",
 ↪  "4", "4", "4~
$ cty          <int> 18, 21, 20, 21, 16, 18, 18, 18, 16, 20, 19, 15,
 ↪  17, 17, 1~
$ hwy          <int> 29, 29, 31, 30, 26, 26, 27, 26, 25, 28, 27, 25,
 ↪  25, 25, 2~
$ fl           <chr> "p", "p", "p", "p", "p", "p", "p", "p", "p",
 ↪  "p", "p", "p~
$ class        <chr> "compact", "compact", "compact", "compact",
 ↪  "compact", "c~
```

For a box-and-whisker plot of highway mileage (`hwy`) vs. car type (`class`), with explicit labels.

```
ggplot(mpg, aes(x = class, y = hwy)) + geom_boxplot() +
  labs(x="type of car", y="highway mileage")
```

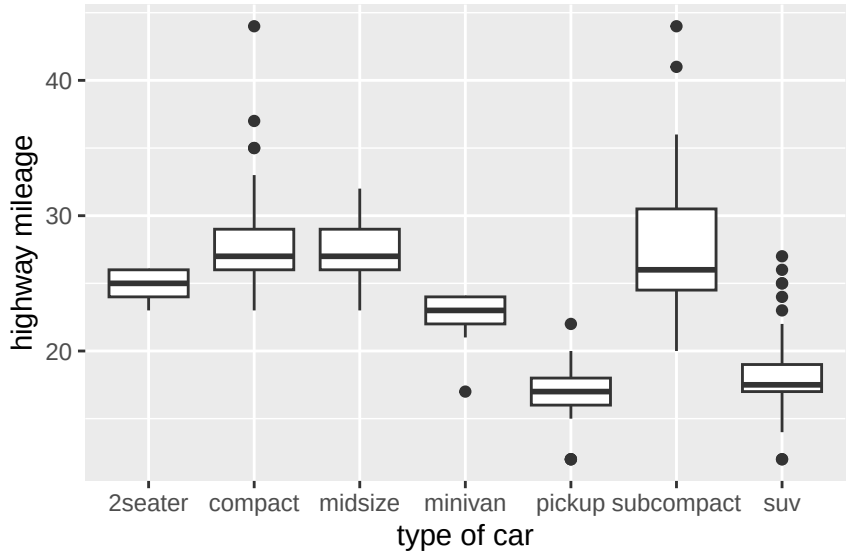

Figure 6.2.: Box-and-whisker plot of highway mileage (hwy) vs. car type (class), with explicit labels. A boxplot shows the distribution of data, by groups.

6.7.0.1. Scatterplots

```
ggplot(mpg, aes(x=cty, y=hwy)) + geom_point() +
  labs(x="city mileage", y="highway mileage")
```

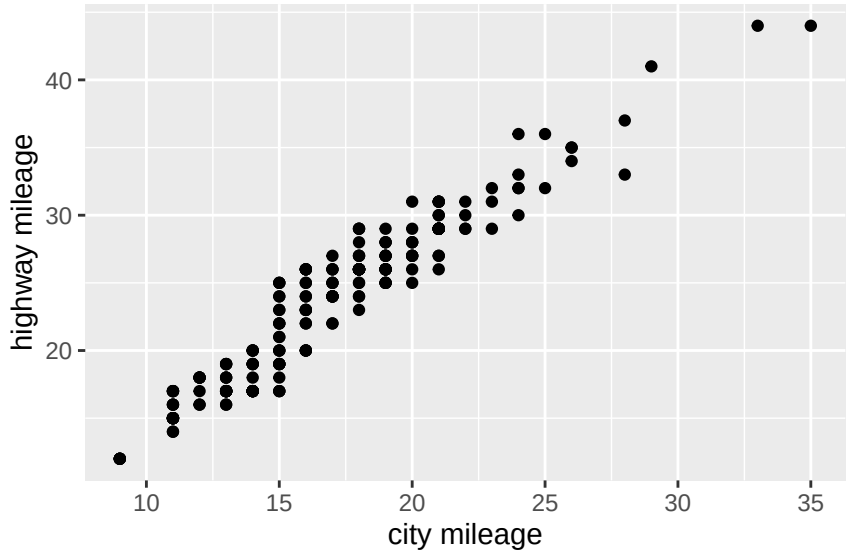

Figure 6.3.: Scatterplot of highway mileage vs. city mileage (miles per gal).

```
ggplot(mpg, aes(x=cty, y=hwy)) + geom_point() +
 ↪  geom_smooth(method="lm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

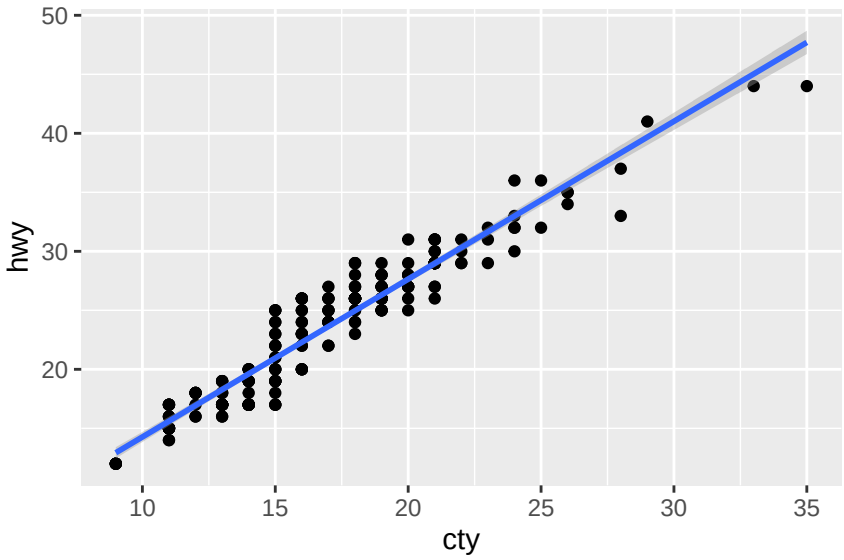

Figure 6.4.: Scatterplot with a fitted straight line, fitting a linear model.

```
## for a scatterplot with a fitted curved line
ggplot(mpg, aes(x=cty, y=hwy)) + geom_point() + geom_smooth()
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

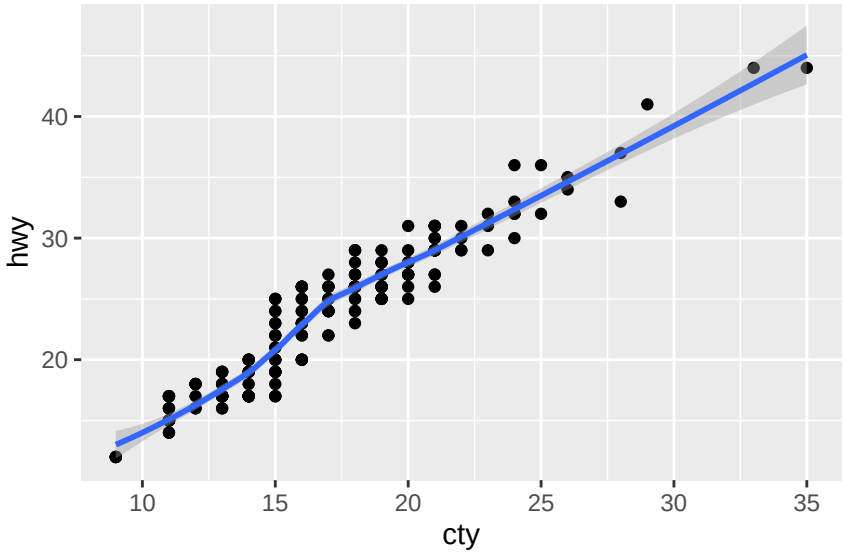

Figure 6.5.: Scatterplot with a flexibly fitted line, using local polynomial regression fitting.

6.7.0.2. Categorical responses

Often we would like to predict whether individuals fall into *one category or another*, such as coin flips of heads vs. tails, success vs. failure, sick vs. healthy, or engaging in extra-pair copulation vs. remaining monogamous. These kind of dichotomous data are called *binomial* data, and we think of these data as zeros and ones $(0, 1)$, perhaps zero for heads and one of tails.

When we model binomial data $(0,1)$, we put these 0s and 1s on the y-axis. Here is a worked example using `mpg` data.

First, create a dichotomous variable (0, 1) indicating whether a car model is front wheel drive or other (rear- or 4-wheel drive). A plot like this puts *probability* on the y-axis. In this case it is the probability that a car with a given mileage is a front-wheel drive car.

Convert a factor or character variable to zeros and ones. Here we convert the category of drive train (rear vs. front wheel drive) into ones for front wheel drive, and zeros for rear wheel drive. We use an "if-else" statement to do that.

```
mpg2 <- mpg %>%
  mutate(
    prob.front.wheel.drive = ifelse(drv=="f", 1, 0)
  )
```

Next we plot the zeros and ones on the y-axis against city mileage on the x-axis. We also add a particular kind of fitted line that shows us the expected probability that a car with a particular city mileage is front wheel drive. This curve is the result of a *logistic regression model* that we use for binomial data. The graph tells us that

```
ggplot(mpg2, aes(cty, prob.front.wheel.drive)) +
  geom_point() +
  geom_smooth(method="glm",
              method.args = list(family = "binomial"))
```

```
`geom_smooth()` using formula = 'y ~ x'
```

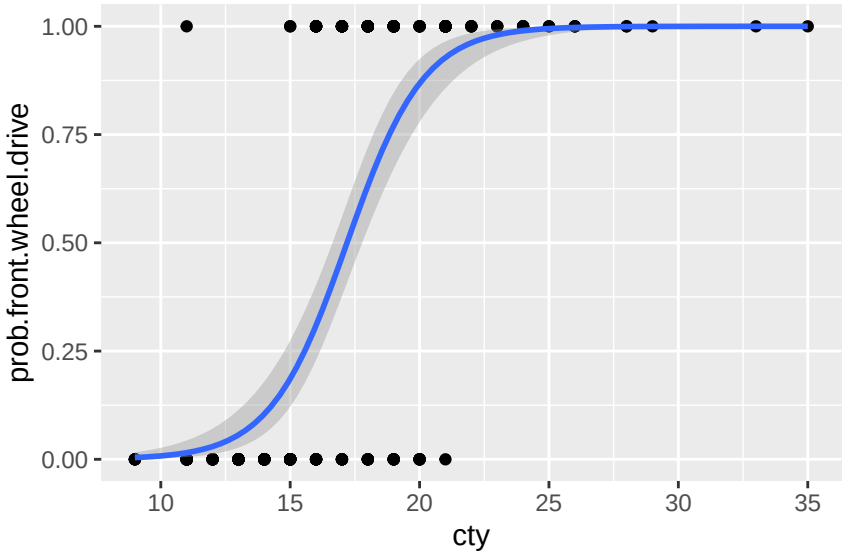

Figure 6.6.: Higher mileage cars tend to have front-wheel drive. Graph shows the probability that a car with a particular city mileage is front wheel drive. The curve is the result of a type of generalized linear model (glm) called *logistic regression* which assumes binomial data.

7. *Density-independent Population Growth: Humans on the rise*

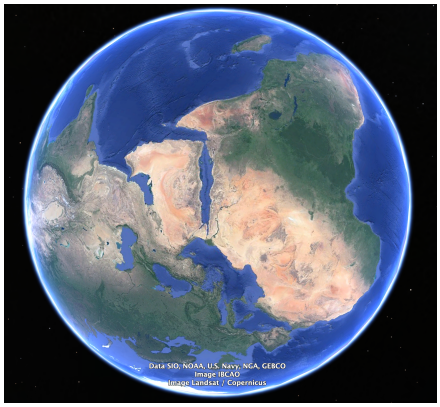

Figure 7.1.: Earth has now entered the Anthropocene. Geologists are debating precisely when that happened.

Reproduction helps define what we call "life", and leads to exponential growth. This chapter introduces this concept and how we quantify it.

Before you start

- **Assumptions about background and guidance:** your `Rwork` project and the ability to read a .csv file into R (Computer setup), and algebra with exponents and natural logarithms (e and ln). If logarithms are rusty, read Using logarithms in the metabolic ecology chapter first. The for-loop is introduced here, so no programming beyond your first script is assumed.

7.1. Goals

In this chapter, you will

- learn the two fundamental principles of population growth,
- do a little math: practice calculating rates of population growth,
- do a little programming: use a "for-loop" to do lots of work very easily,
- determine how the rate of human population growth has changed over time, and
- start to develop a greater understanding and intuition for exponential rates.

In this chapter, we'll use these techniques and data from *Our World in Data* to describe mathematically the growth rate of the world's human population at different points in time over the last 10,000 years.

7.2. Deliverables

Turn in two documents: (i) your R script, and (ii) a .docx or PDF formatted file containing the following:

1. Calculate by hand the exponential growth rate (r) for the years listed in Section 7.5.2. Write out the formula, substitute the values to show you know where they belong, and then finish with the arithmetic to find r.
2. A graph showing per capita growth rate for a time period of your own choosing, from Section 7.6.1.
3. If the human population continued to grow at the rate you calculated for 2020, how large would the population be in 2050? Calculate this and show your work.
4. What assumptions did you make in answering Question 3? Based on results for questions 1 and 2, is it likely that the human population will reach the size that you calculated for 2050? Why or why not? Use **evidence** in the data and the graphs above to justify your explanation.
5. How many people were alive when you were born? How many people will there be at the end of your lifetime? Are you going to live through a doubling of the human population? Why or why not?

7.3. Preparatory Steps

- Read Cohen (1995) in Perusall through our Canvas site. It is also available in our class archive: Cohen (1995)

1. Go to our class data repository, and download `HumanPopData-2023-0101.csv`. Make sure the file remains in the same simple format, "comma separated values" (.csv).

- If you are curious, check Deliverables, above.

7.4. Background

Given sufficient resources, life is autocatalytic – the products of reproduction (offspring) amplify the rate of reproduction, leading to *exponential growth*.

As ecologists, we can state this with the following propositions (Hastings 2011):

1. *A population grows exponentially in the absence of other forces*.
2. *There are forces that can prevent a population from growing exponentially*.

Hastings, A. 2011. *Single Species Population Dynamics and Its Theoretical Underpinnings*. Edited by S. M. Scheiner and M. R. Willig. University of Chicago Press.

This deceptively simple pair of propositions provides a basis for an enormous amount of ecology. They are analogous to Newton's First Law of Motion:

Every body perseveres in its state of being at rest or of moving uniformly straight forward, except insofar as it is compelled to change its state by forces impressed. (*The Principia: Mathematical Principles of Natural Philosophy* (1687), 3rd edition (1726), trans. I. Bernard Cohen and Anne Whitman (1999), Axioms, or Laws of Motion, Law 1, 416.)

Now we would say that *a body remains at rest, or in motion at a constant speed in a straight line, unless acted upon by a force.*

The first proposition of population growth can emerge as the consequence of the following assumptions:

a. all individuals in a population are identical.
b. there is no migration in or out of the population.
c. i. the number of offspring per individual (or the per capita birth and death rates) are constant through time, and (ii) independent of the number of individuals in the population.

Deviations from these assumptions lead to proposition 2, and most of the interesting parts of single species population dynamics (Hastings 2011).

Hastings, A. 2011. *Single Species Population Dynamics and Its Theoretical Underpinnings*. Edited by S. M. Scheiner and M. R. Willig. University of Chicago Press.

However, although propositions 1 and 2 can follow from assumptions a-c, these assumptions are not necessary. For instance, in some cases, individuals don't need to be identical, and we deal with a big exception in the next chapter where we introduce structured population growth. Also, migration is admissible, provided immigration = emigration and that it doesn't alter the rate of change. Nonetheless, other deviations from a and b can have very important consequences for single species population dynamics. We will get to those in later chapters.

Even when populations don't grow exponentially, we often describe population growth using the same mathematical approach. We do that for at least a few of reasons:

- approximations of more complicated dynamics often simplify to exponential growth under reasonable assumptions,
- over short periods of time, deviations to exponential growth still result in patterns that are very similar to exponential growth, and
- we typically don't know precisely what mechanisms limit exponential growth or by how much, and therefore exponential growth is as good a mathematical description as any.

7.4.1. *Population growth of* Homo sapiens*: a case study*

The human population shares many features with other populations. Like other species, humans are distributed unequally around our available habitats, we consume resources and produce waste products, and the size of our population changes over time.

Humans, as we would we recognize ourselves, have been around for ~300,000 years. Since that time a lot has happened to us, and we have done a lot to alter the planet. About 10,000 years ago, we started to use agriculture to raise a lot of our food, in the *agricultural revolution*. Oddly, this seems to have contributed simultaneously to both a decline in life span due to poor health, and yet also an increase in population growth rate and a corresponding increase in population size. The state of the human population is dynamic and undergoing transitions that are in different stages in different places around the world (Lee 2011).

Lee, Ronald. 2011. "The Outlook for Population Growth." *Science* 333 (6042): 569–73. https://doi.org/10.1126/science.1208859.

Here we examine changes in human population size from 10,000 y ago to 2021. The data are estimates, and we have annual estimates with increasing greater frequency, up until 2021.

We will calculate the density-independent per capita growth rate for pairs of years in the data set. We'll do this by written out by hand, and also in R, using code to do everything quickly, efficiently and, importantly, reproducibly, so that you–or anyone–could check your estimatation procedure.

We often use estimates of λ and r to quantify how rapidly a population is growing (or declining) at various points in time. For a population with overlapping generations and reproducing continuously, we can estimate the *per capita* exponential growth rate at a particular point in time (r_t) using the following equation:

$$r_t = \frac{\ln \frac{N_{t+h}}{N_t}}{(t+h) - t} \tag{7.1}$$

where N_t is the size of the population at some starting time point t, and N_{t+h} is the size of the population after some time interval h, and ln is the natural logarithm.

For example, imagine we that in 2020, we had 100 mice and exactly two years later, the population had grown to 150 mice. Here are the values that we would use to calculate r_t:

- $t = 2020$
- $t + h = 2022$
- $h = (t + h) - t = 2022 - 2020 = 2$
- $N_t = 100$
- $N_{t+h} = 150$

We would substitute these values into the above formula (#eq-r) as,

$$\begin{aligned} r_t &= \frac{\ln \frac{150}{100}}{2022 - 2020} \\ &= \frac{\ln 1.5}{2} \\ &= 0.2027 \end{aligned}$$

showing each step of our work. Whenever you need to show how you arrived at an answer (such as in homework, research, or in a job), you should do something like this.

When r_t is positive ($r_t > 0$), this means that the population was growing, and quantifies how *fast* it grew and became larger.

What if we reverse so the we started with $N_t = 150$ and the population declined to $N_{t+h} = 100$?

$$\begin{aligned} r_t &= \frac{\ln \frac{100}{150}}{2021 - 2020} \\ &= \frac{\ln \frac{2}{3}}{2} \\ &= -0.2027 \end{aligned}$$

ending with a negative value ($r_t < 0$), indicating that the population shrank. When $r_t < 0$, that means the population has decreased in size. Sometimes we call that "negative growth".

What do you think we we get if the population didn't change, so that $N_t = N_{t+h} = 100$?

7.4.2. Projecting population size into the future

If we make certain assumptions, we can make predictions about the future. In order to make predictions about the size of the population next year, or in 5, 10, or 20 years, we can extrapolate the current rate of change into the future. To do that, we can use the instantaneous rate of increase, r, and the amount of time, and a solution from calculus.

To predict the size of a population to year t from some starting time, we rely on the rule that,

$$N_t = N_0 e^{rt} \tag{7.2}$$

or equivalently,

$$N_{t+h} = N_t e^{rh}$$

where the parameter r is the per capita instantaneous rate of increase, and t is the amount of time, or time interval. The letter e is the exponential, or Euler's constant. The exponential is the base of the natural logarithm. **The important thing here is the time interval in the exponent (t or h) has to be the time interval from the starting value of N to the projected value of N.**

Figure 7.2.: Common duckweed (*Lemna minor*) populations can grow fast. Credit: https://commons.wikimedia.org/w/index.php?curid=1085476

Imagine we have a population of the small floating plant duckweed (*Lemna minor*), that can double in about two weeks . In that case, $r = 0.35$ new lemna per old lemna per week. Pretend we start with a population of 100 lemna individuals ($N_0 = 100$). How many lemna should we have in three weeks? Show your work (e.g., Figure 7.3).

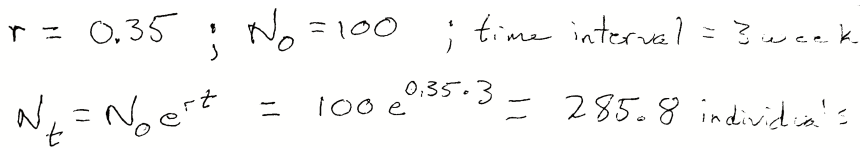

Figure 7.3.: Projecting the size of the duckweed population three weeks into the future.

> **Follow the Logic: Why does this equation exist?**
>
> Equation 7.2 is not a separate fact to memorize — it is a *logical consequence* of Equation 7.1. If you accept that the per capita growth rate r is constant (our assumption), then the projection equation *must* follow. The math does not add new biology; it reveals what our assumptions already imply.
> This is what mathematical reasoning does in ecology: it shows you the consequences you have already committed to when you stated your assumptions. If you change the assumptions — say, by letting r vary with population size — the projection equation changes too. The derivation tells you *why* the equation has the form it does, which means you can adapt it when the biology demands something different. Mathematical reasoning is a form of ecological understanding: following this derivation *is* one of the ways you come to understand how populations grow.

(**Overthinking:** Equation 7.2 is derived from the growth equation Equation 7.1 using integral calculus. It is the integral from 0 to t of Equation 7.1.)

7.5. Methods

1. Make sure your working directory is `Rwork`.
2. Load the necessary packages.

```
library(tidyverse) # for readr, dplyr and others
```

```
-- Attaching core tidyverse packages ------------------------
 ↪  tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------
 ↪  tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to
 ↪  force all conflicts to become errors
```

```
library(patchwork) # to combine ggplot2 figures
```

7.5.1. Load data

Next, we get the data, if you haven't already done so.

1. Go to our class data repository, and download `HumanPopData-2023-01-01.csv`. Make sure the file remains in the same simple format, "comma separated values" (.csv).
2. Put `HumanPopData-2023-01-01.csv` in your working directory, `Rwork`.
3. Open `HumanPopData-2023-01-01.csv` and make note of the first three lines of metadata. When you close the file, do not save any changes, and make sure *not* to save it in a new format, such as ".xlsx" or ".numbers".
4. Continue to make sure the file remains in the same simple format, "comma separated values" (.csv).

Now import the data into R, and *glimpse* the structure of the data frame.

```
d <- read_csv("data/HumanPopData-2023-01-01.csv",
              skip=3 # skip three lines of metadata
              )
```

```
Rows: 259 Columns: 2
-- Column specification
 ↪  --------------------------------------------------------
Delimiter: ","
dbl (2): Year, est.N

i Use `spec()` to retrieve the full column specification for this
 ↪  data.
i Specify the column types or set `show_col_types = FALSE` to quiet
 ↪  this message.
```

```
glimpse(d)
```

```
Rows: 259
Columns: 2
$ Year  <dbl> -10000, -9000, -8000, -7000, -6000, -5000, -4000,
 ↪  -3000, -2000, ~
$ est.N <dbl> 4432266, 5616996, 7242892, 9577918, 13201832,
 ↪  19075771, 28774661~
```

Note that years before the current era (BCE) are negative numbers, while numbers in the current era (CE) are positive.

Let's also graph the data to see if it makes sense.

```
d %>%
  ggplot(aes(x=Year, y=est.N)) +
  geom_point() + geom_line() +
  scale_y_log10() + labs(y="Estimated human population size")
```

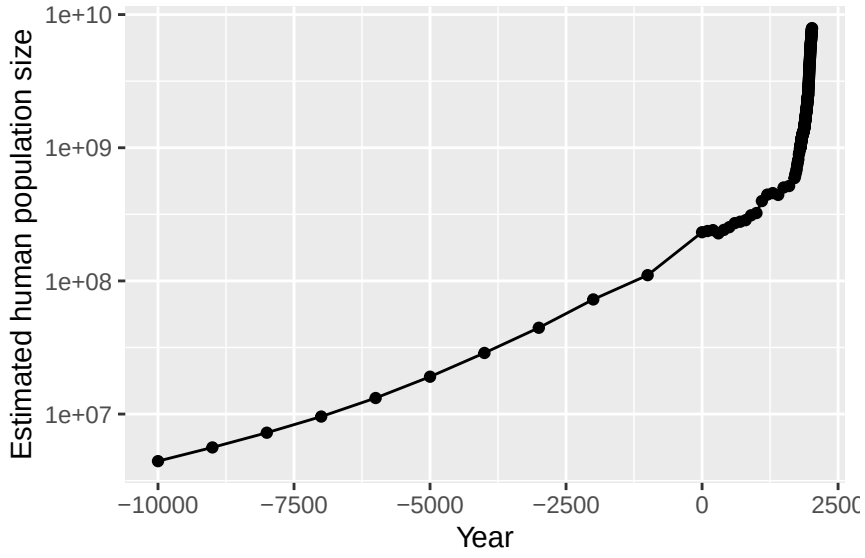

Figure 7.4.: The human population has increased over time. Note the common log scale on the y-axis. Note that years before the current era (BCE) are negative numbers, while numbers in the current era (CE) are positive.

7.5.2. Simple math: Calculations by hand

Doing math and arithmetic by hand is like learning music: you have to do it by hand, even when it feels like copying. This helps it become automatic, which leads to understanding.

Calculate by hand the instantaneous per capita growth for these intervals:

- 10000 to 9000 BCE (or -10000 to -9000),
- 0 to 100 CE, and
- 1960 to 1961 CE.

I recommend letting R help you, but you will need to write it out on paper. Write down these calculations with pen(cil) and paper, and take a picture of it. As with all math and word problems,

- start with a formula, in this case, (#eq-r),
- make the relevant substitutions,

- perform the calculation, and
- thus show your work.

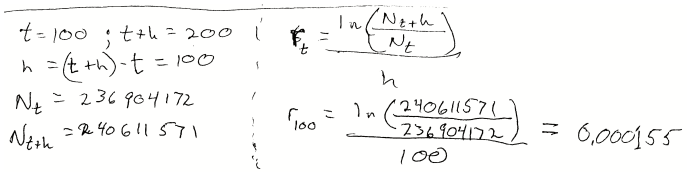

Figure 7.5.: Here I show my work to calculate r for the interval from 100 CE to 200 CE. I used my R script below to extract values from the data set and perform arithmetic.

I would use the following code to give me the data I would need for the interval for the years 100 CE to 200 CE

Specify the times and the interval you want.

```
## Get the right time points and find h
t <- 100
# recall h = (t+h) - t
t.plus.h <- 200
h  <- t.plus.h - t
t; t.plus.h; h
```

```
[1] 100
```

```
[1] 200
```

```
[1] 100
```

Grab the values of N, given the years you specified.

- Find a value of `est.N` when year equals 100
- Find a value of `est.N` when year equals 200

```
## in base R
Nt <- d$est.N[d$Year == t]
Nth <- d$est.N[d$Year == t.plus.h]
Nt; Nth
```

```
[1] 236904172
```

```
[1] 240611571
```

Overthinking: You could use the `tidyverse` syntax to get the same values.

```
## using tidyverse syntax
Nt <-  d %>% # use d
  filter(Year == t) %>%  # use just values in this year
   select(est.N) %>% # select just N
  as.numeric() # convert from a data frame to a numeric variable

Nth <-  d %>% # use d
  filter(Year == t.plus.h) %>%  # ue just values in this year
   select(est.N) %>% # select just N
  as.numeric() # convert from a data frame to a numeric variable
Nt; Nth
```

```
[1] 236904172
```

```
[1] 240611571
```

Do calculations to find r, the instantaneous per capita population growth rate. Note that in R `log()` refers to the natural logarithm, while `log10()` refers to the common logarithm. For more information, type `?log`.

```
# the natural log of the ratio of population sizes
logNth.over.Nt <- log(Nth / Nt)
logNth.over.Nt
```

```
[1] 0.01552817
```

Last, we find r for this particular time step of 1000 y.

```
# finding r
r <- logNth.over.Nt / h
r
```

```
[1] 0.0001552817
```

You can use R as a calculator this way to make your pen and paper work easier.

7.5.3. Programming: Use a for-loop to do all the calculations

A benefit of programming is that we can create a statement to do all our work for use. A "for-loop" is a programming trick to do a repeated process, for instance, *for* each row of a dataset.

Below is code to calculate every value of r in the entire data set, except the last year. We have to exclude the last year because we don't know how fast the population grew from the last year to the subsequent year.

1. First we create an empty vector into which we will put our calculations of r, one at a time.
2. Next, we calculate r for each year, except the last.
3. Finally, we combine our vector of r_t with the original data.

Here we, create an empty vector of missing values. To start, this will have the same number of rows as our data set, `d`.

```
# create an empty vector of missing values.
r <- rep(NA, nrow(d))
```

Now we calculate r **for** each row of data except the last year. Here the letter i stands in for the number of each row.

```
# Do the following steps FOR each row of our data.
# Note we start with row 2, and use row 2 and row 1....
for(i in 2:nrow(d)) {
  # the second, more recent, year
  t.plus.h <- d$Year[i]
  # the previous year
  t <- d$Year[i-1]

  # the time interval in years
  h <- t.plus.h - t

  # the more recent N
  Nth <- d$est.N[i]
  # the previous N
  Nt <- d$est.N[i-1]

  # the natural log ratio
  logNth.over.Nt <- log(Nth/Nt)

  # r, which we put in the appropriate slot, year t
  r[i-1] <- logNth.over.Nt/h

}

# last, let's check the result
glimpse(r)
```

```
 num [1:259] 0.000237 0.000254 0.000279 0.000321 0.000368 ...
```

Finally, we combine this with our original data set. We use `cbind()` which stands for "bind columns".

```
d.r <- cbind(d, r=r)
```

If you examine the last three rows of these data, you'll see that we don't have an estimate of r_t for the last year because we can't calculate it without know the next year's population size.

```
d.r[257:259,]
```

```
    Year      est.N           r
257 2019 7764951040 0.009740210
258 2020 7840952832 0.008678302
259 2021 7909295104          NA
```

From here on out, **we are interested mostly with the growth rate**, r_t, so I will remove the last row of the data set, or keep only the first 258 rows.

```
d.r <- d.r[1:258,]
```

7.6. Changes in human population growth rate

Let's graph our data to examine how the *per capita* growth rate of humans changed since the agricultural revolution. I'll pair them with population size.

```
# population size, using the first data frame
ggplot(data=d.r, aes(x=Year, y=r)) + geom_line() +
  geom_point() + labs(title="Per capita growth rate")
```

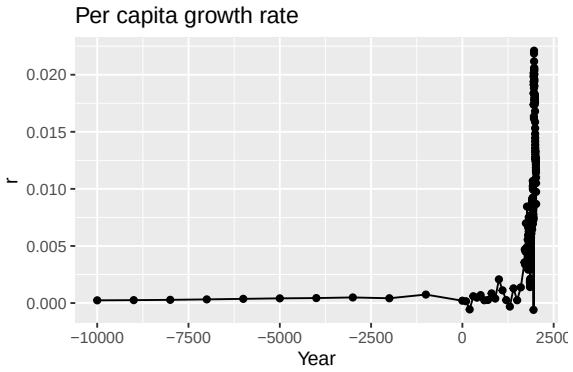

Figure 7.6.: Estimates of per capita growth are positive for nearly every time interval since the agricultural revolution.

This shows us the growth rate didn't change much before the current era. However, it is hard to see what has been going on more recently. Therefore, let's plot only the data starting in year 1.

```
# filter in all years greater than or equal to 1
d.r.y1 <- d.r %>% filter(Year >= 1)

ggplot(data=d.r.y1, aes(x=Year, y=est.N)) + geom_line() +
  geom_point() + labs(title="Population Size")
```

```
ggplot(data=d.r.y1, aes(x=Year, y=r)) + geom_line() +
  geom_point() + labs(title="Per capita growth rate since 1 CE")
```

7.6.1. Your own interval

Make a graph of per capita growth rate for a time period of your own choosing. Select a time that is different than any other that we have shown here, and make it at least 10 years

7.6.2. How does per capita growth rate change with population size?

Large populations differ from small populations in a lot of ways. Large populations might make more waste products and do more damage to their

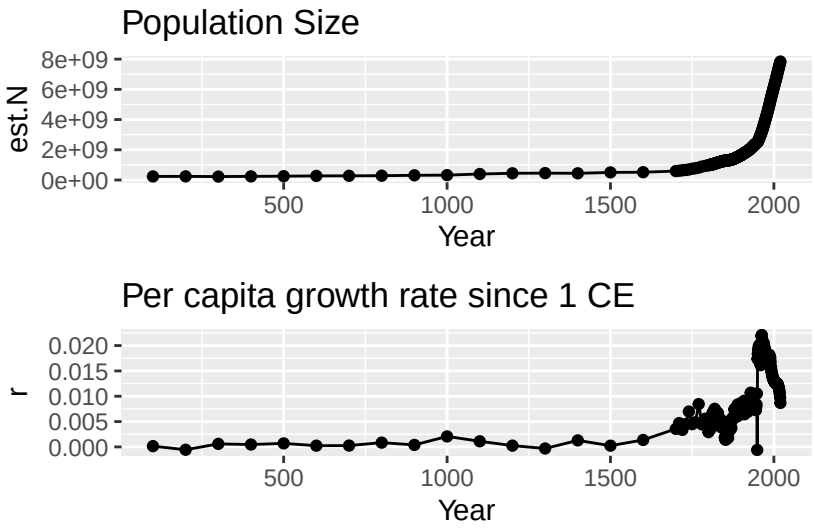

Figure 7.7.: In the current era, per capita growth remains mostly positive, and then rises sharply and begin to fluctuate dramatically after 1500.

own environment, and they also foster more interactions among individuals that might have positive or negative consequences.

Let's make a graph of per capita growth rate as a function of population size.

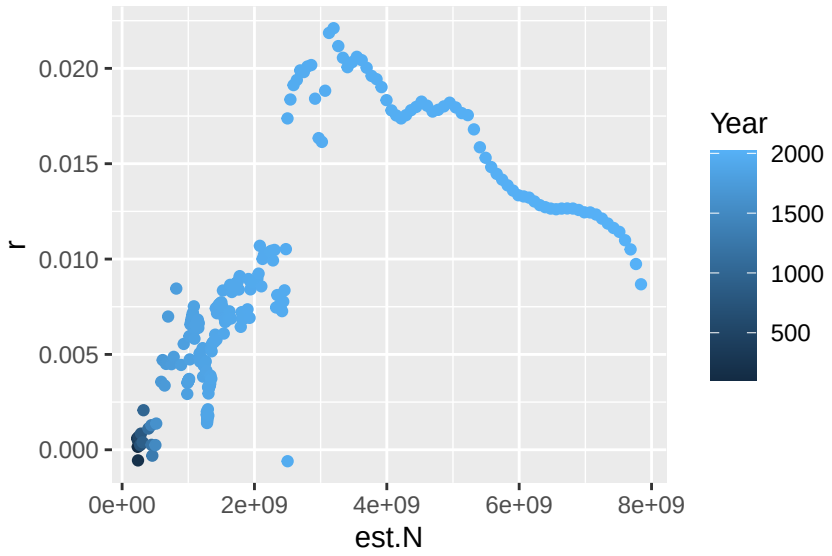

Figure 7.8.: Per capita growth rate vs. population size, since 1 CE after 1500. What do you make of this? :)

7.7. Projecting into the future

Here we use our projection equation from above ($N_t = N_0 e^{rt}$) to make predictions about the future. Imagine a population of 100 individuals and $r = 0.02$ per year, and predict how big that population would become after thirty years. Assume that r is constant over that time period.

$$N_t = N_0 e^{rt} \tag{7.3}$$
$$= 100e^{0.02 \cdot 30} = 182.21 \tag{7.4}$$

When we take this approach, we are assuming that the population *continues* to grow at this rate over the projected time period. That can be a reasonable assumption over short time periods, but not over long time periods. Of course, "short" and "long" are both relative, and depend heavily of the context.

What can we learn from (Figure 7.8)? Over what time periods would it seem reasonable to assume that the human population grows at a constant rate?

8. Structured demography: How a graduate student helped save the loggerhead sea turtle

Figure 8.1.: The loggerhead sea turtle is listed as "vulnerable" by the IUCN 3.1. image: By ukanda - originally posted to Flickr as Loggerhead turtle, CC BY 2.0, https://commons.wikimedia.org/w/index.php?curid=10481668

Before you start

- **Assumptions about background and guidance:** the idea of exponential growth and a population growth rate from Density-independent population growth, and the ability to run a script in your `Rwork` project (Computer setup). No matrix algebra is assumed: R does the eigenanalysis, and this chapter explains what the result means. You will need Table 3 of Crouse et al. (1987).

8.1. Goals

In this chapter, you'll connect sea turtle natural history to a demographic model of population growth and learn how we all learned about how to help sea turtles. Along the way, you'll learn about sea turtle natural history, demography, dynamics and model sensitivity.

If you successfully complete this chapter, you will

- discover key stages in a sea turtles life,
- describe *population structure*,
- use natural history information to parameterize a mathematical model to make predictions about sea turtles,
- distinguish between *transient dynamics* and long-term *asymptotic dynamics*, and
- recommend management options for sea turtles conservation.

8.2. Deliverables

Put the following into a single document:

1. A life cycle graph for the loggerhead population. On your figure, include the numerical elements associated with each transition, based on the original matrix.
2. A table similar to Table 7.1 with all the stages, the provided values of stasis, and growth, and your new values of lambda.
3. Assume you want to maximize long-term population growth rate (λ). For which stage or stages should you invest in protection measures? Explain why, and use your values of lambda as evidence in your answer.

Upload this document and your R script to the appropriate site.

8.3. Preparatory steps

- Read Crouse et al. (1987) in Perusall through our Canvas site. It is also available in our class archive: Cohen (1995)
- This chapter does not use a separate data set, but it does use demographic natural history information in Crouse et al. (1987).
- If you are curious, check Deliverables, above (Section 8.2).

8.4. Demography

Deomgraphy includes the structure of a population which is the relative numbers of individuals of different ages or stages. Deomgraphy also includes the rates that control that structure which are primarily age- and stage-specific birth rates, death rates, and survival. We use this detailed information to make informed predictions about the population's future state, including its overall size and its structure. Therefore, demography also includes the overall rates, including population growth, that emerge from all the detailed age/stage-specific information. The study of human demography helps nations make informed policy decisions that take into account how societies will change over time due to populations that are growing, shrinking, aging or getting younger on average (Lee 2011). The

Lee, Ronald. 2011. "The Outlook for Population Growth." *Science* 333 (6042): 569–73. https://doi.org/10.1126/science.1208859.

study of many rare, threatened or vulnerable species includes *demography* because demography helps us understand and manage populations (Caswell 2001).

Caswell, H. 2001. *Matrix Population Models: Construction, Analysis, and Interpretation*. 2nd ed. Sinauer Associates, Inc., Sunderland, MA, USA.

8.5. Loggerhead sea turtles

The loggerhead sea turtle (*Caretta caretta*) is a widely distributed (i.e. cosmopolitan) species that lives in oceans around the world. The International Union for Conservation of Nature (IUCN, https://www.iucnredlist.org/) lists it as "vulnerable" to extinction, which, sadly, is pretty good these days. The fact that the species is not worse off is in part due to research like that of Crouse et al. (1987). This research by Deborah Crouse, Hal Caswell, and many others is the subject of what you will work with in this assignment.

Crouse, D T, L B Crowder, and H Caswell. 1987. "A Stage-Based Population Model for Loggerhead Sea Turtles and Implications for Conservation." *Ecology* 68: 1412–23.

Sea turtles live in warm oceans around the world and reproduce on sandy beaches. Females return to the same beaches most years to lay eggs in clutches. They come ashore, dig a hole in the sand, lay 10-100 eggs, and cover the nest, often disguising it from terrestrial predators who would like to dig them up to eat them. The female then returns to the sea to feed and ovoid enemies. If the everything goes right, the eggs hatch after incubation, hatchlings emerge together, struggle to the surf, and attempt to avoid the many predators (e.g., gulls, raccoons, crabs, people). Sea turtle hatchlings face many threats, including from humans who eat the eggs, and trucks or off-road vehicles can destroy the nests. In addition, lighting from nearby streets and homes can confuse hatchlings as they may crawl toward the lights as opposed to heading out to sea.

8.5.1. Questions about loggerhead survival

Include answers to these in Deliverables.

1. Consult Table 3 (Crouse et al. 1987).

a. Which two stages have the lowest annual survivorship?
b. Which stages have the highest annual survivorship?

1. Why do you suppose that 1st year remigrants have such low fecundity?

Fishing trawlers, which drag nets for long distances to catch shrimp and fish create another source of mortality in juveniles and adults. The sea turtles get caught in these nets and drown. These dead turtles are referred to as "by-catch", that is, animals killed unintentionally while fishing for other species.

Sea turtle populations have been on the decline for well over a century, and some species are more endangered than others. One common conservation strategy has been to protect the eggs and hatchlings. More recently, some countries have passed laws requiring fishing trawlers (fishing boats dragging huge nets for hours at a time) to use turtle excluders devices (TEDs). These spring-loaded devices fit at the end of a trawler net and will open

up when they receive a big bump from a heavy sea turtle. This allows the turtle to escape, and the fishermen lose only a fraction of their catch.

No one really knew the value of TEDs until Deborah Crouse and others modeled sea turtle populations and showed clearly that (1) saving eggs and hatchlings was not enough, and (2) increasing survival of large turtles (large juveniles and adults) was required to save the turtles from continued declines and extinction.

8.6. Demographic modeling

We use a square matrix to summarize a species' age- or stage-specific demographic rates. A *matrix* is mathematical array of numbers with specified rows and columns. The *dimension* of a matrix is its number of rows and number of columns. A matrix with the same number rows as columns is called a *square matrix*.

The effect of a matrix element on long term asymptotic growth rate (lambda) is known as sensitivity or elasticity, depending on the details of the calculations. There are abstract ways to calculate those using linear algebra, but here we do it the easy and transparent way. We will change annual survivals of particular stages, and then recalculate lambda and r to examine the effect directly.

Crouse, D T, L B Crowder, and H Caswell. 1987. "A Stage-Based Population Model for Loggerhead Sea Turtles and Implications for Conservation." *Ecology* 68: 1412–23.

Crouse et al. (1987) performed a demographic analysis of an endangered sea turtle species, the loggerhead (*Caretta caretta*). Management of loggerhead populations seemed essential for their long term survival, and a popular management strategy had been and still is to protect nesting females, eggs, and hatchlings. The ground breaking work by Crouse[2] and her colleagues compiled data to create a stage-based projection matrix to analyze quantitatively which stages are most important and least important in influencing long-term growth rate. This work led to US Federal laws requiring that US shrimp fishermen use nets that include Turtle Excluder Devices (TEDs).

[2] Crouse was a graduate student at the time—graduate students are the life-blood of modern science, doing cutting edge work and pushing their fields forward.

A *demographic transition matrix* is merely a compact quantitative summary of a species' life history events. It includes the annual probabilities that an individual remains in a particular stage, grows into a later stage, or even regresses back to an "earlier" stage. It also includes the fertility or number of progeny it contributes to the first stage. These stages may be age classes (e.g., 0-5 y), sizes (e.g., 2-4 leaves), or some other major life history stage (e.g., first year breeder).

The transition matrix holds each of the probabilities and fertilities in a way that the column identifies the stage at time t whereas the row identifies the next time period $t + 1$.

In Figure 8.2, the element s_2 is the probability that a juvenile in year 1 survives to remain a juvenile in year 2. The element g_2 represents the probability that a juvenile in year 2 survives and also become a reproductively mature adult in year 2. The element g_1 is the probability that a hatchling

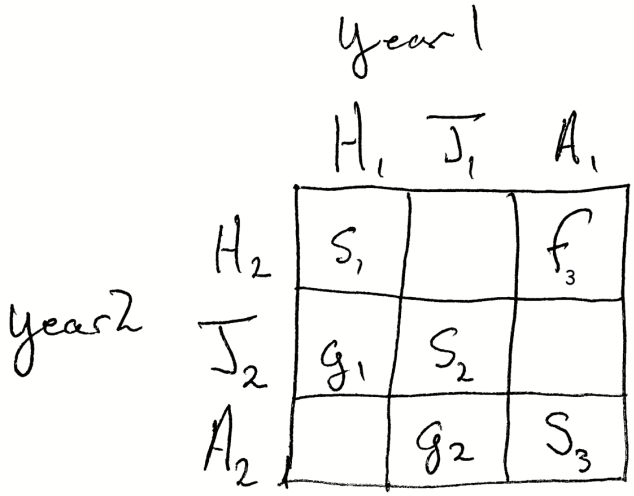

Figure 8.2.: A toy demographic transition matrix, with stages of hatchlings (H), juveniles (J), and adults (A). Elements represent the probability that an individual belonging to the stage in the column stays in or transitions to the stage in the row over a single year. Fecundity (f3) represents the total number of offspring that an individual contributes to the stage in that row.

survives and becomes a juvenile in year 2. The element s_3 is the probability that an adult survives and remains an adult. We refer to the elements on the diagonal (s) as **stasis** and the elements on the off-diagonal (g) as **growth**.

Fecundity or reproduction is seen as adults contributing the to smallest or youngest stage. The element f_3 is the number of hatchlings that occur in year 2 based on the survival and reproduction of the average adult. The element s_1 is the survival of hatchlings. If the stage only exists for one year, then $s_1 = 0$ because, by definition, after one year a hatching is a juvenile, or it is dead.

8.6.1. Connecting life history information to a model

Crouse et al. (1987) gathered data from a variety of sources to estimate loggerhead turtle survival and fertility at different ages and stages. They organized these data in the form of a transition matrix, **A**. They then used this model to evaluate the relative importance of different stage transitions to the population gorwth rate.

Crouse, D T, L B Crowder, and H Caswell. 1987. "A Stage-Based Population Model for Loggerhead Sea Turtles and Implications for Conservation." *Ecology* 68: 1412–23.

On the basis of their analysis, they were able to identify that the survival of juveniles and subadults was the most important determinant of turtle growth rate, under the current circumstances. It was this insight, made possible only because of astructure population model, that led to changes in laws to require the use of turtle excluder devices (TEDs) in fishing and shrimpo trawlers.

In this assignment, you will walk through the types of analyses done by Deborah Crouse and her colleagues. They combed the literature to find information about survival and reproduction by loggerheads and summarized those findings in their tables 1 and 2. They then organized that information into a more con densed form and presented it in Table 3, shown here in Figure 8.3.

Their transition matrix (Table 4):

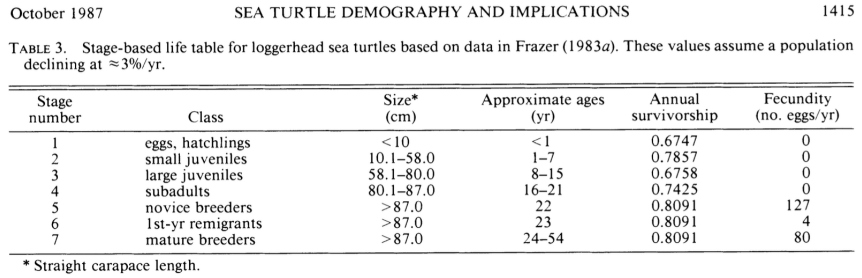

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TABLE 3. Stage-based life table for loggerhead sea turtles based on data in Frazer (1983*a*). These values assume a population declining at ≈3%/yr.

Stage number	Class	Size* (cm)	Approximate ages (yr)	Annual survivorship	Fecundity (no. eggs/yr)
1	eggs, hatchlings	<10	<1	0.6747	0
2	small juveniles	10.1–58.0	1–7	0.7857	0
3	large juveniles	58.1–80.0	8–15	0.6758	0
4	subadults	80.1–87.0	16–21	0.7425	0
5	novice breeders	>87.0	22	0.8091	127
6	1st-yr remigrants	>87.0	23	0.8091	4
7	mature breeders	>87.0	24–54	0.8091	80

* Straight carapace length.

Figure 8.3.: Estimates of stage-specific survival and fecundity for loggerhead sea turtles, from Table 3 in Crouse et al. (1987). Based on information described in Tables 1 and 2 in Crouse et al. (1987).

$$
\begin{array}{c} \\ H \\ J_s \\ J_l \\ sub \\ B_1 \\ B_2 \\ M \end{array}
\begin{pmatrix}
H & J_s & J_l & sub & B_1 & B_2 & M \\
0 & 0 & 0 & 0 & 127 & 4 & 80 \\
0.6747 & 0.7370 & 0 & 0 & 0 & 0 & 0 \\
0 & 0.0486 & 0.6610 & 0 & 0 & 0 & 0 \\
0 & 0 & 0.0147 & 0.6907 & 0 & 0 & 0 \\
0 & 0 & 0 & 0.0518 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0.8091 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0.8091 & 0.8089
\end{pmatrix}
$$

8.6.2. Connecting life histories to demographic matrices

1. Compare Table 3 and Table 4 (above). What are the relationships between survival and fecundity in Table 3, and the diagonal (stasis) and off-diagonal elements (growth) of the matrix in Table 4? How are they related?
2. Why do you think that the diagonal matrix elements (Table 4) not the same are the annual survivorship in Table 3?

To use the information in Table 3 to make predictions about the long-term viability of the population and to make recommendations for the conservation of the population, Crouse and colleagues need to combine these stage-specific demographic rates into a model, and summarize the behavior of that model.

Their first step was to put the above rates into a demographic transition matrix, where columns represent stages in one year, and the rows represent stages in the next year.

8.7. Analysis of a demographic matrix

As we try to understand and appreciate the meaning of sea turtle life history, demography, and survival, we will focus on three characteristics:

- **population growth rate**, λ, is the long term *finite rate of increase* and assumes density-independent growth.
- **stable stage distribution** is the set of expected relative densities of the different stages.

- **elasticies** are the relative importances of the different transitions to determining the finite rate of increase, λ.

We will focus on lambda, and do some experiments playing around with survival and growth.

8.7.1. Population growth rate, λ

Lambda, λ, is the long term *finite rate of increase*.

- It assumes density-independent growth.
- It is a long term prediction which assumes the *population structure* will be constant.
- It is the proportional change in the size of the population from one year to the next.
  - $\lambda = 1.10$ means the population will grow by 10%.
  - $\lambda = 0.95$ means the population will shrink by 5%.

As with all of our work, confirm you are in the correct working directory.

```
getwd()
```

```
[1] "/Users/stevenmh/projects/intro_to_ecology"
```

Fix this, if necessary.

Load needed packages.

```
## for graphics and data manipulation
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------
 ↪  tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------
 ↪  tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to
 ↪  force all conflicts to become errors
```

Two of the techniques we use to analyze a demographic matrix are **simulation**, and a mathematical technique called **eigenanalysis**. Simulation allows us to reveal the dynamics through time, year after year, generation after generation. Eigenanalysis allows us to very quickly quantify average characteristics of the population, including the population growth rate, λ, the stable stage distribution, and the elasticities.

First, we need to create the observed transition matrix.

```
A <- matrix(
  c(0,      0,      0,      0,    127,      4,    80,
    0.6747, 0.7370, 0,      0,      0,      0,     0,
    0,      0.0486, 0.6610, 0,      0,      0,     0,
    0,      0,      0.0147, 0.6907, 0,      0,     0,
    0,      0,      0,      0.0518, 0,      0,      0,
    0,      0,      0,      0,      0.8091, 0,      0,
    0,      0,      0,      0,      0,      0.8091, 0.8089),
nrow=7, ncol=7, byrow=TRUE)
rownames(A) <- colnames(A) <- c("H","J_s","J_l", "sub", "B_1",
 ↪  "B_2", "M")
A
```

```
         H    J_s    J_l    sub      B_1    B_2       M
H   0.0000 0.0000 0.0000 0.0000 127.0000 4.0000 80.0000
J_s 0.6747 0.7370 0.0000 0.0000   0.0000 0.0000  0.0000
J_l 0.0000 0.0486 0.6610 0.0000   0.0000 0.0000  0.0000
sub 0.0000 0.0000 0.0147 0.6907   0.0000 0.0000  0.0000
B_1 0.0000 0.0000 0.0000 0.0518   0.0000 0.0000  0.0000
B_2 0.0000 0.0000 0.0000 0.0000   0.8091 0.0000  0.0000
M   0.0000 0.0000 0.0000 0.0000   0.0000 0.8091  0.8089
```

Then we can calculate lambda. Remember that lambda is just the ratio of population sizes in sequential years ($\lambda = N_{t+1}/N_t$).

```
original_lambda <- Re( eigen(A)$values[1] )
original_lambda
```

```
[1] 0.945031
```

This value, $\lambda = 0.94$, is less than $\lambda = 1$, and therefore indicates a shrinking population. **This shows that the loggerhead sea turtle was predicted to decline.**

What can we do?

8.8. Finding the life history stage that could make the biggest difference

Part of managing rare species is figuring out which part of the population, the age, or **the stage on which to concentrate your efforts.**

A good way to figure out where to focus our effort is to find out which stage could make the biggest difference to the growth rate, λ. If we improve survival of an stage and it helps λ increase above 1, focusing on that stage could be a good strategy.

Your strategy will be to improve the survival in each stage so that individuals in that stage do not die. They either survive and remain in that stage, or they grow into the next stage. That means that with perfect survival, stasis and growth sum to nearly one.

For each stage, we will pretend that we can improve annual survivale 99.9%. Some of our stages are only one year in duration, while others are 7, 8, or 30 years in duration. This makes it a little tricky to calculate our new simulated matrix elements. I will do that part for you.

In each of seven simulations, you will test the effect on growth rate (λ) of improved survival.

8.8.1. Helping hatchlings

For your first experiment, you will test the effect on lambda of improving survival and growth of eggs and hatchlings to 99.9%, or annual survivorship of 0.999. Here are the steps:

1. Copy the old matrix and rename it (see below!).
2. Change the new matrix to include survivorship of 0.999 for the hatchlings (see below!).
3. Calculate lambda for the new matrix (see below).

```
## new matrix
A1 <- A

## Change just the hatchling survival/growth
A1[2,1] <- 0.999
## look at the new matrix
A1
```

```
        H    J_s    J_l    sub      B_1    B_2       M
H   0.000 0.0000 0.0000 0.0000 127.0000 4.0000 80.0000
J_s 0.999 0.7370 0.0000 0.0000   0.0000 0.0000  0.0000
J_l 0.000 0.0486 0.6610 0.0000   0.0000 0.0000  0.0000
sub 0.000 0.0000 0.0147 0.6907   0.0000 0.0000  0.0000
B_1 0.000 0.0000 0.0000 0.0518   0.0000 0.0000  0.0000
B_2 0.000 0.0000 0.0000 0.0000   0.8091 0.0000  0.0000
M   0.000 0.0000 0.0000 0.0000   0.0000 0.8091  0.8089
```

In our new matrix, we see that 99.9% of hatchlings (first column, H) survive to grow into being small juveniles that next year (second row, J_s), that is, $a_{2,1} = 0.999$.

Now calculate lambda of the new matrix.

```
## using Re() extracts just the real part of the eigenvalue, and not
 ↪  the imaginary part
Re( eigen(A1)$values[1] )
```

```
[1] 0.964903
```

This is the new growth rate. Is it greater than 1? Does it suggest that improving hatchling survival will allow this turtle population to grow? How does this compare to the original value of lambda of 0.94? That is, *how much did saving all the hatchlings help the loggerhead population?* You'll need this for deliverables.

8.8.2. Helping small juveniles

Here you will test the effect on lambda of improving survival and growth of small juveniles to 99.9%, or annual survivorship of 0.999. This stage lasts seven years, and so I calculated the new matrix elements for you (see the code).

Here are the steps:

```
## new matrix with NEW NAME, A2
## make sure you start with the ORIGINAL matrix A
A2 <- A

## Change just the stasis and growth for small juveniles
A2[2,2] <- 0.857 # stasis
A2[3,2] <-  0.142 # growth
A2
```

```
         H   J_s    J_l    sub      B_1    B_2       M
H   0.0000 0.000 0.0000 0.0000 127.0000 4.0000 80.0000
J_s 0.6747 0.857 0.0000 0.0000   0.0000 0.0000  0.0000
J_l 0.0000 0.142 0.6610 0.0000   0.0000 0.0000  0.0000
sub 0.0000 0.000 0.0147 0.6907   0.0000 0.0000  0.0000
B_1 0.0000 0.000 0.0000 0.0518   0.0000 0.0000  0.0000
B_2 0.0000 0.000 0.0000 0.0000   0.8091 0.0000  0.0000
M   0.0000 0.000 0.0000 0.0000   0.0000 0.8091  0.8089
```

Now that you have a new matrix, calculate the lambda for this matrix. Do that now by borrowing and editing code from above.

Now calculate the finite rate of increase (λ).

```
Re( eigen(A2)$values[1] )
```

```
[1] 1.03806
```

What does this value of lambda tell you about the effect of increasing survival of small juveniles (if we could do that)?

8.9. Simulating helping other stages

Repeat a similar procedure for each stage, using these values:

Table 8.1.: New values to use in the loggerhead sea turtle transition matrix, changing only one stage at a time. These changes reflect 99.9% annual survival. Stasis values belong on the diagonal, such as `A[5,5] <- 0`, and growth values belong below the diagonal, such as `A[6,5]<-0.999`. Start with the original matrix, make a copy, and then change only one stage at a time.

Stage	Stasis	Growth
H	0.000	0.999
J_s	0.857	0.142
J_l	0.875	0.124
sub	0.833	0.166
B_1	0.000	0.999
B_2	0.000	0.999
M	0.966	0.000

Record the new value of lambda for each separate simulation.

9. *Density-dependence in population growth*

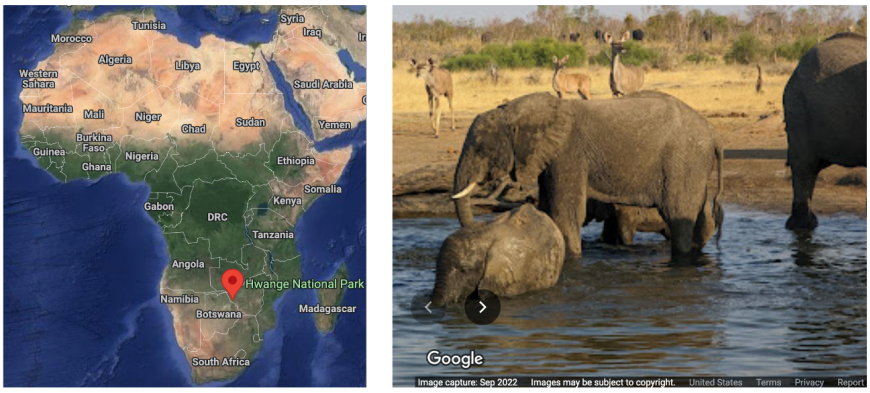


Figure 9.1.: Elephant carrying capacity in Hwange National Park, Zimbabwe, depends strongly on annual rainfall (Chamaillé-Jammes 2008).

Before you start

- **Assumptions about background and guidance:** exponential growth ($dN/dt = rN$) from Density-independent population growth, and comfort reading an equation and rearranging it with algebra. The dN/dt notation is explained here, and you will not need to do any calculus. The for-loop in Assignment 2 works just like the one in Density-independent population growth.

9.1. Goals

In this chapter, you'll learn how we represent crowding in populations. Per capita population growth rate often depends on population density. Along the way, you'll learn we take ideas seriously, and follow them to their logical conclusions.

If you successfully complete this chapter, you will

- extend exponential growth to include the effect of crowding,
- define *density-dependence*,
- distinguish between *transient dynamics* and long-term *asymptotic dynamics*, and
- discover the consequences of crowding by *solving* a rate equation.

9.2. Deliverables

Put your work from section 8.3 (Assignment 1) and section 8.4 (Assignment 2) into one document and turn in by the due date.

9.3. Preparatory steps

- Watch the lecture videos on density-dependent population growth (parts 1 & 2). These videos assume that you watched the preceding population ecology videos.
- Download Chamaille-Jammes (2008) from our article archive
- If you are curious, check Deliverables for this chapter(Section 9.2).

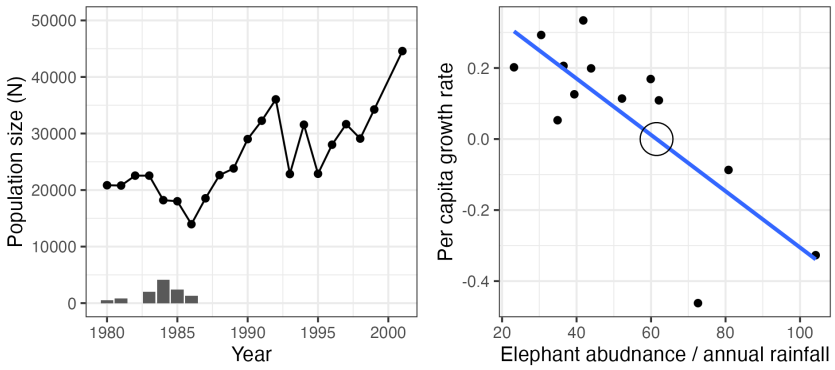


Figure 9.2.: The Hwange elephant population grew dramatically after the end of killing elephants ("culling") that had kept the population size smaller (Chamaillé-Jammes 2008). The carrying capacity (where $r = 0$) appears to depend on the amount of annual rainfall.

9.4. Models as assumptions

Continuous logistic population growth is just a set of assumptions about how a population grows. The rate of change in our population is determined by its birth rate and death rate, and how population size affects these.

The symbol dN/dt means "change in population size (N) per unit change in time (t)".

Recall exponential growth (J-shaped growth) is

$$\frac{dN}{dt} = rN$$

Logistic growth adjusts exponential growth by penalizing it for population size.

$$\frac{dN}{dt} = rN\,(1 - \alpha N)$$

where α is a small number much less than one, and it is the effect of one individual on the growth rate of the population. Thus, as N gets bigger, $\alpha N \to 1$ (approaches 1), and $1 - 1 = 0$.

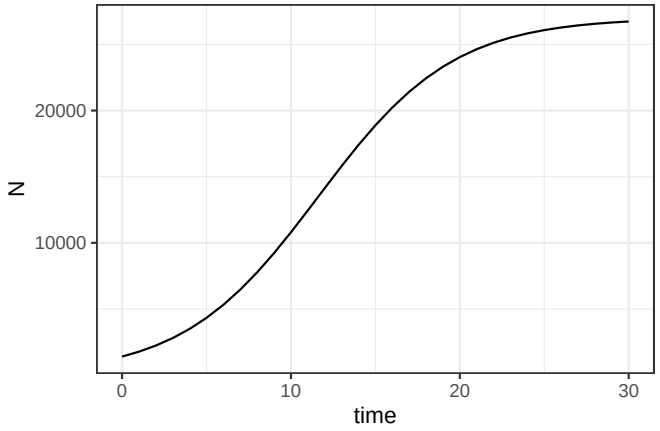


Figure 9.3.: Logistic growth in a hypothetical population (parameters: $r = 0.25$, $\alpha = 1/27000$).

You see that in Figure 9.3, the growth rate slows dramatically as the population size approaches some limit. We call that limit the **carrying capacity** of the population, and it is determined by the effects of individuals on the population growth rate, or α.

9.4.1. Math describes nature: on parameters and their meanings

Often, we can describe a biological process in several different mathematical ways. Density-dependence in logistic growth is one such case.

We can define logistic growth in two ways that are opposite sides of the same coin. Above, we used α to quantify the average effect of one individual on the growth rate of the population. Another common way to define logistic growth is by the consequences of these average individual effects: the carrying capacity which we call K. It turns out that the carrying capacity is the inverse of the effect of an individual, so that

$$K = \frac{1}{\alpha}$$

and as a result we can rewrite logistic growth as

$$\frac{dN}{dt} = rN\left(1 - \frac{N}{K}\right)$$

which is exactly equivalent to the first version we described above. Notice that $K = 1/\alpha$ is not a new assumption — it is a *definition* that follows from the structure of the equation. The two forms of the logistic are mathematically identical; they simply highlight different aspects of the same biology. Understanding the difference between a new assumption and a mathematical consequence of existing assumptions is a skill you will use throughout this book.

9.4.2. Solving a rate equation

We **solve** a rate equation in order to learn something about it's **asymptotic** behavior. By "learn something" we mean figure out what will determine the size of population. By "asymptotic", we mean really, really, long-term. We hope to find out what will eventually happen to the population and what will control it's destiny.

Another way to think about this is that we are seeking to find the **equilibrium state**, the state of the population toward which the population will tend to move. Imagine a marble in a wok or bowl. The marble will tend to move toward the bottom. When it finally settles down and *stops moving*, it is at its equilibrium state.

We solve a rate equation by setting it equal to zero, because **we want to understand what determines when the population stops changing.**

9.4.3. Assignment 1: Use math to understand the consequences of our assumptions.

Now solve for the equilibrium, when $dN/dt = 0$. That is, let

$$0 = rN\,(1 - \alpha N)$$

and solve for N. That is, rearrange to get N on the left side $N =$ and other stuff on the right side, and then simplify. SHOW ALL THE STEPS IN YOUR WORK.

Follow the Logic: How to read a mathematical result

When you solve $0 = rN(1 - \alpha N)$ for N, you will find two solutions. For each one, ask yourself:

- *What does this solution mean biologically?* What state is the population in?
- *Which parameters control it?* What biological processes determine this outcome?
- *Which parameters are absent?* If a parameter does not appear in the equilibrium, that means the long-term outcome does not depend on it. Why not?
- *What happens if I change a parameter?* If α gets bigger, what happens to the equilibrium, and what does that mean for the organisms?

This is the core skill: translating between symbols and organisms. The algebra is the tool; the biology is the point. Mathematical reasoning is not separate from ecological understanding — it *is* one of the ways you come to understand how ecological systems work.

9.5. Discrete difference equations

Sometimes, it is simpler to think about rate equations in a step-wise annual fashion, where each change is a **discrete** step, rather than continuous change.

N (this year) = N (last year) + births - deaths

In this case, the rate of change is a **difference** between N this year versus last year,

N (this year) - N (last year) = ΔN = births - deaths

In this situation, we call this a **difference equation**.

The difference equation for logistic growth is

$$N_t - N_{t-1} = \Delta N = r_d N_{t-1} \left(1 - \alpha N_{t-1}\right)$$

where the number of births and deaths is described by $r_d N_{t-1} \left(1 - N_{t-1}/K\right)$.

Notice an important feature of this equation: the density-dependence $(1 - \alpha N_{t-1})$ depends on the size of the population at one point in time $(t - 1)$, but the population is projected out past this time. As a result, the dampening or slowly effect of density is *delayed*, always lagging behind just a little bit. You will investigate the consequences of this in your exrecise.

We can project the population dynamics using

$$N_t = N_{t-1} + r_d N_{t-1} \left(1 - \alpha N_{t-1}\right)$$

9.5.1. Assignment 2: Explore delayed density-dependence

Here you will explore the consequences delayed density-dependence. In this exercise, you will discover *stable* points in a **stable limit cycle**. These *stable points* are the values of N to which a population returns.

You will also discover mathematical **chaos**. In science, **chaos** is defined as a deterministic, non-repeating pattern that is bounded, and characterized by sensitivity to initial conditions.

In the code below, you will do several things. First, we set up things that won't change,

1. the carrying capacity $(K = 100)$,
2. the number of generations (50).
3. an empty vector that will eventually hold all of our values of N.
4. the initial size of the population. For simplicity, keep this between 40 and 70.

Next we do things to investigate the consequences of delayed density-dependence. Here we investigate what happens when populations grow slowly (small r_d) vs. quickly (large r_d).

You will do these things four times.

1. Select a value of r_d. Pick four different values for rd , one each between
    a. 0.2 and 1.5,
    b. 2.1 and 2.4,
    c. 2.45 - 2.56
    d. 2.7- 3.
2. For your first value of r_d, project the dynamics, graph the time series of the dynamics, and save the graph under a unique name.
3. Repeat for the second, third, and fourth values of r_d.

When you are finished, you should have four graphs that each use a different value of r_d and each should look quite different.

Step 1 Set up things that won't change.

```
## Define K, NO, the number of generations
K <- 100
gens <- 50
## make numeric vector for each N in each generation, plus one more
N <- numeric(gens)

## Start the population out at this number of individuals
N[1] <- 45
```

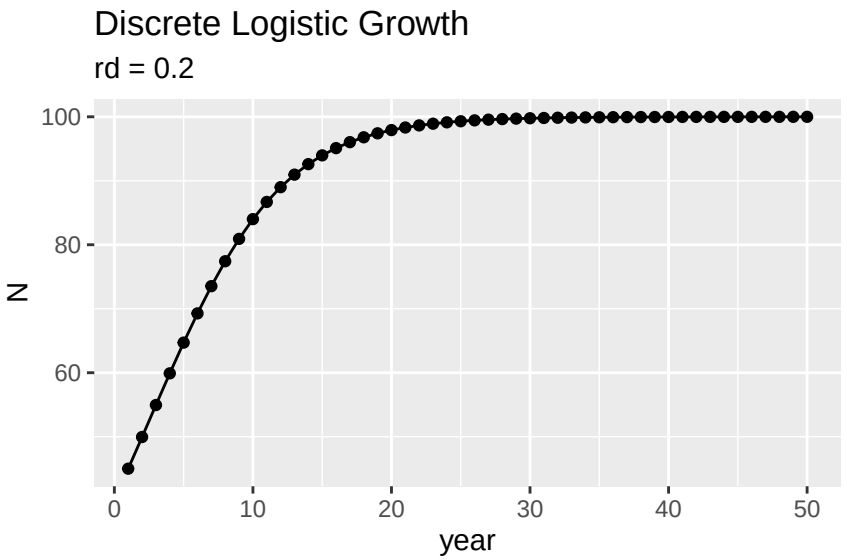
Select a value of r_d and use a for-loop to project the population through time, from one year to the next.

```
##############################################
## THIS IS WHERE YOU PICK THE VALUE OF rd YOU WANT
rd <- 0.2
##############################################


## project the population
for(t in 2:gens) {
  N[t] <- N[t-1] + N[t-1]*rd*(1-N[t-1]/K)
}

# store this result in a data frame
my.N <- data.frame(year=1:gens, N=N)

#for kicks, look at rows 1-3.
my.N[1:3,]
```

```
  year     N
1    1 45.00
2    2 49.95
3    3 54.95
```

Graph the time series of the population dynamics, and **change the name of the file** that you save using `ggsave()`.

```
## graph your data
ggplot(data=my.N, aes(x=year, y=N )) +
  geom_point() +
  geom_line() +
  labs(title = "Discrete Logistic Growth",
       subtitle=paste("rd =", rd))
```




```
## save your graph USING A UNIQUE NAME
ggsave("figs/dynamics-rd-0.2.png", height=5, width=6)
```

Put into one document your four graphs, and a description of how the graphs differ, including

- the relationship between r_d and the year in which the carrying capacity is reached or exceeded,
- the predictability of the dynamics, and
- the number of stable points to which the population returns.

10. *Direct species interactions: understanding mechanisms of invasion in mosquitoes*

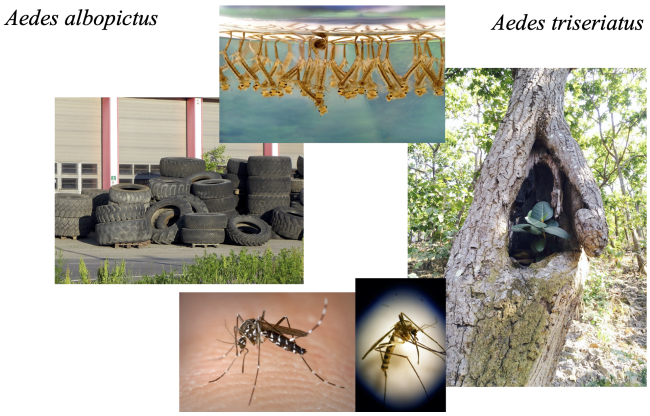


Figure 10.1.: Humans help species move around the planet into new places with species with whom they did not coevolve. Mosquitoes, such as the Asian tiger mosquito (*Aedes albopictus*), wind up in new places, such as North America. Ecologists are trying to understand how they get there and how they persist and spread when they do.

Species interact with each other. Sometimes those interactions are negative and cause population sizes to decline, and we call that **competition**. Other times, those interactions are positive and cause population sizes to increase, and we call that **mutualism**. This chapter explores an example of competition, in which investigators attempted to understand whether a native mosquito, *Aedes triseriatus* would compete with and inhibit the growth and spread of the Asian tiger mosquito, *Aedes albopictus* (Figure 10.1) (Livdahl and Willey 1991).

Livdahl, Todd P., and Michelle S. Willey. 1991. "Prospects for an Invasion: Competition Between *Aedes Albopictus* and Native *Aedes Triseriatus*." *Science* (*New York, N.Y.*) 253 (5016): 189–91. https://doi.org/10.1126/science.1853204.

Before you start

- **Assumptions about background and guidance:** logistic growth and the meaning of α and K from Density-dependence, and algebra for solving two equations at once. If those are new, read Models as assumptions first; it is short. The R code in this chapter is minimal.

10.1. Goals

In this chapter, you will,

- learn a simple rule determining whether can species coexist,
- do a little theory: extend density to include a second species,

- do a little algebra: solve simultaneously two growth equations to determine coexistence criteria, and
- start to develop a greater understanding and intuition for species interactions.

10.2. Deliverables

In one document include these two graphs and answers to these questions:

1. Draw the isoclines for both species in water obtained from tree holes. Add one curved line for population trajectories starting at 5 *A. triseriatus* and 5 *A. albopictus*.
2. Draw the isoclines for both species in water obtained from tires. Add one curved line for the population trajectories starting at 5 *A. triseriatus* and 5 *A. albopictus*.
3. Do the invasion criteria match the figures? Explain.
4. Answer the original question for these two environments: will the native *A. triseriatus* inhibit the establishment of the non-native *A. albopictus*?

10.3. Preparatory steps

- The preceding chapter on density-dependent population growth provides important background
- Read Livdahl and Willey (1991) in Perusall through our Canvas site. It is also available in our class archive: Livdahl and Willey (1991)
- If you are curious, check Deliverables, above (Section 10.2).

10.3.1. Optional background

If you are interested, you can download slides and watch the video on competition (4b) before proceeding.)

10.4. Background

We don't often know for sure what determines the establishment and spread of non-native species in a novel environment, but a variety of mechanisms and factors have been shown to play important roles in some situations, includ ing,

- an environment similar to the home environment, or otherwise hospitable,
- presence of mutualists from which a species benefits,
- an absence of enemies that recognize a species,
- the presence of competitors.

Here we investigate the role of competition in potentially inhibiting the spread of the Asian tiger mosquito (*Aedes albopictus*) outside its native range, in North America (Livdahl and Willey 1991).

Livdahl, Todd P., and Michelle S. Willey. 1991. "Prospects for an Invasion: Competition Between *Aedes Albopictus* and Native *Aedes Triseriatus*." *Science* (*New York, N.Y.*) 253 (5016): 189–91. https://doi.org/10.1126/science.1853204.

Todd Livdahl and Michelle Willey (1991) investigated how intensely *Aedes albopictus* would compete with the North American species *Aedes triseriatus* to try to predict whether the native species would inhibit the spread of the non-native species. Generally, mosquitoes lay eggs on water, which hatch into aquatic larvae that feed on particles in the water. These larvae grow and eventually pupate, and then emerge as flying adults, when the males feed on plant nectar and the females obtain a blood meal. During the larval stage, *if population densities are high, mosquitoes can compete for food resources and potentially limit their own growth and the growth of competitors.* It was this stage during which Livdahl and Willey measure the strength of intraspecific and interspecific competition (Table 10.1).

The version of the two species Lotka-Volterra competition equations we will use are,

$$\frac{dN_1}{dt} = r_1 N_1 \left(1 - \frac{N_1 + \alpha N_2}{K_1}\right)$$

$$\frac{dN_2}{dt} = r_2 N_2 \left(1 - \frac{N_2 + \beta N_1}{K_2}\right)$$

Table 10.1.: Lotka-Volterra growth and competition parameters of *A. triseriatus* and *A. albopictus* in water from tree holes and from tires, based on Livdahl and willey (1991). Values of r and K refer to the species on the respective line. Values for alpha or beta refer to the effect of the competitor. For instance, K = 42.83 is the carrying capacity of A. triseriatus in tree hole water, and alpha = 0.4167 is the effect of A. albopictus on A. triseriatus, in tree hole water.

	r	K	alpha or beta
A. triseriatus (in treehole water)	0.0514	42.8333	0.4167
A. albopictus (in treehole water)	0.0798	53.2000	0.7333
A. triseriatus (in tire water)	0.0591	32.8333	0.8333
A. albopictus (in tire water)	0.0904	45.2000	0.2500

10.5. Invasion criteria

Rather than just applying the invasion criteria from lecture, let's derive them. The reasoning is short and the biological insight is large.

10.5.1. Deriving the invasion criterion

Species 2 can invade a population of species 1 if species 2 can grow when rare — that is, when $N_2 \approx 0$ and species 1 is at its own carrying capacity ($N_1 = K_1$).

Substitute these values into the growth equation for species 2:

$$\frac{dN_2}{dt} = r_2 N_2 \left(1 - \frac{N_2 + \beta N_1}{K_2}\right)$$

1. Set $N_2 \approx 0$ and $N_1 = K_1$, and simplify. What terms drop out? What remains?
2. For $dN_2/dt > 0$ (species 2 is growing), what must be true about βK_1 relative to K_2?
3. State the invasion criterion in a complete sentence: "Species 2 can invade a population of species 1 at equilibrium when..."
4. Now repeat for species 1 invading a population of species 2 at K_2. Substitute $N_1 \approx 0$ and $N_2 = K_2$ into the growth equation for species 1, and derive the condition.

> **Follow the Logic: What the math just told you**
>
> The invasion criterion says that a species can invade when *it limits itself more than the resident limits it*. The math makes this precise: species 2 invades when $K_2 > \beta K_1$, which means species 2's own carrying capacity exceeds the impact the resident has on it (scaled by the competition coefficient).
> This is a biological insight produced entirely by mathematical reasoning. You started with assumptions (the Lotka-Volterra equations), followed the logic (substitution and an inequality), and arrived at an ecological conclusion (the conditions for coexistence). No new data were needed — just careful thinking about what the model already implies. Mathematical reasoning is a form of ecological understanding: the derivation *is* one of the ways you come to understand how species coexist.
> This is also a skill that protects you from errors. An AI can state the invasion criterion, but if it applies it to the wrong model or flips the inequality, you will only catch the mistake if you can re-derive it yourself.

10.5.2. Apply the criteria

Now use the parameter values in Table 10.1 to evaluate coexistence for both environments (tree hole water and tire water). For each environment:

a. Does species 1 (*A. triseriatus*) satisfy its invasion criterion?
b. Does species 2 (*A. albopictus*) satisfy its invasion criterion?

c. What is the predicted outcome — coexistence, competitive exclusion of one species, or competitive exclusion of the other?

11. *Mutualisms: Do ants protect pea plants?*

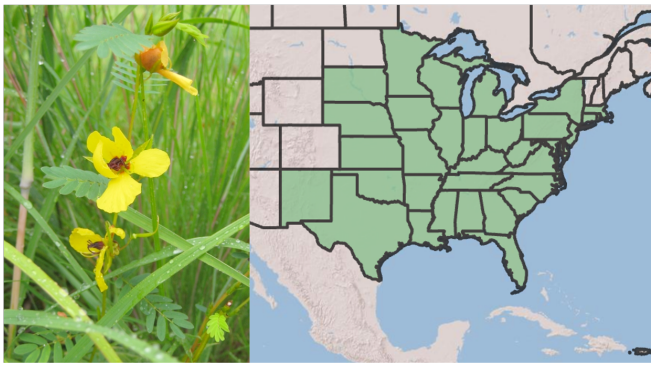


Figure 11.1.: Partridge pea (*Chamaecrista fasciculata*) is a cute little annual plant that is somewhat common throughout much of the eastern United States. It prefers recently disturbed habitats, such as sites following a fire. Credits: Wikipedia, USDA PLANTS Database.

Before you start

- **Time:** about 3 hours (30 min preparation, 45 min hypothesis and predictions, 1–1.5 h analysis and writing).
- **Assumptions about background and guidance:** your `Rwork` project set up as in Computer setup, and the ideas about interaction types (+,+; +,−; −,−) from Direct interactions. See Appendix E for additional assistance with graphing your data.

11.1. Goals

Do animals protect plants? If so, why and under what conditions would they do that? In this chapter, you'll investigate these questions as they apply to partridge pea (*Chamaecrista fasciculata*, Figure 11.1).

In this chapter, you will

- evaluate whether or in what ways species help each other,
- develop testable hypotheses about species interactions,
- generate qualitative and quantitative predictions from your hypothesis,
- learn to recognize *context dependency* (an interaction) in a graph,
- refine your data display expertise, and
- start to develop a greater understanding and intuition for species interactions.

11.2. Deliverables

Three documents:

1. **Your R script**, which has your name, date, and title, and loads packages, imports the data, sets the factor levels, and creates your graph.

2. **Your report**, which must include:

    a. Your hypothesis, and the one treatment you chose to compare with ants (water, AMF, or rhizobia).
    b. Your prediction, in writing.
    c. Your prediction, as a hand-drawn graph.
    d. Your Follow the Logic calculation: your predicted ant effect at each level of your chosen treatment, and whether your hypothesis predicts parallel lines.
    e. Your interaction graph of the data, with the same structure as your sketched prediction.
    f. A paragraph that includes
        i. an interpretation of your results,
        ii. whether your results are consistent with your predictions, and in what ways,
        iii. a short statement about what your results say about the hypothesis, and
        iv. one limitation of using herbivore abundance as a measure of "protection."

3. **Audit the machine.** Below is an AI-generated interpretation of a graph like yours. Using your own graph and what you learned in this chapter, identify **at least three** claims that are wrong, unsupported, or overstated. For each one, quote the claim, explain the problem in one or two sentences, and say what evidence would be needed to support it.

    > *"Ants clearly protected partridge pea plants. The error bars for plants with and without ants do not overlap, which proves that ants significantly reduced herbivore abundance. Because herbivores reduce plant growth, this shows that plants with ants produced more seeds. The effect of ants was the same at both levels of the other treatment, so that treatment does not matter for this mutualism. Mycorrhizal fungi made plants healthier, which is why those plants had fewer herbivores."*

11.3. Preparatory steps

- General background on mutualisms: download the slides and watch the video on mutualisms (4c) before proceeding.

- For a quick summary of the species and experiment in this chapter, watch this five-minute video. (This was presented at the 2021 Annual Conference of the Botanical Society of America by a Miami University graduate student, Laura Fehling.)
- Data: download `ArthropodObservations.csv` (right-click the link and choose "Save Link As...").

    1. Put `ArthropodObservations.csv` in the `data/` folder inside your `Rwork` project.
    2. Keep it in its simple "comma separated values" (.csv) format. Do not open and re-save it in a spreadsheet application.

11.4. Background on partridge pea and friends

Ecological mutualisms are mutually beneficial interactions between species. Ecologists often like to categorize pairwise species interactions using simple plus/minus signs; a mutualism is characterized thus: (+,+). Mutualisms between terrestrial plants and fungi are hundreds of millions of years old and underlie Earth's forests and grasslands. Symbiotic mutualisms between corals (which are animals) and dinoflagellates (a type of alga) allow for the coral reefs that structure Earth's shallow tropical seas. Mutualisms are all around us, and often go underappreciated. This chapter brings our attention to a small annual plant that has mutualisms with four completely different types of organisms, with multiple species within each of those types.

One good way to think about mutualisms is to identify the *types of rewards* that species share with each other. Often those rewards are **resources**, such as when a plant shares sugar with mycorrhizal fungi and the fungus shares phosphorus with the plant. Sometimes a reward is a **service**, as when a bumblebee pollinates a plant by carrying pollen to the stigma of another flower in return for nectar or some of that pollen. In many parts of the world, acacia trees provide ants with hollow thorns in which they can build nests, and ants defend the trees from herbivores of all kinds, from caterpillars to elephants (Bronstein 2015). In this case, the ants provide the *service of defense*, in return for a resource (the physical structure of a hollow thorn).

Bronstein, Judith L., ed. 2015. *Mutualism.* Oxford University Press, Oxford, UK.

Fehling, Laura Sharon. 2022. "Reward Complementarity and Context Dependency in Multispecies Mutualist Interactions in Partridge Pea (Chamaecrista Fasciculata)." PhD thesis, Miami University.

For her thesis at Miami University, Laura Fehling investigated the multi-mutualist interactions of partridge pea (Fehling 2022). Partridge pea (*Chamaecrista fasciculata*, "cam-ee-crist'-a fa-sick'-u-la'-ta") has four mutualist partner groups: arbuscular mycorrhizal fungi, rhizobia bacteria (*Bradyrhizobium* sp.), ants, and pollinators (Figure 11.2).

Chamaecrista fasciculata is an annual legume[3] that is native to the Midwestern and eastern United States, often occurring in open areas or disturbed sites in prairies and old fields (Fehling (2022) and references therein). The *C. fasciculata* mutualist system is ideal for studying multiple mutualist effects given the diversity and manipulability of its partners.

[3] Legumes are a group of plants including peas and beans that typically form rhizobia mutualisms.

Fehling, Laura Sharon. 2022. "Reward Complementarity and Context Dependency in Multispecies Mutualist Interactions in Partridge Pea (Chamaecrista Fasciculata)." PhD thesis, Miami University.

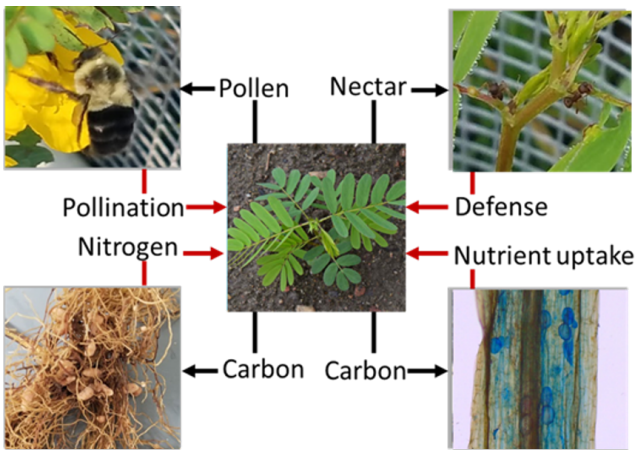


Figure 11.2.: *Chamaecrista fasciculata* mutualisms, showing the exchange of rewards from each partner (left to right, top to bottom: bumblebee, ants, rhizobia nodules, arbuscular mycorrhizal fungi) to the focal plant (grey arrows), and from the plant to the partners (black arrows).

While all partners receive a form of plant carbon, each benefits the plant with a unique nutrition, protection, or transportation reward. These partners further represent diversity in their (i) physical association (symbiotic vs. non-symbiotic), (ii) degree of obligacy, and (iii) specificity to the focal species. Mycorrhizae increase the plant's uptake of nutrients (especially phosphorus) and water, and generally have beneficial effects on plant fitness because the extra nutrients support increased plant growth. Rhizobia form nodules in the roots of plants where they fix nitrogen into a form usable by plants, and the benefits they provide to the plant can vary. *C. fasciculata* depends primarily on bees for pollen transfer; its flowers reward bees with pollen rather than nectar, and it can self-pollinate at very low rates (references in Fehling (2022)).

Fehling, Laura Sharon. 2022. "Reward Complementarity and Context Dependency in Multispecies Mutualist Interactions in Partridge Pea (Chamaecrista Fasciculata)." PhD thesis, Miami University.

In addition to producing showy flowers that attract bees and establishing mycorrhizae and rhizobial nodules, *Chamaecrista fasciculata* also produces nectar on its leaves in order to attract **ants**. At the base of the petiole of each compound leaf, there is a small extrafloral nectary (EFN), which produces nectar to attract ants for defense against herbivory. Unlike in many acacia–ant systems, the actual benefit of ants to the plant is not always clear. Sometimes they seem to help, and sometimes it is not clear whether they do.

Whether this behavior by ants actually helps a plant may depend on several things, including

- the costs to the plants of making the nectar and of getting eaten,
- the costs to the ants of removing herbivores and how effective they are at doing so, as well as the benefits of receiving plant nectar,
- particular environmental conditions that might influence costs, benefits, and herbivore density.

When the outcome of an interaction depends on other conditions like these, ecologists call it **context dependency**. You will look for it in your data.

As you saw in her video (see Preparatory steps), Laura set up an experiment to test how mutualists interact and how the environment alters those interactions. In 2021, she established individual replicate *C. fasciculata* plants **with and without rhizobia, with and without mycorrhizal fungi, with**

and without ants, and under normal watering or drought, for a total of 16 different experimental combinations ($2 \times 2 \times 2 \times 2$). Along with lots of other data that Laura collected, she and her undergraduate field assistant observed and counted herbivorous insects on each plant every week all summer, for two minutes in the morning and two minutes in the afternoon. **You will analyze these data: the total abundance of herbivorous insects on each plant.**

11.5. Your hypothesis and predictions

The data you have available are the **total count of herbivores on each plant**, summed across all of the summer's observation periods, and the combination of **experimental treatments** applied to each plant. Each plant is one row in the data set.

What you now know, based on the above text and preparatory information:

1. About half the plants had ants as normal, while the other half were prevented from having any ants through the use of sticky material applied around the rim and opening of pots (Tanglefoot, The Tanglefoot Company, Grand Rapids, MI).
2. The 16 combinations of experimental treatments (Can you write them all out? Start with "no AMF, no rhizobia, no ants, normal water.")
3. What the mycorrhizal fungi and rhizobia bacteria provide to plants.
4. What ants are thought to do.
5. Your data: the abundance of herbivores found on individual plants that received different treatments.

Questions you might consider:

- Aside from the potential effects of ants, which plants would be the healthiest?
- Which plants might be the most attractive to herbivores?
- Half the plants experienced drought - how might that influence the plants, the ants, and ant behavior?
- How would mycorrhizal fungi (AMF) and rhizobia affect plants, and how might those effects alter ant behavior?
- Not all herbivores are chewers - some feed on sap. Some ants protect and even farm sap-feeding insects such as aphids and treehoppers for their sugary honeydew. How would that change your expectations?
- Ant-exclusion experiments often keep ants off plants with a sticky barrier around the stem. What else might such a barrier keep off the plant?
- What is the relationship between *herbivore abundance* and plant damage, growth, or seed production?

After you have considered the above questions, answer these two questions:

1. Which treatment(s) will create plants that attract the most **herbivores**?
2. Which treatment(s) will create plants that attract the most **ants**?

Use your answers for these two questions to generate a hypothesis about relationships between partridge pea, ants, and herbivores. **Recall that a scientific hypothesis is an idea about how you think the world works.** A hypothesis differs from a prediction: a hypothesis is a mechanism that might operate generally with lots of different organisms, and it is not an observed result in a particular system. A prediction is a measureable outcome.

11.5.1. Your predictions

Use your answers to these two questions to make a specific prediction that you can evaluate with the data we have (herbivore abundance and experimental treatments). To keep things manageable, you will compare plants with and without ants under **just one other treatment**. Choose one: water (normal vs. drought), mycorrhizal fungi (AMF vs. no AMF), or rhizobia (rhizobia vs. no rhizobia). Pick the one your hypothesis says the most about.

Then turn that prediction into a **hand-drawn graph** of what your data would look like if your hypothesized mechanism operated in this partridge pea system.

For example, imagine that I chose water, and answered the questions in this way:

1. Herbivores: I believe that drought-stricken plants will attract more herbivores because i think they may have softer leaves and lower concentrations of chemical defenses.
2. Ants: I believe that well-watered plants will attract more ants because those plants will be able to make more nectar.

Putting those together, I would predict that plants under drought have more herbivores overall, and that ants reduce herbivores much more on well-watered plants than on drought-stricken plants. My sketch would look like Figure 11.3: the effect of ants depends on water.

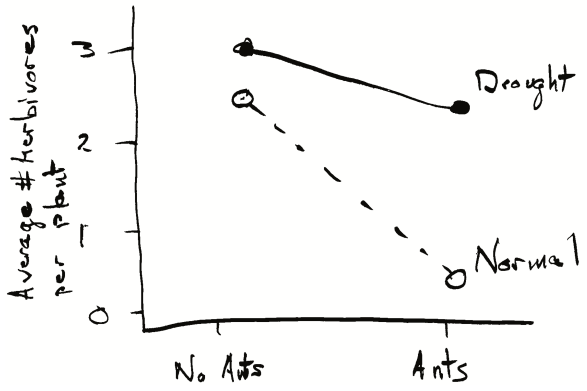


Figure 11.3.: A hand-drawn-style prediction. Drought-stricken plants attract more herbivores, and well-watered plants attract more ants, so ants reduce herbivores mainly under normal watering. Only the pattern matters, not the numbers, so the y-axis has no values.

You will sketch your predictions on paper, take a photo of it, send your photo to your computer, and include the photo in your write-up for the deliverables.

Your hypothesis and graph should look different than mine. What matters is that your graph follows logically from your answers to the two questions.

! Follow the Logic: What does context dependency look like on a graph?

Return to the example in Figure 11.3, where ants protect plants, but less so when plants are drought-stressed. Here are hypothetical mean herbivore counts:

	No ants	Ants
Normal water	2.0	1.0
Drought	3.0	2.5

Step 1. Compute the ant effect in each water treatment. The ant effect is (mean with ants) − (mean without ants).

- Normal: $1.0 - 2.0 = -1.0$ herbivores per plant
- Drought: $2.5 - 3.0 = -0.5$ herbivores per plant

Step 2. Compare the two effects. If ants had the same effect under both conditions, the two differences would be equal. Here, ants remove twice as many herbivores under normal watering. The effect of one factor (ants) *depends on* another factor (water). That is a statistical **interaction**, and it is what ecologists mean by context dependency.

Step 3. Picture it. Put ants on the x-axis and draw one line for each water treatment, connecting the two means. Equal effects give **parallel lines**. Context dependency gives lines that are **not parallel**, and in extreme cases lines that cross.

Think further. Now suppose the means were $2 \to 1$ under normal water and $4 \to 2$ under drought. The *differences* are -1 and -2 (not parallel), but the *ratios* are both 0.5 (ants remove half the herbivores in both cases). Is there context dependency? It depends on whether you think ants remove a fixed *number* or a fixed *fraction* of herbivores. Which seems more biologically reasonable, and why?

Your turn. Before you look at the data, write your predicted means for the four combinations of ants and your chosen treatment, compute the two ant effects by hand, and say whether your hypothesis predicts parallel lines.

11.6. Data analysis

11.6.1. Set up your R session

First, open your `Rwork` project in RStudio by double-clicking `Rwork.Rproj` (or using File > Open Project...). Opening the project automatically sets your **working directory** to the `Rwork` folder, so you do not need `setwd()`.

You can confirm your *current* working directory with this:

```
getwd()
```

The result should end in `Rwork`. If it does not, you probably opened RStudio without opening the project.

Second, use RStudio to create a new file of the type "R script". Save it in the top level of `Rwork`, using a name that reflects this chapter, and end the file name with ".R" or ".r".

Third, add a comment as a brief description of the script, and load any R packages you know you will need.

```
# Chapter 11 Mutualisms: Do ants protect pea plants? ----
# Your name, date

## Load packages ----
library(tidyverse)
# usually just the ggplot2 and dplyr packages within the tidyverse
```

11.6.2. The data

Make sure that your data file is in the `data/` folder of your `Rwork` project and is still just a CSV file.

Open the data file (.csv file) in a text editor or in RStudio (click on it in the Files pane and choose "View File"). Do **not** save the file in a spreadsheet application, because it will try to change the format.

1. With the file open, learn what is on the first several lines. Often, the first line or two will have *metadata*, or information about the data (data about the data).
2. Below the metadata, you will see column headings, separated by commas, because the data are *CSV*, or comma-separated values.
3. Below the column headings are the data, with columns separated by commas.

Close the data file, and *do not save it*. If you save it, chances are good that the application will try to save it in a different format.

Import the data, or *read* the data into R. Here is code for that.

```
## Import data ----
# skip the first 2 lines because they are metadata
d <- read.csv("data/ArthropodObservations.csv", skip = 2)
```

> **Code walkthrough: `read.csv()`**
>
> - `read.csv()` expects a comma-separated file, with column headings on the first line it reads.
> - `"data/ArthropodObservations.csv"` is the path to the file, *relative to your working directory* (`Rwork`). That is why the file must be in `Rwork/data/`.
> - `skip = 2` tells R to ignore the first two lines (the metadata), so that the third line is read as the column headings.
> - `d <-` stores the result as a data frame named `d`.

Look at your data with one of these two:

```
# glimpse(d)
# or
str(d)
```

```
'data.frame':   225 obs. of  5 variables:
 $ AMF     : chr  "no_AMF" "no_AMF" "no_AMF" "no_AMF" ...
 $ Rhizobia: chr  "no_Rhiz" "no_Rhiz" "no_Rhiz" "no_Rhiz" ...
 $ Ants    : chr  "no_Ants" "no_Ants" "no_Ants" "no_Ants" ...
 $ Drought : chr  "Normal" "Normal" "Normal" "Normal" ...
 $ Total   : int  1 1 0 1 1 0 3 1 2 2 ...
```

- `AMF` indicates whether a plant did or did not have arbuscular mycorrhizal fungi.
- `Rhizobia` indicates whether a plant had or did not have symbiotic nitrogen-fixing bacteria.
- `Ants` indicates whether ants did or did not have access to the plant.
- `Drought` indicates whether a plant received normal watering or drought conditions.
- `Total` is the total number of herbivores observed on that plant, summed across all of the summer's observation periods.

Notice that the four treatment columns are *character* (`chr`) variables. R will sort them alphabetically in graphs, which puts "AMF" before "no_AMF" and "Rhiz" after "no_Rhiz", so controls end up in inconsistent places. Convert them to *factors* and put the control level first:

```
## Set treatment levels, control first ----
d <- d |>
  mutate(AMF      = factor(AMF,      levels = c("no_AMF", "AMF")),
         Rhizobia = factor(Rhizobia, levels = c("no_Rhiz", "Rhiz")),
         Ants     = factor(Ants,     levels = c("no_Ants", "Ants")),
         Drought  = factor(Drought,  levels = c("Normal",
          ↪  "Drought")))
```

> **Code walkthrough: `mutate()` and `factor()`**
>
> - `d |> mutate(...)` takes `d` and modifies or adds columns.
> - `factor(AMF, levels = c("no_AMF", "AMF"))` turns the text column into a factor, a categorical variable whose levels have a fixed order. The order in `levels` is the order they will appear on graph axes and legends.
> - `d <-` overwrites the old `d` with the new version.

Now check how many plants are in each combination of ants and your chosen treatment. Here is the example with water; replace `Drought` with `AMF` or `Rhizobia` if you chose one of those.

```
## Count plants per treatment combination ----
count(d, Ants, Drought)
```

```
     Ants Drought  n
1 no_Ants  Normal 60
2 no_Ants Drought 61
3    Ants  Normal 52
4    Ants Drought 52
```

Each group pools plants across the treatments you did not choose. For example, the "Ants, Normal" group includes plants with and without AMF and with and without rhizobia.

11.7. Display the data to evaluate your predictions

Make an **interaction graph** of the data, with the same structure as your sketched prediction (Figure 11.3): ants on the x-axis, one line for each level of your chosen treatment, and the mean number of herbivores per plant on the y-axis. Each mean has an error bar of $\pm$ one standard error. Behind them, faint jittered points show every plant in the data set, so you can see the raw data the means summarize.

Here is the code for the example with water. If you chose AMF or rhizobia, replace `Drought` with `AMF` or `Rhizobia` everywhere it appears.

```
## Interaction graph ----
ggplot(data = d, aes(x = Ants, y = Total, colour = Drought, group =
 ↪  Drought)) +
  geom_point(position = position_jitterdodge(jitter.width = 0.15,
                                             jitter.height = 0,
                                             dodge.width = 0.3),
             alpha = 0.3) +
  stat_summary(fun.data = mean_se,
               position = position_dodge(width = 0.3)) +
  stat_summary(fun = mean, geom = "line",
               position = position_dodge(width = 0.3)) +
  labs(x = NULL, y = "Herbivores per plant")
```

Code walkthrough: the interaction graph

- `aes(x = Ants, y = Total, colour = Drought, group = Drought)` puts ants on the x-axis and gives each level of your chosen treatment its own colour. `group = Drought` tells `ggplot()` which means to connect with a line.
- `geom_point(...)` draws every plant as a faint point (`alpha = 0.3`). `jitter.width = 0.15` spreads the points sideways so they don't sit on top of each other. `jitter.height = 0` matters for counts: it keeps a plant with 2 herbivores plotted at exactly 2.
- `stat_summary(fun.data = mean_se)` calculates the mean and standard error of `Total` for each group and draws a point with an error bar.
- The second `stat_summary(fun = mean, geom = "line")` connects the two means for each level of your chosen treatment. These are the lines you compare with your prediction: parallel or not?
- `position_dodge(width = 0.3)` nudges the colours apart so their error bars don't overlap. All three layers use the same dodge width so that the points, error bars, and lines line up.
- `labs()` sets the axis labels.

Why means and not boxplots? Our data are small whole-number counts, mostly 0, 1, and 2. Many groups have the same median, so medians and boxplots hide differences that the means can show. The jittered points let you see how variable the plants are around each mean, including the few plants with many herbivores that pull some means upward.

Once you have your graph, go back to the Follow the Logic box. Estimate the ant effect at each level of your chosen treatment from your graph, and compare those numbers to your predictions.

Going further (optional): a model for count data

Graphs let you *see* an interaction; a statistical model lets you *estimate* it. Because `Total` is a count, a generalized linear model with a log link is appropriate:

```
m <- glm(Total ~ Ants * Drought, family = quasipoisson, data =
 ↪  d)
broom:tidy(m)
```

- `Ants * Drought` fits the effect of ants, the effect of drought, and their interaction (`AntsAnts:DroughtDrought`). Replace `Drought` with your chosen treatment.
- With a log link, coefficients are on a *multiplicative* scale: `exp(estimate)` is the factor by which the mean count changes. This is the "fixed fraction" view from the Follow the Logic box.

- `quasipoisson` allows the variance to be larger than the mean. Check whether that is true: compare `mean()` and `var()` of `Total` within groups.

12. *Of wolves and moose, on Isle Royale*

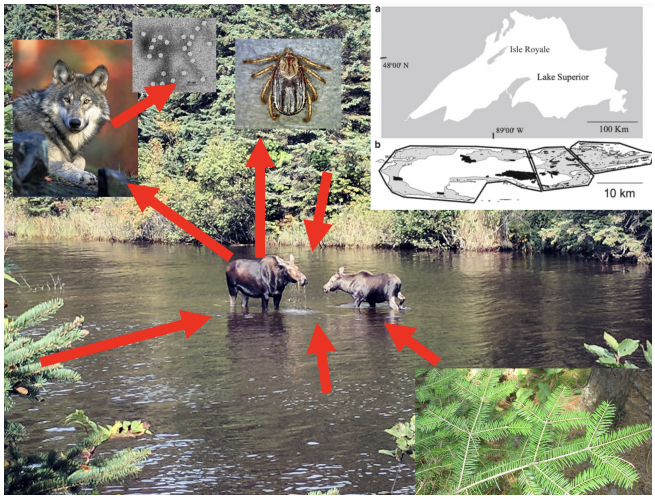


Figure 12.1.: As in other northern forests, on Isle Royale in Lake Superior (upper right), moose feed on vegetation such as aquatic plants, and the tender shoots of both deciduous and coniferous trees like the fir (right) and spruce (left). Isle Royale wolves feed on moose, and parasites can hit both populations pretty hard (parvovirus - wolves; ticks - moose). How the moose and wolf populations affect each other is an open question, and this chapter walks through an attempt to evaluate the role of reciprocal feedbacks between the two populations.

Before you start

- **Assumptions about background and guidance:** your `Rwork` project and the ability to read a .csv file and make a graph with `ggplot()` (Computer setup, Extra-pair copulation), exponential growth from Density-independent population growth, and algebra for solving for an equilibrium. Logistic growth (Density-dependence) helps but is not required. You will install the `deSolve` package; this chapter shows how.

12.1. *Goals*

In this chapter, you will,

- Test the hypothesis that moose and wolves regulate the size of each other's population.
- Use algebra to estimate **parameters** in the mathematical model of predator-prey dynamics.
- Use specific quantitative and qualitative criteria to evaluate how well the model describes the dynamics of the population.

12.2. Deliverables

Turn in two documents.

The first document is your R script.

In the second document, provide

- Phase plane diagram of raw data (x-axis, moose; y-axis, wolves),
- Time series graph of model output (x-axis, years; y-axes, moose and wolves)
- answers to questions 11 and 12, in Predicting population dynamics, below.

12.3. Preparatory steps

1. General background lectures: Download the PDF slides and watch the video on predation and herbivory before proceeding.
2. Data: go to our class data repository, and download `isleRoyaleData.csv`.
    a. Make sure the file remains in the same simple format, "comma separated values" (.csv).
    b. Put `isleRoyaleData.csv` in your working directory, `Rwork`.
    c. Continue to make sure the file remains in the same simple format, "comma separated values" (.csv).
3. Find out what you need for the deliverables.

12.4. Background

We have seen examples of how the distribution and abundance of organisms are influenced by

- the physical environment,
- adaptations to the physical environment,
- the density of individuals in the same population, and
- interactions with other species which may be prey or enemies or mutualists.

Here we take a closer look out how predator-prey interactions affect populations, and precisely how those interactions can be described.

As our example, we focus on the pair-wise predator-prey relationship between moose (*Alces alces*) and wolves (*Canis lupus*) on Isle Royale National Park (USA) in Lake Superior. The moose are herbivores, that is, plant enemies, particularly of balsam fir (*Abies balsamea*) which makes up about 50% of their diet. Aside from severe winter weather and wolves, another enemy of the moose is the winter tick (*Dermacentor albipictus*), which can sap the health of moose and reduce survival. The wolves have an enemy of their

own, canine parvovirus which can kill wolves. In spite of all this complexity, here we investigate the mathematical mechanics of how predators and prey influence each others abundances, and whether we can learn anything about their biology.

When we use mechanistic models in science, we are following the scientific method we may have first been exposed to in K-12 education:

1. observe a pattern in nature.
2. propose an explanation (hypothesis) that can generate predictions.
3. gather data to test the predictions.
4. re-evaluate the hypothesis in light of the data.

A mathematical model of predator and prey *is* a hypothesis about how species interact. You will informally evaluate how well a particular model can explain patterns in our data.

12.5. Details about our data

Our data set comes from censuses of moose and wolf populations on Isle Royale from 1959 to 2011 collected by a number of different researchers over that time (Vucetich and Peterson 2009).

Vucetich, John A., and Rolf O. Peterson. 2009. *Recovery of Gray Wolves in the Great Lakes Region of the United States*. Edited by A. P. Wydeven et al. Springer.

12.6. Your hypothesis and predictions

The data you have available are numbers of moose and wolves each year, 1959-2011.

What you know:

1. predators can sometimes have negative numerical effects on their prey populations.
2. prey can sometimes have positive numerical effects on their predator populations.
3. ratio-dependent and prey-dependent (Lotka-Volterra) mathematical models of predators and prey make different assumptions about whether the killing rate depends on the **ratio** of prey to predators, or simply the **density** of prey.
4. The ratio-dependent predator-prey model predicts that predators will be a constant fraction of the prey population, and does not generate oscillations. In contrast, the Lotka-Volterra predator-prey model predicts that predators are not a constant fraction of the prey population, the populations undergo regular oscillations, and the predator population lags behind the prey population.
5. Your data include the abundances of moose and wolves.

For this chapter, *our hypothesis* is that the prey-dependent Lotka-Volterra predator-prey model is a reasonable description of the mechanisms driving moose-wolf dynamics. This model makes these predictions:

- predators are not a constant fraction of the prey population,
- the populations undergo regular oscillations, and
- the predator population lags behind the prey population.

In order to evaluate the hypothesis you will have to answer about a dozen questions *along the way*, as you go. Don't skip these questions.

These questions will let you estimate the constant **parameters** of the Lotka-Volterra model:

- r, the intrinsic rate of increase.
- a, the attack rate, or the rate at which a predator kills a prey, given the prey density.
- b, the conversion efficiency, or the rate at which predators makes more babies out of all the prey they kills.
- m, per capita mortality rate of the predators.

12.7. Data analysis

First, open R using RStudio, and **set your working directory**. You can do this in two ways:

1. using the Session menu in RStudio, or
2. with code such as this, in which you customize it to your computer:

```
setwd("C:/Users/paltrowg/BIO209W/Rwork")
```

You can confirm or **get** your *current* working directory using this:

```
getwd()
```

```
[1] "/Users/stevenmh/projects/intro_to_ecology"
```

so when I do that, you can see my working directory is the one for this book.

Before you proceed, make sure your working directory is your `Rwork` directory.

Second, use RStudio to create a new file of the type "R script". Save it using a name that reflects this chapter, and end the file name with ".R" or ".r".

Third, add a comment as a brief description of the script, and load any R packages you know you will need.

```
## This script is for chapter XX in Hank's Intro to Ecology primer.

## We always use...
library(tidyverse)
## or usually just ggplot2 and dplyr packages within the tidyverse.
```

In this chapter, we need a new package called `deSolve`. This package solves differential equations and so will let us implement the differential equations of our predator-prey model. First use code below, or the Tools menu in RStudio to install `deSolve`. Don't do both.

```
## For this chapter we also use deSolve
install.packages("deSolve")
```

Then load the package.

```
library(deSolve)
```

12.7.1. Load the data

Make sure that your data file is in your `Rwork` directory. and is still just a CSV file.

Open the data file (.csv file) in a spreadsheet application, a text editor, or in RStudio. Do **not** save it the file the spreadsheet application because it will try to change the format.

1. With the file open, learn what is on the first several lines. Often, the first line or two will have *metadata* or information about the data (data about the data).
2. Below the metadata, you will see column headings, separated by commas, because the data are *CSV* or comma-separated-values.
3. Below the column headings are the data, with columns separated by commas.

Close the data file, and *do not save it*. If you save it, chances are good that the application will try to save it in a different format.

Import the data or *read* the data into R. Here is code for that. You will notice that we use a function that expects a CSV file. You will also notice that we use an *argument* to tell R how many lines of metadata to skip so it reads only the data and its column headings.

```
## NOTE that we skip the two lines because they are metadata
d <- read.csv("isleRoyaleData.csv",
              skip=2)
```

Look at your data with one of these two:

```
# glimpse(d)
# or
str(d)
```

```
'data.frame':   106 obs. of  3 variables:
 $ Year   : int  1959 1960 1961 1962 1963 1964 1965 1966 1967 1968
  ↪  ...
 $ Species: chr  "Moose" "Moose" "Moose" "Moose" ...
 $ N      : int  560 600 625 640 660 700 720 760 900 1050 ...
```

A time series graph of the our data reveal how the populations change over time (Fig Figure 12.2).

```
ggplot(d, aes(Year, N, colour=Species)) + geom_line() +
  facet_grid(Species~., scales = "free") +
  geom_smooth( se=FALSE, span=.2, color="red", linewidth=.5) +
  geom_smooth( se=FALSE, span=.7, color="black", linewidth=.5)
ggsave("figs/MooseWolfTS.png", height=5, width=6)
```

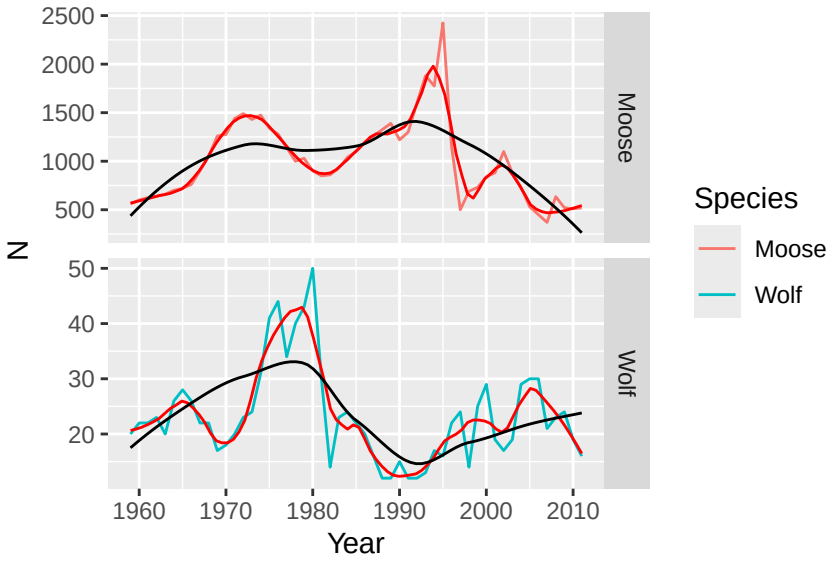


Figure 12.2.: Natural populations undergo fluctuations. Here we smooth the data over different proportions of the time series to reveal different patterns of squiggliness.

Next, let's describe a model of predator and prey interaction. Refer back to the lectures for more information.

12.8. Population growth models

All of our mathematical and conceptual models of populations have the same form:

Rate of change = rate of gains - rate of losses

Sometimes these gains and losses are combined:

Rate of change = net rate of gains

12.8.1. Examples of population growth models

Exponential growth:

$$\frac{dN}{dt} = bN - dN = (b - d)N = rN$$

Logistic growth (density-dependence):

$$\frac{dN}{dt} = rN\left((1 - \frac{N}{K}\right) = rN - \frac{r}{K}N^2$$

Mechanistic models such as there have *parameters* and *state variables*. State variables are the quantities that vary through time, such as population size. In contrast, parameters are constants that determine how fast the state variables change. In our exponential growth equation, $dN/dt = rN$, N is the state variable, and r is the parameter.

Next we create a pair of rate equations where one species' loss is another's gain.

12.9. Predator-prey equations

Maybe the simplest predator prey equation is essentially two-species exponential growth, where exponentially growing prey, N, is partly eaten by exponentially growing predators, P. First the prey,

$$\frac{dN}{dt} = rN - aN$$

where aN is the rate of loss of prey to predators; this is the rate at which predators attack and kill prey. The units of r and d are individuals per individual per unit time.

The predators increase only because they kill and eat prey, aN, and only a fraction of these, b, get converted into new predators,

$$\frac{dP}{dt} = baN - mP$$

and predators die at rate mP. The parameter b is the *efficiency* with which prey are converted into predators, that is, are ingested, assimilated by the predator, and actually lead to growth and reproduction rather than cellular respiration. The units of b are predators per prey per unit time, and the units of m are the same as those of r and d, individuals per individual per unit time.

If we made all these assumptions, and did some algebraic rearrangement, we would learn that predators abundance is predicted to be a constant fraction of prey abundance,

$$P = \frac{ba}{m}N$$

where both populations will grow exponentially as long as $r > d$ (why?).

This model is sometimes referred to as ratio-dependent predation a completely different reason: predation rates can be expressed in terms of the number of prey per predator time number of predators, or

$$aN = a\frac{N}{P}P$$

.

12.10. Lotka-Volterra prey-dependent predation

The most widely taught predator-prey equation was developed independently by Vito Volterra and Alfred Lotka. Here, the predation rate is expressed as a function of prey density rather than the ratio of prey to predators.

> Follow the Logic: What we are about to do, and why
>
> In the sections that follow, you will use the structure of the Lotka-Volterra equations to *extract biological meaning from data*. Each parameter — r, a, b, m — describes a specific biological process. By solving the equilibrium equations and using observed population sizes, you can estimate what these parameters must be if the model is correct. This is mathematical reasoning as ecological understanding: you are using the logical structure of a model to connect data to mechanism. And it gives you a powerful check on plausibility. If your estimated attack rate implies a wolf kills 500 moose per year, something is wrong — either with the model, the data, or your algebra. That judgment requires you to understand what the math means biologically, not just how to solve it.
> An AI can do this algebra. What it cannot do is look at the answer and say "that makes no biological sense for a 40-kg predator on Isle Royale." That judgment is yours.

$$\frac{dN}{dt} = rN - aNP$$
$$\frac{dP}{dt} = baNP - mP$$

In the above equations,

- r is the intrinsic rate of increase.
- a is the attack rate, or the rate at which a predator kills a prey, given the prey density.
- b is the conversion efficiency, or the rate at which predators makes more babies out of all the prey they kills.
- m is per capita mortality rate of the predators.

1. How would you characterize the growth of the prey in the absence of predators? (Pretend $P = 0$ and solve dN/dt)
2. How would you characterize the growth of the predator in the absence of prey? (Pretend $N = 0$, and solve dP/dt).

This model is famous for its oscillations. Do our data show evidence of oscillations? Yes they do. Look at the figures below to estimate how often they undergo a cycle.

```
dw <- pivot_wider(d, names_from=Species, values_from=N)
ggplot(dw, aes(Moose, Wolf, colour=Year)) +
  geom_path(arrow=arrow(length=unit(3, "mm"), angle = 15, type =
   ↪  "open")) +
  labs(subtitle="Each arrow is one annual time step.")
ggsave("figs/MooseWolfPP.png", height=5, width=5.5)
```

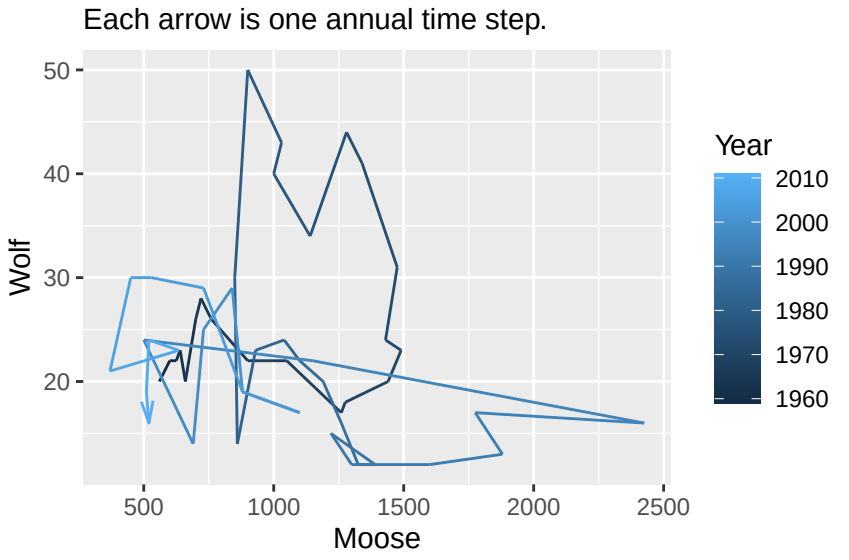


Figure 12.3.: Use this phase plane plot to count how many times the populations undergo a complete counterclockwise cycle.

12.10.1. Equilbria

One thing we can learn from simple models are the *equilibria* they predict. An equilibrium is a point when the populations stop bouncing around. At these points, the populations stop changing - their *growth rates equal zero*. By setting growth rates equal to zero, we can use the model to make predictions about what those equilibria are.

Another name for an equilibrium is an *attractor*. We call it an attractor because the population appears to be attracted to it. If some event pushes the abundance of a population away from an equilibrium, the equilibrium seems to *attract* the population back to it.

Here we solve for the equilibria of the Lotka-Volterra predator-prey model. We set the prey population growth rate equal to zero, and then use algebra to simplify the expression. This is one way we learn the logical consequences of prey growth being zero.

$$0 = rN - aNP$$

If we divide both sides by N, we get

$$0 = r - aP$$

and rearranging we get

$$P = r/a$$

This means that the prey population *growth rate* will be zero when the predator population size is r/a. This happens to be the intrinsic rate of increase divided by the attack rate.

If the prey population stops changing when $P = r/a$, do you think the predator population will stop growth too?

Next, we set the predator population growth rate equal to zero, and then use algebra to simplify the expression.

$$0 = baNP - mP$$

3. Simplify this expression on your own. What do you get? What will N be equal to?

If the prey population stops changing when $P = r/a$ and the predator population stops changing when $N = m/(ba)$, this means that at these two abundances, the population sizes will not change. Births and deaths will continue to occur, and they will perfectly balance each other. These are the expected equilibria for these two populations *under this set of assumptions.* Something different happens when N and P are not at these values.

Follow the Logic: Reading the equilibrium

Look at the equilibrium expressions you just derived. Notice which parameters appear in each one. The equilibrium prey population ($N^* = m/(ba)$) depends on predator parameters — mortality, attack rate, conversion efficiency. The equilibrium predator population ($P^* = r/a$) depends on prey parameters — intrinsic growth rate and attack rate.

This is a non-obvious biological insight that emerges directly from the math: *the long-term size of the prey population is controlled by predator traits, and vice versa.* You would not necessarily guess this from intuition alone. The derivation reveals it.

Now ask: does this make biological sense? Why would the prey equilibrium depend on the predator's mortality rate but *not* on the prey's own intrinsic growth rate? Following the mathematical logic forces you to confront — and resolve — this apparent paradox. That is mathematical reasoning as ecological understanding.

We abbreviate these equilibria as N^* and P^*, which we refer to as "N star" and "P star".

If our populations are regulated *approximately* according to these rules, we might expect that the average abundances of these populations are approximately equal to these values.

4. Find the average or mean population size of the wolf population in our data set. Look at the figure and try to guess.

5. Find the average or mean population size of the moose population in our data set. Look at the figure and try to guess.

Use R to see how close your guess is.

```
View(dw)
mean(dw$Wolf)
mean(dw$Moose)
```

Next we find values for the model parameters. It will help to have just pieces of our data sets:

```
# the number of observations or rows of data
n <- nrow(dw)

# predator and prey numbers for all years except the last.
Pt <- dw$Wolf[-n]
Nt <- dw$Moose[-n]

# predator and prey numbers for all years except the first
Pt1 <- dw$Wolf[-1]
Nt1 <- dw$Moose[-1]
```

12.10.2. Intrinsic rate of increase, r

One way to think about the intrinsic rate of increase of a population, r, is that it is the maximum per capita growth rate. A crude way to estimate that is to find the maximum observed value in our data.

6. Estimate r as the maximum annual per capita rate of increase. Recall that
$$r = \log_e \frac{N_{t+1}}{N_t}$$

```
# each year's r
r <- log(Pt1/Pt)
r
```

```
 [1]  0.09531018  0.00000000  0.04445176 -0.13976194  0.26236426
  ↪  0.07410797
 [7] -0.07410797 -0.16705408  0.00000000 -0.25782911  0.05715841
  ↪  0.10536052
[13]  0.13976194  0.04255961  0.25593337  0.27958486  0.07061757
 ↪  -0.25782911
[19]  0.16251893  0.07232066  0.15082289 -0.51082562 -0.76214005
 ↪  0.49643689
[25]  0.04255961 -0.08701138 -0.09531018 -0.22314355 -0.28768207
 ↪  0.00000000
[31]  0.22314355 -0.22314355  0.00000000  0.08004271  0.26826399
 ↪  -0.06062462
[37]  0.31845373  0.08701138 -0.53899650  0.57981850  0.14842001
 ↪  -0.42285685
[43] -0.11122564  0.11122564  0.42285685  0.03390155  0.00000000
 ↪  -0.35667494
[49]  0.09097178  0.04255961 -0.23361485 -0.17185026
```

```
# the maximum
rMax <- max(r)
rMax
```

```
[1] 0.5798185
```

12.10.3. Attack rate, a

It is worth noting that aN is the rate at which the average wolf kills moose. We will estimate the attack rate, a, by realizing that the predator equilibrium equals r/a, or

$$a = r/P^*$$

where our estimate of P^* is the average size of the wolf population.

7. Estimate a using your above estimates of r and P^*.

```
a <- rMax / mean(dw$Wolf)
a
```

```
[1] 0.0248829
```

12.10.4. Predator mortality rate, m

One way to think about the predator mortality rate, m, is that it is the maximum rate per capita rate of decline. Put another way, it is the minimum per capita population growth rate, and this will be a negative number.

8. Estimate m. Find the minimum r for the wolf population across all years. Let that the absolute value of that minimum be m. Remember that m by itself is a positive number....

```
# mortality, estimated as the minimum log( P[t+1] / P[t] )
m <- abs( min(r) )
m
```

```
[1] 0.7621401
```

12.10.5. Conversion efficiency, b

The predator conversion efficiency is the prey-dependent birth rate of predators. Thus, for each prey killed, an average predator will produce and successfully rear b offspring. Given that only a few females in a population produce pups, and that number is averaged across the entire wolf population, that number is relatively small. However, a single moose is pretty big and so goes a long way.

Here we use our previous estimates of a, m, and N^* to estimate b, where

$$N^* = \frac{m}{ba}$$

and

$$b = \frac{m}{aN^*}$$

9. Use the above expression to estimate b.

```
b <- m / (a * mean(dw$Moose))
b
```

```
[1] 0.03036836
```

12.11. Predicting population dynamics

This is how you tell R what the differential growth equations are.

```
predpreyLV <- function(t, y, p) {
    with( as.list( c(y,p) ), {
      dN <- r*N - a*N*P
      dP <- b*a*N*P - m*P
      return( list( c(dN, dP) ))
    })
}
```

Here you tell R what the parameters, initial abundances, and years are.

```
parms <- c(r = rMax, a=a, b=b, m=m)
y0 <- c(N=560, P=20)
times <- seq(0, 52, by=0.1)
```

Here you use the `deSolve` package to integrate the rate equations over our time points.

```
out <- ode(y=y0, t=times, fun=predpreyLV, parms=parms)
```

Next we rearrange the data, and graph the time series.

```
op <- out %>% as.data.frame() %>%
  pivot_longer(cols=N:P, names_to = "Populations",
                   values_to="Number")

ggplot(op, aes(x=time, y=Number, colour=Populations)) + geom_line()
 ↪  +
      labs(x="Years") + scale_color_discrete(breaks=c("N","P")) +
      facet_grid(Populations~., scales="free")

ggsave("figs/LV_MooseWolfTS.png", height=5, width=6)
```

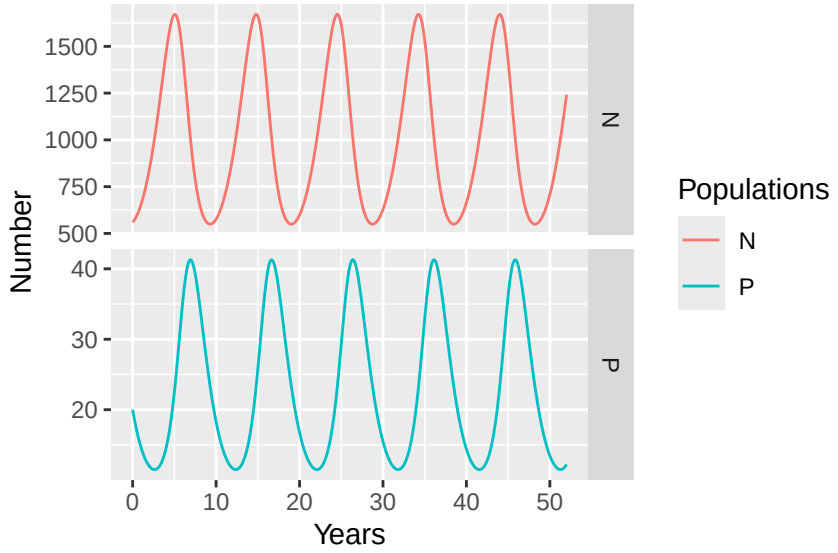


Figure 12.4.: Predicted Lotka-Volterra dynamics based on the Isle Royale moose-wolf data.

10. Once you do this, you should get something that looks like Fig. Figure 12.4, or about 3-6 cycles over 52 years, and the minimum and maximum population sizes should differ tremendously from the observed data. If you get something wackier than that, you've gone astray. If so, *ask for help*.

We can rearrange the data, and graph the *phase plane* plot (Fig. Figure 12.5).

```
outw <- out %>% as.data.frame()

ggplot(outw, aes(N, P, linewidth= -time, color=time)) +
  geom_path() +
  labs(x="Moose population size", y="Wolf population size" )
ggsave("figs/LV_MooseWolfPP.png", height=5, width=6)
```

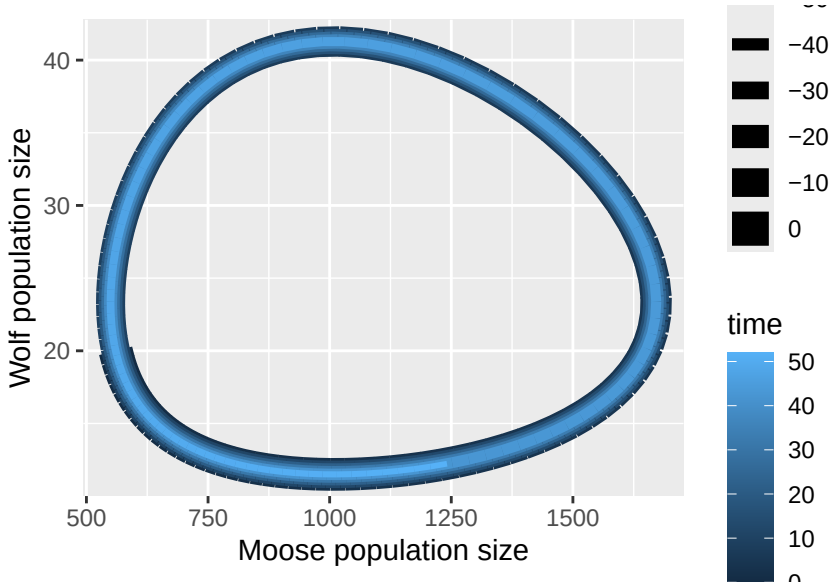


Figure 12.5.: Phase plane diagram of the Lotka-Volterra model of moose wolf dynamics. Once the paramters are fixed and the initial abundances are selected, then the dynamics are completely determined. Initial abundances are moose (560) and wolves (20). The line is black and widest in 1959, and becomes progressively more narrow and blue over time.

Once you are confident that you got something reasonably correct, assess the predicted dynamics.

11. For both the moose and the wolf populations and both the (i) original data and (ii) the Lotka-Volterra model, list the following:

i. the number of cycles,
ii. the heights of the peaks and troughs, and
iii. the **averages**.

Write a sentence or two describing these data in your list.

12. In a short paragraph, explain what you think we can learn from the match (or lack thereof) between the data and the model. Considering only what we have covered in this course, what would you use as an alternative to the Lotka-Volterra model, and how do those compare?

12.12. Discussion questions

How could you test the effect of the parvovirus that began to affect the wolves in 1981?

For the next exam, be able to,

- Explain the meanings of the Lotka-Volterra model parameters.
- Derive the prey and predator isoclines.
- Explain neutral stability.

13. *ForestGEO Community Ecology: Why are there so many tree species, and where?*

Before you start

- **Assumptions about background and guidance:** this chapter uses ideas about counting and comparing populations from the population-growth chapters (a *count* of stems is just population size), and the log-transformation logic from the metabolic ecology chapter (Chapter 15). You can do this chapter without having finished those, but a glance back will help. You should also be able to run a script in your `Rwork` project and make a graph with `ggplot()`; see Computer setup and Displaying data with ggplot2.

13.1. Goals

In this chapter, you will,

- Evaluate the consistency of the latitudinal gradient of species diversity.
- Define species richness, evenness, and diversity, and explain why these are three different things.
- Use a real forest-census data set to measure diversity, and to see why the number you get depends on how hard you looked.
- Develop and evaluate a prediction about how forest diversity changes from the poles to the equator, and confront a real exception to that pattern.
- Practice reading the *shape* of an ecological relationship (rank–abundance and species–accumulation curves) before trusting a single summary number.

Time: ~2–3 h. Most students start the activity in one 80-minute class period and finish the deliverables at home.

13.2. Deliverables

Assemble the following into a single document and submit it online.

1. **Your three hand-drawn predictions** (rank–abundance, species accumulation, richness vs. latitude), photographed, *before* you ran any code.

2. **In your own words (2–3 sentences each):** a. the difference between richness, evenness, and diversity; b. why rarefied richness came out lower than observed richness; c. what `exp(H)` (effective number of species) tells you that raw Shannon H does not.

3. **Reproduce and include** the rank–abundance plot (Figure 13.2) and the global latitude plot (Figure 13.4). For each, write one sentence comparing the real figure to your hand-drawn prediction. Where they differ, explain *why*—this is the point of the assignment.

4. **Build a species–accumulation curve** for BCI and include it. Starter code:

```
accum <- specaccum(BCI)   # cumulative species as plots are
 ↪  added
accum_df <- tibble(plots = accum$sites,
                   richness = accum$richness,
                   sd = accum$sd)

ggplot(accum_df, aes(x = plots, y = richness)) +
  geom_ribbon(aes(ymin = richness - sd, ymax = richness + sd),
              fill = "grey80") +
  geom_line(linewidth = 1) +
  labs(x = "Number of 1-ha plots sampled",
       y = "Cumulative species found") +
  theme_minimal()
```

In one or two sentences, answer: does the curve flatten out by 50 hectares? What does your answer imply about how many tree species the *whole* forest contains versus how many appear in any one plot?

5. **The African exception.** In a short paragraph, propose the explanation you find most convincing for why Korup is less rich than comparable Asian and Neotropical plots, and state one kind of data that would help test your explanation.

6. **Audit the machine.** Below is a paragraph generated by an AI chatbot when asked to summarize this chapter. It sounds authoritative and is *mostly* right, but it contains at least one substantive ecological error. Find the error(s), quote the offending sentence, and correct it using what you learned:

> *"Species richness is the most complete measure of biodiversity because it accounts for both how many species are present and how evenly abundant they are. Because the ForestGEO plots all use the same protocol, a forest's richness is a fixed property, so the 493 species recorded at Korup can be compared directly to the 168 species recorded in a one-hectare plot at BCI. Diversity always increases smoothly toward the equator, with no exceptions."*

This is exactly the skill this whole book is built around: a machine can produce a fluent paragraph in seconds, and only someone who understands the mechanisms can catch where it is subtly—or badly—wrong.

13.3. Preparatory Steps

1. Find out what you need for the deliverables (Section 13.2) **before** you start, including the parts you will sketch by hand and turn in.
2. Watch the video lectures on biodiversity and the latitudinal diversity gradient.
3. Skim (**davies2021?**) for one paragraph on what the ForestGEO network is. You do not need the details—just the idea that many forests around the world have been censused *the same way*. davies2021
4. Install the `vegan` package once (instructions below). Everything else uses packages you already have.

13.4. Background

Walk into a forest in Ohio and you might learn most of the trees in an afternoon: a few oaks, some maples, hickory, beech, ash. Walk into a lowland forest in Panama or Malaysia and you could spend a career and not finish. Why the enormous difference? And how would you even *measure* it fairly, when one forest hands you twenty species and another hands you eight hundred?

To compare forests honestly, ecologists needed forests measured the same way. That is exactly what the **Forest Global Earth Observatory** (**ForestGEO**) provides. Beginning with a single 50-hectare plot on Barro Colorado Island (BCI) in Panama in 1980, the network has grown to dozens of large forest dynamics plots spread across the world's major forests (**davies2021?**). In every plot, field crews follow the same protocol: every free-standing woody stem at least 1 cm in diameter is mapped, tagged, identified to species, and then *re-measured* every five years or so (**anderson-teixeira2015?**). A single 50-ha plot can hold a few hundred thousand individual trees. Because the protocol is identical across continents, a difference in the numbers reflects a difference in the *forests*, not a difference in the *method*. davies2021 anderson-teixeira2015

13.4.1. Three things people mean by "diversity"

"Diversity" is used loosely in conversation, but it bundles together at least three distinct ideas, and confusing them is one of the most common mistakes in ecology.

- **Richness** is simply the *number* of species present. It ignores how common each one is. A forest with 100 species, 99 of them represented

by a single tree, has the same richness as a forest with 100 evenly abundant species.
- **Evenness** describes how equally the individuals are spread among the species. The two forests just described have very different evenness.
- **Diversity** indices (Shannon, Simpson, and others) combine richness and evenness into one number, so that a community is "more diverse" both when it has more species *and* when those species are more evenly abundant.

> **Follow the Logic: why richness alone can fool you**
>
> Imagine two 1-hectare plots. Plot A has 10 species, each with 100 trees. Plot B has 10 species, but one species has 991 trees and the other nine have one tree each. Both have richness = 10. Yet if you walked through Plot B blindfolded and grabbed trees at random, you would almost always grab the same species—it *behaves* like a one-species forest. Richness counts the rare species as fully as the common ones; evenness and diversity indices do not. Before you compute a diversity index, ask yourself which of these three quantities your question is really about.

13.4.2. Richness depends on how hard you looked

Here is the uncomfortable part. Richness is not a fixed property you can read off a forest like its temperature. The more individuals you count, the more species you find, because rare species only show up once you have sampled enough trees to stumble onto them. Census 1 hectare at BCI and you find about 168 tree species; census all 50 hectares and you find roughly 300 (**condit1996?**). The forest did not change—your *sample* got bigger.

condit1996

This is the **species–accumulation** (or species–area) relationship, and it means two numbers are only comparable if they come from the same amount of effort. Comparing "168 species" from a 1-ha plot to "493 species" from a 50-ha plot tells you almost nothing, because the second number had fifty times the chance to find rarities. Ecologists handle this with **rarefaction**: ask how many species you would *expect* if every sample were thinned to the same number of individuals (Gotelli 2001). You will do this below.

Gotelli, N J. 2001. *A {P}rimer of {E}cology*. 3rd ed. Sinauer Associates, Inc.

13.4.3. The latitudinal diversity gradient—and an exception

Across most groups of organisms, diversity rises as you move from the poles toward the equator (**hillebrand2004?**). Tropical forests are the headline example. But the tropics are not uniform. Compared per hectare, lowland forests in the Neotropics and Southeast Asia are staggeringly rich, while African rainforests are consistently *less* rich—a pattern old enough that ecologists once called Africa "the odd man out" (**kenfack2007?**). The 50-ha plot at Korup in Cameroon holds about 493 tree species among more than 329,000 individuals—extraordinary by temperate standards, yet well

hillebrand2004

kenfack2007

below comparable Asian and Neotropical plots censused the same way (**kenfack2007?**). A good theory of diversity has to explain both the broad gradient *and* its exceptions. kenfack2007

13.5. Hypotheses and Predictions

We will pursue two questions, each with its own logic.

Question 1 (within one forest): How is diversity structured at BCI—are species roughly equal in abundance, or are a few common and most rare?

This is an **inductive** question: we will let a large, carefully collected data set reveal a pattern, rather than deducing the pattern from a model first.

Question 2 (across forests): How does tree richness change with latitude across the ForestGEO network?

This one is **deductive**: the latitudinal-gradient hypothesis makes a clear, testable prediction *before* we look at the data—richness should decline as we move away from the equator. When we meet the African exception, we will switch to **abductive** reasoning: given a surprising observation, what is the best explanation?

> **!** Predict before you compute
>
> Before you run a single line of code, sketch three graphs by hand on paper and photograph them for your deliverables:
>
> 1. **Rank–abundance.** Put species *rank* (most abundant = 1) on the x-axis and *abundance* on the y-axis. Draw the shape you expect for a tropical forest.
> 2. **Species accumulation.** Put *number of hectares sampled* on the x-axis and *cumulative species found* on the y-axis. Draw the shape you expect, and mark whether you think it ever flattens out.
> 3. **Richness vs. latitude.** Put *absolute latitude* on the x-axis and *species richness* on the y-axis. Draw the line the latitudinal-gradient hypothesis predicts.
>
> Keep these. The point of the chapter is to compare your hand-drawn predictions to what the data actually do—and to be able to *explain* any mismatch.

13.6. Get R ready for work

We use the `vegan` package, the standard toolkit for community ecology in R, together with the tidyverse and `broom`. `vegan` ships with the **BCI** data set: counts of tree species in fifty 1-hectare plots from the Barro Colorado Island census—real data, already in R, so nothing can break or go offline.

```
## Run this ONCE, then comment it out.
install.packages("vegan")
```

```
library(vegan)      # community ecology: diversity, rarefaction,
 ↪  accumulation
library(tidyverse)  # data wrangling and ggplot2
library(broom)      # tidy model output

data(BCI)           # load the Barro Colorado Island plot-by-species
 ↪  table
```

```
dim(BCI)            # rows = 1-ha plots, columns = species
```

```
[1]  50 225
```

```
BCI[1:5, 1:6]       # a corner of the table: counts of trees, by
 ↪  species, by plot
```

```
  Abarema.macradenia Vachellia.melanoceras Acalypha.diversifolia
1                  0                     0                     0
2                  0                     0                     0
3                  0                     0                     0
4                  0                     0                     0
5                  0                     0                     0
  Acalypha.macrostachya Adelia.triloba Aegiphila.panamensis
1                     0              0                    0
2                     0              0                    0
3                     0              0                    0
4                     0              3                    0
5                     0              1                    1
```

> **Walk me through this data table**
>
> Each **row** is one 1-hectare plot (there are 50). Each **column** is one tree species (there are 225 in this curated subset). Each **cell** is the *number of individual trees* of that species counted in that plot. A community data table in this "sites-by-species" layout is the workhorse of community ecology—almost everything below is just a clever summary of these counts. If you understand that one cell is "how many trees of species *j* I found in plot *i*," you understand the whole table.

13.7. Measuring diversity at BCI

13.7.1. Richness, plot by plot

`specnumber()` counts the species present (abundance > 0) in each plot.

```
richness <- specnumber(BCI)      # one richness value per 1-ha plot
summary(richness)                # how many species per hectare?
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  77.00   86.00   91.00   90.78   94.00  109.00
```

```
tibble(richness = richness) |>
  ggplot(aes(x = richness)) +
  geom_histogram(binwidth = 5, fill = "forestgreen", colour =
   ↪  "white") +
  labs(x = "Species per 1-ha plot", y = "Number of plots") +
  theme_minimal()
```

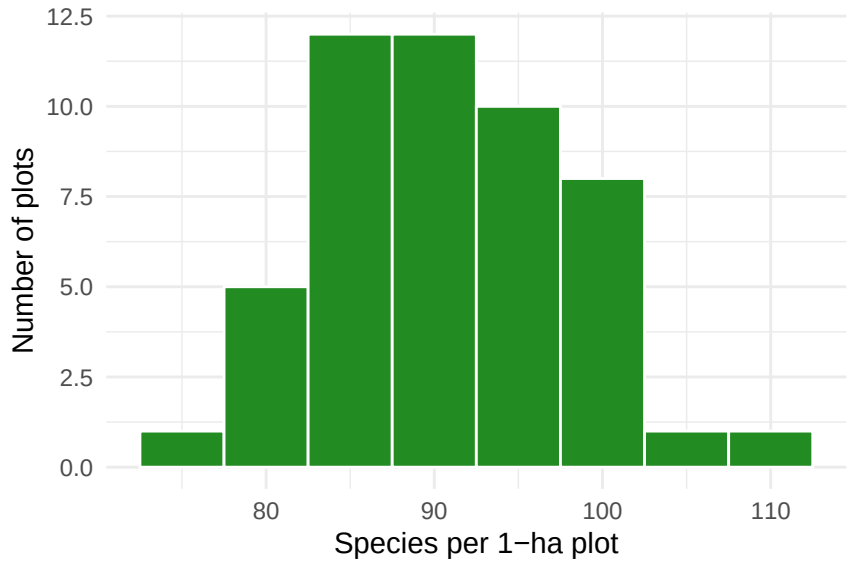


Figure 13.1.: Distribution of tree species richness across the fifty 1-ha BCI plots.

Notice that even within a single 50-ha forest, 1-ha richness varies from plot to plot. Diversity is patchy at small scales.

13.7.2. Evenness and the rank–abundance curve

To *see* evenness, pool all 50 plots, sort species from most to least abundant, and plot abundance against rank. This is a **Whittaker (rank–abundance) plot**.

```
pooled <- colSums(BCI)                      # total trees of each
 ↪  species, all plots

rank_ab <- tibble(species = names(pooled), abundance = pooled) |>
  filter(abundance > 0) |>
  arrange(desc(abundance)) |>
  mutate(rank = row_number())

ggplot(rank_ab, aes(x = rank, y = abundance)) +
```

```
  geom_line(colour = "grey40") +
  geom_point(size = 1) +
  scale_y_log10() +
  labs(x = "Species rank (1 = most abundant)",
       y = "Total abundance (log scale)") +
  theme_minimal()
```

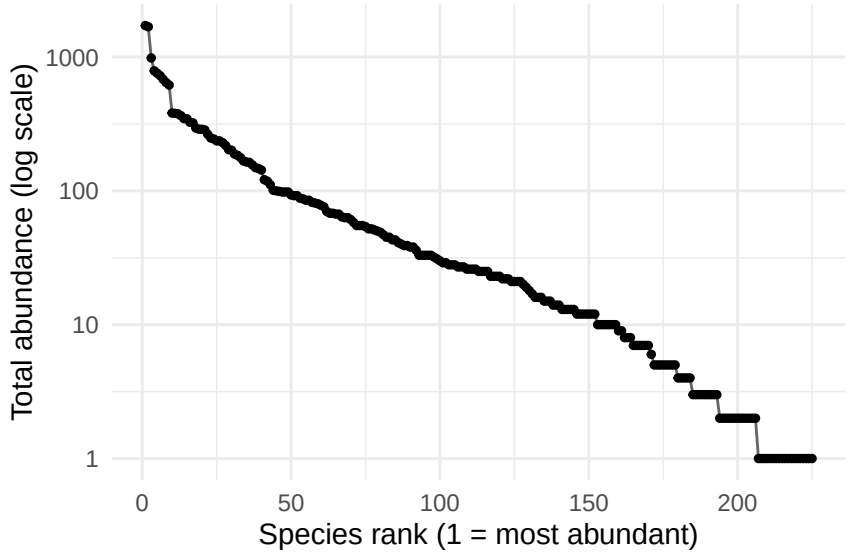


Figure 13.2.: Rank–abundance (Whittaker) plot for pooled BCI trees. A few species are very common; a long tail is rare.

Follow the Logic: why a log scale, and why the long tail matters

On a log y-axis, a straight-ish declining line means abundance falls by a roughly *constant factor* with each step down the ranking—the 10th species is some fixed fraction of the 1st, the 100th a fixed fraction of the 10th. The long, flat tail on the right is the army of **rare** species, each with only a handful of individuals. That tail is the reason richness keeps climbing as you sample more, and the reason a single common-species walk through the forest gives a badly biased impression of diversity. Compare this real curve to the one you drew by hand.

13.7.3. One number: Shannon and Simpson diversity

```
shannon <- diversity(BCI, index = "shannon")   # accounts for
 ↪  richness + evenness
simpson <- diversity(BCI, index = "simpson")   # prob. two random
 ↪  trees differ

tibble(shannon = shannon, simpson = simpson) |>
  summarise(across(everything(), list(mean = mean, sd = sd)))
```

```
# A tibble: 1 x 4
  shannon_mean shannon_sd simpson_mean simpson_sd
```

```
         <dbl>      <dbl>        <dbl>      <dbl>
1         3.82      0.234        0.959     0.0257
```

```
## The "effective number of species": how many EQUALLY-common
 ↪  species
## would give the same Shannon diversity?
effective_species <- exp(shannon)
summary(effective_species)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  14.04   42.27   46.99   46.66   52.94   58.99
```

> **Follow the Logic: what `exp(H)` means**
>
> Shannon's H is in awkward units (it's an entropy). Exponentiating it converts H into an **effective number of species**: the number of *equally abundant* species that would produce the diversity you observed. If a plot has 90 real species but `exp(H)` is only 35, that is the data telling you "this forest *behaves* like 35 evenly common species, because the other ~55 are rare." That single translation—from an abstract index back into 'number of species'—is why effective numbers are now preferred over raw indices.

13.7.4. Rarefaction: comparing fairly

Plots differ in how many trees they contain, so a plot with more stems gets more chances to accumulate species. Rarefaction asks: at a *common* number of individuals, how many species would we expect?

```
stems <- rowSums(BCI)                # total trees per plot
min_stems <- min(stems)              # thin every plot to this many
rarefied <- rarefy(BCI, sample = min_stems)

tibble(observed = richness, rarefied = rarefied) |>
  summarise(mean_observed = mean(observed),
            mean_rarefied = mean(rarefied))
```

```
# A tibble: 1 x 2
  mean_observed mean_rarefied
          <dbl>         <dbl>
1          90.8          83.0
```

Why does the rarefied richness come out lower than the observed richness? Make sure you can answer that in your own words before moving on—it is the whole reason rarefaction exists.

13.7.5. Does richness just track the number of stems?

If richer plots are simply the ones with more trees, that is worth knowing. We can fit a quick linear model and read it with `broom`.

```
model_df <- tibble(richness = richness, stems = rowSums(BCI))

fit <- lm(richness ~ stems, data = model_df)

tidy(fit)      # slope, intercept, and their uncertainty
```

```
# A tibble: 2 x 5
  term        estimate std.error statistic  p.value
  <chr>          <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept) 91.9       10.2        8.97  7.85e-12
2 stems       -0.00260    0.0238    -0.109 9.13e- 1
```

```
glance(fit)    # r.squared: how much of richness variation stems
 ↪  explain
```

```
# A tibble: 1 x 12
  r.squared adj.r.squared sigma statistic p.value    df logLik   AIC
   ↪  BIC
      <dbl>         <dbl> <dbl>     <dbl>   <dbl> <dbl>  <dbl> <dbl>
       ↪  <dbl>
1  0.000249       -0.0206  7.08    0.0120   0.913     1  -168.  342.
 ↪  347.
# i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

```
ggplot(model_df, aes(x = stems, y = richness)) +
  geom_point() +
  geom_smooth(method = "lm", se = TRUE, colour = "darkorange") +
  labs(x = "Stems per 1-ha plot", y = "Species per 1-ha plot") +
  theme_minimal()
```

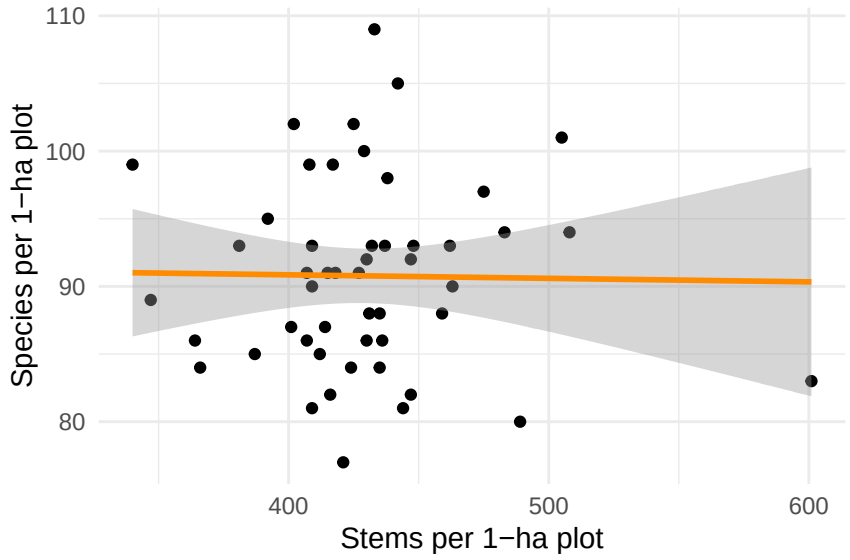


Figure 13.3.: Plot-level richness vs. number of stems, with a fitted line.

> **Walk me through `tidy()` and `glance()`**
>
> `lm()` fits the line. Instead of reading the dense default `summary()`, `broom` gives us two tidy tables. `tidy(fit)` returns one row per coefficient—the slope tells you how many extra species you get per additional stem, and its `p.value` tells you whether that slope is distinguishable from zero. `glance(fit)` returns one row for the whole model; the key column is `r.squared`, the fraction of plot-to-plot richness variation explained by stem number. If `r.squared` is low, then richness is *not* just a stem-count artifact—something ecological is going on.

13.8. Comparing forests across the globe

Now the cross-continental question. Below is a small table of ForestGEO (and sister) plots, with the species richness reported for each and the plot area it came from. The numbers are drawn from the published site descriptions and papers cited at the end of the chapter.

```
sites <- tribble(
  ~site,             ~country,    ~continent,      ~abs_latitude,
   ↪  ~area_ha, ~species,
  "SCBI",            "USA",       "North America",          38.9,
   ↪  25.6,       65,
  "BCI",             "Panama",    "Neotropics",              9.2,
   ↪  50.0,      300,
  "Korup",           "Cameroon",  "Africa",                  5.1,
   ↪  50.0,      493,
  "Pasoh",           "Malaysia",  "Asia",                    3.0,
   ↪  50.0,      800,
  "Yasuni",          "Ecuador",   "Neotropics",              0.7,
   ↪  50.0,     1150
)

sites
```

```
# A tibble: 5 x 6
  site   country  continent     abs_latitude area_ha species
  <chr>  <chr>    <chr>                <dbl>   <dbl>   <dbl>
1 SCBI   USA      North America         38.9    25.6      65
2 BCI    Panama   Neotropics             9.2    50       300
3 Korup  Cameroon Africa                  5.1    50       493
4 Pasoh  Malaysia Asia                    3      50       800
5 Yasuni Ecuador  Neotropics             0.7    50      1150
```

> **A trap built into this table**
>
> These richness numbers come from **different plot areas** (25.6 ha vs. 50 ha). You just spent half a chapter learning that richness depends

on sampling effort. So before you compare these numbers, ask: is it fair to compare a 25.6-ha count to a 50-ha count? For the cleanest comparison, ecologists use *1-hectare* richness, where the published values are roughly: BCI **168**, Pasoh **497**, Yasuni **670** species per hectare (**condit1996?**). Carry that caveat into every conclusion you draw below.

condit1996

```
ggplot(sites, aes(x = abs_latitude, y = species,
                  colour = continent, size = area_ha)) +
  geom_point(alpha = 0.8) +
  geom_text(aes(label = site), vjust = -1, size = 3, show.legend =
   ↪  FALSE) +
  labs(x = "Absolute latitude (degrees from equator)",
       y = "Species richness (in plot)",
       colour = "Continent", size = "Plot area (ha)") +
  theme_minimal() # + scale_y_log10()
```

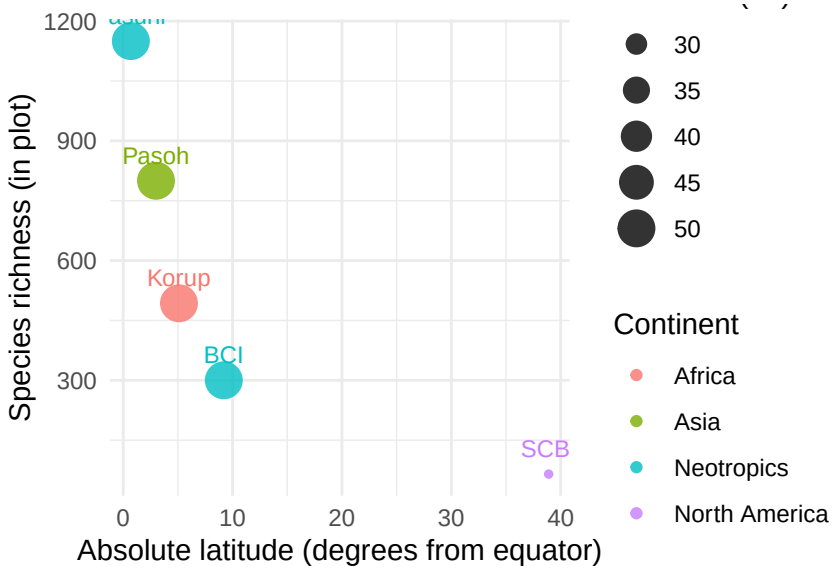


Figure 13.4.: Tree species richness vs. absolute latitude across forest plots. Point size shows plot area; colour shows continent. Mind the area caveat.

Two things should jump out. First, the broad **latitudinal gradient**: the temperate plot (SCBI, Virginia) has a tiny fraction of the richness of the tropical plots—exactly what the hypothesis predicted, and, ideally, what you drew by hand. Second, the **exception**: among the tropical plots, the African site (Korup) is clearly less rich than the Asian and Neotropical sites, even after accounting for area. The gradient is real, but it is not the whole story.

Follow the Logic: from deduction to abduction

The latitudinal-gradient hypothesis *deduced* a prediction (richness falls with latitude) and the data largely confirmed it. But Korup is a surprise the hypothesis did not predict. Now you reason **abductively**: what is the best explanation for a depauperate African tropics?

Candidate explanations in the literature include a drier, more variable climate history (Pleistocene contractions of African rainforest), a smaller total area of lowland tropical forest on the continent, and higher past extinction. You do not have to settle this—you have to recognize that a good observation generates the *next* hypothesis, and to be able to say what data would help you choose among these explanations.

13.9. References

This chapter cites (**davies2021?**), (**anderson-teixeira2015?**), (**condit1996?**), (**kenfack2007?**), (**hillebrand2004?**), Gotelli (2001), and the `vegan` package (Oksanen et al. 2026). Full references appear in the book's reference list once these keys are in your `.bib`.

davies2021 anderson-teixeira2015 condit1996 kenfack2007 hillebrand2004

Gotelli, N J. 2001. *A {P}rimer of {E}cology*. 3rd ed. Sinauer Associates, Inc.

Oksanen, Jari, Gavin L. Simpson, F. Guillaume Blanchet, et al. 2026. *Vegan: Community Ecology Package*. Manual. https://doi.org/10.32614/CRAN.package.vegan.

14. GBIF: Why are there so many species, and where?

Before you start

- **Assumptions about background and guidance:** this chapter treats a *count* of individuals as a population size (the population-growth chapters) and reuses the log-transformation logic from the metabolic ecology chapter (Chapter 15). You can do this chapter on its own, but a glance back will help. You should also be able to run a script in your `Rwork` project and make a graph with `ggplot()` (Computer setup, Displaying data with ggplot2). The data come live from GBIF, so you will need an internet connection; see Accessing public databases for more on GBIF.

14.1. Goals

In this chapter, you will,

- Define species richness, evenness, and diversity, and explain why they are three different things.
- Pull real biodiversity data **live** from a public scientific data repository (GBIF) and measure diversity with it.
- Develop and evaluate a prediction about how diversity changes from the poles to the equator—and confront a real exception.
- See, in real data, why the number of species you "find" depends on how hard people looked, and why that makes raw richness comparisons dangerous.

Time: ~3 h. Most students start the activity in one 80-minute class period and finish the deliverables at home. You need an internet connection.

14.2. Deliverables

Assemble the following into a single document and submit it online.

1. **Your three hand-drawn predictions** (richness vs. latitude; the three equatorial countries; richness vs. effort), photographed *before* you ran any code.

2. **In your own words (2–3 sentences each):** a. the difference between richness, evenness, and diversity; b. why GBIF richness reflects sampling effort, and what rarefaction does about it; c. why comparing whole countries confounds latitude with area.

3. **Reproduce and include** the country-richness figure (Figure 14.1) and the gradient figure (Figure 14.3). For each, write one sentence comparing the real figure to your hand-drawn prediction, and *explain* any mismatch.

4. **The African anomaly.** Using the `records` column and your rarefaction result, argue whether Cameroon's low palm richness is mostly *biology* or mostly *under-sampling*. State one additional piece of evidence that would strengthen your case.

5. **Oaks vs. palms.** Re-run the country analysis for the oak family, `name_backbone(name = "Fagaceae")$usageKey`, across the same countries. Do oaks show the *same* latitudinal pattern as palms? In two or three sentences, explain what it means that "the latitudinal gradient" can point different directions for different groups.

6. **Audit the machine.** Below is a paragraph an AI chatbot produced when asked to summarize this chapter. It is fluent and mostly right, but contains at least one substantive error. Quote the offending sentence(s) and correct them:

   > *"Species richness is the most complete measure of biodiversity because it captures both how many species live somewhere and how evenly abundant they are. Because GBIF aggregates records from around the world, the number of species it reports for a country is an accurate, effort-independent count of the species that live there. Diversity always increases smoothly toward the equator, so any tropical country will be more diverse than any temperate one."*

Catching where a confident-sounding summary is wrong—about effort, about what richness measures, about exceptions to the gradient—is the whole point of learning the mechanisms instead of trusting the output.

7. **Cite your data.** GBIF results change daily, so reproducibility requires a citable snapshot. Read the one-paragraph note below and state, in one sentence, why a real publication should use `occ_download()` rather than the live `occ_count()` calls we used for teaching.

> **Citing GBIF properly.** The quick `occ_count()`/`occ_search()` calls here are perfect for learning but are not citable snapshots. For research, `rgbif`'s `occ_download()` workflow asks GBIF to assemble your exact records and issues a **DOI** that pins the data and credits every contributing dataset. Always cite that DOI plus GBIF itself (**gbif?**) and the `rgbif` package (**rgbif?**).

14.3. Preparatory Steps

1. Find out what you need for the deliverables (Section 14.2) **before** you start, including the parts you will sketch by hand and turn in.
2. Watch the video lectures on biodiversity and the latitudinal diversity gradient.
3. Open gbif.org in a browser. Search for "Arecaceae" (the palm family) and look at the occurrence map. You are about to query that same database from R.
4. Install the `rgbif` and `vegan` packages once (instructions below).

14.4. Background

Walk into a forest in Ohio and you could learn most of the trees in an afternoon: a few oaks, some maples, hickory, beech, ash. Walk into a lowland forest in Colombia or Malaysia and you could spend a career and not finish. Why the enormous difference—and how would you measure it *fairly*, when one place hands you twenty species and another hands you eight hundred, and nobody counted either one the same way?

14.4.1. A gold standard, and why we usually can't use it

davies2021

The cleanest way to compare forests is to measure them identically. That is what the **Forest Global Earth Observatory (ForestGEO)** does: in each of its large plots, every woody stem at least 1 cm in diameter is mapped, tagged, and identified, then re-measured every few years (**davies2021?**). Effort is held constant, so a difference in the numbers is a difference in the *forests*.

rgbif

The catch is that there are only a few dozen such plots on Earth, and the data often require formal requests. Most of what humanity knows about *where* species live is not tidy plot censuses—it is hundreds of years of messy, uneven observations: museum specimens, herbarium sheets, naturalists' notebooks, and now millions of smartphone photos. The **Global Biodiversity Information Facility (GBIF)** gathers those records into one searchable repository of more than two billion occurrences (**rgbif?**). It is gloriously real, gloriously global, and gloriously *biased*—and learning to work with it honestly is one of the most valuable data skills in modern ecology.

14.4.2. Three things people mean by "diversity"

- **Richness** is the *number* of species present. It ignores how common each one is.
- **Evenness** describes how equally individuals are spread among species.

- **Diversity** indices (Shannon, Simpson) combine richness and evenness into a single number.

> **Follow the Logic: why richness alone can fool you**
>
> Two forests each have 10 species. In the first, each species has 100 trees. In the second, one species has 991 trees and the other nine have one tree each. Both have richness = 10. But if you walked the second forest blindfolded and grabbed trees at random, you would almost always grab the same species—it *behaves* like a one-species forest. Richness counts a species seen once exactly as much as a species seen a thousand times; evenness and diversity indices do not. Before you compute anything, decide which of the three you actually care about.

14.4.3. Richness depends on how hard you looked

Here is the uncomfortable part, and it is the heart of this chapter. Richness is not a fixed property you can read off a place like its temperature. The more you look, the more species you find, because rare species only appear once you have sampled enough to stumble onto them. In a ForestGEO plot, "how hard you looked" is fixed by protocol. In GBIF, it is wildly uneven: a well-studied national park in Europe may have a hundred thousand records, while an equally rich forest in central Africa has a few hundred—not because one is poorer, but because more people with more cameras have walked the first one. Comparing raw richness between them measures *researcher effort* at least as much as it measures *biology*. Ecologists deal with this using **rarefaction**: asking how many species you would expect if every place were sampled to the same number of records (Gotelli 2001). You will do this below, with real data.

Gotelli, N J. 2001. *A {P}rimer of {E}cology*. 3rd ed. Sinauer Associates, Inc.

14.4.4. The latitudinal diversity gradient—and an exception

hillebrand2004

Across most groups of organisms, diversity rises from the poles toward the equator (**hillebrand2004?**). Palms are a textbook case: there are essentially no wild palms in Canada and thousands in the equatorial tropics. But the tropics are not uniform. Mainland tropical **Africa** is strikingly *palm-poor* compared to equally wet, equally warm forests in tropical Asia and the Americas—a real biogeographic puzzle usually attributed to Africa's drier, more turbulent climate history (**kissling2012?**). A good theory of diversity has to explain both the broad gradient *and* its exceptions.

kissling2012

14.5. Hypotheses and Predictions

We pursue three questions, each with its own logic.

Q1 (across continents): At roughly the *same* latitude, do equally tropical forests on different continents have the same palm richness? — an **inductive** look for a pattern.

Q2 (across latitudes): Does palm richness decline as we move away from the equator? — a **deductive** test of the latitudinal-gradient hypothesis.

Q3 (the catch): How much of the "richness" we measure is really just *sampling effort*? — the question that decides whether Q1 and Q2 mean anything. When Africa breaks the pattern in Q1, we will reason **abductively**: given a surprising observation, what is the best explanation?

> ! Predict before you compute
>
> Before you run any code, sketch three graphs by hand and photograph them for your deliverables:
>
> 1. **Richness vs. latitude.** Absolute latitude on the x-axis, palm species richness on the y-axis. Draw the line the gradient hypothesis predicts.
> 2. **Three equatorial countries.** Bars for Colombia, Malaysia, and Cameroon (all near the equator). Draw how tall you expect each bar to be.
> 3. **Richness vs. effort.** Number of records on the x-axis, species found on the y-axis. Draw the shape you expect, and mark whether it ever flattens.
>
> Keep these. The chapter's payoff is comparing them to what the data do—and *explaining* the mismatches.

14.6. Get R ready for work

```
## Run ONCE, then comment out.
install.packages(c("rgbif", "vegan"))
```

```
library(rgbif)      # live access to the GBIF biodiversity
 ↪  repository
library(vegan)      # diversity, rarefaction
library(tidyverse)  # wrangling and ggplot2
library(broom)      # tidy model output
```

First we look up the taxonomic key GBIF uses for the palm family. Every query below filters on this key, so we are always asking specifically about palms.

```
palm_key <- name_backbone(name = "Arecaceae")$usageKey
palm_key
```

```
[1] "7681"
```

> The `#| cache: true` on the data chunks means the book stores GBIF's answers so it can rebuild without re-querying every time. GBIF grows constantly, so if you re-run these with the cache cleared, **your numbers will differ slightly from the printed ones.** That is not an error—it is what a living database looks like.

14.7. Q1 — Same latitude, different continents

We ask GBIF how many palm species have been recorded in each of several countries. Three of them—Colombia, Malaysia, Cameroon—sit at almost the same distance from the equator but on three different continents. We add Madagascar (an African island famous for endemic palms) and two temperate countries for contrast.

The modern `occ_count()` returns one row per species when we facet on `speciesKey`, so the *number of rows* is the species richness.

```
countries <- tribble(
  ~country,      ~iso, ~continent,      ~abs_latitude,
  "Colombia",    "CO", "S. America",            4.0,
  "Malaysia",    "MY", "Asia",                  4.0,
  "Cameroon",    "CM", "Africa",                6.0,
  "Madagascar",  "MG", "Africa (island)",      19.0,
  "United States","US","N. America",           38.0,
  "United Kingdom","GB","Europe",              54.0
)

palm_richness <- function(iso) {
  occ_count(facet = "speciesKey", facetLimit = 99999,
            taxonKey = palm_key, country = iso) |> nrow()
}

palm_records <- function(iso) {
  occ_count(taxonKey = palm_key, country = iso)
}

countries <- countries |>
  mutate(richness = map_int(iso, palm_richness),
         records  = map_dbl(iso, palm_records))

countries |> select(country, continent, abs_latitude, richness,
 ↪  records)
```

```
# A tibble: 6 x 5
  country        continent       abs_latitude richness records
  <chr>          <chr>                  <dbl>    <int>   <dbl>
1 Colombia       S. America                 4      383  212888
2 Malaysia       Asia                       4      507   14381
```

```
3 Cameroon       Africa                  6       55    2056
4 Madagascar     Africa (island)        19      238   12141
5 United States  N. America             38      626   75350
6 United Kingdom Europe                 54       84     709
```

> **Walk me through this code**
>
> `name_backbone()` translates the name "Arecaceae" into the numeric key GBIF stores it under. `occ_count(facet = "speciesKey", ...)` asks GBIF to break its count down by species and hand back one row per species; piping that to `nrow()` counts the species = richness. The second helper, with no facet, returns the plain total number of *records* (our measure of effort). `map_int()` and `map_dbl()` just run those helpers once per country and collect the answers into new columns. Everything is one live request per country—no files, no downloads.

```
countries |>
  mutate(country = fct_reorder(country, richness)) |>
  ggplot(aes(x = country, y = richness, fill = continent)) +
  geom_col() +
  coord_flip() +
  labs(x = NULL, y = "Recorded palm species (GBIF)", fill =
   ↪  "Region") +
  theme_minimal()
```

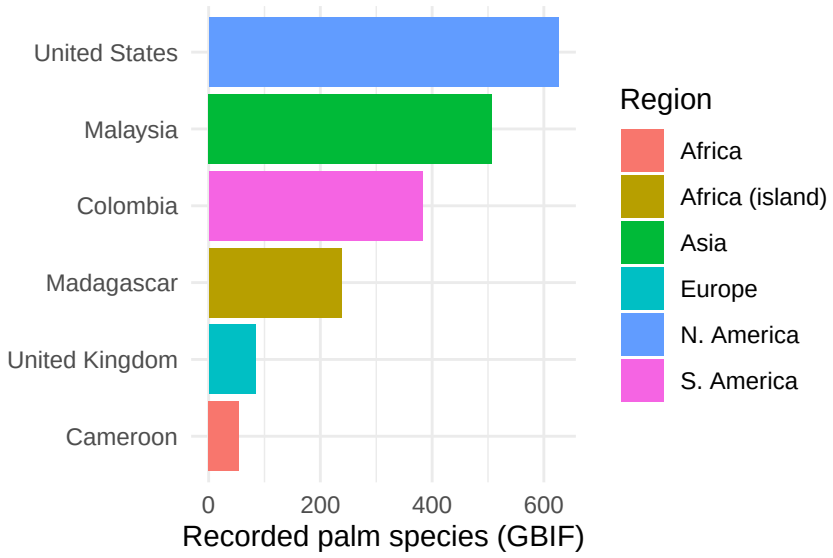


Figure 14.1.: Recorded palm species richness by country. The three left-hand bars sit at nearly the same latitude but on different continents.

Two things should stand out. The temperate countries have almost no wild palms, as the gradient predicts. And among the three equatorial countries, the African mainland (Cameroon) is far poorer than tropical Asia or South America—even though all three are wet, warm, and near the equator. That is the African anomaly, live.

> **Follow the Logic: before you trust that African bar**
>
> Cameroon's bar is short. Two very different stories could produce it: (1) central Africa really does have fewer palm species, or (2) far fewer botanists have uploaded records from Cameroon than from Malaysia or Colombia, so we simply haven't *found* its palms yet. Look at the `records` column you just printed. If Cameroon also has far fewer records, you cannot yet tell biology from effort. This is exactly why the next two sections exist.

14.8. Q3 — How much is effort? Rarefaction

Plot richness against effort directly. If richer countries are mostly the better-sampled ones, the points will climb together.

```
ggplot(countries, aes(x = records, y = richness, label = country)) +
  geom_point(size = 2) +
  geom_text(vjust = -0.8, size = 3) +
  scale_x_log10() +
  labs(x = "Number of GBIF records (log scale)", y = "Recorded palm
   ↪  species") +
  theme_minimal()
```

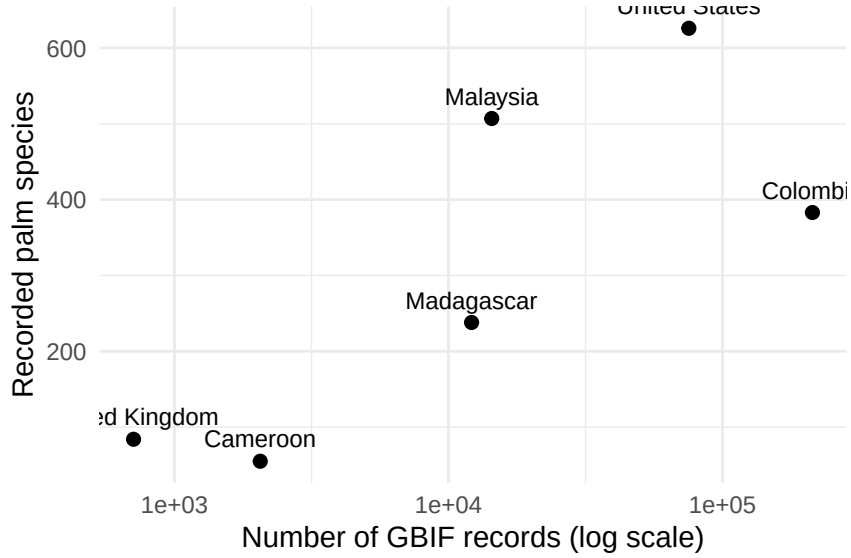


Figure 14.2.: Recorded richness rises with sampling effort (number of records). Raw richness is partly a map of where people have looked.

To compare *fairly*, we **rarefy**: download a sample of actual occurrence records for two equatorial countries, count records per species, and ask how many species each would show if thinned to the same number of records.

```
get_species_counts <- function(iso, n = 3000) {
  occ_search(taxonKey = palm_key, country = iso,
             hasCoordinate = TRUE, limit = n)$data |>
```

```
    filter(!is.na(species)) |>
    count(species, name = "n")
}

co <- get_species_counts("CO")   # Colombia
cm <- get_species_counts("CM")   # Cameroon
```

```
## Build named count vectors and rarefy both to a common sample
 ↪  size.
co_vec <- setNames(co$n, co$species)
cm_vec <- setNames(cm$n, cm$species)

common_n <- min(sum(co_vec), sum(cm_vec))

tibble(
  country        = c("Colombia", "Cameroon"),
  raw_species    = c(length(co_vec), length(cm_vec)),
  rarefied_species = c(rarefy(co_vec, sample = common_n),
                       rarefy(cm_vec, sample = common_n))
)
```

```
# A tibble: 2 x 3
  country  raw_species rarefied_species
  <chr>          <int>            <dbl>
1 Colombia         102             82.8
2 Cameroon          46             46
```

> **Follow the Logic: what rarefaction does and does not fix**
>
> Rarefaction puts two samples on equal footing for *number of records*, so a difference that survives rarefaction is unlikely to be pure effort. But it cannot fix everything: if collectors in one country ignored common roadside palms and chased rare ones, the *sample* is biased no matter how many records it has. Rarefaction equalizes quantity, not quality. State this limit in your write-up.

14.9. Q2 — The gradient, done right

Comparing whole countries confounds latitude with *area*: Brazil is enormous and the UK is small, and bigger areas hold more species regardless of latitude. To isolate latitude, query equal-sized boxes ($10° \times 10°$) marching down the Americas, so area is held roughly constant and only latitude changes.

```
boxes <- tribble(
  ~lon,  ~lat,
  -110,   50,    # W Canada / N US
  -100,   35,    # S US
  -100,   20,    # Mexico
```

```
   -75,    5,    # Colombia
   -70,  -10,    # W Brazil / Peru
   -60,  -25     # S Brazil / Paraguay
)

box_wkt <- function(lon, lat, half = 5) {
  sprintf("POLYGON((%f %f,%f %f,%f %f,%f %f,%f %f))",
          lon - half, lat - half,  lon + half, lat - half,
          lon + half, lat + half,  lon - half, lat + half,
          lon - half, lat - half)
}

box_richness <- function(lon, lat) {
  occ_count(facet = "speciesKey", facetLimit = 99999,
            taxonKey = palm_key, geometry = box_wkt(lon, lat)) |>
             ↪  nrow()
}

boxes <- boxes |>
  mutate(abs_latitude = abs(lat),
         richness = map2_int(lon, lat, box_richness))

boxes
```

```
# A tibble: 6 x 4
    lon   lat abs_latitude richness
  <dbl> <dbl>        <dbl>    <int>
1  -110    50           50        3
2  -100    35           35       60
3  -100    20           20      101
4   -75     5            5      384
5   -70   -10           10      173
6   -60   -25           25       60
```

```
ggplot(boxes, aes(x = abs_latitude, y = richness)) +
  geom_point(size = 2) +
  geom_smooth(method = "lm", se = TRUE, colour = "darkorange") +
  labs(x = "Absolute latitude (degrees from equator)",
       y = "Palm species in box") +
  theme_minimal()
```

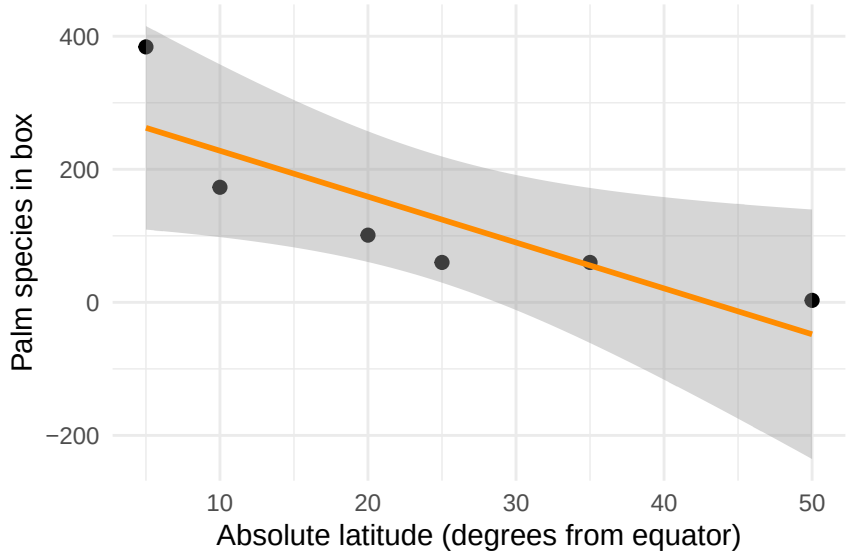


Figure 14.3.: Palm species richness in equal-sized 10°×10° boxes vs. absolute latitude, with a fitted line.

```
fit <- lm(richness ~ abs_latitude, data = boxes)
tidy(fit)      # slope: species lost per degree of latitude
```

```
# A tibble: 2 x 5
  term         estimate std.error statistic p.value
  <chr>           <dbl>     <dbl>     <dbl>   <dbl>
1 (Intercept)    297.       64.3       4.61 0.00993
2 abs_latitude    -6.90      2.26     -3.06 0.0378
```

```
glance(fit)    # r.squared: how much of the variation latitude
 ↪  explains
```

```
# A tibble: 1 x 12
  r.squared adj.r.squared sigma statistic p.value    df logLik   AIC
   ↪  BIC
      <dbl>         <dbl> <dbl>     <dbl>   <dbl> <dbl>  <dbl> <dbl>
       ↪  <dbl>
1     0.700         0.625  83.6      9.34  0.0378     1  -33.9  73.7
 ↪  73.1
# i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

Walk me through the box query and the model

`box_wkt()` writes the corners of a square as a text "polygon" in the format GBIF expects (longitude latitude pairs, closed back to the start). `box_richness()` hands that polygon to `occ_count()` and counts species the same way we did per country. Because every box is the same size, differences in richness can't be blamed on area. `lm()` fits a straight line; `tidy(fit)` reports its slope (how many palm species disappear per degree you move from the equator) and `glance()` reports `r.squared`, the share of box-to-box variation latitude accounts for.

The slope should be negative: palm richness falls as you leave the equator, exactly as the latitudinal-gradient hypothesis predicts—and now with area controlled, so you can believe it.

14.10. References

> This chapter cites (**davies2021?**), (**hillebrand2004?**), Gotelli (2001), (**kissling2012?**), the GBIF repository (**gbif?**), and the `rgbif` package (**rgbif?**). Full references appear in the book's reference list once these keys are in your `.bib`.

davies2021
hillebrand2004
kissling2012
gbif
rgbif
Gotelli, N J. 2001. *A {P}rimer of {E}cology*. 3rd ed. Sinauer Associates, Inc.

15. *Metabolic Ecology: How do body size and temperature affect nutrient cycling?*

(a) Gizzard shad (b) Three reservoirs

15.1. *Acknowledgement*

Vanni, Michael J., and Jessica A. Gephart. 2011. "Metabolic Ecology: How Do Body Size and Temperature Affect Nutrient Cycling Rates?" *Teaching Issues and Experiments in Ecology* 7: 1386–97.

This chapter is an R version of Vanni and Gephart (2011). I am grateful to Mike and Jessica and the Ecological Society of America for making the data and the article available.

Before you start

- **Assumptions about background and guidance:** your `Rwork` project and the ability to read a .csv file and make a scatterplot with `ggplot()` (Computer setup, Extra-pair copulation), and the idea of a straight line with a slope and an intercept. Logarithms and linear regression are both explained in this chapter (see Using logarithms), so neither is assumed. For more on figures, see Displaying data with ggplot2.

15.2. *Goals*

In this chapter, you will explore how two fundamental properties of organisms—body size and temperature—control rates of nutrient cycling. Along the way, you will test specific quantitative predictions of the **Metabolic Theory of Ecology** (MTE), and discover how well those predictions hold up under real field conditions.

If you successfully complete this chapter, you will be able to

- explain how metabolic rates scale with body size (allometry),
- explain how metabolic rates depend on temperature (Q_{10}),
- use log-transformations to linearize power-law relationships,
- fit and interpret linear regressions in R,
- evaluate the support for specific quantitative hypotheses, and
- reason about how body size and temperature act simultaneously on a biological rate.

15.3. Deliverables

Compile the following into one document and submit by the due date. Turn in a Word document or PDF with the following:

1. Log-log scatter plots of N and P excretion vs. body mass for each temperature.
2. Confidence intervals of the allometric scaling exponent (z) for both N and P excretion, and for temperature sensitivity (E_a).
3. Address these questions:

    a. How well do the predictions of the Metabolic Theory of Ecology hold up for nutrient excretion by fish under field conditions? What specific factor or factors might cause deviations from MTE predictions in the field?
    b. Suppose a lake warms by 2°C due to climate change. Based on your E_a estimate, by how much would individual excretion rates change?
    c. If warming also shifts the fish population toward smaller body sizes (as has been observed in some systems), how would that interact with the temperature effect on total nutrient flux?

15.4. Preparatory steps

1. General background: Read the sections below on allometry and temperature dependence. If your instructor has assigned background readings or lectures on metabolic ecology, complete those first.
2. Data: go to our class data repository, and download `fish_nutrient_excretion.csv`.

    a. Put `fish_nutrient_excretion.csv` in your `data` directory (inside `Rwork`).

3. Find out what you need for the deliverables (Section 15.3).

15.5. Background

15.5.1. The Metabolic Theory of Ecology: Why body size and temperature matter

The metabolic rates of organisms—plants, microbes, and animals—are strongly influenced by body size and body temperature (Gillooly et al. 2001; Brown et al. 2004). Large organisms obviously have higher metabolic rates than small ones, when rates are expressed on a *per individual* basis. And because biochemical reactions are temperature-dependent, most metabolic rates increase with body temperature, at least up to some upper tolerance limit.

Gillooly, James F., James H. Brown, Geoffrey B. West, Van M. Savage, and Eric L. Charnov. 2001. "Effects of Size and Temperature on Metabolic Rate." *Science* 293 (5538): 2248–51. https://doi.org/10.1126/science.1061967.

Brown, James H., James F. Gillooly, Andrew P. Allen, Van M. Savage, and Geoffrey B. West. 2004. "Toward a Metabolic Theory of Ology." *Ecology* 85 (7): 1771–89. https://doi.org/10.1890/03-9000.

Understanding how biological rates scale with size and temperature is important for several reasons. There is enormous variation in body size among Earth's life forms, from tiny bacteria to great whales and giant sequoia trees. Even within a single species, body size can vary dramatically. Knowledge of how rates vary with size tells us how individuals of different sizes use resources, process materials like nutrients and energy, and interact with other species. Similarly, because organisms experience a wide range of temperatures—both naturally and increasingly due to human activity—understanding temperature effects helps us predict how organisms and ecosystems may respond to environmental change.

The **Metabolic Theory of Ecology** (MTE) provides mechanistic explanations for these relationships (West et al. 1997; Brown et al. 2004; Banavar et al. 2014). The theory describes how diffusion and diffusion networks in an individual organism cause body size alone to exert a strong and predictable effect on whole-organism metabolic rate. This theory also describes how the temperature regulation of biochemical processes at the molecular level has a similarly profound and predictable influence on whole-organism metabolic rate. Notably, and in contrast to most ecological theory, MTE makes predictions about the quantitative effects of body size and temperature metabolic rate.

West, G B, J H Brown, and B J Enquist. 1997. "A General Model for the Origin of Allometric Scaling Laws in Biology." *Science* (*New York, N.Y.*) 276: 122–26.

Brown, James H., James F. Gillooly, Andrew P. Allen, Van M. Savage, and Geoffrey B. West. 2004. "Toward a Metabolic Theory of Ology." *Ecology* 85 (7): 1771–89. https://doi.org/10.1890/03-9000.

Banavar, Jayanth R, Todd J Cooke, Andrea Rinaldo, and Amos Maritan. 2014. "Form, Function, and Evolution of Living Organisms." *Proceedings of the National Academy of Sciences of the United States of America* 111: 3332–37. https://www.ncbi.nlm.nih.gov/pubmed/24550479.

At the base of all the explanation is the geometry of the resource distribution system (vascular system, simple 3D diffusion). All organisms take in limiting resources and have to distribute those resources to each part of each cell in the body. The key point is that *the larger the organism, the greater the portion of the resources are in transit at any instant in time*. This leads to an increasingly inefficient system, in which the metabolism of larger organisms has to run more slowly per unit resource:

15.5.2. Allometry: how rates scale with body size

At a constant temperature, the effect of body size on metabolic rate can be described with a simple power law:

$$B = B_0 M^z \tag{15.1}$$

where B is the organism's metabolic rate (e.g., micromoles of nitrogen excreted per hour), B_0 is a normalization constant, z is the **allometric scaling exponent**, and M is the organism's body mass.

> Follow the Logic: Why log-transform?
>
> Equation 15.1 is a *power law*—the dependent variable (B) is a power function of the independent variable (M). Power laws are curved on arithmetic axes, which makes them hard to interpret visually and statistically. But watch what happens when we take the logarithm of both sides:
>
> $$\log B = \log B_0 + z \cdot \log M \tag{15.2}$$
>
> This is the equation of a straight line! The slope is z (the allometric exponent) and the intercept is $\log B_0$. This is why ecologists plot metabolic data on log-log axes: it turns a curved relationship into a linear one, and the slope tells us directly how metabolic rate scales with body size.
> This is a case where the mathematics *is* the ecological reasoning. The log-transformation isn't just a trick to make a pretty graph—it reveals the fundamental scaling relationship and lets us estimate the parameter (z) that captures how biology works.

Allometric theory and extensive empirical data show that for many biological rates, the increase in metabolic rate is less than proportional to the increase in body mass. Mathematically, this means $b < 1$, sometimes called **negative allometry**. The **Metabolic Theory of Ecology** (MTE) makes a very specific prediction: $b = 0.75$ (West et al. 1997; Brown et al. 2004; Banavar et al. 2014). This prediction arises from theory about how resource distribution networks (like circulatory systems) scale with body size.

To get a feel for why $b < 1$ matters, consider this: an elephant (5×10^6 g) weighs about 500,000 times more than a mouse (10 g). If $b = 0.75$, the elephant's metabolic rate is only about 19,000 times greater than the mouse's—proportionally much less than the difference in body mass. Per gram, the mouse burns hotter.

Thus, Larger organisms can process more resources per unit time ($B = aM^{3/4}$), but do so less and less efficiently ($\frac{B}{M} = aM^{-1/4}$) due to resources in transport.

However, the prediction that $b = 0.75$ is somewhat controversial. Some data suggest that the exponent is closer to 1.0 for many organisms (e.g., Reich et al. 2006), and others argue that $b \approx 0.75$ only when averaged across many species but varies considerably among them (Isaac and Carbone 2010).

15.5.3. Temperature dependence and the Arrhenius Equation

In addition to the role of body size, MTE incorporates the effects of temperature. Gillooly et al. (2001) showed that metabolic rate can be predicted

West, G B, J H Brown, and B J Enquist. 1997. "A General Model for the Origin of Allometric Scaling Laws in Biology." *Science (New York, N.Y.)* 276: 122–26.

Brown, James H., James F. Gillooly, Andrew P. Allen, Van M. Savage, and Geoffrey B. West. 2004. "Toward a Metabolic Theory of Eology." *Ecology* 85 (7): 1771–89. https://doi.org/10.1890/03-9000.

Banavar, Jayanth R, Todd J Cooke, Andrea Rinaldo, and Amos Maritan. 2014. "Form, Function, and Evolution of Living Organisms." *Proceedings of the National Academy of Sciences of the United States of America* 111: 3332–37. https://www.ncbi.nlm.nih.gov/pubmed/24550479.

Reich, Peter B., Mark G. Tjoelker, Jose-Luis Machado, and Jacek Oleksyn. 2006. "Universal Scaling of Respiratory Metabolism, Size and Nitrogen in Plants." *Nature* 439 (7075): 457–61. https://doi.org/10.1038/nature04282.

Isaac, Nick J. B., and Chris Carbone. 2010. "Why Are Metabolic Scaling Exponents so Controversial? Quantifying Variance and Testing Hypotheses." *Ecology Letters* 13 (6): 728–35. https://doi.org/10.1111/j.1461-0248.2010.01461.x.

Gillooly, James F., James H. Brown, Geoffrey B. West, Van M. Savage, and Eric L. Charnov. 2001. "Effects of Size and Temperature on Metabolic Rate." *Science* 293 (5538): 2248–51. https://doi.org/10.1126/science.1061967.

from both body mass and body temperature:

$$B = B_0 M^z e^{-E_a/(kT)} \tag{15.3}$$

where e is the base of the natural logarithm, Ea is the activation energy of metabolic enzymes, k is Boltzmann's constant (8.617×10^{-5} eV K^{-1}), and T is temperature in Kelvin.[4]

[4] Recall Kelvin degrees = Celsius degrees + 273.15

In the late 1800s, Svante Arrhenius described the power law relation for the temperature dependence of chemical reaction rates; you may have encountered this in your chemistry class. This was further quantified and explained using first principles of entropy and probability by Ludwig Boltzmann.[5] Boltzmann showed how the emergent property of "temperature" emerges from the thermal energy of individual particles. Later, Max Planck further explicated this quantity, referring to the specific quantity as k, and then named it after Boltzmann. You will also encounter Boltzmann's constant in our last chapter, on global heating.

[5] Boltzmann's work on this and on *entropy* have connections throughout science, including Planck's constant and blackbody radiation, and even the meaning of time.

15.5.4. Nutrient excretion by fish

Most experimental tests of MTE have been done in the lab, where activity, temperature, and diet can be controlled and we can measure *basal metabolic rate* (i.e. base metabolic rate, at rest). But organisms in nature are often active, experience variable temperatures, and are feeding—they display *active metabolic rates.* Far fewer studies have tested whether MTE predictions hold under field conditions, where diet, activity, and other factors add noise.

Excretion rate is a type of whole-organism metabolic rate. It is an output of many metabolic processes, and it represents the quantity of nutrient released after maintenance and growth needs are met.

This output from the metabolic processes of many individuals has important consequences (Sterner et al. 2002). This excretion of nitrogen (N) and phosphorus (P) by fish is ecologically important because these nutrients, released in dissolved form that algae can readily use, can fuel a significant fraction of primary production in lakes (Vanni et al. 2006).

Sterner, R. W., J. J. Elser, and P. M. Vitousek. 2002. *Ecological Stoichiometry: The Biology of Elements from Molecules to the Biosphere*. Princeton University Press, Princeton, New Jersey, USA.

Vanni, Michael J., Anna M. Bowling, Elizabeth M. Dickman, et al. 2006. "Nutrient Cycling by Fish Supports Relatively More Primary Production and Lake Producitivty Increases." *Ecology* 87 (7): 1696–709. https://doi.org/10.1890/0012-9658(2006)87%5B1696:NCBFSR%5D2.0.CO;2.

15.6. Using logarithms

It is common to express these relationships on a logarithmic scale, and we do this for two reasons. First, it expresses the relations on a relative in which multiplicative differences are shown as additive and our brains can understand additive differences more eaasily. Second, using a logarithmic scale linearizes the relations and allows us to more easily estimate the parameters of interest: z (allometric scaling) and E_a (temperature sensitivity).

Taking the logarithms of Equation 15.3, we have

$$\ln(B) = \ln(B_0) + z\ln(M) - E_a\frac{1}{kT}$$

where k is Boltzmann's constant and T is temperature in degrees Kelvin. Below, we will start with the raw data and create new variables of $\ln(B)$, $\ln(M)$, and $1/(kT)$.

15.7. Hypotheses, our data, and predictions

Once we know what data we have available to us, we can use our hypotheses to make specific predictions about those data. Remember that a *scientific hypothesis* should be a general explanation of a process or pattern in nature, anbd *prediction* is a logical deduction that follows when we apply that general explanation to a particular system.[6]

[6] Recall that a *scientific theory* is a very strongly supported and even more general set of explanations that provides framework to organize our well-supported ideas and helps us generate new hypotheses.

So, logic follows in this manner:

The MTE describes the ecological consequences of body size and temperature. Given that theory, we can hypothesize that 1. because the metabolic rate of larger bodied organisms scale to the 3/4 power of size, fish nutrient excretion rate should also scale in the same manner; and 1. because the metabolic rate of warmer organisms scale with temperature by a factor of $E_a/(kT)$ where $E_a \approx 0.23$, fish nutrient excretion rate should also scale in the same manner.

What data do we have?

The data we will analyze come from field studies of nutrient excretion by gizzard shad (*Dorosoma cepedianum*) in three Ohio lakes (Figure 15.1b) (Schaus et al. 1997; Higgins et al. 2006). We have excretion data for 200 individual fish. These 200 fish are a tiny fraction of the fish in these lakes, but they represent a useful sample from which we can extrapolate.

Schaus, M. H., M. J. Vanni, T. E. Wissing, M. T. Bremigan, J. E. Garvey, and R. A. Stein. 1997. "Nitrogen and Phosphorus Excretion by Detritivorous Gizzard Shad in a Reservoir Ecosystem." *Limnology and Oceanography* 42 (6): 1386–97. https://doi.org/10.4319/lo.1997.42.6.1386.

Higgins, Karen A., M. J. Vanni, and María J. González. 2006. "Detritivory and the Stoichiometry of Nutrient Cycling by a Dominant Fish Species in Lakes of Varying Productivity." *Oikos* 114 (3): 419–30. https://doi.org/10.1111/j.2006.0030-1299.14745.x.

Individual fish were captured, placed in filtered lake water, and investigators measured the change in (i) dissolved N and (ii) P concentration over a fixed period of time. The variables in the data set are:

- `temp_C` — water temperature (°C) experienced by the fish
- `mass_g` — wet body mass (grams)
- `N_u.h` — nitrogen excretion rate (μmol N per fish per hour)
- `P_u.h` — phosphorus excretion rate (μmol P per fish per hour)

15.7.1. Predictions

Given our hypotheses and the data we have, we *predict* that gizzard shad nutrient excretion rate will scale with the 3/4 power of gizzard shad body mass, and by a factor of by a factor of E_a with the inverse of temperature $1/T$.

You will use multiple linear regression to evaluate those predictions.

15.8. Getting R ready for work

Open R using RStudio, and load the packages we'll need. Note that we will use a new package, `broom`. You will need to install `broom` if you hadn't done so for some other exercise or your own research.

To install `broom`, you can use the *Tools* drop-down menu in RStudio, or run this code:

```
install.packages("broom")
```

Now load `broom`, and `tidyverse` if you have not already done so.

```
library(tidyverse)
library(broom)
```

Now read in the data and take a look.

```
# Note we skip the first 8 lines because they are 'metadata',
# which is 'data about data` or information about the variables.

fish <- read.csv("data/fish_nutrient_excretion.csv", skip=8)
glimpse(fish)
```

```
Rows: 200
Columns: 5
$ Fish_ID <int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15,
 ↪  16, 17, 18,~
$ temp_C  <dbl> 16.4, 16.4, 16.4, 16.4, 16.4, 16.4, 16.4, 16.4,
 ↪  16.4, 16.4, 16~
$ mass_g  <dbl> 4.50, 5.40, 5.75, 5.80, 5.95, 9.20, 10.90, 11.10,
 ↪  11.65, 11.80~
$ N_u.h   <dbl> 2.45, 10.67, 14.89, 21.69, 20.14, 25.25, 31.93,
 ↪  9.40, 37.01, 3~
$ P_u.h   <dbl> 1.71, 2.16, 1.66, 1.49, 2.14, 3.21, 1.99, 2.69,
 ↪  2.48, 3.91, 2.~
```

```
summary(fish)
```

```
    Fish_ID          temp_C          mass_g           N_u.h
 Min.   :  1.00   Min.   :16.40   Min.   :  2.32   Min.   :  2.45
 1st Qu.: 50.75   1st Qu.:18.50   1st Qu.: 21.93   1st Qu.: 62.88
 Median :100.50   Median :23.10   Median : 46.00   Median :119.34
 Mean   :100.50   Mean   :22.54   Mean   : 49.16   Mean   :150.75
 3rd Qu.:150.25   3rd Qu.:25.50   3rd Qu.: 65.28   3rd Qu.:213.14
 Max.   :200.00   Max.   :29.40   Max.   :210.94   Max.   :524.29
     P_u.h
 Min.   : 0.590
```

```
 1st Qu.: 4.975
 Median : 9.950
 Mean   :12.304
 3rd Qu.:17.273
 Max.   :50.660
```

You should have 200 rows and 5 columns. The fish range in mass from about 2 g to over 200 g, and temperatures range from about 16°C to 29°C.

What are 16°C to 29°C in Fahrenheit degrees? (The conversion factor is $Y^\circ\,(F) = \frac{180}{100} X^\circ\,\mathrm{C} + 32$)

15.9. Exploring the data

We already know that we will want to create a few new variables based on the raw data, including the logs of excretion rates, mass and the inverse of temperature in Kelvin. Below we do that.

First, here is the value of Boltzmann's constant, in units of electron volts per degree Kelvin

```
boltzmann_k <- 8.617333262e-5
```

Now we transform our variables.

```
## Add log-transformed columns to the data
fish <- fish %>% #start with our fish data set, and
  mutate(log_mass = log(mass_g), # create ln(mass)
         log_N = log(N_u.h), # ln(N excretion)
         log_P = log(P_u.h), # ln(P excretion)
         temp_K = 273.15 + temp_C, # degrees Kelvin
         inv.kT = 1/(boltzmann_k * temp_K) # inverse scaled Kelvin
  )
```

Before testing any hypotheses, let's look at the data. Always plot your data before you analyze it.

15.9.1. Raw-scale plots

```
ggplot(fish, aes(x = mass_g, y = N_u.h)) +
  geom_point(alpha = 0.5) +
  labs(x = "Body mass (g)",
       y = expression(paste("N excretion (", mu, "mol ", fish^{-1},
        ↪  " ", hr^{-1}, ")")),
       title = "N excretion vs. body mass") +
  theme_minimal()
```

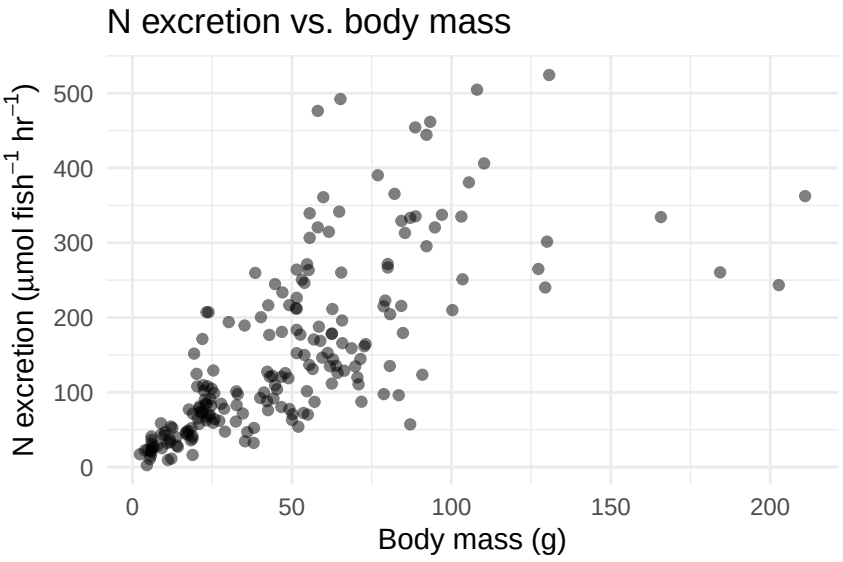


Figure 15.2.: Nitrogen excretion rate vs. body mass on arithmetic (raw) axes.

Make a similar plot for P excretion rate. What do you notice about the shape of these relationships? They are curved—larger fish excrete more, but the relationship is not a straight line.

15.9.2. Log-transformed plots

Now let's see what happens when we log-transform both axes, as suggested by the allometric equation (Equation 15.2).

```
ggplot(fish, aes(x = log(mass_g), y = log(N_u.h))) +
  geom_point(alpha = 0.5) +
  labs(x = expression(log[10]~"(body mass, g)"),
       y = expression(log[10]~"(N excretion, " * mu * "mol " *
        ↪  fish^{-1} * " " * hr^{-1} * ")"),
       title = "Allometric scaling of N excretion") +
  theme_minimal()
```

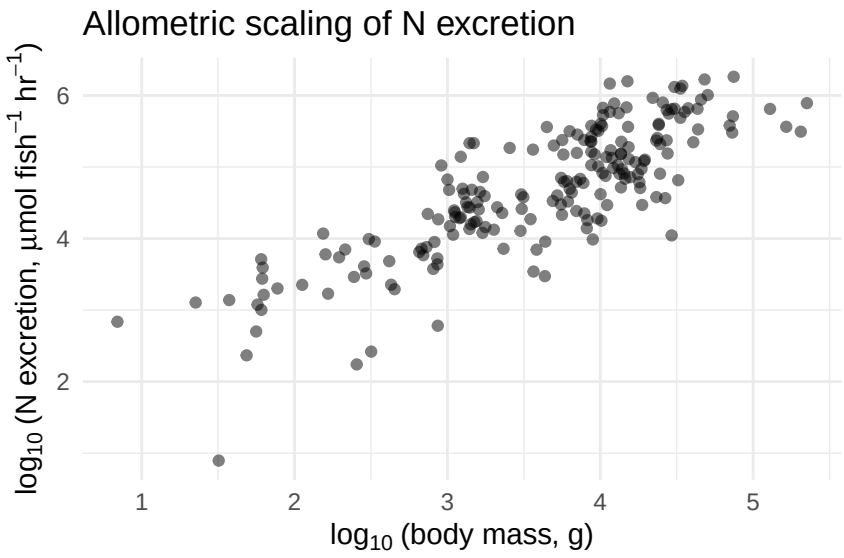


Figure 15.3.: Log nitrogen excretion rate vs. log body mass. If the allometric model holds, this should be approximately linear.

That looks much more linear! The log-transformation did exactly what Equation 15.2 predicted. Make the same plot for P excretion.

MTE also suggests that temperature might influence excretion rates. Here we plot separate rate vs. mass graphs for each temperature:

```
ggplot(fish, aes(x = log_mass, y = log_N)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", se = TRUE) +
  facet_wrap(~temp_C) +
  labs(x = expression(log~"(body mass, g)"),
       y = expression(log~"(N excretion)"),
       title = "Allometric scaling of N excretion") +
  theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'
```

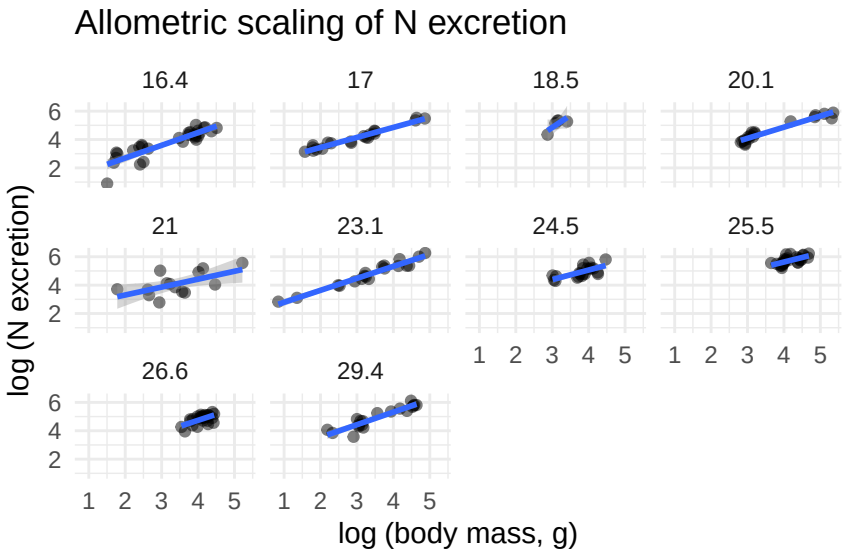


Figure 15.4.: Linear regression of log N excretion on log body mass, for each temperature.

15.9.3. Allometric and temperature scaling of excretion rates

Here you need to estimate the allometric scaling exponent (z) and the energy of activation (aka temperature sensitivity) for both N and P excretion rates. Multiple regression will let us address a key question how we can how do you deal with the fact that excretion rates were measured at different temperatures?

There is more than one reasonable approach. Think about this before reading on. One approach is to pool all the data together and fit a single multiple regression. Another is to fit separate regressions for each temperature and then compare slopes. You could try both and think about what you learn from each approach.

15.9.3.1. Mulltiple linear regression

Now fit a linear regression for log N excretion vs. log mass and inverse temperature.

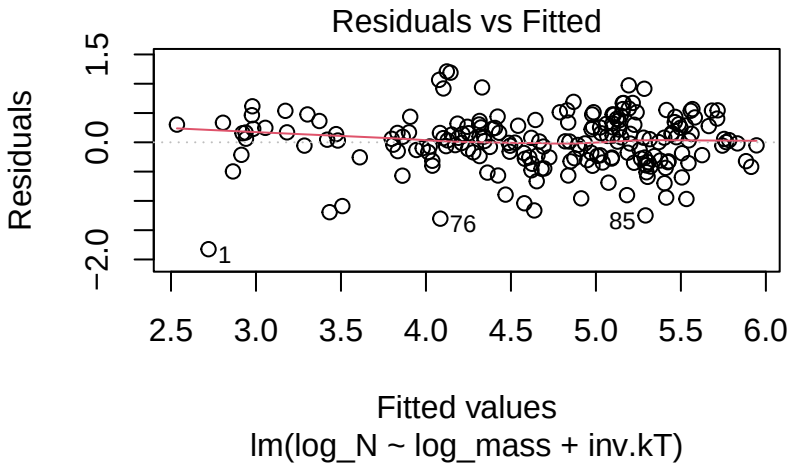
```
## Fit the regression
m_N_all <- lm(log_N ~ log_mass + inv.kT, data = fish)
```

If it ran without error, you are ready for the next step.

Next step: check your assumptions

All statistical models have builit-in assumptions. The regression we ran here assumes, among other things, that the relationships are linear and the unexplained noise in the data are normally distributed, both above and below the expected or fitted values. Here we examin that graphically.

```
plot(m_N_all, which=1)
```



If these assumptions are met, then the *residual* unexplained noise is just as likely to be positive and negative for any fitted value. And that is essentially what we see. Thus these assumptions appear to be met.

15.9.3.2. Evaluating the predictions

Recall that our equation is now

$$\ln(B) = \ln(B_0) + z\ln(M) - E_a\frac{1}{kT}$$

and our regression will estimate z for log mass, and E_a for inverse scaled temperature. Here are the 95% *confidence intervals* on those.

A *confidence interval* (CI) is essentially a *procedure*, rather than a thing. A 95% CI is a procedure wherein if you repeated the experiment many, many times, the CIs would capture the true estimate about 95% of the time.

Here are the confidence intervals for z (for log mass) and E_a (for inverse scaled temperature).

```
## View the results
# ignore the intercept
confint(m_N_all)
```

```
                2.5 %     97.5 %
(Intercept) 11.4522604 21.4184325
log_mass     0.7062991  0.8718949
inv.kT      -0.4957793 -0.2478470
```

15.9.3.3. ***Do the same*** *for P excretion.*

15.9.4. Thinking bigger: ecosystem consequences

We have focused on excretion by individual fish. But remember the bigger picture (Figure 15.1a): gizzard shad are extremely abundant in many Midwestern lakes. They consume sediment detritus, extract what they need, and excrete the rest as dissolved N and P that algae can immediately use.

Consider this: if you know the allometric scaling exponent and E_a, and you know the size distribution and temperature regime of a lake, you could *predict* the total nutrient flux from the fish population to the water column. This is exactly the kind of scaling-up that makes metabolic ecology so powerful—and so important for understanding ecosystem function.

16. An Introduction to the Carbon Cycle and Global Heating

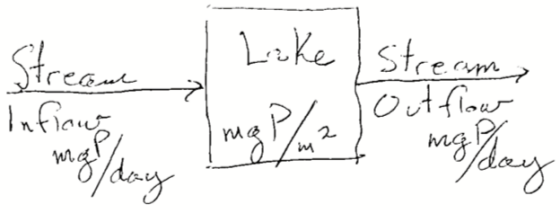
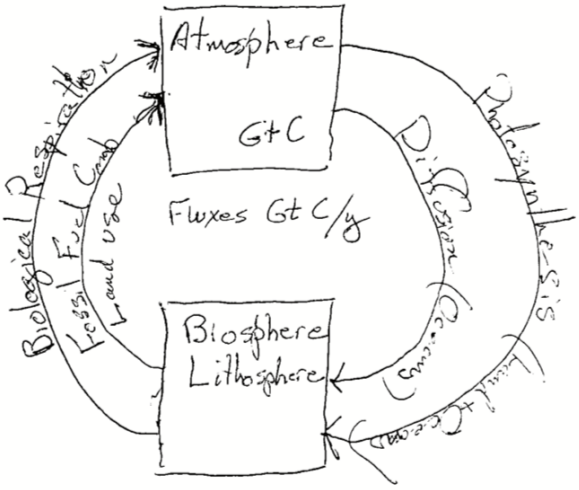


Figure 16.1.: Ecosystems comprise pools (compartments) and fluxes in and out of those pools. Ecosystem ecologists typically study ecosystems by tracking a particular chemical element from compartment to compartment. Note the units are consistent throughout an ecosystem. In the above example of a lake, fluxes of phosphorus are milligrams per day in or out. At the global scale, we can estimate fluxes of carbon in gigatonnes per year.



Before you start

- **Assumptions about background and guidance:** algebra and the idea of a rate of change (an amount per unit time); this chapter builds everything else from pencil-and-paper budgets. If dN/dt notation is new, Models as assumptions introduces it. For the simulations, you will need your `Rwork` project (Computer setup) and the `deSolve` package, which is also used in Of wolves and moose.

16.1. Goals

In this chapter, you will provide partial answers for these questions:

1. Why do we have this much carbon in the atmosphere?
2. How does including the physical process of diffusion of alter predictions about future atmospheric carbon?
3. What is the connection between atmospheric CO_2 and temperature, and how does that relationship affect our appreciation of Earth's current and future status?

To address these questions, we will use pencil and paper and R to,

1. build a simple ecosystem budget with steady fluxes and changing pools,
2. create a dynamic ecosystem in which the fluxes change, depending on the amount in pools, and
3. investigate how matter can relate dynamically to energy.

One of the main the points of doing these activities is to help to develop an intuition for *systems thinking*. A system is "a group of interacting, interdependent parts that form a complex whole" and ecosystem ecology is a central subdisicpline of ecology and the one that has contributed the most to our understanding of climate change and ecosystem services.

The other main point of these activities is to reinforce simple ideas related to the global carbon cycle and global warming.

16.2. Deliverables

16.2.1. Part 1

Two documents:

The first document is your R script (just the R script - do not save it in any other format).

The second document should provide the following written responses and quantities you calculated in R:

1. Quantities from the simple budget, including units:
    a. the annual flux
    b. the size of the pool in years 2 and 25.
2. Quantities from the simple dynamic model, including units:
    a. the graph of atmospheric carbon dynamics.
    b. solution to the dynamic rate equation (i.e. what is A^* predicted to be at equilibrium?). Show your work.
3. State what your solution tells us should happen to A^* if e gets smaller and also if it gets larger.

4. Compare and contrast the simple budget and the simple dynamic system, in terms of how they represent a system, and in what they predict about future atmospheric carbon levels.
5. Explain what the simple dynamic system would predict about the ocean acidification due to fossil fuel emissions?
6. Warmer water holds lower concentrations of gases such as oxygen and carbon dioxide than does colder water. Explain how this might alter predictions about ocean acidification assuming the atmosphere and oceans continue to warm. If you know a little about chemistry, include anything you know about the effect of temperature on the dissociation of acids.

16.2.2. Part 2

Include two documents; the first is your R script.

In the second document, provide results for each experiment (1-3). For each experiment include a graph of CO2 concentration and a graph of observed and equilibrium temperatures, as well as these data:

- A, atmospheric CO2 concentration at the start and end of your simulation.
- K, modeled "observed" or contemporaneous temperature (deg C) at the start and end of your simulation.
- K_eq, equilibrium temperature (deg C) at the start and end of your simulation.
- $K_eq - K$, calculated difference between the contemporaneous and equilibrium temperatures at the start and end of your simulation.

At the end, use your data and figures to **tell the story** of how our activities alter the planet. Be sure to mention how we have not yet experience the consequences of greenhouse gases that we have already produced.

In sum, your one document should include results of three experiments and then a short narrative or story explaining what your simulations show.

16.3. Preparatory steps

1. General background lectures: Download and watch the video on the carbon cycle before proceeding.
2. We will not need data for this chapter.
3. This chapter will ask you to complete some work by hand (without R) and turn that in in class. Find out what you need for the deliverables parts 1 (Section 16.2.1) and 2 (Section 16.2.2).
4. Get some paper and pencil.
5. Get R ready for work - see the next section.

16.4. Get R ready for work

Open R using RStudio, and **set your working directory**. You can do this in two ways:

1. using the Session menu in RStudio, or
2. with code such as this, in which you must customize it to your computer!

```
setwd("C:/Users/chenjoe/BIO209W/Rwork")
```

You can confirm or **get** your *current* working directory using this:

```
getwd()
```

```
[1] "/Users/stevenmh/projects/intro_to_ecology"
```

so when I do that, you can see my working directory is the one for this book.

Before you proceed, make sure your working directory is your `Rwork` directory.

Next, use RStudio to create a new file of the type *R script*. Save it using a name that reflects this chapter, and end the file name with ".R" or ".r".

Third, add a comment as a brief description of the script, and load any R packages you know you will need.

```
## This script is for chapter XX in Hank's Intro to Ecology primer.

## We always use...
library(tidyverse)
## or usually just ggplot2 and dplyr packages within the tidyverse.
## In this chapter we also need
library(deSolve)
# for solving or integrating differential equations
```

16.5. Background on ecosystems

An ecosystem is a biological community of interacting organisms and their physical environment. Ecosystem ecologists study an ecosystem by tracking a particular chemical element such as carbon (C) or nitrogen (N) through the environment. We conceive of the environment in terms of compartments that contain the element (a.k.a. pools, stocks) and movement between compartments (fluxes, inflows, outflows).

Examples of ecosystem budgets:

- **Phosphorus budget of a lake ecosystem.** Phosphorus (P) flows in via a stream coming into the lake, and discharges in a stream leaving the lake.

- **Carbon budget of Earth's atmosphere.**: Carbon enters as CO_2 via biological respiration and fossil fuel combustion and exits via uptake in photosynthesis in land plants and aquatic algae, and also via diffusion into oceans.
- **Energy budget of mountain stream.** Energy is captured from sunlight via photosynthesis in aquatic algae and falls in as dead leaves, and exits via molecules of CO_2 respired by organisms and as organic molecules and particles flowing flowing downstream.

The ecosystem paradigm connects energy and matter to the evolution and ecology of life. Ecosystem scientists focus on the flow of energy or matter through all parts of the environment, especially as those flows are mediated by living organisms.

16.6. A simple budget provides one explanation for why we have so much carbon in the atmosphere

Why do we have so much carbon in the atmosphere? One answer to that question comes from simply tallying up the inputs and the outflows. In this sense, it is a lot like our household budget.

Ecosystem budgets are simple[7], at least conceptually. An ecosystem budget is just like your household budget or bank account: if more money goes in than goes out, you accumulate money. In contrast, if you spend more money than you bring in, your wealth declines.

[7] A *zeroth order* flux is one in which the rate depends on the amounts in zero pools - it doesn't depend on pools.

An ecosystem budget simply keeps track of the amount of a chemical element, such as nitrogen or carbon, that, is in a compartment, goes into a compartment, and leaves a compartment, over a specified time interval such as a year. The **size** of a pool or compartment is the amount of an element in that pool.

16.6.1. Rate of change

For a simple ecosystem carbon budget, the *rate of change* of the size of a pool is the **difference** between the input and the output over a specified time interval,

$$\Delta \text{ Pool} = \text{Inputs} - \text{Outputs}$$

For instance, consider the carbon budget for Earth's atmosphere from the 1990's that we described in lecture. We had these inputs and outputs (units are petagrams, 10^{15} g):

- Inputs to the atmosphere
  - diffusion from the ocean (respiration, 91 Pg/y)
  - River outgassing (1 Pg/y)

  - Fire (4 Pg/y)
  - Cement production (0.1 Pg/y)
  - Fossil fuel (7.6 Pg/y)
  - Land use change (1.5 Pg/y)
  - Terrestrial respiration (mostly soil microbes 56 Pg/y)

These total to $91 + 1 + 4 + 0.1 + 7.6 + 1.5 + 56 = 161$ Pg C / y.

- Outputs from the atmosphere
  - diffusion into the ocean (photosynthesis, and carbonic acid formation 92 Pg/y)
  - Weathering (0.2 Pg/y)
  - Land sink (2.8 Pg/y)
  - NPP on land (60 Pg/y)

These total to $0.2 + 2.8 + 60 = 161$ Pg C / y.

At the time, the **atmosphere held 762 Pg C** mostly in the forms of carbon dioxide, carbon monoxide, and methane (CO_2, CO, CH_4).

Given this budget, how much carbon would be in the atmosphere,

- one year later?
- two years later?
- fifty years later?

To figure this out with a simple budget, we simply start with the pool, add the inputs and subtract the outputs.

$$\text{Atmosphere}_{t+1} = \text{Atmosphere}_t + \text{inputs} - \text{outputs}$$

Here we have inputs into the atmosphere.

```
or <- 91 # diffusion out of the ocean, due to respiration of
 ↪  microbes
ro <- 1 # river outgassing (diffusion)
fi <- 4 # fire
cp <- 0.1 # cement production
ff <- 7.6 # fossil fuel combustion (1998)
lu <- 1.5 # land use change
tr <- 56 # terrestrial respiration
inputs <- or + ro + fi + cp + ff + lu + tr
inputs
```

```
[1] 161.2
```

Here we have outputs from the atmosphere mostly into plants on land and into the oceans where most of it is taken up by phytoplankton, kelp, seagrasses, and corals. A small amount of it remains in the water and makes it more acidic.

```
npp.o <- 92 # net primary production in the ocean (diffusion)
we <- 0.2 # Weathering of carbonate rock
ls <- 2.8 # land sink (burial)
npp.l <- 60 # net primary production on land

outputs <- npp.o + we + ls + npp.l
outputs
```

```
[1] 155
```

The difference between inputs and outputs is *net flux*:

```
net.flux <- inputs - outputs
net.flux # in Pg C per year
```

```
[1] 6.2
```

At that time, Earth had a pool of carbon in the atmosphere of about 762 Pg C.

```
# the pool
At <- 762
```

So, what would it have one year later? It would have the pool plus the net flux.

```
At.plus.1 <- At + net.flux
At.plus.1
```

```
[1] 768.2
```

What about two years later? What about 3 or 25 or 75 years later? If we choose to assume that these rates won't change, the answer is easy, but only because we are working with a simple budget. We simply add the annual flux for each year, starting at year 0.

$$\text{Atmosphere}_t = \text{Atmosphere}_0 + \text{net flux} + \text{net flux} + ... + \text{net flux}$$

That is equivalent to multiplying the annual flux by the number of years, t.

$$\text{Atmosphere}_t = \text{Atmosphere}_0 + \text{net flux} \cdot t$$

Now it is your turn to do that next for two years and for 25 years.

```
## write code two years and for 25 years
##
```

Note that as long as these fluxes remain constant the system will continue to change at this rate. There is no equilibrium or *balance of nature*.

16.7. Simple dynamics emerge from processes like diffusion, and complicate prediction

How does including the physical process of diffusion of alter predictions about future atmospheric carbon? A simple budget is a great starting place, but it ignores a lot of biological and physical reality. Here we add one piece of that reality: diffusion of CO_2 between the air and water.

In the above example, carbon went into the atmosphere at a constant rate and left at a constant rate. What if a flux depends on the amount in the pool? It's actually very common for a flux to depend on the amount in one or more pools. Here are some examples:

- The flux of water leaving a bathtub depends on the height of water in the bathtub.
- The flux of radiant heat from the body of a basking lizard depends on the temperature (or amount of heat energy) of the lizard.
- Rate of diffusion of CO2 into the oceans depends on the concentration in the atmosphere, due to a concentration gradient.
- Rate of carbon capture by plants depends on the amount of vegetation (compare a potted plant in a parking lot to an intact forest).

16.7.1. A first order flux

A *first order* flux is one in which each term in the rate equation depends on no more than *one* pool.

Imagine we have bathtub that we are filling with water, and that the drain has a leak. Just like with our simple budget above, rate of change in the depth (or height) of the water is

$$\Delta\text{height} = \text{inflow} - \text{outflow}.$$

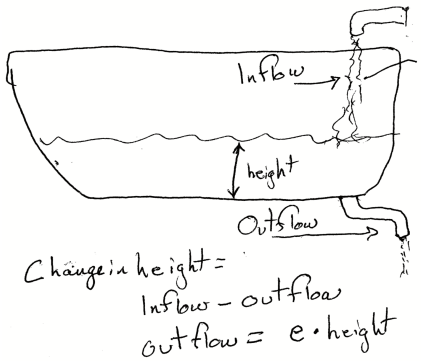


Figure 16.2.: Filling a bathtub with a leaky drain.

Complication: while the inflow is fixed by the faucet, the outflow is determined by two things: how big the leak is, and how deep the water is or

the height of the water in the tub. The deeper the water, the more pressure there is that pushes the water out faster. We can approximate that as

$$\text{Outflow} = e \cdot \text{height}$$

where e is a constant that depends on the size of the leak. As the height decreases, the outflow rate will decrease. With this detail, the rate equation is now,

$$\Delta\text{height} = \text{inflow} - e \cdot \text{height}$$

and when the change in the height slows to zero ($\Delta\text{height} = 0$), that means the depth will stop changing. That happens when,

$$\text{inflow} = e \cdot \text{height}.$$

We can *solve for height* to figure out what the height will be. Rearranging, we get,

$$\text{height} = \frac{\text{inflow}}{e}.$$

We refer to this height as an equilibrium because it is predicted to not change.

Let's see this operate in an ecosystem.

16.7.2. Earth's atmosphere as a first order equation

Let's assume that at first approximation, the diffusion of CO_2 from the atmosphere into the ocean is a first order process - that it is *proportional to the amount* of CO_2 in the atmosphere. We can represent that in the differential rate equation

$$\frac{dA}{dt} = I - eA$$

where A is the amount of CO_2 in the atmosphere, I is a constant flux, and e is the diffusion rate into the ocean.

Note that *all* of the uptake by the ocean is via diffusion, because it has to leave the air and move into the water in the process of diffusion. However, most of what moves into the water is captured via photosynthesis, by phytoplankton[8], coral symbionts[9], and kelp. The CO_2 that is *not* captured this way **increases in concentration and acidifies the ocean**, as

$$CO_2 + H_2O \rightarrow H_2CO_3 \rightleftharpoons HCO_3^- + H^+$$

.

If we assume that in the ocean, photosynthesis is balanced by respiration, then we can pretend that the difference leads to ocean acidification. Using data presented above, that means that

[8] floating single-celled algae

[9] dinoflagellates *Symbiodinium*

```
respiration <- 91 # Pg
photosynthesis <- 91 # Pg
carbonic_acid <- 1 #Pg
```

We will estimate the approximate value of e using the net carbon flux.

$$F_{oa} = \text{photosynthesis}$$

If the net flux into the atmosphere were $I = 6.2$ Pg C/y, and the current diffusion rate out of the atmosphere is $e = 0.008/\text{y}$[10], **what would happen?** What is the predicted equilibrium and how fast would it get there?

[10] Units for e are Pg C lost per Pg C in the atmosphere per y; the Pg/Pg cancels out, leaves "per year" or 1/y.

Let's ask R to show us the dynamics. First, we make an R function with the rate equation.

```
## simple first order equation.
## start
atmosphere1 <- function(t,y,p){
  with( as.list(c(y, p)), {

    ## this is the bit to recognize
    dA.dt <- I - e*A

    return(list( c(dA.dt),
                 ## convert gigatonnes C to CO2 equivalent mass
                 CO2_global_mass = A * (12 + 2*16)/12,
                 ## convert gigatonnes C to CO2 equivalent ppm
                 CO2_ppm = A / 2.13
    ) )
  })
}
## end
```

Next, we tell R what parameters to use, and the initial amount of carbon in the atmosphere at time zero.

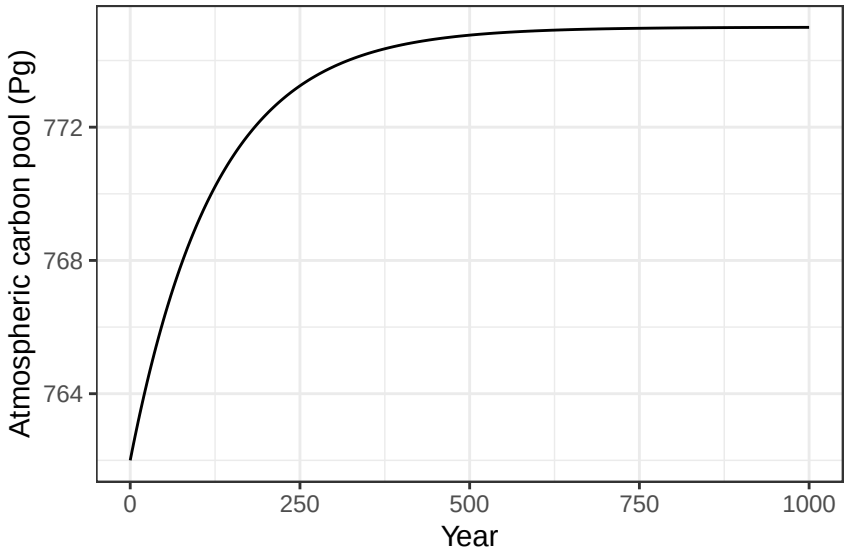
```
# Constant influx, and diffusion rate
p1 <- c(I=6.2, e=0.008)
# starting atmospheric carbon pool
y0 <- c(A=762);
```

Next we tell R the time points we want it to show us. Let's get annual estimates for 1000 years.

```
# from zero to 100, by steps of one year
t <- seq(from=0, to=1000, by=1)
# integrate the rate equation for all time points,
# starting with y0 at time zero, using our function and parameters.
output <- ode(y=y0, times=t, func=atmosphere1, parms=p1)
```

Next we plot the dynamics of the atmospheric carbon. I have hidden the graphic in order to allow you to create it on your own.

```
# Finally, we plot the output.
out_df <- as.data.frame(output)
firstorderA <- ggplot(out_df, aes(time, A)) + geom_line() +
  labs(y="Atmospheric carbon pool (Pg)",
       x="Year") +
  theme_bw()
firstorderA
```




```
ggsave("figs/firstOrderAtmosphere.png", width=5, height=4)
```

Next, solve the rate equation directly by setting $\frac{dA}{dt} = 0 = I - eA$ and solving for A^*, the equilibrium.

$$A^* = ?$$

What does the above solution tell us should happen if e gets smaller? gets larger?

16.8. Linking two different ecosystem processes shows lagged effects

In this section we connect the carbon cycle to an entirely different set of fluxes, those of heat, or radiant **energy**.

What is the connection between atmospheric CO_2 and temperature, and how does that relationship affect our appreciation of Earth's current and future status? Most of us don't care much about the amount of CO_2 in the atmosphere (per se), but we care *a lot* about how CO_2 and other greenhouse gases trap and re-radiate heat. In this section, we quantify this connection.

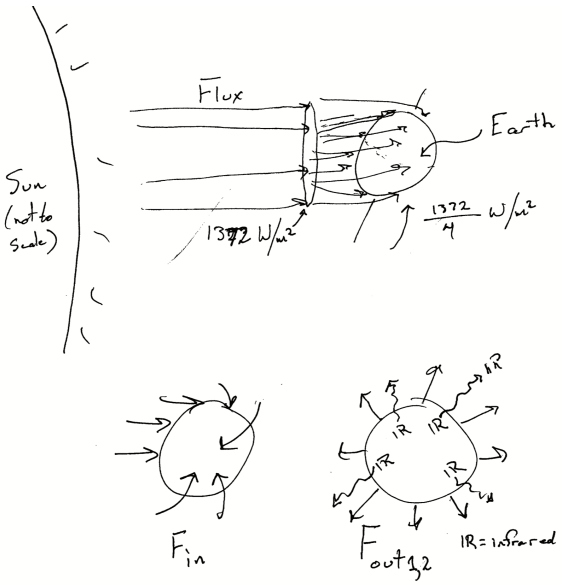


Figure 16.3.: Earth is approximately a black body absorbing and emitting radiant energy. Straight-ish and curved lines indicate high energy, shortwave radiation. Wiggly lines indicate lower energy infrared radiation emitted from the warmed surface of Earth. Greenhouse gases (not illustrated here), would slow the escape of infrared radiation, by absorbing and re-emitting it.

Nearly all of the heating of Earth's atmosphere occurs when warmed land or ocean surface emits infrared radiation that is then captured by greenhouse gases which then re-radiate this energy. This all starts when our sun radiates energy that travels through space (Fig. Figure 16.3) and eventually encounters Earth's atmosphere. That incoming radiation is high energy short-wave radiation which penetrates and *passes through the atmosphere*, and hits Earth's surface. Most of this energy is absorbed, and then re-emitted as long-wave infrared radiation which we can experience as heat. It is this infrared radiation that is trapped by greenhouse gases prior to eventually exiting, back out into space (not shown in Fig. Figure 16.3).

Greenhouse gases re-radiate the heat they absorb. In doing so, they heat up Earth's surface even more. Because of that, Earth's surface never has a chance to really cool down, because it is always being re-warmed by the atmosphere that it made warm in the first place. The energy is bouncing back and forth between the atmosphere and the Earth's surface (land and oceans). It also bounces back and forth between layers of the atmosphere. The energy will *eventually* find its way back into outer space, but not before it has bounced around a lot.

The flux of solar radiation (Fig. Figure 16.3) coming at Earth is about $\Omega = 1372$ W/m^2[11]. However, because Earth is round, that energy get spread out and each square meter would receive only about a fourth of that, or $\Omega/4$, *if* it all reached the surface. Thus, the effective incoming flux is $F_{in} = \Omega/4$.

[11] a watt is a energy per second, technically one Joule per second, J/s

Unlike the incoming flux, the outgoing flux comprises two basic types of radiation (Fig. Figure 16.3). About 30% the incoming radiation is *reflected* back out into space, unchanged. It bounces off of clouds and snow and ice, and we refer to this reflectance as **albedo**. Thus, part of the outgoing flux is $F_{out1} = a \cdot 1372/4$, where a is for *albedo* which is about 0.3.

The second part of the outgoing flux is the infrared radiation that first

bounces around the atmosphere before it makes its way out into the universe (Fig. Figure 16.3). The rate rate which that happens is proportional to the Stefan-Boltzmann constant[12], σ, which quantifies radiation emitted from a perfect black body in relation to it's temperature. Needless to say, we won't get into *why* it happens at that rate. What we *will* say is that the second part of the outgoing energy flux is $F_{out2} = \sigma T^4$, where T is temperature in degrees Kelvin, and sigma is the Stefan-Boltzmann constant, $\sigma \approx 5.67 \times 10^{-8}$. Or, at least it would be *if we ignored greenhouse gases*.

[12] This constant derives from Plank's constant, the speed of light in a vacuum, Boltzmann's constant relating temperature to kinetic energy, and of course, π, the ratio of the diameter of a circle to its circumference. Well, of course it does....

All together, we might approximate Earth's energy flux as

$$\Delta E_E = F_{in} - F_{out1} - F_{out2} = \frac{\Omega}{4} - a\frac{\Omega}{4} - \sigma T^4$$

if we ignore the effect of greenhouse gases.

If we want to express temperature at equilibrium in terms of fluxes, we set $\Delta E_E = 0$ and solve for T,

$$F_{in} = F_{out1} + F_{out2}$$

and using algebra to rearrange, and we get

$$T^4 = (1 - a)\,\frac{\Omega}{4\sigma}$$

or,

$$T = \left[(1 - a)\,\frac{\Omega}{4\sigma}\right]^{1/4}$$

where T is degrees Kelvin.

If we calculate the predicted temperature, what would we get? Here is code that can help you answer that.

```
Omega <- 1372
sigma <- 5.67e-8
albedo <- 0.3

temp_K <- ( (1-albedo) * Omega/(4*sigma) )^(1/4)
temp_K
```

16.8.1. Adding greenhouse (heat-trapping) gases

As we mentioned, the above approximation ignores the effect of greenhouse gases. Thousands of scientists around the world are attempting to model Earth's energy balance, including the dynamics and effects of greenhouse gases that trap outgoing radiation. That research is well beyond the scope of this class, but we will approximate the relationship.

We will approximate the effect of greenhouse gases by merely adjusting the Stefan-Boltzmann constant by an amount related to the amount of CO_2 in

the atmosphere, A^g, where A is the CO_2 concentration in parts per million (~420 ppm in 2023), and g is what I call *Hank's greenhouse constant*. This gives us $F_{out2} = \sigma A^g T^4$.

We can now rewrite the predicted atmospheric temperature as

$$T = \left[(1 - a) \frac{\Omega}{4} \frac{A^g}{\sigma} \right]^{1/4} + \epsilon$$

where ϵ is statistical error or residual.

16.8.1.1. Intuition about $T = F(A)$

Let's use the above equation to reinforce our understanding of how atmospheric temperature is a function[13] of the amount of carbon compounds in the atmosphere ($T = F(A)$).

[13] We can represent any function with 'F()', so "some function of X" could be written as F(X).

First, we re-inject meaning into the equation:

- $(1 - a)$: **fraction** of incoming radiation initially absorbed.
- $\Omega/4$: **incoming radiation**.
- σ: scales the relationship between radiant energy and temperature as well as the **radiant loss** of heat energy escaping the atmosphere into the universe.
- A: **CO_2 concentration** in the atmosphere.
- g: scaling constant that quantifies the **heat trapping** relation between CO_2 in parts per million (ppm) and temperature.

Re-writing the equation, we have

$$\text{temp} = \text{fraction} \times \text{incoming rad} \times \frac{\text{CO}_2 \text{ heat trapping}}{\text{radiant loss}}$$

Hank's greenhouse constant g is just a statistically-fitted constant to draw a smooth curve through known temperatures that researchers have shown would occur at CO_2 concentrations of $(0, 280, 450$ ppm$)$. These concentrations reflect zero, pre-industrial, and near future CO_2 predictions. These concentrations have been shown to result in predicted temperatures 255, 288, and 290 deg K.

Note that we can represent heat trapping gases in the atmosphere as "CO2-equivalents", which represent all of the carbon based molecules that help trap heat, especially carbon dioxide (CO2) and methane (CH4). These relate directly to the total amount of carbon in the atmosphere. The total mass of carbon in gigatonnes is about 2.13 times the CO2 concentration in ppm (parts per million). The total mass of carbon dioxide (CO2) in gigatonnes is about 7.82 times the CO2 concentration in ppm (source).

Hank's greenhouse constant is related to the **radiative forcing** of greenhouse gases. Radiative forcing is the potential for a system to undergo energy change.

Using the fitted expectation of $g = 0.0849$ gives us the following:

```
## graphing the curve of temperature vs. CO2
g <- 0.08488
Omega <- 1372
sigma <- 5.67e-8
albedo <- 0.3

# function to describe the relationship
predict.warming <-
  function(x, g, Omega, sigma, albedo){
    ((1-albedo)*Omega/(4*sigma)*x^g )^(1/4)
    }

base <- ggplot() + lims(x=c(0, 500), y=c(250,300))
base +
  geom_function(fun = predict.warming,
                      args = list(g = g, Omega, sigma, albedo),
                n=5001) +
  annotate(geom="point", x=c(0, 280, 450),
           y=c(250, 288, 290) ) +
  labs(
    x="Carbon dioxide (ppm)",
      y="Predicted temperatures",
  ) +
  theme_bw()
```

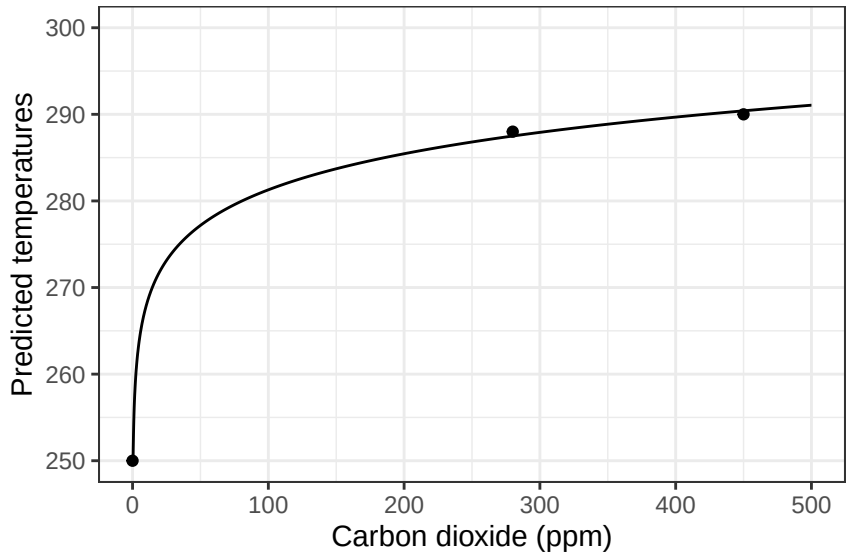


Figure 16.4.: Temperatures predicted from first principles and using Hank's greenhouse constant, g, which adjusts for observed and predicted temperatures, based on actual CO2 and temperature data and extremely complex and well-tested models used in IPCC reports.

16.8.2. Why is all this math important?

All this math is important for several reasons.

First, we want to estimate both the expected temperatures now, and also what the temperature *would become even if we stop emitting any greenhouse gases today*.

Second, we all need practice understanding quantitative approximations. Also known as "back of the envelope calculations", these rough estimates give us approximate information about our world so that we can make timely decisions. We cannot wait for answers with ten sig-figs of precision. Here are some important questions to which there is no single answer, but for which approximations can help:

- What is my relative risk of dying if I drive 500 miles to my wife's house vs. if I flew?
- What is the probability that this stranger I am with will give me a disease?
- If I add $25 dollars to my IRA every month, what would that be worth in 20 years?
- If the planet warms by 2 deg C, how big of a change is that, relative to its usual temperature?
- If I get a positive HIV test result, what is the probability that I am actually infected with HIV?

We live better lives when we are comfortable with good quantitative approximations.

Third, and finally, it's just nerdy-fun to realize that this is really how the planet works!

16.8.3. Expected temperatures now, and also what the temperature would become

One important reason that we are creating rate equations is because *what we observe now is not what we will get in the future*. Solving rate equations over time shows us change through time, not just the end result.

Because the amount of atmospheric CO_2 is constantly increasing, **the temperature has never caught up to what it would become** if CO_2 weren't steadily increasing.

Here is an analogy. You put a pot of cold water on the stove. You turn the burner to "high" and the gas ignites or the coils start to glow red. The burner is emitting heat, but the water is not yet boiling. In this analogy, the water is the atmosphere and the burner is the increasing greenhouse gases. The temperature difference between the burner and the water is the **radiative forcing**. On Earth, the greenhouse gases are accumulating and trapping lots of heat energy, but **Earth has not yet warmed up as much as it would given the current levels of greenhouse gases**.

So, now we know there is a lag or delay between greenhouse gas concentrations and atmsopheric temperatures. What we DON'T know is how *big* is this lag between CO_2 concentration and temperature actually is. Is the lag only a day, or year, or is it longer, like a decade or a century? Quantifying the *importance* of the lag is the kind of thing that only a quantitative model can tell us.

16.9. *Simulations:* in silico *experiments*

In order to explore these connections between greenhouse gases and global temperatures, and the lag between them, we will do several simulations to examine the consequences of our assumptions. In each case, we will track atmospheric carbon concentration and temperature to 2100. This interval of less than 80 years is likely to be about the life span of you, or your children if you have any, or your young nieces and nephews.

1. Business as usual anthropogenic carbon emissions.
2. Keep atmospheric CO2 at the 2023 level by reducing anthropogenic inputs to $I = 0.5$.
3. Reduce anthropogenic carbon emissions to 1980 levels (338 ppm) by reducing anthropogenic inputs to $I = 0.357$.

For each simulation, you will run the simulation, make a graph of its outputs, and extract certain data from the output data sets, specifically,

- atmospheric CO2 concentration at the start and end of your simulation.
- modeled "observed" or contemporaneous temperature (deg C) at the start and end of your simulation.
- equilibrium temperature (deg C) at the start and end of your simulation.
- calculated difference between the contemporaneous and equilibrium temperatures at the start and end of your simulation.

Remember that the *equilibrium temperature* is the temperature to which the atmosphere will eventually rise if kept at a particular CO2 concentration. In contrast the modeled "observed" or *contemporaneous temperature* is the temperature that we are predicted to experience at any moment in time, given the changing CO2 concentrations and background uptake rates.

To do these simulations, we need a set of rate equations that incorporate everything we have covered above. Here they are, in one R function. Copy and paste this into your R script, and run it so that R creates this function in its working memory.

```
Earth1 <- function(t,y,p){
  with( as.list(c(y, p)), {
    ## hypothesized relation of carbon inputs and outputs for
    ## the atmosphere
    ## constant input I, diffusion loss out eA
    ## CO2 rate equation, units CO2 / year
    dA.dt <- I + (1-A/280)

    ## incoming radiant energy
    F.in <- Omega/4
    ## outgoing radiant energy
    F.out <- a*Omega/4 + (sigma/A^g)*K^4
```

```
    ## heat energy rate equation
    dK.dt <- (F.in - F.out)/lag

    ## What the temperature WOULD be,
    ## without a lag
    Equilibrium_K = ( (1-a) * Omega / (4*(sigma)) * A^g )^(1/4)

    ### Have R return these results to us
    return(list( c(dA.dt, dK.dt) ,
                 K_eq = Equilibrium_K
    ) )
  })
}
```

For all our simulations, we'll simulate the years from 2023 to 2100. We will consider 2023 to be time zero ($t = 0$), and project every week until 2100 ($t = 77$).

```
t <- seq(0, 77, 1/52)
```

Next, we specify initial state of the system, including carbon concentration and temperature.

```
y0 <- c(A=420, # CO2 equivalents in ppm in 2023
        K=289 # degrees Kelvin in 2023
        )
```

Finally, and as always, make sure that you have loaded functions we will need, as you did for the beginning of this chapter.

```
library(tidyverse) # data wrangling and graphics
library(deSolve) # for numerical integration of ODE models
```

16.9.1. *Experiment 1: Business as usual*

Here we simulate some of the situations we discussed above. First, let's assume *business as usual*, that our net greenhouse emissions continue as they have been. Those emissions result in a net increase of about 2.1 ppm ($I = 2.1$). We specify particular constants or *parameters* for our equations. We create a vector of those here.

```
parameters1 <- c(Omega = 1372, # incoming solar radiative forcing
       a=0.3, # albedo
       sigma = 5.67e-8, # Stefan-Boltzmann constant for radiative
        ↪  loss
       g = 0.08488, # greenhouse constant
       I = 2.1, # increase in CO2 ppm per year
       lag = 50 # lag time to partial temp equilibration
)
```

Next we project the dynamics of atmospheric carbon and temperature, starting at a particular concentration and temperature (`y0`), for particular times in the future (`t`), using our system of equations (`Earth`), and particular values of each parameter (`parameters1`, including `I=2.1`).

```
output1 <- ode(y=y0,
               times=t,
               func=Earth1,
               parms=parameters1)
head(output1)
```

```
          time        A        K     K_eq
[1,] 0.00000000 420.0000 289.0000 289.9797
[2,] 0.01923077 420.0308 289.0012 289.9802
[3,] 0.03846154 420.0615 289.0025 289.9806
[4,] 0.05769231 420.0923 289.0037 289.9811
[5,] 0.07692308 420.1231 289.0050 289.9815
[6,] 0.09615385 420.1538 289.0062 289.9820
```

The first six lines of data show us daily predicted changes in atmospheric carbon (A), temperature (K), and what the temperature would be at equilibrium *if the temperature could catch up to the increasing CO2 concentrations (K_eq).

What is the equilibrium temperature for 2023, and how much higher is it compared to the current temperature?

Here are the last six lines of the data set, leading up to the year 2100.

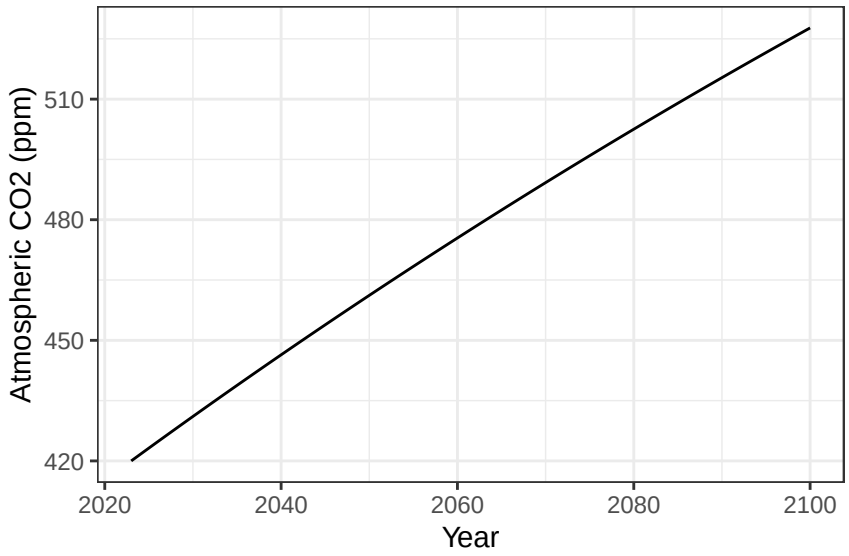
```
tail(output1)
```

```
           time        A        K     K_eq
[4000,] 76.90385 527.5948 291.1446 291.3865
[4001,] 76.92308 527.6182 291.1449 291.3868
[4002,] 76.94231 527.6416 291.1452 291.3871
[4003,] 76.96154 527.6649 291.1455 291.3874
[4004,] 76.98077 527.6883 291.1459 291.3876
[4005,] 77.00000 527.7117 291.1462 291.3879
```

Based on these data, what is the equilibrium temperature for 2100, and how much higher is it compared to theobserved temperature predicted for 2100?

Use this code to make a graph of this output.

```
p1.1 <- output1 %>% as.data.frame() %>%
  mutate(
    Year = time + 2023
  ) %>%
  ggplot(aes(x=Year, y=A)) +
  geom_line() + labs(y="Atmospheric CO2 (ppm)") +
  theme_bw()
p1.1
```

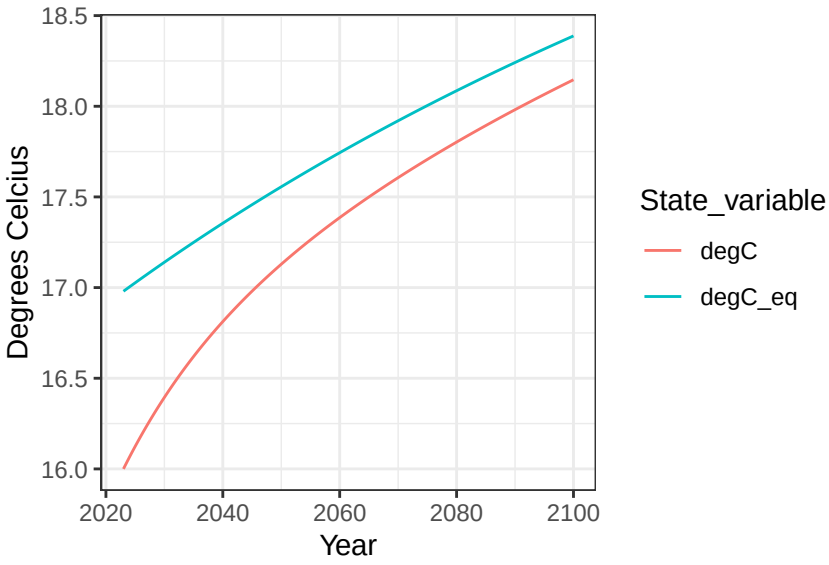



```
ggsave("figs/Exp1_CO2.png", plot=p1.1)
```

```
Saving 4.5 x 3 in image
```

Now make a graph of temperatures.

```
out.long1 <- output1 %>% as.data.frame() %>%
  select(time, K, K_eq) %>%
  transmute(
    Year = time + 2023,
    degC = K -273,
    degC_eq = K_eq -273
  ) %>%
  pivot_longer(cols=-Year, names_to = "State_variable",
               values_to="Value")

p2.1 <- ggplot(out.long1, aes(x=Year, y=Value,
 ↪  color=State_variable)) +
  geom_line() +
  labs(y="Degrees Celcius")  +
  theme_bw()
p2.1
```

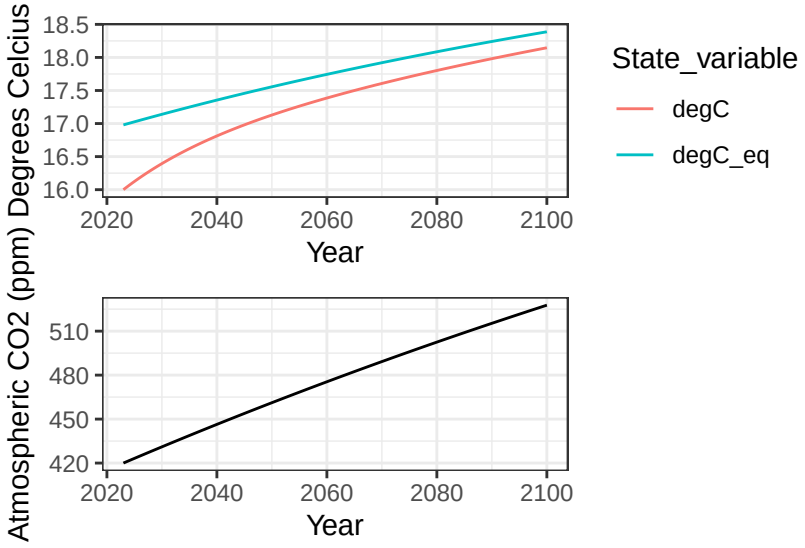



```
ggsave("figs/Exp1_temps.png", plot=p2.1)
```

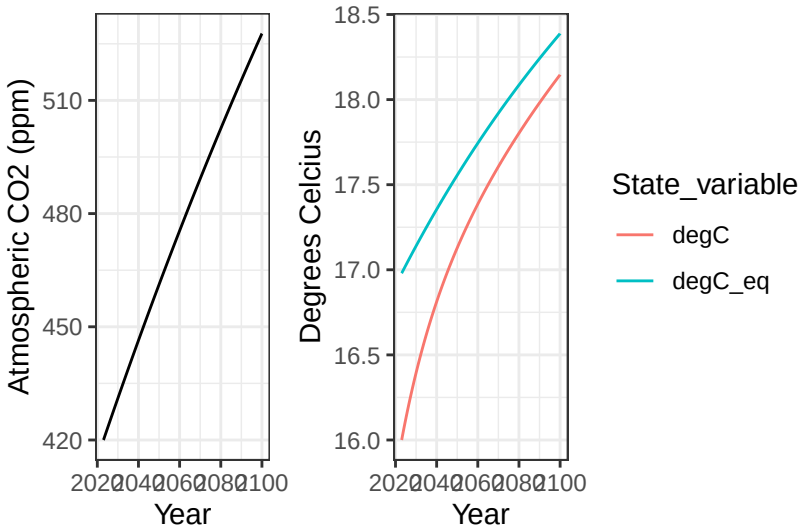
```
Saving 4.5 x 3 in image
```

If you want to put two graphs together, I recommend you install the `patchwork` package, and then load it.

```
# eval: false
# uncomment and run
# install.package("patchwork", dep=TRUE)

# after installation, load it:
library(patchwork)

# use it.
# try
p2.1 / p1.1
```



```
# or
p1.1 + p2.1
```




```
ggsave("figs/Exp1_CO2andK.png", height=4, width=8)
```

16.9.2. *Experiment 2: Keep CO2 at 2023 levels (420 ppm)*

Here we make one change: $I = 0.5$ in the parameters so that we keep CO2 at 420 ppm.

```
parameters2 <- c(Omega = 1372, # incoming solar radiative forcing
       a=0.3, # albedo
       sigma = 5.67e-8, # Stefan-Boltzmann constant for radiative
        ↪  loss
       g = 0.08492, # greenhouse constant
       I = 0.5, # increase in CO2 ppm per year
       lag = 50 # lag time to partial temp equilibration
)
```

Rerun the model, keeping everything else the same.

```
output2 <- ode(y=y0, times=t, func=Earth1, parms=parameters2)
tail(output2)
```

```
           time   A        K     K_eq
[4000,] 76.90385 420 289.9911 289.9972
[4001,] 76.92308 420 289.9911 289.9972
[4002,] 76.94231 420 289.9911 289.9972
[4003,] 76.96154 420 289.9911 289.9972
[4004,] 76.98077 420 289.9911 289.9972
[4005,] 77.00000 420 289.9911 289.9972
```

Use this code to make a graph of this output.

```
p1.2 <- output2 %>% as.data.frame() %>%
  mutate(
    Year = time + 2023
  ) %>%
  ggplot(aes(x=Year, y=A)) +
  geom_line() + labs(y="Atmospheric CO2 (ppm)") +
  theme_bw()
p1.2
ggsave("figs/Exp2_CO2.png", plot=p1.2)
```

Now make a graph of temperatures.

```
out.long2 <- output2 %>% as.data.frame() %>%
  select(time, K, K_eq) %>%
  transmute(
    Year = time + 2023,
    degC = K -273,
    degC_eq = K_eq -273
  ) %>%
  pivot_longer(cols=-Year, names_to = "State_variable",
               values_to="Value")

p2.2 <- ggplot(out.long2, aes(x=Year, y=Value,
 ↪  color=State_variable)) +
  geom_line() +
  labs(y="Degrees Celcius")  +
  theme_bw()
p2.2
ggsave("figs/Exp2_temps.png")
```

16.9.3. Experiment 3: Reduce CO2 to 1980 levels (338 ppm)

Here we make one change: $I = 0.357$ in the parameters so that we begin to drive atmospheric CO2 concentration down toward 1980 levels (338 ppm).

```
parameters3 <- c(Omega = 1372, # incoming solar radiative forcing
       a=0.3, # albedo
       sigma = 5.67e-8, # Stefan-Boltzmann constant for radiative
        ↪  loss
       g = 0.08492, # greenhouse constant
       I = 0.357, # increase in CO2 ppm per year
       lag = 50 # lag time to partial temp equilibration
)
```

Rerun the model, keeping everything else the same.

```
output3 <- ode(y=y0, times=t, func=Earth1, parms=parameters3)
tail(output3)
```

```
            time        A        K     K_eq
[4000,] 76.90385 410.3837 289.8742 289.8547
[4001,] 76.92308 410.3816 289.8742 289.8546
[4002,] 76.94231 410.3795 289.8742 289.8546
[4003,] 76.96154 410.3774 289.8741 289.8546
[4004,] 76.98077 410.3754 289.8741 289.8545
[4005,] 77.00000 410.3733 289.8741 289.8545
```

Use this code to make a graph of this output.

```
p1.3 <- output3 %>% as.data.frame() %>%
  mutate(
    Year = time + 2023
  ) %>%
  ggplot(aes(x=Year, y=A)) +
  geom_line() + labs(y="Atmospheric CO2 (ppm)") +
  theme_bw()
p1.3
ggsave("figs/Exp3_CO2.png", plot=p1.3)
```

Now make a graph of temperatures.

```
out.long3 <- output3 %>% as.data.frame() %>%
  select(time, K, K_eq) %>%
  transmute(
    Year = time + 2023,
    degC = K -273,
    degC_eq = K_eq -273
  ) %>%
  pivot_longer(cols=-Year, names_to = "State_variable",
               values_to="Value")

p2.3 <- ggplot(out.long3, aes(x=Year, y=Value,
 ↪  color=State_variable)) +
  geom_line() +
  labs(y="Degrees Celcius")  +
  theme_bw()
p2.3
ggsave("figs/Exp3_temps.png", plot=p2.3)
```

17. *Funding Ecological and Environmental Science: Following the Money*

...in which we discover who pays for ecology, and why that answer keeps changing.

Before you start

- **Assumptions about background and guidance:** your `Rwork` project (Computer setup) and comfort reading in data and making line graphs with `ggplot()` (Extra-pair copulation, Displaying data with ggplot2). No background in economics or government budgets is assumed; inflation adjustment is explained here. Part 1 pulls data live from a government API, so you will need an internet connection; Accessing public databases has more on working with live data sources.

17.1. Goals

In this chapter, you will

- Identify the major federal agencies–and one major federal-to-state funding stream–that pay for ecological and environmental research.
- Use R to pull live budget data directly from a government API, rather than a file someone else already cleaned for you.
- Construct and compare time series of funding across agencies and across a fundamentally different funding *mechanism* (formula-based state apportionments).
- Distinguish nominal dollars from inflation-adjusted ("real") dollars, and explain why the distinction matters when you compare years.
- Distinguish an agency's *entire* budget from the slice of it that actually funds research–and see, concretely, how much bigger the difference can be for some agencies (NASA, USDA) than others (NSF).
- Find and explain research that interests you; describe the scientific motivation for the research and explain how the investigators used or plan to use the research award.
- Evaluate what recent (2025-2026) federal budget changes mean for the pipeline of ecological research, and for your own options as an early-career scientist.

17.2. Deliverables

Turn these in to the relevant Canvas assignment before the due date. Use the `ggsave()` code in each section below to produce a PNG file for each figure. Each one lands in your `figs` folder; insert those PNGs into your document from there.

1. A single document (.docx) containing:
    i. Your version of Figure 17.1 and Figure 17.2 (nominal and inflation-adjusted, all six federal agencies).
    ii. Your version of Figure 17.3, Figure 17.4, and Figure 17.5 (research-only obligations, department and sub-agency views, and the total-vs-research comparison).
    iii. Your class's version of Figure 17.6 (all states your classmates collected).
    iv. Your short write-up from Part 3: the title, PI, and institution of the award you chose, and your two-question answer explaining its scientific motivation and how the funding was or will be used.
    v. Written answers to the nine questions in Questions to answer, below.
2. Your R script, named `funding_science_YourUniqueID.R`. This must be a `.R` file, not a Word document.

17.3. Background

Ecological and environmental science does not fund itself. Someone decides, every year, how many dollars go toward understanding populations, ecosystems, and environmental change–and those decisions shape which questions get asked, which species and places get studied, and who gets to be a scientist. If you take classes in university science departments, or plan to do research, apply for a grant, or simply understand why a particular topic is (or is not) well studied, it helps to know where the money for research comes from.

In the United States, most federal money for ecological and environmental research flows through a handful of agencies, each with a different mission.

- The **National Science Foundation (NSF)** is the primary funder of basic, curiosity-driven ecological research at universities. Within NSF, the Directorate for Biological Sciences, and, inside that, the Division of Environmental Biology (DEB), funds most of the population, community, and ecosystem ecology done at U.S. universities.
- The **Environmental Protection Agency (EPA)**, mainly through its Office of Research and Development, funds applied research on pollution, toxicology, and environmental health. Its competitive extramural program, **Science to Achieve Results (STAR)**, has historically funded

university research connecting environmental quality to human and ecological health.

- The **U.S. Department of Agriculture** (**USDA**) funds applied ecology two ways: through the National Institute of Food and Agriculture (NIFA), which makes competitive grants to university researchers (for example, the Agriculture and Food Research Initiative and the McIntire-Stennis forestry program), and through its own **Forest Service Research and Development** branch, which conducts forest and rangeland ecology research directly–often at USDA's own experimental forests–rather than by funding outside investigators.
- The **National Oceanic and Atmospheric Administration** (**NOAA**), part of the Department of Commerce, funds fisheries, coastal, and climate-related ecological research.
- The **Department of Energy** (**DOE**) funds a large research portfolio, but most of it is about energy technology–fossil fuels, nuclear power, renewables–not ecology. The slice that funds ecosystem, watershed, and climate science sits in a single office inside DOE's Office of Science, **Biological and Environmental Research** (**BER**); later in this chapter we isolate BER's own numbers to see how small a share of DOE's total research budget it really is.
- The **National Aeronautics and Space Administration** (**NASA**) funds Earth-observing satellites and remote sensing research that ecologists use to track land cover, wildfire, drought, and ecosystem change–a smaller, less obvious slice of an agency best known for astronauts and telescopes.
- The **U.S. Fish and Wildlife Service** (**FWS**), an agency inside the **Department of the Interior** (**DOI**)–alongside the U.S. Geological Survey, the Bureau of Land Management, and the National Park Service–funds and conducts wildlife research directly. FWS also plays a second, very different role you will meet in Part 2: it is the agency that administers the formula-based apportionments that send hunting- and fishing-tax revenue to state wildlife agencies.
- The **U.S. Geological Survey** (**USGS**), also inside DOI, is–by dollars obligated to research–the single largest source of federal ecological field science: climate, ecosystems, wildlife disease, and water research, carried out mostly by USGS's own scientists rather than through grants to universities.
- **Private foundations** fund ecological and environmental science too, outside the federal appropriations process entirely. The Gordon and Betty Moore Foundation is one of the largest funders of marine and environmental science in the country; the David and Lucile Packard Foundation and the Pew Charitable Trusts fund ocean conservation and environmental research as well. Foundation funding is a much smaller pool than federal funding in total, but it can move faster and more flexibly, since a foundation sets its own priorities rather than waiting on an annual congressional appropriation.

Scientists, teams of scientists, and universities compete with each other for this funding via *competitive grants*.

1. In response to Requests for Proposals (RFPs), scientists write extensive proposals for carefully crafted research projects and submit them.
2. The agencies review all of the submitted proposals, relying extensively on review panels made up of other scientists, typically from universities.
3. Based largely on the recommendations of these review panels, the agency chooses a small fraction to fund. These awards are driven by the number and quality of proposals and, critically, by Congress's annual appropriations decisions.

Federal money also reaches ecological work at the *state* level, but often through a completely different mechanism: not a competed, peer-reviewed grant, but a **formula-based apportionment**. The clearest example is the pair of laws that fund most state fish and wildlife agencies: the **Pittman-Robertson Wildlife Restoration Act (1937)** and the **Dingell-Johnson/Wallop-Breaux Sport Fish Restoration Act (1950)**. Both work the same way: the federal government levies an excise tax on hunting and fishing equipment, and the U.S. Fish and Wildlife Service apportions that money to every state automatically, using a formula based on land area and the number of licensed hunters or anglers–no competition, no proposal, no peer review. State wildlife agencies use these funds to support a great deal of applied wildlife and fisheries ecology, including much of the work that eventually shows up in conservation plans. In contrast to the competitive process described above, state fish and wildlife agencies are funded through a legitimate *entitlement*, which is driven by consumer purchases on hunting and fishing equipment and licenses, and a fixed formula.

A word of caution about all of these numbers: an agency's *total budget* is not the same thing as its *research budget*. NASA's budget includes crewed spaceflight and space station operations; USDA's includes crop insurance and nutrition assistance programs; EPA's includes pollution cleanup and regulatory enforcement. None of that is research. The federal government does track a "research and development obligations" figure for each agency that is meant to exclude exactly this kind of non-research spending–we will use it later in this chapter to see how much of each agency's total budget is actually research.

Why this matters right now. Federal science funding has been unusually volatile the last two years, which makes this a good moment to look at the data rather than rely on impressions. A few reported, verifiable facts, current as of mid-2026:

- NSF's FY2026 appropriation came in at about $8.75 billion–close to its FY2024 level–after the Trump administration had proposed cutting it by roughly 55%, to $3.9 billion; Congress pushed back and did not adopt that cut. (See the AIP FYI summary and the Congressional Research Service report on NSF's FY2026 appropriations.)
- Even so, in spring 2025 NSF terminated roughly 1,750 already-active grants, worth a combined $1.4 billion, as part of a federal review of research spending led by the Department of Government Efficiency

(DOGE). These were grants that were already awarded and for which the research had already begun. Terminating them resulted in a tremendous waste of time and money that had already been spent. Terminations were concentrated in STEM education and the social and behavioral sciences, with the biological sciences affected less, in the single digits as a percentage of active awards. (See Eos, "NSF Cuts Hundreds of Science Program Budgets".) DOGE-driven terminations and office closures were not limited to NSF–see the note on obligations vs. outcomes later in this chapter, where NOAA, USGS, and FWS examples are discussed in more detail.

- EPA's FY2026 budget request proposed a substantial overall reduction, and in spring 2025 EPA leadership directed staff to cancel a number of active research grants, including some STAR grants. (See the CRS brief on EPA's FY2026 budget request and reporting in *Science*.)

These are contested political decisions, and reasonable people disagree about whether they are good policy. Our job in this chapter is not to take a side, but to look directly at the numbers and let the data show us what actually happened, and to keep watching–this is a fast-moving topic and today's figures will keep changing. The Ecological Society of America maintains a running Federal Budget Tracker if you want to follow along after this course ends.

17.4. Activity

17.4.1. Set up

If you have not already done so, create a script file for this assignment, `funding_science_YourUniqueID.R`, in your `Rwork` project.

Install and load the packages we'll need. `jsonlite` lets R talk directly to a web API and turn the response into a data frame; you have already used `tidyverse` and `googlesheets4` in earlier chapters.

```
install.packages("jsonlite", repos = "https://cloud.r-project.org")
```

```
## load packages
library(jsonlite)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------
 ↪  tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------
 ↪  tidyverse_conflicts() --
```

```
x dplyr::filter()  masks stats::filter()
x purrr::flatten() masks jsonlite::flatten()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to
 ↪  force all conflicts to become errors
```

```
library(googlesheets4)
```

17.4.2. Part 1: Federal agency budgets, straight from the source

Rather than downloading a file that someone else already cleaned, we are going to pull data directly from USAspending.gov, the federal government's official spending transparency site. USAspending.gov has a public web **API** (Application Programming Interface): instead of a web page built for a human to read, an API endpoint is a URL that returns structured data–here, in a format called JSON–for a program like R to parse directly. You build the URL yourself (below, we paste an agency's toptier code into it), visit it the same way you'd visit any web address, and the server sends back an answer instead of a page of formatted text. USAspending's API requires no login or password, which is typical for public government data. Because it is *live*, your numbers will reflect whatever the most recent fiscal year is when you run the code, which is a nice feature, but it also means your plot may look slightly different than a classmate's if you run this weeks apart.

Every federal agency has an identifying **toptier code**. Here are six agencies relevant to ecological and environmental science:

Agency	Toptier code	Why it matters to ecologists
National Science Foundation (NSF)	049	Main funder of basic academic ecology (Division of Environmental Biology)
Environmental Protection Agency (EPA)	068	Applied environmental and human-health research (STAR grants)
U.S. Department of Agriculture (USDA)	012	Agricultural and forest ecology (NIFA)
U.S. Department of Energy (DOE)	089	Ecosystem and climate science, via its Biological and Environmental Research program–a small slice of DOE's total, isolated later in this chapter

Agency	Toptier code	Why it matters to ecologists
U.S. Department of the Interior (DOI)	014	Parent agency of USGS and the U.S. Fish and Wildlife Service
National Aeronautics and Space Administration (NASA)	080	Earth-observation satellites and remote sensing

Notice that **NOAA, FWS, USGS, and the Forest Service are missing**. USAspending only assigns toptier codes to top-level, cabinet-style agencies; NOAA is a *sub-agency* inside the Department of Commerce, FWS and USGS are *sub-agencies* inside the Department of the Interior, and the Forest Service is a *sub-agency* inside USDA. Their dollars are already folded into their parent department's toptier total in the table above–DOI's total includes FWS, USGS, the Bureau of Land Management, and the National Park Service all lumped together, and USDA's includes the Forest Service along with NIFA and everything else USDA does. This is a real feature of how the federal government is organized, not a limitation of our code–and it is exactly why, later in this chapter, we will switch to a different data source that reports each of these sub-agencies on its own.

We will write a small function that asks USAspending.gov for one agency's budget history, and then apply that function to all six agencies at once.

```
## a named vector: the names are labels we choose, the values are
 ↪  USAspending's codes
agency_codes <- c(NSF  = "049",
                  EPA  = "068",
                  USDA = "012",
                  DOE  = "089",
                  DOI  = "014",
                  NASA = "080")

## a function that, given one toptier code, returns that agency's
## budgetary resources for every fiscal year USAspending has on file
get_budget <- function(code) {
  url <- paste0("https://api.usaspending.gov/api/v2/agency/", code,
                "/budgetary_resources/")
  result <- fromJSON(url)          # ask the API, and parse its JSON
 ↪  response
  result$agency_data_by_year       # this is the part we actually
 ↪  want: one row per fiscal year
}
```

Now we apply that function to every agency code in `agency_codes`, and stick the six results together into one data frame, with a new column (`agency`) recording which agency each row came from.

```
## map() applies get_budget() to each element of agency_codes;
## bind_rows(..., .id="agency") stacks the six results and labels
 ↪  each
budget_list <- map(agency_codes, get_budget)
budget_data <- bind_rows(budget_list, .id = "agency")

## take a look at what we got
names(budget_data)
```

```
[1] "agency"                      "fiscal_year"
[3] "agency_budgetary_resources"  "agency_total_obligated"
[5] "agency_total_outlayed"       "total_budgetary_resources"
[7] "agency_obligation_by_period"
```

```
head(budget_data)
```

```
  agency fiscal_year agency_budgetary_resources
   ↪  agency_total_obligated
1    NSF        2027                         NA
 ↪  NA
2    NSF        2026                10223664068
 ↪  6550550600
3    NSF        2025                10738133526
 ↪  9885079238
4    NSF        2024                10751753768
 ↪  9746030086
5    NSF        2023                11074706436
 ↪  9861960524
6    NSF        2022                10217493512
 ↪  9523708321
  agency_total_outlayed total_budgetary_resources
1                    NA                         NA
2            8053031182               1.618120e+13
3           10153840633               1.326370e+13
4            9614870757               1.224855e+13
5            9201120908               1.188986e+13
6            8328424648               1.140981e+13
```

```
1
 ↪  NULL
2                    2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12007312,
 ↪  305843689, 544135092, 729869844, 958970606, 1431143778,
 ↪  2122419362, 2802516096, 4672418782, 6550550600
3 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 259157056, 794313662,
 ↪  1050845347, 1516591847, 1932680742, 2217339136, 2592569683,
 ↪  3473544082, 5314345721, 8368173707, 9885079238
4 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 348155303, 778081228,
 ↪  1338132973, 1619581610, 2240834442, 2744463181, 3252556366,
 ↪  4209667874, 5921507757, 8466376552, 9746030086
5  2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 240428993, 690189001,
 ↪  995167687, 1281071903, 1925374299, 2459931484, 3293996282,
 ↪  4204697559, 5923684535, 7866112064, 9861960524
```

```
6 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 316195091, 827514797,
 ↪  1218195618, 1690650294, 2104712341, 2638100357, 3096507876,
 ↪  4284895627, 5787601542, 8563093406, 9523708321
```

You should see several columns you didn't ask for by name–the API sends back everything it has for that agency and year, not just one number. The ones we'll use in this chapter:

- `fiscal_year`: the federal fiscal year (October-September) the row describes.
- `agency_budgetary_resources`: the total funds available for this agency to spend that year–new appropriations, plus any unspent balances carried over from prior years. This is the number we'll plot as each agency's "budget."
- `agency_total_obligated`: how much of that the agency has legally committed to spend–signed a contract, made a grant award–whether or not the money has actually gone out the door yet.
- `agency_total_outlayed`: how much has actually been paid out in cash. Outlays typically lag obligations, sometimes by years, because a multi-year grant or construction contract gets paid out gradually as the work happens.
- `total_budgetary_resources`: not specific to this agency–this is the whole federal government's budgetary resources for that year, included so you can see roughly what share of total government resources one agency controls.

Let's plot all six agencies together, using `agency_budgetary_resources`.

```
p_agency_budgets <- ggplot(data = budget_data,
       aes(x = fiscal_year, y = agency_budgetary_resources / 1e9,
           color = agency)) +
  geom_line(linewidth = 1) +
  geom_point() +
  labs(x = "Fiscal year",
       y = "Budgetary resources (billion $, nominal)",
       color = "Agency",
       title = "Federal budgets for science-relevant agencies")
p_agency_budgets
```

```
Warning: Removed 6 rows containing missing values or values outside
 ↪  the scale range
(`geom_line()`).
```

```
Warning: Removed 6 rows containing missing values or values outside
 ↪  the scale range
(`geom_point()`).
```

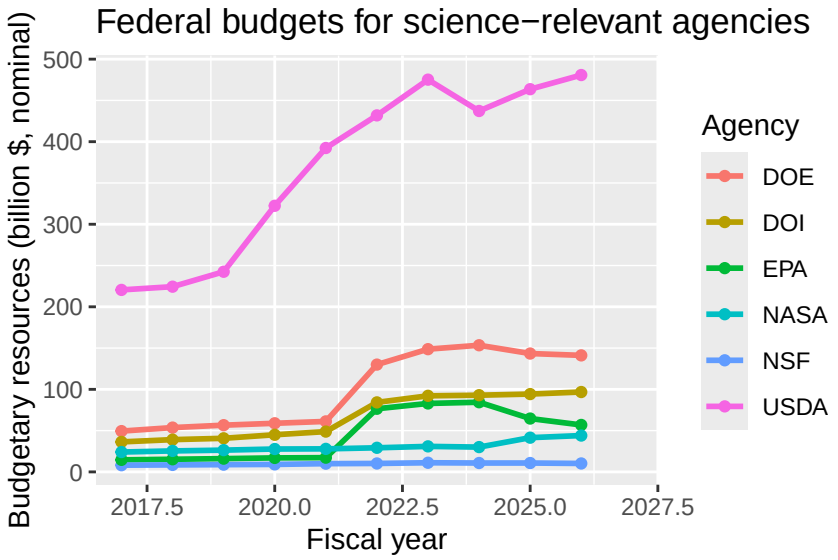


Figure 17.1.: Total budgetary resources for six federal agencies relevant to ecological and environmental science, in nominal (not inflation-adjusted) dollars.

```
## Save a PNG file to your working directory--this is the figure to
## insert into your deliverable document.
ggsave("figs/fig-agency-budgets.png", plot = p_agency_budgets,
       width = 7, height = 5)
```

Refer to your version of Figure 17.1. A couple of things to keep in mind:

- These numbers are the agency's *entire* budget–not just the part that funds ecological research. NSF, for example, also funds physics, computer science, and engineering. Think of this as an upper bound on what could be available for ecology, not a direct measure of ecology funding.
- These are **nominal dollars**: the actual number of dollars appropriated that year, with no adjustment for inflation. A dollar in 2010 bought more than a dollar today. If an agency's budget looks flat over ten years in nominal dollars, its *purchasing power* has actually shrunk.

17.4.3. Adjusting for inflation

To compare dollars from different years fairly, we need to convert everyone's budget into the purchasing power of a single reference year. We'll use the **Consumer Price Index** (**CPI-U**), the standard cost-of-living measure published monthly by the U.S. Bureau of Labor Statistics.

Rather than fetching this from an API, we'll use a small table you keep directly in your script. CPI values are revised only rarely and change slowly, so a short, clearly-sourced table is actually the more reliable choice here–it has no dependency on a web service being available, and everyone in the class gets exactly the same numbers.

```
## Annual average CPI-U (all items, U.S. city average), 1982-84 =
 ↪  100.
## Source, 2017-2025: U.S. Bureau of Labor Statistics,
##
 ↪  https://www.bls.gov/regions/mid-atlantic/data/consumerpriceindexannualandsemiann
## Note: the 2025 average is BLS's own preliminary figure--October
 ↪  2025 data
## could not be collected during a lapse in appropriations that
 ↪  year.
## Source, 2026: BLS has not published a 2026 annual average
 ↪  yet--the year
## isn't over. Rather than leave 2026 blank, we use a *projected*
 ↪  value:
## the Federal Reserve Bank of Philadelphia's Survey of Professional
## Forecasters (SPF), Q2 2026 release, put median expected headline
 ↪  CPI
## inflation for 2026 at 3.5% (Q4-over-Q4 basis):
##
 ↪  https://www.philadelphiafed.org/surveys-and-data/real-time-data-research/spf-q2-
## We applied that growth rate to the 2025 annual average (321.943 *
 ↪  1.035
## = ~333.2) to get a rough 2026 estimate. This is explicitly an
 ↪  estimate,
## not an official BLS figure--replace it with the real annual
 ↪  average
## once BLS publishes it (early the following year).
## Stored in data/cpi.csv; add a row there each time you re-teach
 ↪  this
## chapter, replacing last year's projection with the real BLS
 ↪  figure and
## adding a new projected row for the new current year.
cpi <- read_csv("data/cpi.csv")
```

```
Rows: 10 Columns: 2
-- Column specification
 ↪  --------------------------------------------------------
Delimiter: ","
dbl (2): fiscal_year, cpi_u

i Use `spec()` to retrieve the full column specification for this
 ↪  data.
i Specify the column types or set `show_col_types = FALSE` to quiet
 ↪  this message.
```

Federal fiscal years run October through September, while CPI is published by calendar year; matching a fiscal year to the calendar year with the same number is a common simplification, not an exact correspondence–worth knowing, not worth losing sleep over in an introductory exercise.

Now we join this table to our budget data by `fiscal_year`, and re-express every year's budget in a single reference year's dollars–here, 2025.

```
## the CPI value for our reference year
base_cpi <- cpi$cpi_u[cpi$fiscal_year == 2025]

budget_real <- budget_data %>%
  left_join(cpi, by = "fiscal_year") %>%
  ## multiplying by (base year's CPI / that year's CPI) rescales
   ↪  every
  ## year's dollars into 2025 purchasing power
  mutate(budget_real_2025usd = agency_budgetary_resources *
   ↪  (base_cpi / cpi_u))
```

Notice that the current fiscal year, 2026, uses the *projected* CPI value described above rather than an official BLS annual average–BLS won't publish that until early 2027. Everything downstream (the inflation-adjusted figure below, and any questions you answer about 2026) is therefore built on an estimate for that one year, not a final number. `left_join()` would fill any genuinely missing year with `NA`; we keep the `filter(!is.na(...))` step below as a safeguard in case a future year is missing from `data/cpi.csv` before you've had a chance to add a projection for it.

```
p_agency_budgets_real <- budget_real %>%
  filter(!is.na(budget_real_2025usd)) %>%
  ggplot(aes(x = fiscal_year, y = budget_real_2025usd / 1e9, color =
   ↪  agency)) +
  geom_line(linewidth = 1) +
  geom_point() +
  labs(x = "Fiscal year",
       y = "Budgetary resources (billion 2025 $)",
       color = "Agency",
       title = "Federal budgets for science-relevant agencies,
        ↪  inflation-adjusted")
p_agency_budgets_real
```

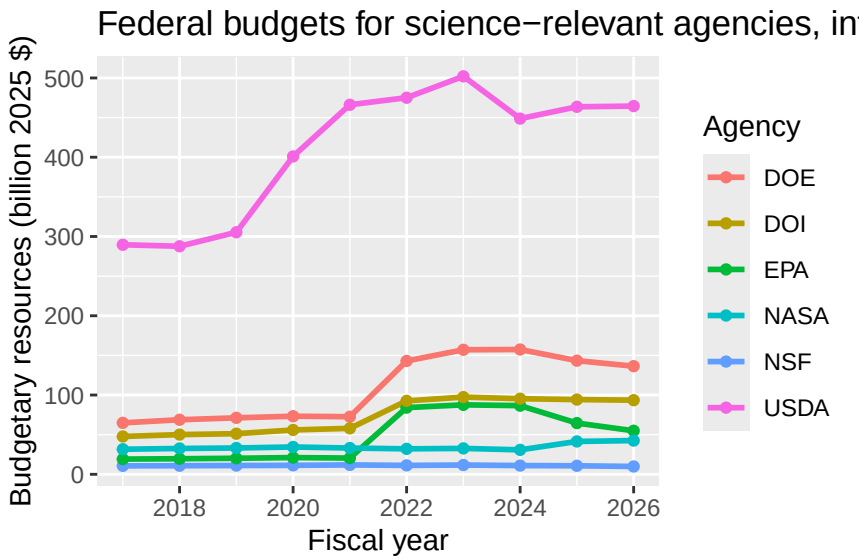


Figure 17.2.: The same six agency budgets as Figure 17.1, expressed in constant 2025 dollars.

```
## Save a PNG file to your working directory--this is the figure to
## insert into your deliverable document.
ggsave("figs/fig-agency-budgets-real.png", plot =
 ↪  p_agency_budgets_real,
       width = 7, height = 5)
```

Compare Figure 17.1 and Figure 17.2 side by side. Any agency whose nominal line looked flat should now show a gentle *decline* in Figure 17.2–that's inflation quietly eroding purchasing power even when Congress technically kept the dollar amount "the same."

17.4.4. How much of that is actually research?

Everything so far has used each agency's *entire* budget, which for some agencies is a poor stand-in for "money available for ecological research." NASA's total budget includes the International Space Station and crewed spaceflight. USDA's includes crop insurance and nutrition assistance. EPA's includes cleanup and enforcement. None of that funds research.

The federal government has an office that exists specifically to separate research spending from everything else: NCSES, the National Center for Science and Engineering Statistics (part of NSF), conducts an annual **Survey of Federal Funds for Research and Development**. Every year, each agency reports which of its budget accounts count as "R&D obligations" under a common federal definition, and NCSES compiles the results. The distinction is not cosmetic–NCSES's own documentation notes, for example, that it reclassified NASA's Space Operations account (which includes the Hubble Space Telescope and crewed missions) out of NASA's R&D total. The same logic keeps USDA's crop insurance and SNAP out of USDA's R&D total. Usefully for us, NCSES also reports sub-agencies separately from their parent department–which is how we can finally see NOAA on its own apart from the rest of Commerce, and FWS, USGS, and the Forest Service each on their own apart from the rest of DOI and USDA.

This survey is not a live API–it is a set of tables NCSES publishes once a year, usually as Excel and PDF files, at ncses.nsf.gov/surveys/federal-funds-research-development. That makes it a good second example of when a small, well-sourced table you maintain yourself (like the CPI table above) is more practical than trying to script a live connection to it.

```
## Federal R&D OBLIGATIONS by agency--NOT each agency's entire
 ↪  budget, but the
## slice NCSES classifies as research and development (plus R&D
 ↪  facilities/
## equipment, i.e. "R&D plant"). In millions of dollars.
##
## IMPORTANT: nine of these ten rows are OBLIGATIONS, but DOE_BER is
 ↪  not.
## Source (NSF, EPA, USDA, NOAA, NASA, DOE, DOI, FWS, USGS, USFS):
 ↪  NCSES Survey
```

```
## of Federal Funds for Research and Development, Table 3, multiple
 ↪  survey
## years:
 ↪  https://ncses.nsf.gov/surveys/federal-funds-research-development.
## This is OBLIGATIONS: a legal commitment the agency made that year
 ↪  (a grant
## awarded, a contract signed)--the same "obligated" concept as
## agency_total_obligated back in Part 1. It is NOT the same as an
 ↪  OUTLAY
## (cash actually paid out, which can lag obligations by months or
 ↪  years),
## and it is NOT the same as an APPROPRIATION (the budget authority
 ↪  Congress
## granted, which an agency can carry into future years without
 ↪  fully
## obligating it the same year).
## Source (DOE_BER only): NCSES does not break DOE down by program
 ↪  office, so
## this row comes directly from DOE Office of Science's own
 ↪  congressional
## budget justifications for Biological and Environmental Research
 ↪  (BER).
## Those documents report ENACTED APPROPRIATIONS, not obligations--a
 ↪  related
## but different number (see above). Treat the DOE_BER row as "what
 ↪  Congress
## gave BER to work with," not "what BER legally committed," and
 ↪  keep that
## distinction in mind when comparing it to the other nine rows in
## @fig-rd-only-subagencies. See
## https://science.osti.gov/budget/Budget-by-Program/BER-Budget for
 ↪  the source
## documents by year.
## Stored in data/rd_only.csv. NCSES (and DOE) revise their own
 ↪  estimates in
## the following year's release (a "preliminary" figure becomes
 ↪  "final"), so a
## few of these will differ slightly from whatever is on the source
 ↪  website
## when you read this--that is normal for survey/budget data, not a
 ↪  mistake.
## Update/extend data/rd_only.csv when you re-teach the chapter.
rd_only <- read_csv("data/rd_only.csv")
```

```
Rows: 77 Columns: 3
-- Column specification
 ↪  --------------------------------------------------------
Delimiter: ","
chr (1): agency
dbl (2): fiscal_year, rd_obligations

i Use `spec()` to retrieve the full column specification for this
 ↪  data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet
 ↪  this message.
```

All of the dollar figures in `rd_only` (fiscal years, not calendar years–see the CPI section above for why that distinction matters) are **obligations**, except `DOE_BER`, which is an **appropriation**, for the reasons explained in the code comment above. Obligations and appropriations are close cousins but not the same thing, and neither one is the same as an *outlay*–Part 1 already introduced all three: `agency_budgetary_resources` (roughly, the appropriation), `agency_total_obligated`, and `agency_total_outlayed`.

That distinction matters more than usual right now. An obligation is a real, legally binding commitment–but a legal commitment is not a guarantee. Under the current administration, a substantial number of already-obligated federal research grants have been terminated after the fact, and some research offices have been downsized or closed outright, largely at the direction of the Department of Government Efficiency (DOGE) and newly appointed agency leadership. This is not unique to NSF (mentioned in Background above): NOAA canceled leases on research facilities, including its National Centers for Environmental Prediction building in College Park, MD and its Radar Operations Center in Norman, OK, alongside broad workforce cuts–reported at roughly 10% of staff as of early 2025, with some former officials warning the total could reach half the agency. USGS's Ecosystems Mission Area–the program behind avian-influenza monitoring, invasive-species tracking, and dozens of other applied wildlife and ecosystem research efforts across 16 research centers–was funded at about $293 million in FY2025 and proposed for complete elimination in FY2026; USGS's total workforce is projected to shrink from about 7,870 employees (FY2024) to roughly 5,150 (FY2026). FWS's own staff fell from 9,957 to 8,179 between 2024 and May 2025 (about 18%), with concrete on-the-ground effects–at one California refuge, wetland habitat went unflooded for an entire winter because no staff remained to operate the water-control structures. Government-wide, DOGE had terminated nearly 16,000 federal grants worth an estimated $49 billion as of January 2026. Some of this is being contested in court–several federal judges have ruled specific terminations unlawful and blocked them–but litigation moves slowly, and many terminated grants and closed offices remain terminated and closed while cases proceed. (See DataCenterDynamics on NOAA facility and staffing cuts, E&E News on USGS's Ecosystems Mission Area, CalMatters on FWS refuge staffing, and reporting on DOGE's grant-termination totals and legal challenges.)

The practical lesson for this chapter: in most years, treating an obligated dollar as money that funded real research was a safe simplification. For FY2025 and FY2026 specifically, it is not–some of the dollars counted as "obligated" in this chapter's data were later clawed back. Keep that in mind especially when you get to FWS's and USGS's most recent numbers below.

Look closely at FWS's and USGS's last two rows: FWS drops from $48 million in FY2024 to $6 million in FY2025 (an 87% cut), and USGS drops from $803 million in FY2023 to $378 million in FY2024 (more than half).

Both are inside DOI, and together they explain most of the decline you may already have noticed in DOI's whole-department line in Figure 17.1. Some of this reflects the obligations-vs-outcomes gap just described; some of it (as you'll see if you investigate further) also reflects one-time infrastructure-law funding from 2022-2023 working its way through the system on its own schedule, unrelated to any single year's political decisions. Untangling exactly how much of a given year's change comes from which cause is a real, current, and genuinely open research question–not something this chapter can answer for you. FWS's formula-funded apportionments to the states (Part 2) come from a completely different pot of money (excise tax revenue, not annual appropriations)–that contrast is worth keeping in mind once you get to Part 2.

DOE's row tells a different kind of story, and it needs a slightly different fix. NOAA, FWS, USGS, and the Forest Service are missing from Figure 17.1 and Figure 17.3 because they are *sub-agencies* with no toptier code of their own–their dollars are folded into a parent department's total. DOE is not like that: it is already a toptier agency, and its row in Figure 17.3 is a real, complete R&D total. The problem is what that total contains. NCSES's R&D figure for DOE bundles energy-technology research–fossil fuels, nuclear power, renewables, DOE's national labs doing all of the above–together with DOE's actual ecological and environmental science. Splitting those apart means going one level deeper than NCSES goes, into DOE's own Office of Science budget, to the **Biological and Environmental Research (BER)** program specifically. BER's numbers (`DOE_BER` below) are consistently a small fraction of DOE's whole R&D total–well under 10% most years–which tells you something about what most of DOE's research budget actually buys.

Eleven agencies is a lot of colors for one line graph to hold at once–past six or seven categories, a line chart gets hard to read no matter how careful the color choices are. Rather than force all eleven into a single crowded plot, we'll split them the same way the data itself is organized: toptier departments and agencies (the six you already have live budgets for) in one figure, and the five agency- and program-level drilldowns–NOAA, FWS, USGS, and the Forest Service (true sub-agencies), plus DOE BER (a program office inside an agency that is already toptier)–in a second.

```
p_rd_departments <- rd_only %>%
  filter(agency %in% c("NSF", "EPA", "USDA", "DOE", "DOI", "NASA"))
   ↪  %>%
  ggplot(aes(x = fiscal_year, y = rd_obligations / 1000, color =
   ↪  agency)) +
  geom_line(linewidth = 1) +
  geom_point() +
  labs(x = "Fiscal year", y = "R&D obligations (billion $,
   ↪  nominal)",
       color = "Agency",
       title = "Research-only funding, toptier agencies")
p_rd_departments
```

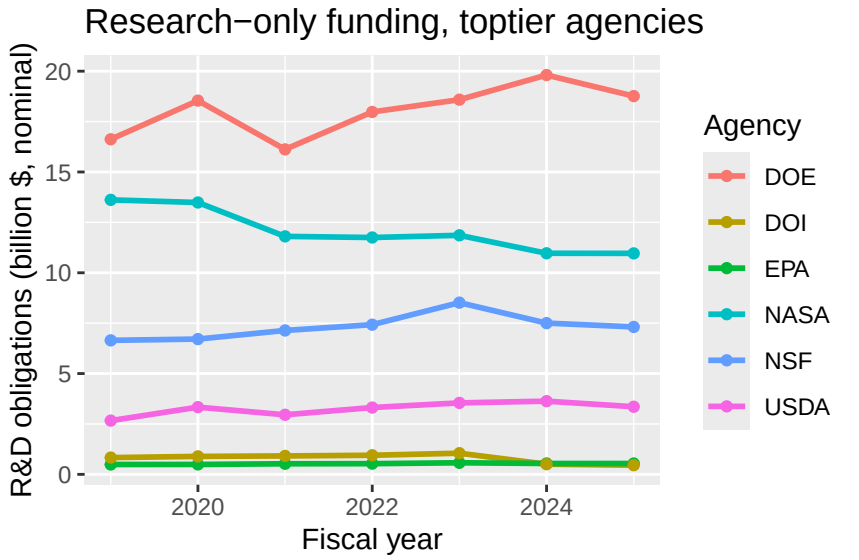


Figure 17.3.: Research and development obligations only (not total agency budget), for the six toptier agencies, FY2019-2025.

```
## Save a PNG file to your working directory--this is the figure to
## insert into your deliverable document.
ggsave("figs/fig-rd-only-departments.png", plot = p_rd_departments,
       width = 7, height = 5)
```

```
p_rd_subagencies <- rd_only %>%
  filter(agency %in% c("NOAA", "FWS", "USGS", "USFS", "DOE_BER"))
   ↪ %>%
  ggplot(aes(x = fiscal_year, y = rd_obligations, color = agency)) +
  geom_line(linewidth = 1) +
  geom_point() +
  labs(x = "Fiscal year", y = "R&D obligations (million $,
   ↪ nominal)",
       color = "Agency",
       title = "Research-only funding, sub-agencies and DOE BER")
p_rd_subagencies
```

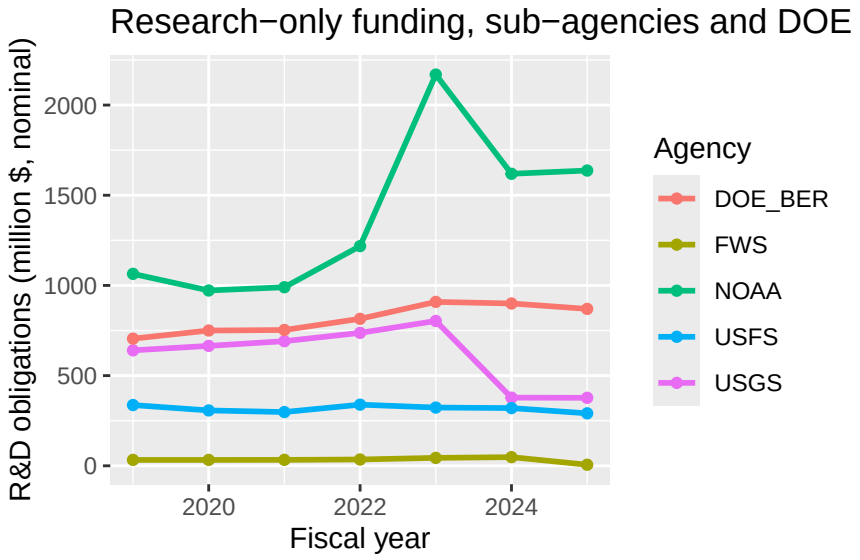


Figure 17.4.: Research and development obligations only, for NOAA, FWS, USGS, and the Forest Service (sub-agencies USAspending can't show on their own) and DOE BER (a program office isolated from DOE's much larger, mostly non-ecological R&D total), FY2019-2025.

```
## Save a PNG file to your working directory--this is the figure to
## insert into your deliverable document.
ggsave("figs/fig-rd-only-subagencies.png", plot = p_rd_subagencies,
       width = 7, height = 5)
```

Notice NASA switches rank between the two budget figures: it is one of the biggest budgets in Figure 17.1, but a much smaller share of its money counts as research once we exclude everything else NASA does. In Figure 17.4, notice how much larger USGS's research funding is than FWS's or the Forest Service's–despite USGS rarely making headlines the way FWS (endangered species) or the Forest Service (wildfire) do–and compare DOE BER's line to DOE's full R&D line in Figure 17.3: BER is a similar size to USGS, but it is a much smaller piece of its own parent agency's total than USGS is of DOI's.

Now let's make that contrast explicit, by putting each agency's *entire* budget next to its *research-only* obligations for a single year.

```
compare_year <- 2024

total_fy <- budget_data %>%
  filter(fiscal_year == compare_year) %>%
  transmute(agency, amount = agency_budgetary_resources / 1e6,
            category = "Entire agency budget")

research_fy <- rd_only %>%
  filter(fiscal_year == compare_year) %>%
  transmute(agency, amount = rd_obligations,
            category = "Research (R&D) obligations only")

p_total_vs_research <- bind_rows(total_fy, research_fy) %>%
  ggplot(aes(x = agency, y = amount / 1000, fill = category)) +
  geom_col(position = "dodge") +
  labs(x = "Agency", y = "Billion $",
       fill = NULL,
       title = paste("How much of each agency's FY", compare_year,
        ↪  "budget is research?"))
p_total_vs_research
```

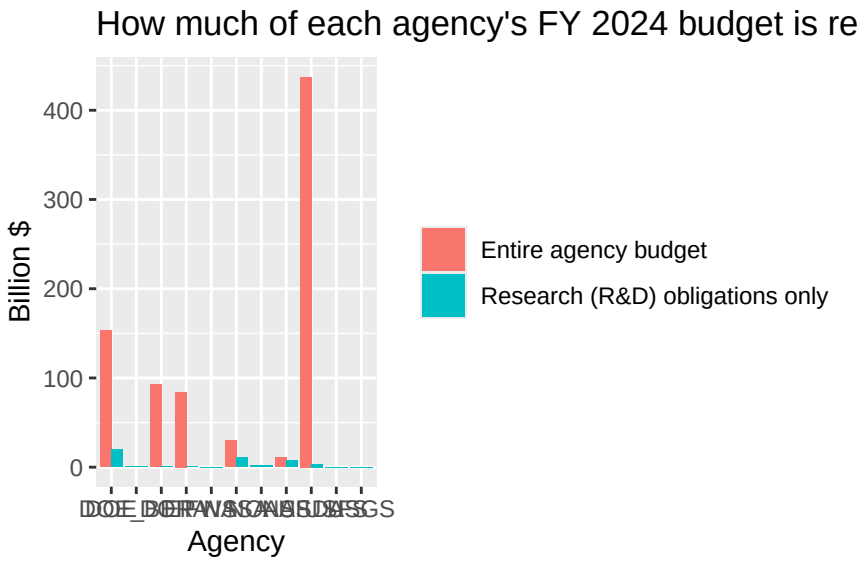


Figure 17.5.: Total agency budget (USAspending) versus research-only obligations (NCSES, or DOE's own budget justifications for DOE BER) for fiscal year 2024. NOAA, FWS, USGS, the Forest Service, and DOE BER have no bar for 'entire agency budget': the first four are not toptier agencies (see above), and DOE BER is a program office, not an agency–DOE's own 'entire agency budget' bar is what its BER bar should be compared against.

```
## Save a PNG file to your working directory--this is the figure to
## insert into your deliverable document.
ggsave("figs/fig-total-vs-research.png", plot = p_total_vs_research,
       width = 7, height = 5)
```

Figure 17.5 is the figure that directly answers "how much of NASA's or USDA's budget is really research"–look at how much shorter the research-only bar is than the entire-budget bar for those two agencies, compared to NSF, where the two bars are close to each other.

17.4.5. Part 2: A different funding mechanism–state wildlife agencies

Now compare that competitive, congressionally-appropriated funding to a formula-based stream: the Pittman-Robertson (Wildlife Restoration) and Dingell-Johnson (Sport Fish Restoration) apportionments that the U.S. Fish and Wildlife Service sends to every state's fish and wildlife agency every year.

Work in pairs, and coordinate with the rest of the class so that no two groups pick the same state.

1. Go to our class Google Sheet for **state wildlife funding** (ask your instructor for the link if it is not already posted). Enter the name of the U.S. state you will investigate, and add your name so no one else picks the same one.
2. Go to the Partner with a Payer funding-sources page, which the U.S. Fish and Wildlife Service maintains as the public source for current and historic Wildlife and Sport Fish Restoration apportionment data by state and year.

3. Go to the Funding Sources tab, and look up your selected state's **Wildlife Restoration** apportionments for each of the five most recent fiscal years available. Place your cursor over the column of your state, and write down the dollar amount for that year. Do this for each of the five most recent years.
4. Enter each year, program, and dollar amount as its own row in the class Google Sheet, along with your state and your uniqueID.
5. Repeat this process for **Sport Fish Restoration.**

Once the whole class has entered data, read the shared sheet into R the same way you did in the tree-cover assignment.

```
d_state <- read_sheet("PASTE_YOUR_CLASS_GOOGLE_SHEET_URL_HERE")
```

```
p_state_funding <- ggplot(data = d_state,
       aes(x = fiscal_year, y = apportionment / 1e6, color = state))
        ↪  +
  geom_line(linewidth = 1) +
  geom_point() +
  facet_wrap(~program) +
  labs(x = "Fiscal year", y = "Apportionment (million $)", color =
   ↪  "State")
p_state_funding
```

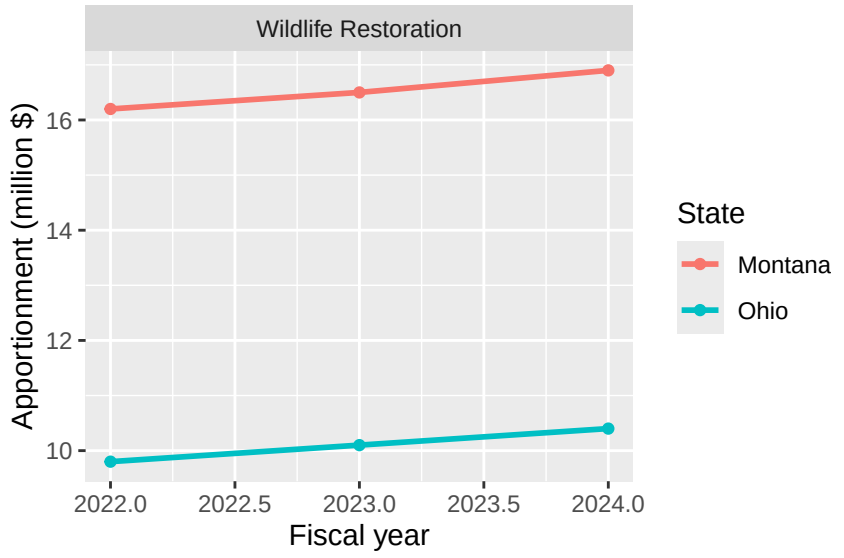


Figure 17.6.: Wildlife Restoration apportionments for a sample of states. Your class figure will include every state your classmates collected.

```
## Save a PNG file to your working directory--this is the figure to
## insert into your deliverable document.
ggsave("figs/fig-state-funding.png", plot = p_state_funding,
       width = 7, height = 5)
```

17.4.6. Part 3: Find and explain one funded research award

Everything so far has been about dollar totals. This part is about a single, real research award–so that "funding" stops being an abstract number and becomes an actual project someone is doing.

NSF exposes every award it has ever made through its own public Award Search API. Unlike USAspending, this lets you search by keyword, so you can find awards on a specific research topic rather than an entire agency's totals.

```
## pick a keyword related to something you're curious about--
## try your own before settling on "biodiversity"
url <-
 ↪  "https://api.nsf.gov/services/v1/awards.json?keyword=biodiversity"
nsf_result <- fromJSON(url)

## the awards themselves are nested inside the response--explore its
 ↪  structure
## before assuming you know what's in it
str(nsf_result, max.level = 3)
```

```
List of 1
 $ response:List of 2
  ..$ award   :'data.frame':    25 obs. of  63 variables:
  .. ..$ abstractText       : chr [1:25] "This Research
   ↪  Infrastructure Improvement (RII) EPSCoR Research Fellows
   ↪  project provides a fellowship to an Assi"| __truncated__ "This
   ↪  Research Infrastructure Improvement (RII) EPSCoR Research
   ↪  Fellows project provides a fellowship to an Assi"|
   ↪  __truncated__ "Nontechnical\r\n\r\nAntarctic organisms persist
   ↪  under extreme conditions lethal to most life on Earth: months
   ↪  o"| __truncated__ "The full inventory of living beings on
   ↪  Earth, underpins human health, economic prosperity,
   ↪  environmental resili"| __truncated__ ...
  .. ..$ activeAwd          : chr [1:25] "true" "true" "true" "true"
   ↪  ...
  .. ..$ agency             : chr [1:25] "NSF" "NSF" "NSF" "NSF" ...
  .. ..$ awardAgencyCode    : chr [1:25] "4900" "4900" "4900" "4900"
   ↪  ...
  .. ..$ awardee            : chr [1:25] "UNIVERSITY OF HAWAII"
   ↪  "UNIVERSITY OF KENTUCKY RESEARCH FOUNDATION, THE" "NEVADA
   ↪  SYSTEM OF HIGHER EDUCATION" "THE ROCKEFELLER UNIVERSITY" ...
  .. ..$ awardeeAddress     : chr [1:25] "2425 CAMPUS RD SINCLAIR RM
   ↪  1" "500 S LIMESTONE" "2215 RAGGIO PKWY" "1230 YORK AVE" ...
  .. ..$ awardeeCity        : chr [1:25] "HONOLULU" "LEXINGTON"
   ↪  "RENO" "NEW YORK" ...
  .. ..$ awardeeCountryCode : chr [1:25] "US" "US" "US" "US" ...
  .. ..$ awardeeDistrict    : chr [1:25] "01" "06" "02" "12" ...
  .. ..$ awardeeDistrictCode: chr [1:25] "HI01" "KY06" "NV02" "NY12"
   ↪  ...
```

```
.. ..$ awardeeName        : chr [1:25] "University of Hawaii"
 ↪  "University of Kentucky Research Foundation" "Nevada System of
 ↪  Higher Education, Desert Research Institute" "Rockefeller
 ↪  University" ...
.. ..$ awardeePhone       : chr [1:25] "8089567800" "8592579420"
 ↪  "7756737300" "2123278309" ...
.. ..$ awardeeStateCode   : chr [1:25] "HI" "KY" "NV" "NY" ...
.. ..$ awardeeZipCode     : chr [1:25] "968222247" "405260001"
 ↪  "895121095" "100656307" ...
.. ..$ cfdaNumber         : chr [1:25] "47.083" "47.083" "47.078,
 ↪  47.083" "47.074, 47.083" ...
.. ..$ date               : chr [1:25] "08/17/2026" "03/31/2026"
 ↪  "09/14/2026" "09/14/2026" ...
.. ..$ dirAbbr            : chr [1:25] "O/D" "O/D" "O/D" "BIO" ...
.. ..$ divAbbr            : chr [1:25] "OIA" "OIA" "OPP" "IIC" ...
.. ..$ estimatedTotalAmt  : chr [1:25] "295906" "235654"
 ↪  "18462872" "11876081" ...
.. ..$ expDate            : chr [1:25] "12/31/2028" "12/31/2028"
 ↪  "09/30/2031" "09/30/2029" ...
.. ..$ fundAgencyCode     : chr [1:25] "4900" "4900" "4900" "4900"
 ↪  ...
.. ..$ fundProgramName    : chr [1:25] "EPSCoR RII: EPSCoR
 ↪  Research Fe" "EPSCoR RII: EPSCoR Research Fe" "Mid-scale RI -
 ↪  Track 1" "Cross-BIO Activities, Mid-scale RI - Track 1" ...
.. ..$ fundsObligated     :List of 25
.. ..$ fundsObligatedAmt  : chr [1:25] "295906" "235654"
 ↪  "18462872" "11876081" ...
.. ..$ histAwd            : chr [1:25] "false" "false" "false"
 ↪  "false" ...
.. ..$ id                 : chr [1:25] "2624662" "2531827"
 ↪  "2535692" "2535312" ...
.. ..$ initAmendmentDate  : chr [1:25] "08/17/2026" "03/31/2026"
 ↪  "09/14/2026" "09/14/2026" ...
.. ..$ latestAmendmentDate: chr [1:25] "08/17/2026" "04/07/2026"
 ↪  "09/14/2026" "09/14/2026" ...
.. ..$ managingPec        : chr [1:25] "196Y00" "196Y00" "511600"
 ↪  "108Y00" ...
.. ..$ orgCodeDir         : chr [1:25] "01000000" "01000000"
 ↪  "01000000" "08000000" ...
.. ..$ orgCodeDiv         : chr [1:25] "01060000" "01060000"
 ↪  "01150000" "08240000" ...
.. ..$ orgLongName        : chr [1:25] "Office of the Director"
 ↪  "Office of the Director" "Office of the Director" "Directorate
 ↪  for Biological Sciences" ...
.. ..$ orgLongName2       : chr [1:25] "Office of Integrative
 ↪  Activities" "Office of Integrative Activities" "Office of
 ↪  Polar Programs" "Infrastructure, Institutes, & Centers" ...
.. ..$ orgUrl             : chr [1:25]
 ↪  "https://www.nsf.gov/od/oia" "https://www.nsf.gov/od/oia"
 ↪  "http://www.nsf.gov/od/" "https://www.nsf.gov/bio" ...
.. ..$ parentUeiNumber    : chr [1:25] "UALXDJ3K6393" ""
 ↪  "MV1JFXA4S621" "" ...
.. ..$ pdPIName           : chr [1:25] "Lenore Pipes" "Joseph R
 ↪  Burger" "Alison E Murray" "Erich D Jarvis" ...
```

```
.. ..$ perfAddress        : chr [1:25] "1800 EAST-WEST Road" "371
 ↪  Jane Stanford Way" "2215 RAGGIO PKWY" "1230 YORK AVE" ...
.. ..$ perfCity           : chr [1:25] "HONOLULU" "Stanford"
 ↪  "RENO" "NEW YORK" ...
.. ..$ perfCountryCode    : chr [1:25] "US" "US" "US" "US" ...
.. ..$ perfDistrict       : chr [1:25] "01" "16" "02" "12" ...
.. ..$ perfDistrictCode   : chr [1:25] "HI01" "CA16" "NV02" "NY12"
 ↪  ...
.. ..$ perfLocation       : chr [1:25] "University of Hawaii"
 ↪  "Stanford University" "Nevada System of Higher Education,
 ↪  Desert Research Institute" "Rockefeller University" ...
.. ..$ perfStateCode      : chr [1:25] "HI" "CA" "NV" "NY" ...
.. ..$ perfZipCode        : chr [1:25] "968222318" "943055008"
 ↪  "895121095" "100656399" ...
.. ..$ pi                 :List of 25
.. ..$ piEmail            : chr [1:25] "lpipes@hawaii.edu"
 ↪  "robbieburger@uky.edu" "alison@dri.edu"
 ↪  "ejarvis@rockefeller.edu" ...
.. ..$ piFirstName        : chr [1:25] "Lenore" "Joseph" "Alison"
 ↪  "Erich" ...
.. ..$ piId               : chr [1:25] "270121464" "269966667"
 ↪  "269686515" "269677919" ...
.. ..$ piLastName         : chr [1:25] "Pipes" "Burger" "Murray"
 ↪  "Jarvis" ...
.. ..$ poEmail            : chr [1:25] "andjohns@nsf.gov"
 ↪  "andjohns@nsf.gov" "wambrose@nsf.gov" "sraghava@nsf.gov" ...
.. ..$ poName             : chr [1:25] "Andrea Johnson" "Andrea
 ↪  Johnson" "William Ambrose" "Sridhar Raghavachari" ...
.. ..$ poPhone            : chr [1:25] "7032925164" "7032925164"
 ↪  "7032928048" "7032924845" ...
.. ..$ primaryProgram     :List of 25
.. ..$ progEleCode        : chr [1:25] "196Y00" "196Y00" "108Y00"
 ↪  "727500, 108Y00" ...
.. ..$ program            : chr [1:25] "EXP PROG TO STIM COMP RES"
 ↪  "Biotechnology, EXP PROG TO STIM COMP RES" "Artificial
 ↪  Intelligence (AI), INTERDISCIPLINARY PROPOSALS, ANTARCTIC
 ↪  ORGANISMS & ECOSYST, MID-SIZE INFRASTRUCTU"| __truncated__
 ↪  "Artificial Intelligence (AI), INTERDISCIPLINARY PROPOSALS,
 ↪  MID-SIZE INFRASTRUCTURE, Biotechnology, Midscale Res"|
 ↪  __truncated__ ...
.. ..$ progRefCode        : chr [1:25] "9150" "8038, 9150" "075Z,
 ↪  4444, 5111, 7716, 8038, 8812, 9150" "075Z, 4444, 7716, 8038,
 ↪  8812" ...
.. ..$ publicAccessMandate: chr [1:25] "1" "1" "1" "1" ...
.. ..$ startDate          : chr [1:25] "01/01/2027" "01/01/2027"
 ↪  "10/01/2026" "10/01/2026" ...
.. ..$ title              : chr [1:25] "EPSCoR Research Fellows:
 ↪  NSF: Using Ancient DNA From Ocean Sediments to Forecast
 ↪  Ecosystem Shifts" "EPSCoR Research Fellows: NSF: Linking
 ↪  Numerical and Metabolic Theories of Life History" "Research
 ↪  Infrastructure: Mid-scale RI-1 (M1:IP) Genome Infrastructure
 ↪  catalyzing next Generation Antarctic Data"| __truncated__
 ↪  "Mid-scale RI-1 (M1: DA): Designing Scalable Community-driven
 ↪  Infrastructure to Support Biodiversity Genomics (SCIBG)." ...
```

```
.. ..$ transType          : chr [1:25] "Standard Grant" "Standard
 ↪  Grant" "Cooperative Agreement" "Cooperative Agreement" ...
.. ..$ ueiNumber          : chr [1:25] "NSCKLFSSABF2"
 ↪  "H1HYA8Z1NTM5" "MV1JFXA4S621" "LHGDNJMZ64Y1" ...
.. ..$ piMiddeInitial     : chr [1:25] NA "R" "E" "D" ...
.. ..$ coPDPI             :List of 25
..$ metadata:List of 3
.. ..$ offset    : int 0
.. ..$ rpp       : int 25
.. ..$ totalCount: int 10000
```

If "biodiversity" doesn't interest you, or returns awards that feel too broad, try a keyword closer to a specific subfield instead. Some options, across a few areas of ecology:

- **Population ecology:** population dynamics, demography, life history evolution, density dependence, metapopulation
- **Community ecology:** species interactions, competition, predator-prey, community assembly, invasive species
- **Food webs:** food web, trophic cascade, trophic ecology, consumer-resource dynamics, bioaccumulation
- **Ecosystem structure and function:** nutrient cycling, primary productivity, carbon cycling, ecosystem services, biogeochemistry

Each award record includes fields like `title`, `piFirstName`/`piLastName` (the Principal Investigator), `awardeeName` (the PI's institution), `fundsObligatedAmt` (the award's dollar value), `startDate`, and–usefully for this part–`abstractText`, a plain-language summary the investigator wrote explaining what the project will do.

```
## look at the list of award titles to help you pick one that
 ↪  interests you
nsf_result$response$award$title
```

```
 [1] "EPSCoR Research Fellows: NSF: Using Ancient DNA From Ocean
 ↪  Sediments to Forecast Ecosystem Shifts"
 [2] "EPSCoR Research Fellows: NSF: Linking Numerical and Metabolic
 ↪  Theories of Life History"
 [3] "Research Infrastructure: Mid-scale RI-1 (M1:IP) Genome
 ↪  Infrastructure catalyzing next Generation Antarctic Data,
 ↪  Access, Tools and Analytics"
 [4] "Mid-scale RI-1 (M1: DA): Designing Scalable Community-driven
 ↪  Infrastructure to Support Biodiversity Genomics (SCIBG)."
 [5] "STAR: Theory and Software for Detecting Speciation with
 ↪  Periodic Gene Flow"
 [6] "Collaborative Research: Research Infrastructure: Biodiversity
 ↪  On Demand (BOND): A Web Platform for Discovering, Comparing,
 ↪  and Integrating Biodiversity Databases"
 [7] "EAGER: Project BASELINE - Building Faculty Capacity via
 ↪  Biodiversity and Community Partnerships"
 [8] "Collaborative Research: Research Infrastructure: Biodiversity
 ↪  On Demand (BOND): A Web Platform for Discovering, Comparing,
 ↪  and Integrating Biodiversity Databases"
```

```
 [9] "Collaborative Research: Research Infrastructure: Biodiversity
  ↪  On Demand (BOND): A Web Platform for Discovering, Comparing,
  ↪  and Integrating Biodiversity Databases"
[10] "BIO-AI: STAR: IUSC: Morphological Barcoding, a novel, machine
 ↪  learning-based approach to solving the taxonomic impediment"
[11] "Collaborative Research: ULTRA-Data: Microbial Mat Community
 ↪  Structure and Metabolic Potential Changes Following an Eruptive
 ↪  Event at Axial Seamount"
[12] "CAREER: Empirical tests of biogeographic theory: island
 ↪  mammals in the Gulf of Maine"
[13] "Collaborative Research: The role of belowground resource
 ↪  partitioning in tree biodiversity-ecosystem function
 ↪  relationships"
[14] "NSF-SNSF: Biodiversity-inspired lake modeling linking
 ↪  processes from cells to ecosystems"
[15] "Collaborative Research: Paleoaltimetry of the South-Central
 ↪  Tibetan Plateau Using Triple Oxygen Isotope Measurements"
[16] "Collaborative Research: GCR: A new scientific framework for
 ↪  understanding and valuing ecosystem services at scale"
[17] "Collaborative Research: Paleoaltimetry of the South-Central
 ↪  Tibetan Plateau Using Triple Oxygen Isotope Measurements"
[18] "Collaborative Research: GCR: A new scientific framework for
 ↪  understanding and valuing ecosystem services at scale"
[19] "Collaborative Research: The role of belowground resource
 ↪  partitioning in tree biodiversity-ecosystem function
 ↪  relationships"
[20] "Postdoctoral Fellowship: EAR-PF: Tracing Coral-Algal Symbiosis
 ↪  Across Geologic Time Using Nitrogen Isotopes"
[21] "Collaborative Research: ULTRA-Data: Microbial Mat Community
 ↪  Structure and Metabolic Potential Changes Following an Eruptive
 ↪  Event at Axial Seamount"
[22] "Collaborative Research: Uncovering the role of planktic
 ↪  foraminifera in marine snow formation and ocean carbon export"
[23] "Collaborative Research: GCR: A new scientific framework for
 ↪  understanding and valuing ecosystem services at scale"
[24] "Integrating Genomics, Physiology, and Ecology to Predict
 ↪  Intertidal Ecosystem Resilience"
[25] "Collaborative Research: The role of belowground resource
 ↪  partitioning in tree biodiversity-ecosystem function
 ↪  relationships"
```

```
## pick one (change the row number to whichever award you chose)
my_award <- nsf_result$response$award[1, ]
my_award$title
```

```
[1] "EPSCoR Research Fellows: NSF: Using Ancient DNA From Ocean
 ↪  Sediments to Forecast Ecosystem Shifts"
```

```
my_award$abstractText
```

```
[1] "This Research Infrastructure Improvement (RII) EPSCoR Research
 ↪  Fellows project provides a fellowship to an Assistant Professor
 ↪  and training for a postdoctoral fellow at the University of
 ↪  Hawai‘i at Mānoa. This work is conducted in collaboration with
 ↪  Dr. Brice Semmens at the Scripps Institution of Oceanography.
 ↪  Through the fellowship, the PI will investigate how marine life
 ↪  reacted to past environmental changes to better predict how
 ↪  oceans will respond to future environmental conditions.
 ↪  Integrating computer science, statistics, and marine biology,
 ↪  the project will extract historical DNA preserved in deep-sea
 ↪  sediment cores and develop new software to identify highly
 ↪  degraded genetic fragments. These historical records will inform
 ↪  predictive models to forecast future shifts in ocean ecosystems.
 ↪  The award will benefit society by bringing advanced
 ↪  oceanographic forecasting capabilities to Hawai‘i and expanding
 ↪  the STEM workforce through hands-on training for students and
 ↪  researchers in data science and computational
 ↪  biology.\r\n\r\nThis project will establish a novel
 ↪  computational method to reconstruct historical marine
 ↪  biodiversity and forecast ecosystem responses to environmental
 ↪  shifts. The primary intellectual contribution is a damage-aware
 ↪  phylogenetic placement algorithm that computationally rescues
 ↪  highly degraded sedimentary ancient DNA (sedaDNA) without
 ↪  relying on complete reference genomes. Methodologically, the PI
 ↪  will extract sedaDNA from refrigerated Pacific basin sediment
 ↪  cores using optimized target-capture protocols. To address
 ↪  chemical degradation, the PI will embed position-dependent
 ↪  substitution matrices into conditional likelihood calculations
 ↪  to accurately identify and place sequence reads. These
 ↪  historical baselines will then parameterize phylogenetically
 ↪  informed Bayesian Joint Species Distribution Models to forecast
 ↪  community-level shifts under future projected environmental
 ↪  scenarios. The project will substantially improve research
 ↪  infrastructure at the home institution by establishing a new,
 ↪  localized marine paleogenomics capability and advancing the PI’s
 ↪  professional trajectory. These activities are fully integrated
 ↪  with workforce expansion, providing specialized
 ↪  cross-disciplinary training for a postdoctoral researcher and
 ↪  local students, and incorporating the generated open-source
 ↪  software into graduate computer science curricula. This project
 ↪  is supported by the EPSCoR Research Infrastructure Improvement
 ↪  Program: EPSCoR Research Fellows (ERF). The ERF program
 ↪  supports early- and mid-career investigators in eligible
 ↪  jurisdictions to develop collaborations at the nation’s private,
 ↪  government or academic research institutions.\r\n\r\nThis award
 ↪  reflects NSF's statutory mission and has been deemed worthy of
 ↪  support through evaluation using the Foundation's intellectual
 ↪  merit and broader impacts review criteria."
```

```
## the total dollar amount NSF obligated to this award
my_award$fundsObligatedAmt
```

```
[1] "295906"
```

Read the abstract of the award you picked. In your own words–not copied from the abstract–write a short paragraph explaining:

1. What scientific question or problem motivated this research?
2. How did (or will) the investigators use the award–what will the money actually pay for (fieldwork, equipment, student support, lab analysis, and so on)?

Notice that `fundsObligatedAmt` is a single total, not an itemized budget. NSF does not publish a line-item breakdown of how much of an award goes to salaries, equipment, travel, or overhead–that level of detail is part of the confidential budget justification submitted with the original proposal, and it stays between NSF and the awardee institution. The total award amount is public; the itemized budget is not. That is exactly why question 2 above asks you to *infer* what the money likely paid for from the abstract, rather than look it up.

NSF's API documentation describes additional parameters (date ranges, award amount thresholds, and more) that you can use to narrow your search–worth a look if this topic becomes useful for your own research question later in the semester.

17.5. Overthinking: how much of NSF is really DEB?

Everything in this chapter so far has treated NSF as one number–all of Figure 17.1, Figure 17.2, and Figure 17.3 show NSF's *entire* budget, several billion dollars a year covering physics, computer science, engineering, and every other field NSF funds, not just ecology. Background mentioned that the part of NSF most directly relevant to this course is one specific division: the **Division of Environmental Biology** (**DEB**), inside the Directorate for Biological Sciences (BIO). This section isolates DEB's own numbers, the same way the chapter earlier isolated DOE BER from DOE's total.

```
## NSF's Division of Environmental Biology (DEB) budget--not NSF's
 ↪  total
## budget, and not even all of BIO's budget, but the one division
 ↪  that most
## directly funds population, community, and ecosystem ecology. In
 ↪  millions
## of dollars.
## Source: NSF's own Congressional Budget Justification, Biological
 ↪  Sciences
## (BIO) directorate chapter, which breaks BIO's budget down by
 ↪  division.
## Unlike DOE (which reports clean enacted appropriations for BER
 ↪  every
## year), NSF's public budget documents do not give DEB a
 ↪  consistent,
## fully-enacted multi-year series--figures are a mix of "Actual,"
 ↪  "Base
```

```
## Plan," and "Request," and some years are marked "TBD" in the
 ↪  document
## available when that year's chapter was written. The `status`
 ↪  column
## below tells you which is which; do not treat all rows as equally
 ↪  solid.
## FY2021 Actual:
 ↪  https://nsf-gov-resources.nsf.gov/about/budget/fy2023/pdf/69_fy2023.pdf
## FY2023 Actual, FY2025 Request:
 ↪  https://nsf-gov-resources.nsf.gov/files/66_fy2025.pdf
## Stored in data/deb_budget.csv. Replace the FY2025 row with the
 ↪  enacted
## figure once it's final, and add newer years as NSF publishes
 ↪  them.
deb_budget <- read_csv("data/deb_budget.csv")
```

```
Rows: 3 Columns: 3
-- Column specification
 ↪  --------------------------------------------------------
Delimiter: ","
chr (1): status
dbl (2): fiscal_year, deb_budget_millions

i Use `spec()` to retrieve the full column specification for this
 ↪  data.
i Specify the column types or set `show_col_types = FALSE` to quiet
 ↪  this message.
```

```
p_deb_budget <- ggplot(deb_budget,
       aes(x = factor(fiscal_year), y = deb_budget_millions, fill =
        ↪  status)) +
  geom_col() +
  labs(x = "Fiscal year", y = "DEB budget (million $)", fill =
   ↪  "Status",
       title = "NSF's Division of Environmental Biology (DEB)")
p_deb_budget
```

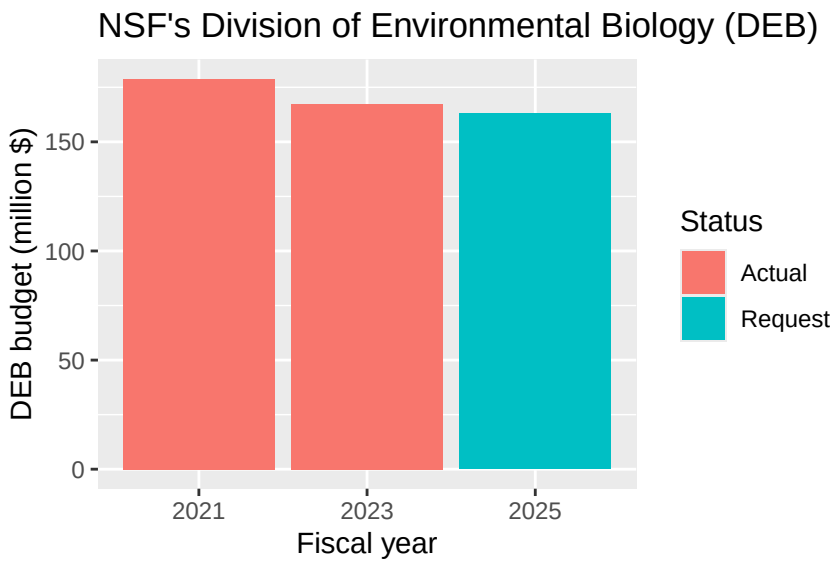


Figure 17.7.: NSF's Division of Environmental Biology (DEB) budget–the part of NSF most directly funding population, community, and ecosystem ecology. FY2025 is a request, not yet a final enacted figure; see the `status` column.

```
## Save a PNG file to your working directory.
ggsave("figs/fig-deb-budget.png", plot = p_deb_budget, width = 7,
 ↪  height = 5)
```

For scale: DEB's entire annual budget–somewhere around $160-180 million–is a small slice of BIO's roughly $800-950 million, which is itself a small slice of NSF's total budget of several billion dollars in Figure 17.1. If your own interests are specifically in population, community, or ecosystem ecology, DEB, not NSF's headline number, is the figure that actually describes the funding available to you.

17.6. Questions to answer

1. Looking at Figure 17.1, which agency's nominal budget grew the most over the years you plotted? Which shrank, or grew the least?
2. NSF's FY2026 appropriation was reported as roughly flat compared to FY2024, yet in the months before that NSF terminated over 1,700 already-active grants worth about $1.4 billion. How can an agency's total annual budget stay flat while individual grants it already promised to pay get cancelled? What does this tell you about the difference between an agency's *annual appropriation* and the multi-year *obligations* it has already made?
3. How does the funding mechanism for your assigned state's Wildlife Restoration/Sport Fish Restoration apportionment differ from how NSF or EPA award money? Based on your figure, does the state funding stream look more or less stable year-to-year than the federal agency budgets in Part 1? Does that match what you would predict from how each mechanism works?
4. Pick one agency and compare its line in Figure 17.1 to its line in Figure 17.2. Does adjusting for inflation change the story you would

tell about that agency's funding trend? Is there a year where the nominal and real dollar amounts tell noticeably different stories?

5. Look at Figure 17.5. For NASA and USDA, the research-only bar is much shorter than the entire-budget bar; for NSF, the two bars are close together. Explain why, in terms of what each agency actually spends most of its money on. Why don't NOAA, FWS, USGS, or the Forest Service have an "entire agency budget" bar at all?
6. FWS's own R&D obligations (Figure 17.4) dropped about 87% from FY2024 to FY2025, but the Wildlife Restoration/Sport Fish Restoration apportionments you collected in Part 2 come from earmarked excise tax revenue, not FWS's regular appropriated budget. Does your class's state-level data in Figure 17.6 show anything similar happening to the apportionments over the same period? What does that suggest about which of FWS's funding streams is more insulated from short-term budget cuts?
7. Compare Figure 17.4 to Figure 17.1. USGS obligates far more research funding than FWS or the Forest Service, yet DOI as a whole (which contains all three) is one of the smaller lines in Figure 17.1. What does that tell you about the relationship between an agency's public visibility and the size of its research budget?
8. If you were choosing a research area for graduate school today, would any of the trends you found here factor into your decision? Why or why not?
9. DOE's row in Figure 17.3 is DOE's entire R&D total–energy-technology research (fossil fuels, nuclear power, renewables) together with its ecological and environmental science. Compare that line to DOE BER's line in Figure 17.4. Roughly what fraction of DOE's total research funding actually goes to ecological/environmental science? Does that fraction surprise you, given how DOE is usually described in the news?

A. Community structure

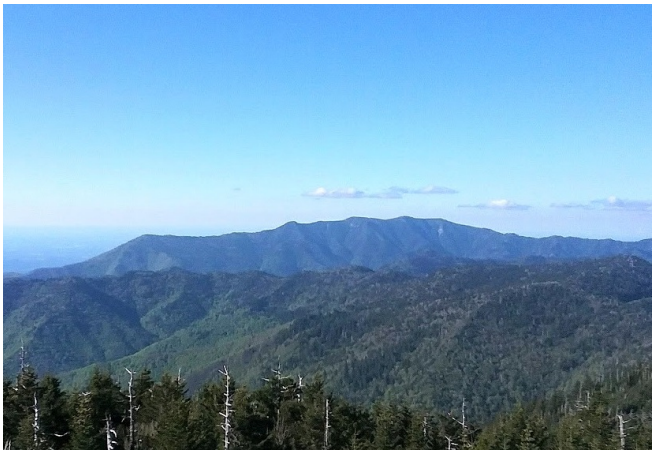


Figure A.1.: Mt. Le Conte in Tennessee, U.S.A., sits in what was originally part of the Cherokee Nation, prior to their forced removal in the 1830's as part of the *Trail of Tears*. It is now the heart of the Great Smoky Mountains National Park. The Smokies are part of the Blue Ridge geologic and physiographic province of the Appalachian Mountains, which extend from Georgia northward into southeastern Canada. It's tree communities cover a variety of elevations, aspects, and moisture regimes, which were described in Robert Whittaker opus, Vegetation of the Great Smoky Mountains (1956, *Ecol. Mongraphs* 26:1-80.)

A.1. Goals

In this chapter, you will,

- represent multivariate community data with quantitative summaries and graphical displays.
- Describe patterns of diversity, including evenness and richness.

A.2. Preparatory steps

1. General background lectures: Download the PDF slides and watch the video on community structure and diversity before proceeding.
2. Data: go to our class data repository, and download `WHITTAKER_TABLE_3.csv`.
    a. Make sure the file remains in the same simple format, "comma separated values" (.csv).
    b. Put `WHITTAKER_TABLE_3.csv` in your working directory, `Rwork`.
    c. Continue to make sure the file remains in the same simple format, "comma separated values" (.csv).
3. Find out what you need for the deliverables.

A.3. Background

We have seen examples of how the size of populations are influenced by the physical environment and interactions with other species which may be prey or enemies or mutualists. In this section, we describe how those populations may change in concert with each other and so constitute **communities**.

A.4. DRAFT

A.5. *Do not use for BIO 209W until the following is shortened and edited.*

title: "Community Structure With Data From Whittaker (1956)" format: pdf date: 2023-11-26 author: Hank Stevens —

A.6. Community structure

This is a super short introduction to working with multivariate data in which we have lots of species of a similar type (e.g., plant community, fungal community) at multiple sites or samples.

Make a note of the new functions or commands we introduce.

A.6.1. Background

Make sure you have read Primer secs. 14-14.2, and seen Hank's lecture on community structure.

A great reference for all this is: McCune, B. and J. Grace with contribution from D. Urban. 2002. Analysis of Ecological Communities. MJM Software, Gleneden Beach, Oregon. 304 pages.

Dr. Mike Palmer has a great website on ordination, The Ordination Website: https://ordination.okstate.edu/

A good introduction to ordination in `vegan`, written by the first author of the `vegan` package, is available is you Google "Oksanen ordination vegan".

Next load relevant data analysis packages. The new one is `vegan`, for vegetation analysis.

```
library(vegan) # multivariate analysis
library(tidyverse) # data wrangling and graphics
```

Table A.1.: Small subset of data from Whittaker (1956) Fig. 4 & Table 3, reflecting a moisture gradient.

moisture	Halesia	Pinus	Quercus	Tilia	Tsuga
1	5	0	0.0	29	20.0
2	8	0	0.0	11	22.0
3	4	0	0.0	9	34.0
4	1	0	0.0	1	62.0
5	9	0	0.0	14	18.0
6	13	0	2.0	3	0.1
7	3	0	1.0	0	0.1
8	1	0	8.0	0	1.0
9	1	1	24.0	0	0.0
10	0	4	10.0	0	0.0
11	0	54	0.1	0	0.0
12	0	49	0.0	0	0.0

Let's use the following data (Table A.1), which is a small subset of Table 3 from Robert Whittaker's gradient analysis of forests in the Smoky Mountains (Whittaker 1956).

```
d <- data.frame(
  moisture = 1:12,
  Halesia = c(5,8,4,1,9,13,3,1,1, 0,0,0),
  Pinus = c(0,0,0,0,0,0,0,0,1,4,54,49),
  Quercus = c(0,0,0,0,0,2,1,8,24,10,.1,0),
  Tilia = c(29,11,9,1,14,3, 0,0,0,0,0,0),
  Tsuga = c(20,22,34,62,18,.1,.1,1, 0,0,0,0)
)
d
```

Sometimes simply rearranging a table can help turn it into a graphic. Let's group species by how much moisture they prefer; *Tilia* and *Tsuga* seem to like the wettest sites (Table A.2).

```
d2 <- d %>% select(moisture, Tilia, Tsuga, Halesia, Quercus, Pinus)

# if the above generates an error message, it may be because
# of 'select()' refers to the wrong R package. If you get an errror,
 ↪
# try referring specifically to a particular package with the
 ↪  following code.
# Uncomment the following and use this instead.
# d <- d %>% dplyr::select(moisture, Tilia, Tsuga, Halesia, Quercus,
 ↪  Pinus)

d2
```

Now let's create a line graph of abundances along this gradient.

Table A.2.: Reorganizing Table 1 to comunicate better.

moisture	Tilia	Tsuga	Halesia	Quercus	Pinus
1	29	20	5	0	0
2	11	22	8	0	0
3	9	34	4	0	0
4	1	62	1	0	0
5	14	18	9	0	0
6	3	0	13	2	0
7	0	0	3	1	0
8	0	1	1	8	0
9	0	0	1	24	1
10	0	0	0	10	4
11	0	0	0	0	54
12	0	0	0	0	49

First, rearrange the data into a long format.

```
yl <- d2 %>%
  pivot_longer(cols=Tilia:Pinus, values_to="abundance",
    ↪  names_to="species")
head(yl)
```

```
# A tibble: 6 x 3
  moisture species abundance
     <int> <chr>       <dbl>
1        1 Tilia          29
2        1 Tsuga          20
3        1 Halesia         5
4        1 Quercus         0
5        1 Pinus           0
6        2 Tilia          11
```

Now plot abundances by moisture, where each species get's its own color and line type (Figure A.2). In addition, we will scale the y-axis by square root, in order to avoid hiding the least common species.

```
ggplot(data=yl, aes(x=moisture, y=abundance,
                color=species, linetype=species)) +
  geom_line() +
  scale_y_sqrt() +
  labs(y="Abundance", x="Moisture index")
```

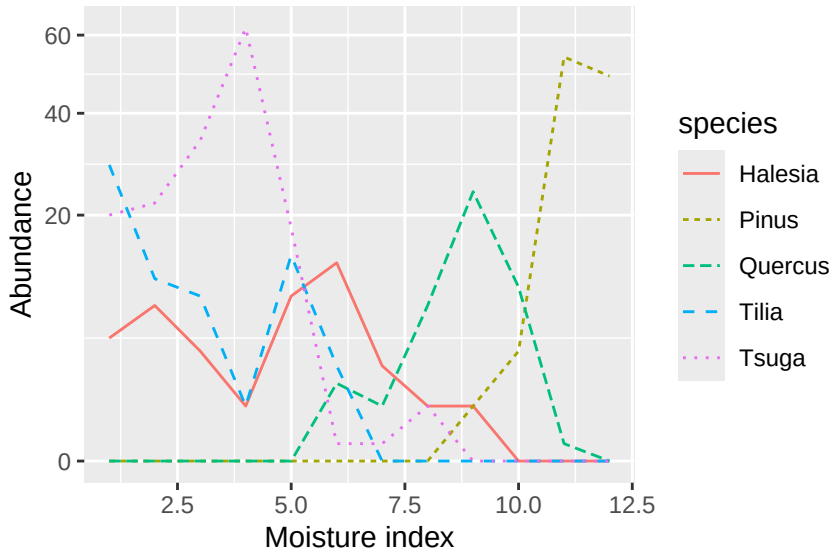


Figure A.2.: Tree species abundances along a moisture gradient in the Smoky Mountains of North Carolina, USA (Whittker 1956). Y-axis scaled by square root transformation.

A.7. Dissimilarities between sites

Let's calculate differences among sites, but first we should ask ourselves some questions:

1. Should we treat all sites equally?
2. Would we like to consider all species equally, or let the abundances dictate importance?
3. Should we exclude rare species that might appear in samples by chance alone?
4. Do we have an a priori hypothesis or is this primarily exploratory?

The answer to the first (sites equal?) is often a qualified "yes", especially if there might be artifactual differences among sites. Therefore, we often make sure that the totals among sites are adjusted to be equal.

The answer to the second (species equal?) is often a qualified "yes", if we are interested in community composition and structure, rather than only the abundant species driving carbon fluxes. Indeed, above we graphed the *square roots* of species abundances to help us see the patterns in the less common species.

The answer to the third (rare species?) is often "yes" because we would like to avoid unnecessary stochastic noise in our data. Some species may be uncommon but also habitat generalists; in these cases, they are not responding to anything we are interested in, but rather just present or absent by chance alone. These species would add noise to our analysis and not inform of us of anything interesting. Many authors report that they "removed taxa that appeared in fewer than X samples", where X is a small number of samples relative to their total. Nonetheless, let us assume that we have already removed such species, and so the answer here is "no", because we already have.

Last, let's assume that our analyses are exploratory because this is not a designed experiment.

Given these answers, we can proceed.

A.7.1. Treating sites and species equally

A common data transformation is the "Wisconsin" transformation, which is a double standardization method made popular by botanists at the University of Wisconsin. Here, species (in columns) are first standardized by their own maxima, and then sites (in rows) are standardized by site totals.

```
Y <- d2 %>% select(-moisture)
Yw <- wisconsin(Y)
# totals of each site (rows)
apply(Yw, MARGIN=1, sum)
```

```
 [1] 1 1 1 1 1 1 1 1 1 1 1 1
```

```
# resulting maxima of each species (cols)
apply(Yw, MARGIN=2, max)
```

```
    Tilia     Tsuga   Halesia   Quercus     Pinus
0.5857558 0.8997613 0.8420735 0.9128739 1.0000000
```

Note that because we re-weighted the sites after weighting the species, the species got adjusted again.

If we wanted to adjust only the species, we could do the following.

```
# Using a function in vegan
decostand(Y, method="max", MARGIN=2)
```

```
        Tilia       Tsuga    Halesia     Quercus      Pinus
1  1.00000000 0.322580645 0.38461538 0.000000000 0.00000000
2  0.37931034 0.354838710 0.61538462 0.000000000 0.00000000
3  0.31034483 0.548387097 0.30769231 0.000000000 0.00000000
4  0.03448276 1.000000000 0.07692308 0.000000000 0.00000000
5  0.48275862 0.290322581 0.69230769 0.000000000 0.00000000
6  0.10344828 0.001612903 1.00000000 0.083333333 0.00000000
7  0.00000000 0.001612903 0.23076923 0.041666667 0.00000000
8  0.00000000 0.016129032 0.07692308 0.333333333 0.00000000
9  0.00000000 0.000000000 0.07692308 1.000000000 0.01851852
10 0.00000000 0.000000000 0.00000000 0.416666667 0.07407407
11 0.00000000 0.000000000 0.00000000 0.004166667 1.00000000
12 0.00000000 0.000000000 0.00000000 0.000000000 0.90740741
```

```
# Doing the same thing in base R
apply(Y, 2, function(x) x/max(x))
```

```
          Tilia       Tsuga    Halesia     Quercus      Pinus
 [1,] 1.00000000 0.322580645 0.38461538 0.000000000 0.00000000
 [2,] 0.37931034 0.354838710 0.61538462 0.000000000 0.00000000
 [3,] 0.31034483 0.548387097 0.30769231 0.000000000 0.00000000
 [4,] 0.03448276 1.000000000 0.07692308 0.000000000 0.00000000
 [5,] 0.48275862 0.290322581 0.69230769 0.000000000 0.00000000
 [6,] 0.10344828 0.001612903 1.00000000 0.083333333 0.00000000
 [7,] 0.00000000 0.001612903 0.23076923 0.041666667 0.00000000
 [8,] 0.00000000 0.016129032 0.07692308 0.333333333 0.00000000
 [9,] 0.00000000 0.000000000 0.07692308 1.000000000 0.01851852
[10,] 0.00000000 0.000000000 0.00000000 0.416666667 0.07407407
[11,] 0.00000000 0.000000000 0.00000000 0.004166667 1.00000000
[12,] 0.00000000 0.000000000 0.00000000 0.000000000 0.90740741
```

Sometimes we want to just scale species a little differently using the logarithm or square root to keep species rank abundances less disturbed, but reduce the importance of the really common species. Square root (or fourth root) is useful, because zeroes are not a problem. We could do this:

```
# Use square roots of our data
Ysr <- apply(Y, 2, function(x) (x^0.5))
Ysr
```

```
         Tilia     Tsuga  Halesia   Quercus    Pinus
 [1,] 5.385165 4.4721360 2.236068 0.0000000 0.000000
 [2,] 3.316625 4.6904158 2.828427 0.0000000 0.000000
 [3,] 3.000000 5.8309519 2.000000 0.0000000 0.000000
 [4,] 1.000000 7.8740079 1.000000 0.0000000 0.000000
 [5,] 3.741657 4.2426407 3.000000 0.0000000 0.000000
 [6,] 1.732051 0.3162278 3.605551 1.4142136 0.000000
 [7,] 0.000000 0.3162278 1.732051 1.0000000 0.000000
 [8,] 0.000000 1.0000000 1.000000 2.8284271 0.000000
 [9,] 0.000000 0.0000000 1.000000 4.8989795 1.000000
[10,] 0.000000 0.0000000 0.000000 3.1622777 2.000000
[11,] 0.000000 0.0000000 0.000000 0.3162278 7.348469
[12,] 0.000000 0.0000000 0.000000 0.0000000 7.000000
```

Now let's calculate distances or dissimilarities among sites, using Euclidean distances with the raw data.

```
Y.euc.raw <- vegdist(Y, method="euclid")
round(Y.euc.raw,1)
```

```
      1    2    3    4    5    6    7    8    9   10   11
2  18.4
3  24.4 12.8
4  50.6 41.8 29.3
5  15.7  5.1 17.5 46.6
6  33.8 23.9 35.6 63.1 21.5
7  35.2 25.0 35.1 61.9 23.5 10.5
8  35.8 26.0 35.3 61.5 24.8 13.8  7.3
9  42.8 35.1 42.7 66.5 34.1 25.3 23.1 16.1
10 37.2 28.0 37.0 62.9 26.8 16.1 10.3  4.7 14.4
11 64.7 59.9 64.6 82.2 59.3 55.7 54.1 54.6 58.1 51.0
12 60.6 55.4 60.4 79.0 54.8 50.8 49.1 49.7 53.7 46.1  5.0
```

We see the lower triangle matrix shows all distances among different species. The upper half is redundant and so we don't bother to show those.

Now we do the same thing, but with the square-root transformed abundances, and we compare the two in a scatterplot (Figure A.3).

```
Y.euc.sqrt <- vegdist(Ysr, method="euclid")
plot(Y.euc.raw, Y.euc.sqrt )
```

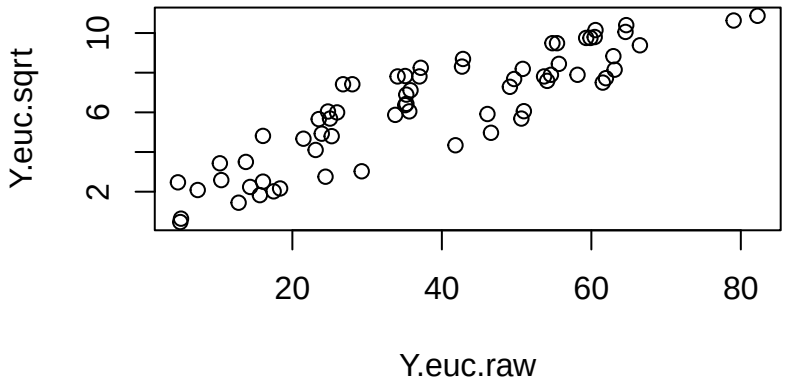


Figure A.3.

Now calculate other dissimilarities.

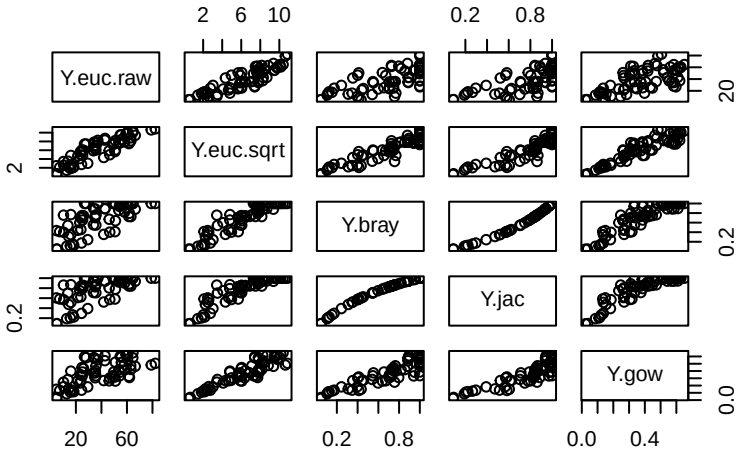
```
Y.bray <- vegdist(Ysr, method="bray")
Y.jac <- vegdist(Ysr, method="jaccard")
Y.gow <- vegdist(Ysr, method="gower")
```

And now combine into a data frame.

```
d.dist <- data.frame(
  Y.euc.raw, Y.euc.sqrt, Y.bray, Y.jac, Y.gow
  )
# in the event the above returns an error message,
# uncomment and use the following:
# d.dist <- data.frame(
#   Y.euc.raw = as.numeric(Y.euc), Y.bray = as.numeric(Y.bray),
#   Y.jac = as.numeric(Y.jac), Y.gow = as.numeric(Y.gow)
#   )
```

Finally, let's compare these visually using the scatterplots of all dissimilarity measures in a *scatterplot* matrix.

```
pairs(d.dist)
```



We can calculate the Spearman rank-order correlations among these to see how similar they are.

```
cor(d.dist, method="spearman")
```

```
           Y.euc.raw Y.euc.sqrt    Y.bray     Y.jac     Y.gow
Y.euc.raw  1.0000000  0.8889492 0.7021381 0.7021381 0.6582403
Y.euc.sqrt 0.8889492  1.0000000 0.9018327 0.9018327 0.8942726
Y.bray     0.7021381  0.9018327 1.0000000 1.0000000 0.8927829
Y.jac      0.7021381  0.9018327 1.0000000 1.0000000 0.8927829
Y.gow      0.6582403  0.8942726 0.8927829 0.8927829 1.0000000
```

We see that the Euclidean distances using raw data differ the most from the other dissimilarities. This is not surprising as all the others use square-root transformed data.

A.8. Ordination - a huge topic

Ordination is a dimension reduction technique that attempts to glean emergent simple patterns in multivariate data. Think of it as a 2-D shadow of a 3-D (or 4-D, or 1000-D) object.

Ordination may be constrained by covariates or unconstrained. We typically constrain an ordination when we have an a priori hypothesis. Here we will will use only one method, and it will be unconstrained.

A.8.1. Unconstrained ordination

Among the oldest is principal components analysis (PCA). This is a good method when zeroes are very rare in your data, and variables are symmetrically distributed. It can be good for quantitative traits, where all species have some numerical value for each traits. The method is lousy for community data, especially along long gradients.

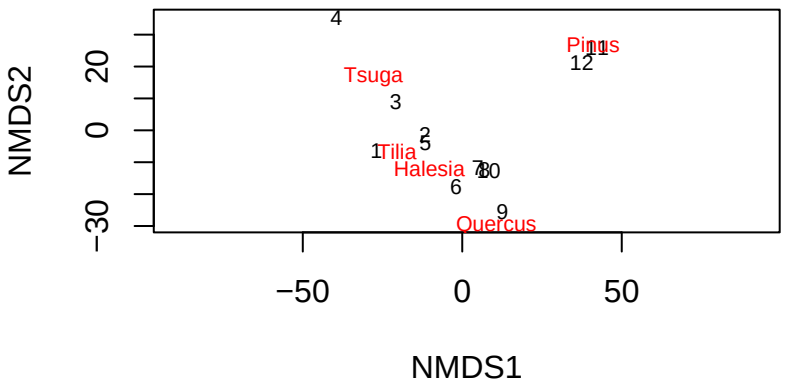
Non-metric multidimensional scaling: Let's use this algorithm, which tries to render fewer dimensions from more. However, we will first do it

porrly, by not transforming the data and using a distance measure that is rarely recommended.

```
m1 <- metaMDS(Y, distance="euc",
              autotransform = FALSE,
              k=2, # number of dimensions
              trace=0 # silence simulation progress info
)
m1
```

```
Call:
metaMDS(comm = Y, distance = "euc", k = 2, autotransform = FALSE,
 ↪  trace = 0)

global Multidimensional Scaling using monoMDS

Data:     Y
Distance: euclidean

Dimensions: 2
Stress:     0.03038792
Stress type 1, weak ties
Best solution was repeated 4 times in 20 tries
The best solution was from try 3 (random start)
Scaling: centring, PC rotation
Species: expanded scores based on 'Y'
```

```
plot(m1, type="t")
```

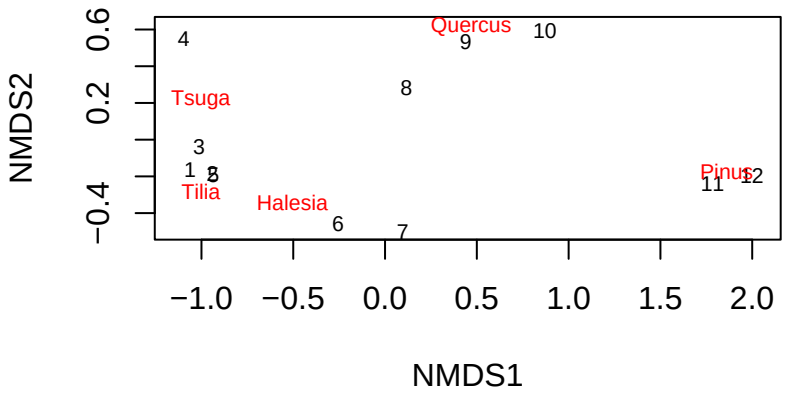


Now we do it better.

```
m2 <- metaMDS(Y, distance="bray",
              autotransform = TRUE,
              k=2, # number of dimensions
              trace=0 # silence simulation progress info
)
m2
```

```
Call:
metaMDS(comm = Y, distance = "bray", k = 2, autotransform = TRUE,
 ↪  trace = 0)

global Multidimensional Scaling using monoMDS

Data:     wisconsin(sqrt(Y))
Distance: bray

Dimensions: 2
Stress:     0.01534085
Stress type 1, weak ties
Best solution was repeated 1 time in 20 tries
The best solution was from try 18 (random start)
Scaling: centring, PC rotation, halfchange scaling
Species: expanded scores based on 'wisconsin(sqrt(Y))'
```

```
plot(m2, type="t")
```



A.9. And now with a larger data set

These data (this data set) is arranged by species in rows, and sites in columns. Usually we do analyses of sites by having sites in rows and species in columns.

```
dw <- read.csv("data/WHITTAKER_TABLE_3.csv")
dw[is.na(dw)] <- 0
# transpose the data, but exclude the first column of species names

dt <- t(dw[,-1])
# use the species names as column names
colnames(dt) <-  dw$SP
```

Here we don't transform data, and use Euclidean distances.

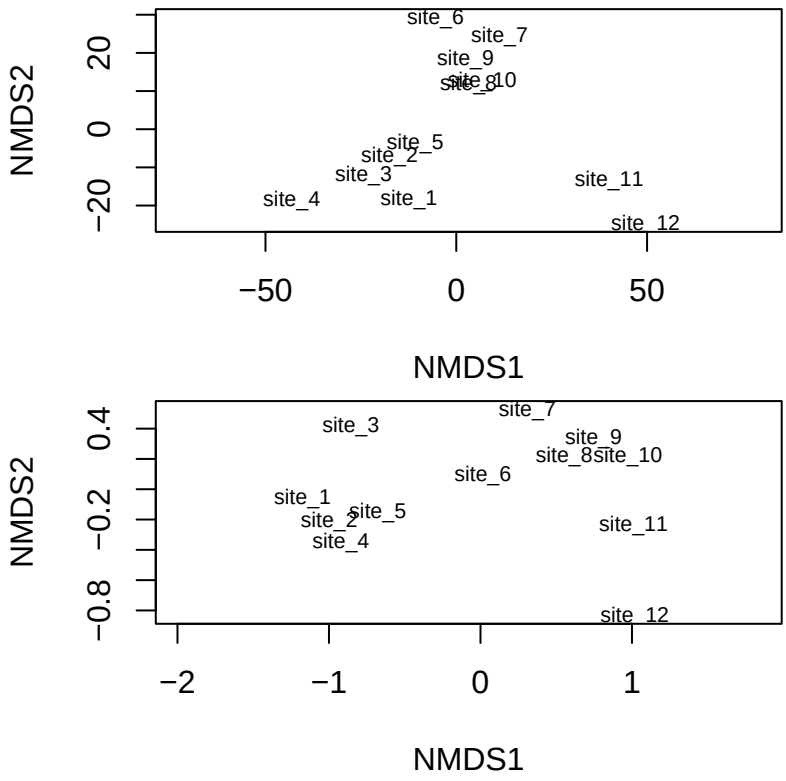
```
m.whit1 <- metaMDS(dt, distance="euc",
              autotransform = FALSE,
              k=2, # number of dimensions
              trace=0 # silence simulation progress info
)
```

Here we transform data, and use Bray-Curtis distances.

```
m.whit2 <- metaMDS(dt, distance="bray",
              autotransform = TRUE,
              k=2, # number of dimensions
              trace=0 # silence simulation progress info
)
```

Here we plot these both. Can you see differences?

```
plot(m.whit1, display="sites", type="t")
plot(m.whit2, display="sites" , type="t")
```



B. From This Book to Your Résumé: A Skills Crosswalk

...in which we translate what you just did in R into words a hiring manager will recognize–without overselling it.

B.1. Goals

In this appendix, you will

- See, chapter by chapter, exactly which R tools and quantitative tasks you were exposed to and practiced.
- Learn the standard industry name for each skill–the words recruiters and applicant-tracking systems actually search for.
- Get worked examples of turning a chapter assignment into a résumé bullet that is specific *and* honest about what you actually did.
- Learn how to talk about this experience credibly in an interview, without claiming more expertise than you have.

B.2. What this course is, and isn't

This is book is for an ecology course, not a data-science bootcamp, and this appendix is not asking you to pretend otherwise. You are not being trained as a professional data analyst. What you *are* getting, most chapters, is real exposure to what practicing ecologists actually do with data and models: querying a live database, fitting a model, making a forecast, building a figure–using R code that is mostly provided to you, which you run, adapt in small ways, and interpret.

That is genuinely useful, real experience. It is not the same thing as designing an analysis from scratch or understanding the statistical theory behind it. Running a Generalized Additive Mixed Model (GAMM) and correctly reading its forecast is not the same as understanding *why* a GAMM works the way it does–that second thing is graduate-level material. Both are true at once: you did something real and specific, and you should not overstate what it means.

The rest of this appendix is about naming what you did accurately–specific enough to be credible, honest enough to survive a follow-up question in an interview.

B.3. The core skills you're already building

Two things recur across nearly every chapter, as the Preface mentions: wrangling data with **tidyverse**-style functions (`read_csv`, `subset`, `mutate`, and similar), and visualizing it with `ggplot2`. By the time you finish this book you will have done both dozens of times, on dozens of different real datasets. That repetition–not any single chapter–is what actually makes those two skills worth naming: you can honestly say you've practiced them across a whole semester of varied, real data.

The table below focuses on what's *different* about each chapter, on top of that recurring foundation. A small number of chapters (noted in the table) don't use `ggplot2` or tidyverse wrangling at all.

B.4. How to use this appendix, honestly

For each chapter you complete, find its row below. The **distinctive skill** column names what made that chapter different–in addition to the recurring wrangling/visualization work above.

When you turn any of this into a résumé bullet or interview answer, favor verbs that match what you actually did:

- **Honest:** *used*, *ran*, *practiced*, *interpreted*, *worked with*, *applied* [*a provided method*].
- **Overclaiming, unless you genuinely wrote the method yourself rather than adapting provided code:** *built*, *engineered*, *developed*, *designed*.

You do not need to list all 19 chapters on a résumé. Pick 3-5 that showcase different kinds of tasks (for example: one API/live-data chapter, one modeling/simulation chapter, one statistics chapter), and write one honest, specific bullet for each rather than a long, generic list.

B.5. Chapter-by-chapter skills crosswalk

Chapter	Core skills (recurring)	R tools used	Distinctive task for this chapter
01. Computer Setup and First R Script	`ggplot2` viz (wrangling not yet introduced)	base R, `ggplot2`	Reproducible scripting and environment setup

Chapter	Core skills (recurring)	R tools used	Distinctive task for this chapter
Thinking about Thinking about Concepts	Wrangling + `ggplot2` viz	`tidyverse`, `ggplot2`, `patchwork`	Bayesian statistical reasoning; causal diagram (DAG) construction
02. Trees and People	`ggplot2` viz	`googlesheets4`, `ggplot2`, `patchwork`	Web-based/crowd-sourced data collection; socioeconomic-environmental analysis
03. Global Temperature Anomalies	Wrangling + `ggplot2` viz	`tidyverse`, `mgcv` (GAMM), `ggplot2`	Ran and interpreted a provided time-series forecasting model (generalized additive mixed model, GAMM); communicated uncertainty
Traits, Physiology, and Plant Form	Wrangling + `ggplot2` viz	`dplyr`, `tidyr`, `MASS`, **BIEN** (live trait database API)	Queried a live scientific database (BIEN); reshaped data (long-to-wide); ran a provided null-model simulation
05. Extra-Pair Copulation	Wrangling + `ggplot2` viz	`tidyverse`, `ggplot2` (linear and logistic regression)	Ran and interpreted logistic regression for a binary outcome
06. Density-Independent Population Growth	Wrangling + `ggplot2` viz	`tidyverse`, `patchwork`, base R loops	Wrote a basic for-loop; interpreted exponential growth

Chapter	Core skills (recurring)	R tools used	Distinctive task for this chapter
07. Structured Demography (Loggerhead Turtles)	**Neither**–base R and linear algebra only, no tidyverse or `ggplot2`	base R (`matrix`, `eigen`), `diagram`	Ran and interpreted a provided matrix population model (eigenanalysis) for a conservation question
08. Density-Dependence in Population Growth	`ggplot2` viz (light wrangling)	`deSolve`/`primer` (ODE solvers), `ggplot2`	Ran a provided differential-equation model; explored nonlinear/chaotic dynamics
09. Direct Species Interactions	Wrangling (`dplyr`) only–base R graphics, not `ggplot2`	`knitr`, `dplyr`, base R graphics	Worked through a provided analytical model parameterized from the literature
10. Mutualisms	Wrangling + `ggplot2` viz	`tidyverse`, `ggplot2` (faceted boxplots)	Interpreted a factorial experimental design
11. Predator-Prey (Wolves and Moose, Isle Royale)	Wrangling + `ggplot2` viz	`tidyverse`, `deSolve::ode`, `ggplot2`	Ran a provided differential-equation model against real long-term census data
13. ForestGEO Community Ecology	Wrangling + `ggplot2` viz	`vegan`, `broom`, `tidyverse`	Calculated and compared community diversity metrics (Shannon/Simpson) and rarefaction
13. GBIF Biodiversity Data	Wrangling + `ggplot2` viz	**`rgbif`** (live REST API), `vegan`, `purrr`	Queried a live biodiversity API (GBIF); reasoned about sampling bias

Chapter	Core skills (recurring)	R tools used	Distinctive task for this chapter
15. Metabolic Ecology	Wrangling + `ggplot2` viz	`tidyverse`, `broom`, base R (`lm`)	Ran and interpreted a log-linear regression and diagnostic checks
16. Carbon Cycle and Global Heating	Wrangling + `ggplot2` viz	`deSolve::ode`, base R (`nls`), `tidyverse`	Ran provided dynamical (ODE) and curve-fitting models of a biogeochemical cycle
Funding Ecological Science	Wrangling + `ggplot2` viz	`jsonlite` (live REST APIs), `tidyverse` (joins), `googlesheets4`	Queried live government APIs and joined several real data sources into one comparison

B.6. Which chapters best showcase which kind of exposure

If you want to lead with a specific kind of experience on a résumé, these are your strongest, most concrete pieces of evidence:

Live API integration / working with real-world data feeds – Funding Ecological Science (USAspending.gov, NSF Award Search), GBIF Biodiversity Data (`rgbif`), Traits and Physiology (BIEN). These are the most immediately recognizable to a non-academic employer: you queried a live government or scientific database, not a cleaned CSV someone handed you.

Data wrangling and reshaping ("tidy data") – Funding Ecological Science (joining agency budgets, CPI, and state apportionment data from different sources), Traits and Physiology (long-to-wide reshaping), ForestGEO/GBIF (standardizing across sites).

Exposure to statistical modeling – Global Temperature Anomalies (GAMM/time-series forecasting), Extra-Pair Copulation (logistic regression), Metabolic Ecology (multiple regression and diagnostics). "Exposure to" is the honest frame here–you ran and interpreted these models; you were not asked to derive or fully understand the underlying theory.

Exposure to simulation and dynamical/mathematical modeling – Density-Dependence, Predator-Prey, and Carbon Cycle chapters

(differential-equation modeling with `deSolve`); Structured Demography (matrix/eigenanalysis).

Data visualization – present in most chapters (see the core-skills note above), but Global Temperature Anomalies, Predator-Prey, and Thinking about Concepts push furthest into multi-panel, publication-style `ggplot2` figures.

Comfortable working outside `ggplot2`/tidyverse – Structured Demography and Direct Species Interactions use base R graphics and linear algebra instead. Worth mentioning too: it shows you're not tied to one toolkit.

B.7. Worked examples: chapter to résumé bullet

Each of these names the real tool and the real task, without claiming you designed the method yourself:

Funding Ecological Science → "Used R (`jsonlite`, `tidyverse`) to query live federal budget APIs (USAspending.gov, NSF Award Search) and join them with inflation and state-apportionment data, producing agency-by-agency funding comparisons."

GBIF Biodiversity Data → "Used R (`rgbif`) to query a live biodiversity database (GBIF), assembling species-occurrence data across a latitudinal gradient and applying provided rarefaction and regression methods to examine a sampling-effort question."

Predator-Prey (Isle Royale) → "Used a provided differential-equation model (`deSolve`, R) to fit 60+ years of empirical wolf-moose population data, and evaluated how well a classical ecological theory (Lotka-Volterra predator-prey dynamics) matched the real-world pattern."

Structured Demography (Loggerhead Turtles) → "Used a provided stage-structured matrix population model (eigenanalysis, R) to identify which life stages most influence loggerhead sea turtle population growth, and interpreted the results for a conservation-management question."

Global Temperature Anomalies → "Used a provided generalized additive mixed model, or GAMM (`mgcv`, R) to forecast global temperature trends with quantified uncertainty, and communicated the results through a publication-style figure."

Compare these to a version that oversells: "Built a forecasting engine" or "developed a predictive model" implies you designed the method. "Used a provided model to…" is still specific and still names real tools and real tasks–it just doesn't claim ownership you don't have.

B.8. Being honest about what you know

A few rules of thumb for staying credible, especially in an interview:

- If someone asks a follow-up question you can't answer–"why a GAMM instead of a linear model?", "how does the eigenanalysis actually work?"–the honest answer is some version of: *"That was the method used in the course. I understand what it produced and why it mattered for the question we were asking, but I haven't studied the underlying theory in depth."* That is a completely normal, credible thing for an undergraduate to say, and it holds up far better than getting caught overselling.
- What you *can* honestly claim is broader than any single technique: a semester of hands-on, repeated practice reading, cleaning, visualizing, and interpreting real data, across many different real-world sources and problems. That pattern–not mastery of any one method–is the actual transferable skill, and it's a genuinely useful thing to be able to say.

B.9. Building a portfolio, not just a bullet list

A résumé line is words on a page; a public, working example is proof–and it's also a natural way to show, rather than claim, the level of understanding you actually have. Your regular assignments in this course are turned in as a Word document plus a separate R script–a format that works well for grading, but isn't something you can link to as a live, running example.

This book itself is written entirely in **Quarto**–a tool that combines code, narrative text, and output (figures, tables) into a single, reproducible, publishable document. You have not been asked to use Quarto for your regular assignments, and that's a reasonable choice for a first ecology course. But if you want a portfolio piece, learning just enough Quarto to rebuild 1-2 of your best analyses as short documents–with your own explanation, in your own words, of what the code does and what you do and don't understand about it–is a real, additional skill, and a more honest way to demonstrate your actual level of understanding than a résumé bullet alone:

1. Pick 1-2 analyses you understood well enough to explain to someone else.
2. Rebuild each as a short Quarto (`.qmd`) document combining your code, your figure, and your own written explanation–including, where relevant, an honest note about what you're still fuzzy on.
3. Render them as a small Quarto website (a few lines of `_quarto.yml` configuration) and publish it for free (e.g., GitHub Pages, Quarto Pub).
4. Link that site from your résumé and LinkedIn profile instead of–or alongside–the bullet points above.

B.10. Quick-reference skill glossary

Pair any of these with an honest verb (*used*, *practiced*, *interpreted*, *worked with*) rather than an ownership verb (*built*, *engineered*, *developed*), unless you genuinely wrote the method yourself:

- REST API integration / live data querying
- JSON parsing
- Data wrangling / data cleaning
- Tidy data / data reshaping
- Exposure to statistical modeling (regression, GAMM, logistic regression)
- Exposure to differential equation modeling / dynamical systems simulation
- Exposure to matrix modeling / linear algebra (eigenanalysis)
- Bayesian statistical reasoning
- Causal inference / causal diagrams (DAGs)
- Data visualization (`ggplot2`, publication-quality graphics)
- Reproducible research / literate programming (Quarto)–only if you build the optional portfolio site above; regular assignments don't currently provide this
- Version control (if you use git/GitHub for your assignments)

C. The Changing Landscape of Skills in Ecology

What should you learn when the machine already "knows" so much?

C.1. Background

This expands on similar content in the preface.

You are learning ecology at a moment when artificial intelligence can write passable R code, summarize journal articles, and define terms like "carrying capacity" faster than you can open a textbook. That fact changes what it means to be well-trained in ecology. It does not, however, reduce the amount you need to learn — it shifts *what* you need to learn.

This chapter is addressed to both students and instructors. Its goal is to lay out a practical framework for thinking about which skills matter more than ever in an era of large language models (LLMs) like Claude, ChatGPT, and their successors, and which tasks can be responsibly shared with these tools. We focus on quantitative ecology because that is what this book is about, but the ideas extend broadly.

The central argument is this: **AI's fluency with ecological terminology makes it *more* important, not less, that you develop deep understanding of ecological mechanisms, model assumptions, and data interpretation.** A machine can generate a plausible-sounding paragraph about density dependence in thirty seconds. Only a well-trained ecologist can recognize when that paragraph is subtly wrong.

C.2. A framework: durable skills vs. AI-augmentable tasks

Not all skills are equally vulnerable to automation. A useful way to think about your education is to distinguish between two categories.

Durable human skills are things that require biological intuition, judgment under uncertainty, and the ability to connect mathematical abstractions to the living world. These are hard-won, slow to develop, and not reliably performed by current AI. They include:

- *Ecological intuition* — the ability to walk into a forest, a marsh, or a dataset and notice what is ecologically interesting or surprising.
- *Mechanistic reasoning* — understanding *why* a model behaves the way it does, not just *that* it produces a particular output.
- *Mathematical reasoning* — the ability to follow the formal logic of a derivation, so that you can verify results, understand why an equation predicts what it does, and catch errors that sound plausible but are wrong.
- *Model assumption literacy* — knowing what a model assumes, when those assumptions are reasonable, and what happens when they are violated.
- *Data literacy* — the judgment to know whether data are adequate to answer a question, what biases they carry, and what they cannot tell you.
- *Multi-scale synthesis* — connecting processes at the scale of an individual organism to patterns at the scale of a landscape or a globe.

AI-augmentable tasks are things that AI tools can do reasonably well right now, and will likely do better over time. These include:

- Summarizing the main findings of a journal article.
- Generating boilerplate R code for standard analyses (e.g., fitting a linear model, making a scatterplot).
- Looking up definitions of ecological terms.
- Drafting outlines or initial structure for a written argument.
- Translating between programming languages.

Here is the key insight: **AI-augmentable does not mean unimportant.** You still need to understand what a linear model does and why, even if you let an AI draft the code. The danger is not that AI does these tasks — it is that you might skip the understanding that makes you capable of evaluating whether the AI did them *correctly.*

C.3. Why AI makes ecological understanding more important

This point deserves emphasis, because it is counterintuitive. You might think: if AI can write my R code and summarize my readings, I can spend less time on those things and more time on… something else. But consider what actually happens.

Suppose you ask an AI to write R code that simulates logistic population growth. The AI will almost certainly produce something that runs. But will it use the continuous-time or discrete-time model? Will it handle the time lag that makes the discrete model capable of oscillations and chaos? Will it set up the parameters in a biologically meaningful range? If you do not know enough to answer these questions, you cannot evaluate the code. You are flying blind.

Or suppose you ask an AI to explain why two competing species cannot coexist indefinitely. The AI will probably give you something that sounds like the competitive exclusion principle. But it might gloss over the specific conditions required, or fail to mention that the principle applies to a particular model with particular assumptions, and that real communities are far more complicated. The explanation will *sound* right. Whether it *is* right depends on what you, the human ecologist, know.

> **!** The Plausibility Trap
>
> AI-generated text about ecology is often *plausible but wrong* in ways that require ecological training to detect. The better the AI gets at sounding like an ecologist, the more you need to actually *be* one.

This is what we call the **plausibility trap**. AI systems are trained to produce text (and code) that looks like text (and code) produced by experts. They are not trained to be correct about the natural world. They have no field experience, no memory of watching a population crash, no intuition about whether a parameter value makes biological sense. They are pattern-matching engines — powerful ones, but without grounding in physical and biological reality.

C.4. Six categories of skills to develop

Let us get specific. Below are six categories of durable skills, each illustrated with examples from quantitative ecology. These are the skills this textbook is designed to build, and the ones you should invest in most heavily.

C.4.1. 1. Mechanistic reasoning

Mechanistic reasoning is the ability to explain *why* an ecological pattern exists by tracing it back to underlying biological processes. It is the difference between saying "the population oscillates" and saying "the population oscillates *because* the discrete-time logistic model builds in a one-generation time lag between when crowding occurs and when its effects on reproduction are realized, and when r is large enough, this lag causes the population to overshoot and undershoot K repeatedly."

AI can describe patterns. It can even parrot mechanistic explanations it has seen in training data. What it cannot do reliably is generate mechanistic reasoning about novel situations — situations where you are looking at a dataset or a model output that nobody has analyzed before. That is what ecologists do.

How to practice it. Throughout this book, when you are asked to run a simulation or examine data, *always* ask yourself: *Why does this pattern exist? What mechanism produces it?* Do not settle for describing what you see. Explain what causes it.

C.4.2. 2. Mathematical reasoning as ecological understanding

Ecology uses mathematics to turn assumptions into predictions. When we write $dN/dt = rN(1-\alpha N)$ and solve for equilibrium, we are not doing math for its own sake — we are asking "what controls the long-term fate of this population?" and letting algebra answer the question. The derivation *is* the ecological reasoning, expressed in a language precise enough to check.

Put simply: **mathematical reasoning is a form of ecological understanding** — not a prerequisite for it, not a supplement to it, but one of the ways you actually come to understand how ecological systems work. When a student derives $K = 1/\alpha$ and grasps that carrying capacity is the reciprocal of the per-capita competitive effect, they have learned something about ecology that no amount of reading prose — whether written by a textbook or an AI — will give them. The derivation is not an obstacle on the way to understanding. It *is* the understanding.

This matters enormously in an AI era. An AI can solve $0 = rN(1-\alpha N)$ and tell you $N^* = 1/\alpha$. But can you verify that? Can you see *why* the equilibrium depends on α and not on r? Can you explain what that means biologically — that the long-term population size depends on how strongly individuals affect each other, not on how fast they reproduce? If you can follow the two lines of algebra, these insights are obvious. If you cannot, you are taking the AI's word for it.

The same applies at every level of model complexity in this book. When you derive the invasion criterion for two-species competition ($K_2 > \beta K_1$), you are discovering — through logic — the conditions under which a species can persist. When you estimate predator-prey parameters from moose and wolf population data, you are using the structure of an equation to extract biological meaning. These are not exercises in calculation. They are exercises in understanding.

The risk of skipping the math is not that you get a wrong number. It is that you lose the ability to tell whether a number is wrong. An AI can produce an answer that looks right. Only someone who can follow the formal logic can confirm that it *is* right — or catch where it isn't.

How to practice it. When this book asks you to solve an equation or work through a derivation, do not skip to the answer. Follow each step and ask: *what biological fact does this step reflect?* When you reach the result, ask: *which parameters appear, which don't, and what does that tell me about what matters in this system?*

C.4.3. 3. Model assumption literacy

Every ecological model is a set of assumptions about how the world works. The exponential growth model assumes unlimited resources and no density dependence. The logistic model assumes that per capita growth rate declines linearly with population size. The Lotka-Volterra competition model

assumes that the effect of one competitor on another is constant regardless of density.

These assumptions are not fine print. They are the *substance* of the model. When an assumption is violated, the model's predictions can be wildly wrong. Understanding which assumptions matter and when they might be violated is one of the most important skills in ecology.

> **A Useful Habit**
>
> Before running any model, write down its key assumptions in your own words. Then ask: *Which of these assumptions could be wrong for the system I am studying, and how would that change the predictions?*

AI tools are particularly poor at this. They can list assumptions from a textbook, but they cannot walk outside and check whether density dependence in your study population really is linear, or whether the competitive effect of species A on species B really is constant. That judgment requires you.

C.4.4. 4. Prediction before computation

One of the most effective learning strategies in quantitative ecology is to *make a prediction before you run the code.* What do you think will happen when you double the intrinsic growth rate? What do you expect the graph to look like when you increase carrying capacity? What will happen to coexistence if you make one species a much stronger competitor?

If you write down your prediction and it turns out to be wrong, you have learned something valuable: there is a gap between your mental model and the mathematical model. That gap is where real understanding develops.

AI eliminates this step if you let it. You can ask an AI what will happen, and it will tell you, and you will nod and move on without ever experiencing the discomfort of being wrong. *Resist this.* Your mistakes are the raw material of learning.

How to practice it. Every time this textbook asks you to run a simulation or modify a parameter, pause. Write down, in one sentence, what you predict will happen and *why.* Then run the code and compare. If you were wrong, figure out why.

C.4.5. 5. Data literacy and judgment

Data literacy goes beyond knowing how to make a scatterplot or compute a mean. It includes:

- Recognizing what kinds of variation are present in a dataset (measurement error, natural variation, sampling bias).
- Understanding what a dataset *cannot* tell you because of how it was collected.

- Knowing when a sample size is too small to support a conclusion.
- Distinguishing correlation from causation with genuine understanding rather than just reciting the phrase.
- Recognizing when an apparent pattern might be an artifact of how the data were processed.

These judgments are context-dependent and require understanding of both the statistical tools and the biological system. AI can compute a p-value, but it cannot tell you whether the study design that produced the data makes that p-value meaningful.

How to practice it. When you encounter data in this book, do not just analyze them. *Interrogate* them. How were they collected? What could go wrong? What alternative explanations exist for the patterns you see?

C.4.6. 6. Multi-scale synthesis

Ecology spans scales from molecules to the biosphere. A well-trained ecologist can connect processes at different scales: how individual behavior gives rise to population dynamics, how population dynamics shape community composition, how community composition affects ecosystem function.

This kind of synthesis requires holding multiple frameworks in your head simultaneously and recognizing when processes at one scale constrain or enable processes at another. AI can retrieve information about any one of these scales, but it struggles to make the cross-scale connections that produce genuine ecological insight.

How to practice it. As you move through this book from individual ecology to population dynamics to community ecology to ecosystems, keep asking: *How does what I learned in earlier chapters connect to what I am learning now?*

C.5. Learn to distinguish between understanding and doing

Writing code that plots your data is different from writing code that simulates density-dependent growth. The first you check by looking at the plot — did it do what I wanted? The second you check by interrogating the assumptions you built into the model. This distinction matters for how you think about AI and coding in this course.

Code that generates data displays, apps, or games is code that you check primarily via output. You look at or use the output and ask, did the code do what I wanted it to do? If not, attempt to troubleshoot and try again.

In contrast, code that generates mechanistic and statistical models is code that you have to understand and check primarily by evaluating assumptions that you tried to manifest in your code.

Mechanistic models are simply tools to turn assumptions into predictions, so you really need to understand how you are building those assumptions.

The output of your mechanistic model constitutes your predictions. You need to trust those predictions because they follow directly from the assumptions that you made. If the predictions seem off to you, then you need to reconsider the assumptions you made. There are three types of situations to consider:

1. *Do you have a simple coding error*? Does your code do what you intended? E.g., did you use Boolean logic correctly, or create input to a function that merely wasn't in the right sequence?
2. *Does your math match your ecology*? How you use mathematical functions to manifest your assumptions about the biological/ecological/physical features of the world can be tricky and problems fall into two buckets: sometimes the math doesn't do what you think it does, and other times the ecological assumptions have different consequences than you thought they would.
    a. *Are your mathematical assumptions correct*? There are some convenient mathematical operations and forms that you can collect in your toolbox: functions that put a ceiling or floor on values, functions that create positive feedback loops and negative feedback loops, functions that add and those that subtract; 0th, 1st, and 2nd order products of variables.
    b. *Are your ecological assumptions correct*? This is the interesting situation, where your choices of mathematical functions or parameter estimates may or may not map onto reality.
        i. If your ecological assumptions are wrong, that is great! You're learning something important about the world.
        ii. If your ecological assumptions are correct but the outcome is unexpected, that is great — you are discovering a consequence of your assumptions that your intuition hadn't predicted.

This is where mathematical reasoning becomes your most important verification tool. If an AI generates model code, you can check whether it runs (easy) and whether the output looks reasonable (harder). But the deepest check is whether the mathematical structure matches the biology you intended. Does the code implement $1 - N/K$ or $1 - \alpha N$? Those are equivalent — but only if $\alpha = 1/K$. If you can't follow that equivalence, you can't tell whether the AI's implementation matches your biological intention. The formal logic is the bridge between "the code runs" and "the code is right."

How to practice it. When you encounter a place in this book where you need to use code, do not just copy some code nor ask AI for it. Ask *why am I writing code*?

1. Is it to make a graph that displays data clearly?
2. Is it to use mathematics to represent some biological or physical process?

3. Is it to perform a statistical test in which I have to understand the assumptions of that test to apply it correctly?

If you answer "yes" to no. 1, then using AI is *relatively* safe, but you can probably be most effective if you can work directly with any code AI spits out for you.

If you answer "yes" to 2 or 3 above, then you really need to understand what you are doing. You need to know *not only* that two cans of beer plus two cans of beer add up to four cans of beer, but what will happen at the party once people start drinking it.

C.6. Using AI responsibly in this course

None of the above means you should avoid AI entirely. AI tools can be genuinely useful in your ecological education, but they are useful in the way a calculator is useful in a math course: they can save time on mechanical tasks, but they cannot replace the thinking that makes the tasks meaningful.

Here are some guidelines for responsible use.

Good uses of AI in ecology coursework:

- Debugging R code that throws an error you do not understand.
- Getting a quick refresher on R syntax (e.g., "How do I add a legend to a ggplot?").
- Generating a first draft of an outline that you then revise substantially.
- Asking for alternative explanations of a concept you have already studied but find confusing.

Risky uses of AI in ecology coursework:

- Asking AI to explain an ecological concept you have not yet studied yourself. (You will not be able to evaluate the answer.)
- Using AI-generated code without understanding every line of it.
- Asking AI to interpret your results for you. (Interpretation requires the biological context that you, not the AI, possess.)
- Asking AI to write your discussion or analysis. (This is where your own ecological thinking is supposed to develop.)

> ⚠ The Code Comprehension Rule
>
> If you use AI to help write R code, you should be able to explain what every line does and *why* it is there. If you cannot, you have not learned what you need to learn.

C.7. A note for instructors

If you are teaching with this book, the AI landscape is shifting under your feet. Here are some principles that may help.

Design assignments that require judgment, not just output. If an assignment can be completed by pasting the prompt into ChatGPT and submitting the result, the assignment is testing the wrong thing. Assignments should require students to make predictions, evaluate assumptions, connect models to specific biological systems, and explain *why* a result makes ecological sense.

Make AI use transparent, not forbidden. Blanket bans on AI are difficult to enforce and send the message that AI is cheating rather than a tool. Instead, consider requiring students to document when and how they used AI, and to reflect on what they learned (or failed to learn) by using it. The goal is to develop students who can use AI wisely, not students who can pretend they do not use it.

Foreground assumptions explicitly. When teaching any model, spend time on what the model assumes and what would happen if those assumptions were violated. Use callout boxes, discussion questions, and activities that force students to engage with assumptions as substantive ecological content, not as disclaimers to be skipped.

Require prediction before computation. Build the "predict first" step into your workflow explicitly. Have students write predictions in a worksheet, a shared document, or their lab notebooks before they run code. This simple intervention creates the productive struggle that drives learning.

Use AI failures as teaching moments. Ask an AI to explain an ecological concept, and then have students identify what it got wrong. This is a great exercise in critical thinking and assumption literacy — and it demonstrates, concretely, why human expertise matters.

C.8. The skills that will define your career

The ecology profession is changing. Ecologists who graduate in the next decade will work alongside AI tools that can process satellite imagery, classify species from audio recordings, and generate code for complex statistical analyses. The value of a human ecologist will not lie in being able to do these things manually. It will lie in being able to:

- Ask the right questions about the natural world.
- Design studies and models that can actually answer those questions.
- Recognize when a tool (AI or otherwise) is giving you the wrong answer.
- Connect mathematical results to biological reality.
- Communicate ecological understanding to people who need it — policymakers, land managers, communities.

These are the skills this book is designed to develop. The AI tools will change. The need for ecologists who understand the living world will not.

C.9. Deliverables

1. **Reflection (individual, ~300 words).** Choose one of the six durable skill categories (mechanistic reasoning, mathematical reasoning, model assumption literacy, prediction before computation, data literacy, or multi-scale synthesis). Describe a specific moment in your education — in this course or another — when you were forced to use this skill. What made it difficult? What did you learn? How would the experience have been different if you had used AI to skip the hard part?

2. **AI evaluation exercise (pairs).** Pick an ecological concept covered in a previous chapter of this book (e.g., exponential growth, logistic growth, competitive exclusion, density dependence). Ask an AI to explain this concept in 2–3 paragraphs. Then, working with your partner, write a critical evaluation of the AI's response:

    a. What did the AI get right?
    b. What did it get wrong, or what important nuance did it miss?
    c. What assumptions or caveats were absent from the AI's explanation?
    d. Would a student who read only the AI's response develop a correct understanding of the concept? Why or why not?

3. **Assumption audit (individual).** Choose one model from this course (e.g., the discrete-time logistic model, the Lotka-Volterra competition model). List its key assumptions. For each assumption, write one sentence explaining a real ecological situation in which that assumption would be violated, and one sentence explaining what consequence the violation would have for the model's predictions. Do not use AI for this exercise.

4. **Code classification (individual).** Choose three coding tasks from a previous chapter of this book. For each one, classify it as primarily "check via output" (you evaluate it by looking at what it produces) or "check via assumptions" (you evaluate it by examining the ecological and mathematical logic built into the code). Write a sentence or two explaining your reasoning for each classification, and describe what you would need to understand about the ecology to evaluate the code properly.

5. **Math as verification (individual).** Ask an AI to derive the equilibrium of a model from this course (e.g., the continuous logistic, or the invasion criterion for Lotka-Volterra competition). Copy the AI's derivation. Then, working on paper, do the derivation yourself step by step. Compare: Did the AI skip any steps? Did it make any errors?

Did it explain what each step means biologically, or only algebraically? Write a paragraph reflecting on whether you could have caught an error without doing the derivation yourself.

D. Accessing Public Ecological Databases

…in which we collect, in one place, every live database this book has already sent you to query – plus a map of where to look next.

D.1. Goals

In this appendix, you will

- Find, in one place, every live public database used elsewhere in this book, with the R package (or API) that reaches it, a minimal worked query, and a note on what it costs to use.
- Learn a few habits that apply to *any* live data source: how to avoid hammering a server, when to save a local copy, and how to cite data that changes from one day to the next.
- Get a short, honest map of other public ecological databases worth knowing about, for a class project, an independent study, or just curiosity, beyond the ones this book puts you through directly.

This appendix does not re-teach any chapter. It is a reference – a place to come back to when you remember "there was a package for that" but not which chapter, or when you want to point a new project at a database this book never used.

D.2. Why bother with live databases?

Every dataset in this book could have been handed to you as a clean `.csv`. Several of them are not, on purpose. Ecology today runs on databases that update continuously – new occurrence records, new budget lines, new trait measurements – and a large part of learning to do ecological data analysis is learning to reach into one of these moving targets and pull out exactly what you need, in a form you can trust and reproduce. A `read_csv()` teaches you to wrangle a table. An `occ_count()` or a `fromJSON()` call teaches you that too, plus something a static file cannot: how to ask a real, living system a specific question.

D.3. Quick reference

Database	What it holds	R access	Used in	Needs a key?
Google Sheets	Any spreadsheet a class or collaborator is actively adding to	`googlesheets4::read_sheet()`	02. Tools and People	No (browser sign-in the first time)
BIEN (Botanical Information and Ecology Network)	Plant trait, range, and checklist data for the Americas	`BIEN` package (`BIEN_trait_*()`, `BIEN_list_*()`)	04. Traits, Physiology, and Plant Form	No
GBIF (Global Biodiversity Information Facility)	2+ billion species occurrence records worldwide	`rgbif` package (`occ_count()`, `occ_search()`, `occ_download()`)	14. GBIF Biodiversity Data	No for search/count; free account for `occ_download()`
USAspending.gov	Federal agency budgets, obligations, and outlays	`jsonlite::fromJSON()` on the REST API	15. USDA Funding Ecological Science	No
NSF Award Search	Individual NSF grant records, including funded amounts	`jsonlite::fromJSON()` on the REST API	15. USDA Funding Ecological Science	No

None of the five require you to register for an API key just to follow along with this book. That is not universal – some of the databases in the "where to look next" section below do require one – so get in the habit of checking a package's documentation for an `Authentication` or `API key` section before you assume a query will just work.

D.4. 1. Crowdsourced data: Google Sheets

Package: `googlesheets4`

Chapter 02 has an entire class contributing rows to the same live spreadsheet, and `read_sheet()` pulls whatever is currently in it, including rows that were not there yesterday.

```
library(googlesheets4)

d <- read_sheet(
  "https://docs.google.com/spreadsheets/d/YOUR_SHEET_ID/edit",
  sheet = 1
)
```

The first time you run this in a session, R opens a browser tab and asks you to sign in to a Google account; after that it caches the authorization. If the sheet is set to "anyone with the link can view," you can also call `gs4_deauth()` first to skip sign-in entirely for read-only access.

> **When to use a live sheet vs. a csv**
>
> A Google Sheet is the right tool when the data are still being *collected* – a class survey, a shared field log. Once collection is finished, export a static snapshot (File > Download > csv) into `data/` so your analysis does not silently change if someone edits the sheet later.

D.5. 2. Plant trait data: BIEN

Package: `BIEN`

Chapter 04 queries BIEN for whole-plant height and leaf area across a species list, then reshapes the result from long to wide.

```
library(BIEN)

BIEN_trait_list()  # see every trait name BIEN recognizes

my_species <- BIEN_list_country(country = "Costa
 ↪  Rica")$scrubbed_species_binomial

my_traits_data <- BIEN_trait_traitbyspecies(
  species = my_species,
  trait   = c("whole plant height", "leaf area")
)
```

`BIEN_trait_list()` is worth running before you commit to a trait name – BIEN's trait vocabulary is specific (e.g., "whole plant height", not "height"), and a mismatched string returns zero rows rather than an error. There are equivalent `BIEN_list_state()`, `BIEN_trait_traitbygenus()`, and `BIEN_trait_traitbyfamily()` functions for narrower or broader species lists.

D.6. 3. Biodiversity occurrences: GBIF

Package: `rgbif`

Chapter 14 uses GBIF to compare species richness across countries and across a latitudinal gradient, entirely through live queries – no files downloaded.

```
library(rgbif)

palm_key <- name_backbone(name = "Arecaceae")$usageKey

richness_by_country <- occ_count(
  taxonKey   = palm_key,
  facet      = "speciesKey",
  facetLimit = 99999,
  country    = "CR"
)

effort_by_country <- occ_count(taxonKey = palm_key, country = "CR")
```

`name_backbone()` translates a taxonomic name into GBIF's internal numeric key. `occ_count(facet = "speciesKey", ...)` returns one row per species, so its row count is species richness; the same call without `facet` returns total records, a rough proxy for sampling effort. `occ_search()` returns the actual occurrence records (coordinates, dates, collector) rather than just counts.

> **!** GBIF data are biased, and citing them takes an extra step
>
> GBIF is assembled from museum specimens, herbaria, and citizen-science apps, so coverage tracks where people (and phone cameras) are, not where organisms are. Treat any pattern you find as "difference in recorded occurrences" until you have reason to think it reflects a real ecological signal.
> The quick `occ_count()`/`occ_search()` calls above are fine for exploring, but are not a citable, reproducible snapshot – GBIF's holdings change daily. For any real analysis, use `occ_download()` instead: it asks GBIF to assemble your exact record set and issues a DOI that pins the data and credits every contributing dataset. Cite that DOI, GBIF itself (**gbif?**), and the `rgbif` package (**rgbif?**).

rgbif

D.7. 4. Federal spending data: USAspending.gov

Package: `jsonlite` (talking to a REST API directly – there is no dedicated USAspending R package)

The Funding Ecological Science chapter builds agency budget comparisons this way:

```
library(jsonlite)

code <- "049"  # NSF's toptier agency code
url <- paste0("https://api.usaspending.gov/api/v2/agency/", code,
 ↪  "/")
result <- fromJSON(url)
```

`fromJSON()` fetches the URL and parses the response straight into an R list/data frame – no separate download step. USAspending.gov needs no key and has no rate limit aggressive enough to matter for classroom use, but every agency has its own numeric "toptier code" (NSF is `049`), and some ecologically relevant agencies (NOAA, USGS, USFS, FWS) are *sub-agencies* without their own toptier code, so they have to be reached through their parent department instead.

D.8. 5. Grant-level data: NSF Award Search

Package: `jsonlite`, same pattern as above, different host

```
url <-
 ↪  "https://api.nsf.gov/services/v1/awards.json?keyword=biodiversity"
nsf_result <- fromJSON(url)
```

This returns individual award records, including `fundsObligatedAmt` – the funded amount for that specific grant. That is public; the PI's detailed line-item budget within the award is not, so do not expect (or need) more granularity than the award total.

D.9. Practical habits for working with live data

Save a snapshot when the data need to stay fixed, not when they are just large. The book's own convention (see `01-computer-setup.qmd`) is that hand-built or one-time reference datasets live in `data/` and get read back in with `read_csv()`, rather than being rebuilt from a live source every render. Apply the same logic to your own project: if you need the numbers to be identical every time you re-run your script – for a report, a manuscript, a comparison over time – pull once, `write_csv()` the result into `data/`, and read *that* back in from then on. If you genuinely want "whatever is current right now," call the live API every time and skip the local copy.

Respect rate limits, even generous ones. None of the five databases above will complain about the handful of queries a course assignment makes. That changes fast if you write a loop that queries thousands of times without pausing. If you are ever iterating over a long list (many species, many grid cells, many years), check the API's documentation for a stated rate limit and

add a short `Sys.sleep()` between calls if one is not built into the package already.

An API key, when one is required, is not optional and not something to hardcode. A few databases below (eBird, Movebank) require a free registration before you can query them at all. Store the key as an environment variable (an `.Renviron` file, never committed to git) rather than typing it directly into a script – the same reason you would not paste a password into code that might end up on GitHub.

Cite the data, not just the package. `library(rgbif)` gets you the records; it does not by itself credit the people who collected and digitized them. Most of these databases either issue a citable DOI for a specific pull (GBIF's `occ_download()`) or ask for a standard attribution string (check the database's own "how to cite" page). Get in the habit of copying that citation into your script or report at the same time you pull the data, not weeks later when you can no longer remember exactly what you queried.

D.10. Where to look next

The five databases above are the ones this book actually puts you through. They are nowhere near the whole landscape. If you are looking for data for a class project, an independent study, or your own curiosity, these are worth knowing exist. Function names below are illustrative of what each package is *for*, not verified worked examples the way the sections above are – read that package's own documentation before you build an assignment or a project around one.

- **NEON (National Ecological Observatory Network)** – standardized, continent-scale sensor and field data (climate, soil, vegetation, organisms) from ~80 U.S. sites. R access: the `neonUtilities` package.
- **iNaturalist** – citizen-science species observations with photos, often used alongside or instead of GBIF (which actually ingests research-grade iNaturalist records itself). R access: the `rinat` package, or GBIF directly.
- **eBird** – bird sightings, one of the largest citizen-science datasets in existence. R access: the `auk` or `rebird` packages; querying eBird's own API requires a free key.
- **Movebank** – GPS and satellite tracking data for individual tagged animals. R access: the `move` package; many datasets require the data owner's permission to download.
- **NOAA climate and ocean data** – weather station records, sea surface temperature, and related long-term monitoring data. R access: the `rnoaa` package.
- **USA National Phenology Network (USA-NPN)** – crowdsourced plant and animal phenology observations (leaf-out, flowering, migration) submitted through the Nature's Notebook program, nationwide. R access: the `rnpn` package, an actively maintained client for USA-NPN's own API.

- **LTER Network and the Environmental Data Initiative (EDI)** – curated, citable datasets from the U.S. Long Term Ecological Research sites and beyond, each with its own DOI. Two different entry points depending on what you want: `lterdatasampler` bundles a small set of real LTER datasets already cleaned up for teaching, ready to load with no API calls at all; `EDIutils` is a live API client for the full EDI repository (every published LTER dataset, not just the teaching subset), for when you know roughly what you are looking for and want to query it directly.
- **Global Population Dynamics Database (GPDD)** – long time series of population abundance across thousands of animal and plant populations worldwide, a classic source for population-ecology questions (population cycles, extinction risk, density dependence). Access has changed: the old `rgpdd` package talked to a database API that no longer exists and is now deprecated – do not install it. Instead, download the data directly as csv files from its permanent home at the KNB (Knowledge Network for Biocomplexity) repository, DOI `10.5063/F1BZ63Z8`, and read them in with `read_csv()` like any other static dataset. A useful reminder that "public database" sometimes means "a DOI-cited file," not an API.
- **RMBL Phenology Project** (rmblphenologyproject.weebly.com) – worth knowing as a contrast to the big federated databases above: a single, long-running field program at the Rocky Mountain Biological Laboratory in Gothic, Colorado, tracking flowering phenology in permanent plots for 150+ plant species since 1973, native bee communities since 2009, and climate/animal-arrival records since 1974. There is no live API or R package – this is one research group's monitoring program, not a data service. Specific datasets from it have been published, tied to individual papers, through RMBL's own data catalog (rmbl.org) and through general-purpose repositories like OSF and Dryad. If a project idea points you toward a single well-known field site or long-term study rather than a global database, this is the kind of place you end up: search the study's own site or the paper's data-availability statement for a repository link and a DOI, rather than expecting a package.
- **Dryad and other data repositories** – data deposited alongside individual published papers, useful when you want the exact dataset behind a specific result rather than a whole live database.

D.11. A note on why some of this book's chapters use live queries and others use `data/`

You may notice this book is inconsistent, on purpose: some chapters query a live database every time they render, and others read a fixed `.csv` from `data/`. That is not sloppiness – it reflects the same choice described in "Practical habits" above, made deliberately for each chapter. A chapter built around teaching *how to query a live source* (BIEN, GBIF, USAspending, NSF)

leaves the query live, so what you see is really happening in real time. A chapter that needed a small, specific, one-time dataset (inflation figures, a curated budget table) built it once and saved it to `data/`, so the numbers in the book do not drift out from under the prose describing them. When you design your own project, make that same choice deliberately, not by default.

E. Displaying Data with ggplot2

...in which we keep a short list of habits and recipes for making figures that are easy to read and honest about the data.

E.1. Goals

In this appendix, you will

- Use a handful of rules of thumb that fix most problems in student figures.
- Pick a sensible plot type from the kinds of variables you have.
- Copy and adapt a small set of `ggplot2` recipes for the plots you will make most often.
- Run through a checklist before you hand in or present any figure.

All examples use `CO2`, a data set that comes with R: CO_2 uptake by the grass *Echinochloa crus-galli* from two origins (Quebec, Mississippi), chilled or not, across a range of ambient CO_2 concentrations. Type `?CO2` to read about it.

E.2. Rules of thumb

1. **Show the data.** If you can plot the raw observations, do. Add means, lines, or boxes *on top of* the points, not instead of them.
2. **Put the response on the y-axis** and the variable you think drives it on the x-axis.
3. **Label every axis with the variable and its units**, e.g., "CO2 uptake (umol m^-2 s^-1)", never `uptake`.
4. **Use color, shape, or panels only to show a variable.** If color does not mean something, leave it out.
5. **Use colorblind-safe colors** (e.g., `scale_color_viridis_d()`), and do not rely on red vs. green.
6. **Order categories meaningfully** (by treatment logic, or by the size of the response), not alphabetically by default.
7. **Avoid** 3-D effects, pie charts, two different y-axes on one plot, and bar charts of means with no data shown.
8. **Make text big enough to read** from the back of a room: `theme_bw(base_size = 14)` or larger for slides.

E.3. Which plot?

x variable	y variable	Start with	ggplot2 geoms
none	one continuous	histogram or density	`geom_histogram()`, `geom_density()`
categorical	continuous	points + box or mean	`geom_jitter()`, `geom_boxplot()`, `stat_summary()`
continuous	continuous	scatterplot	`geom_point()`, `geom_smooth()`
time	continuous	line + points	`geom_line()`, `geom_point()`
categorical	counts	bar chart	`geom_bar()` (counts), `geom_col()` (values you supply)

A third variable goes to `color =`, `shape =`, or `facet_wrap()`.

E.4. The ggplot2 template

Every plot has the same three parts: the **data**, the **aesthetic mapping** (`aes()`, which says which column goes to which axis or color), and one or more **geoms** (the marks drawn). Everything else is optional polish.

```
ggplot(data = CO2, aes(x = conc, y = uptake)) +
  geom_point()
```

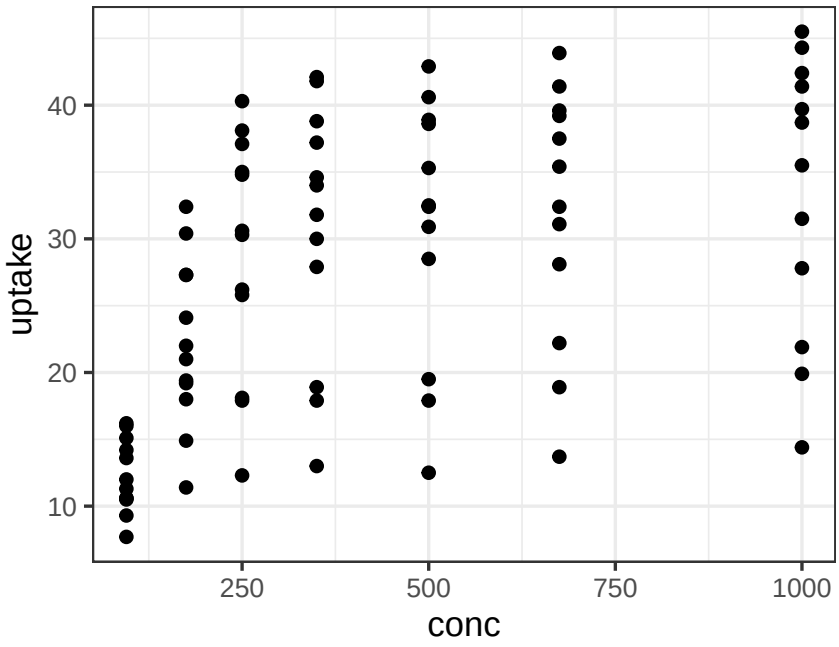


Figure E.1.: The minimal ggplot: data, mapping, geom.

Remember that `aes()` is for things that come *from a column*. A fixed setting goes outside it: `geom_point(color = "blue")`, not `geom_point(aes(color = "blue"))`.

E.5. Recipes

E.5.1. Continuous x and y, with groups

```
ggplot(CO2, aes(x = conc, y = uptake, color = Treatment)) +
  geom_point(alpha = 0.7) +
  geom_smooth(se = FALSE, method = "loess", formula = y ~ x) +
  facet_wrap(~ Type) +
  scale_color_viridis_d(end = 0.8) +
  labs(x = "Ambient CO2 (mL/L)",
       y = expression(CO[2]~uptake~(mu*mol~m^-2~s^-1)),
       color = NULL)
```

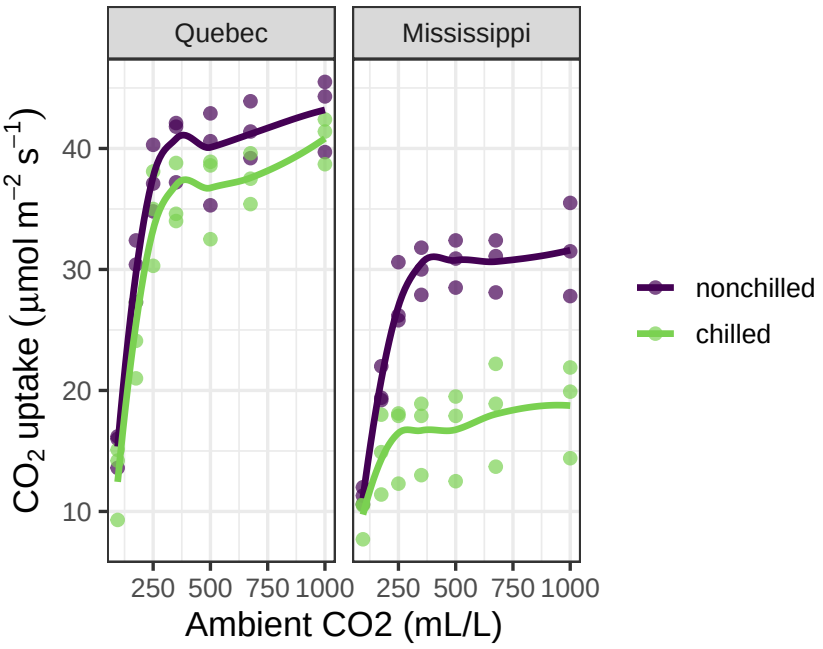


Figure E.2.: Points for the data, a smoother for the trend, color for a grouping variable, and panels for a second one.

- `alpha` makes overlapping points visible.
- `geom_smooth(method = "lm")` gives a straight line instead of a curve.
- `facet_wrap(~ Type)` splits the plot into panels that share axes, which is usually easier to compare than many colors on one panel.

E.5.2. Small multiples: `facet_wrap()` and `facet_grid()`

Panels ("small multiples") are often clearer than piling many colors onto one plot. Each panel uses the same axes, so the eye compares shapes rather than decoding a legend.

`facet_wrap()` takes **one** grouping variable and wraps its panels into rows and columns. Here each of the 12 plants gets its own panel:

```
ggplot(CO2, aes(x = conc, y = uptake)) +
  geom_line() +
  geom_point() +
  facet_wrap(~ Plant, ncol = 4) +
  labs(x = "Ambient CO2 (mL/L)", y = "CO2 uptake (umol m^-2 s^-1)")
```

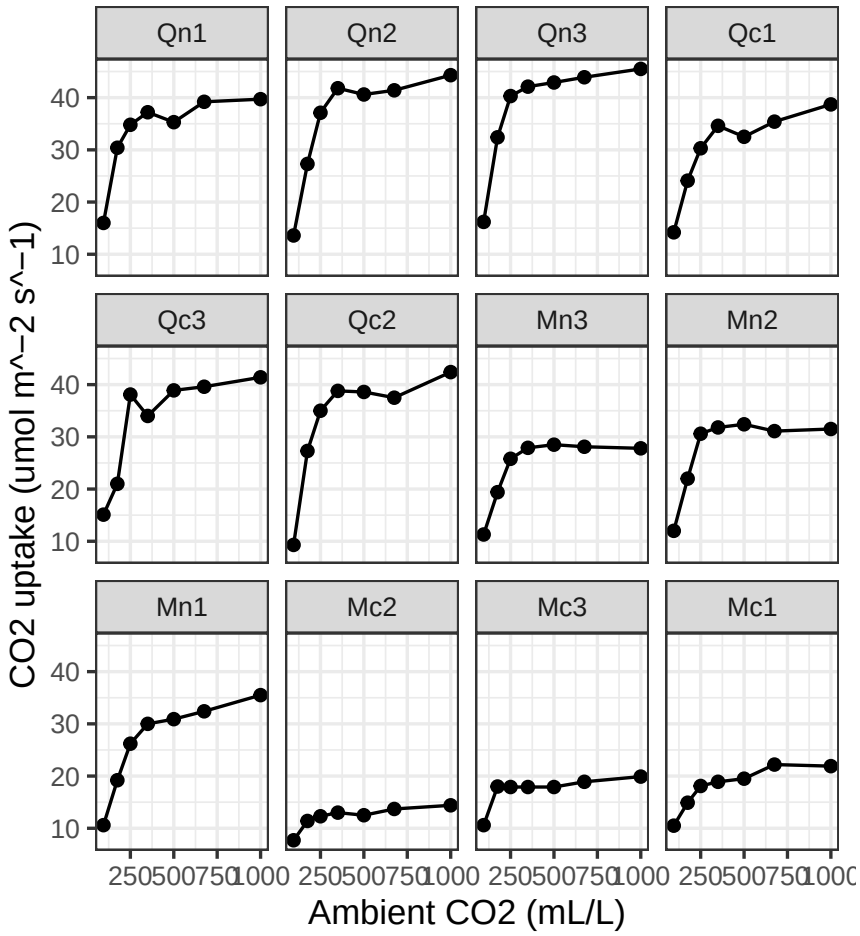


Figure E.3.: facet_wrap(): one panel per plant, arranged in 4 columns.

`facet_grid()` crosses **two** grouping variables: rows by one, columns by the other. This is the natural layout for a two-factor experiment:

```
ggplot(CO2, aes(x = conc, y = uptake, group = Plant)) +
  geom_line(color = "grey40") +
  geom_point() +
  facet_grid(Type ~ Treatment) +
  labs(x = "Ambient CO2 (mL/L)", y = "CO2 uptake (umol m^-2 s^-1)")
```

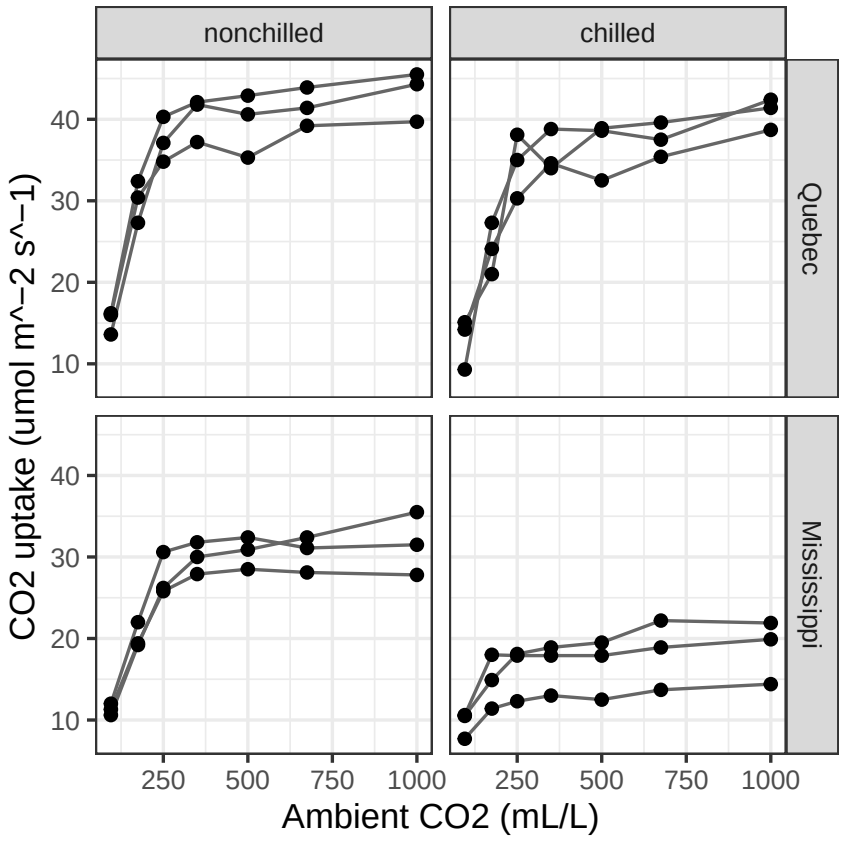


Figure E.4.: facet_grid(): rows are plant origin, columns are treatment. Each line is one plant.

- Read `facet_grid(rows ~ cols)` as "rows by `Type`, columns by `Treatment`." Use `. ~ Treatment` (or `cols = vars(Treatment)`) for columns only.
- `group = Plant` tells `geom_line()` to draw one line per plant instead of zig-zagging through every point in a panel.
- Panels share axes by default, which is usually what you want for comparison. Use `scales = "free_y"` only when the groups differ so much in size that shared axes hide the pattern, and say so in the caption.
- If panel labels are cryptic codes, fix the factor levels in the data (or use `labeller = label_both` to show "Type: Quebec") rather than explaining them in the caption.

E.5.3. Categorical x, continuous y

```
co2_high <- subset(CO2, conc == 1000)  # one measurement per plant

ggplot(co2_high, aes(x = Treatment, y = uptake)) +
  geom_jitter(width = 0.1, height = 0, alpha = 0.4) +
  stat_summary(fun.data = mean_se, color = "red", size = 0.8) +
  facet_wrap(~ Type) +
  labs(x = NULL, y = "CO2 uptake (umol m^-2 s^-1)")
```

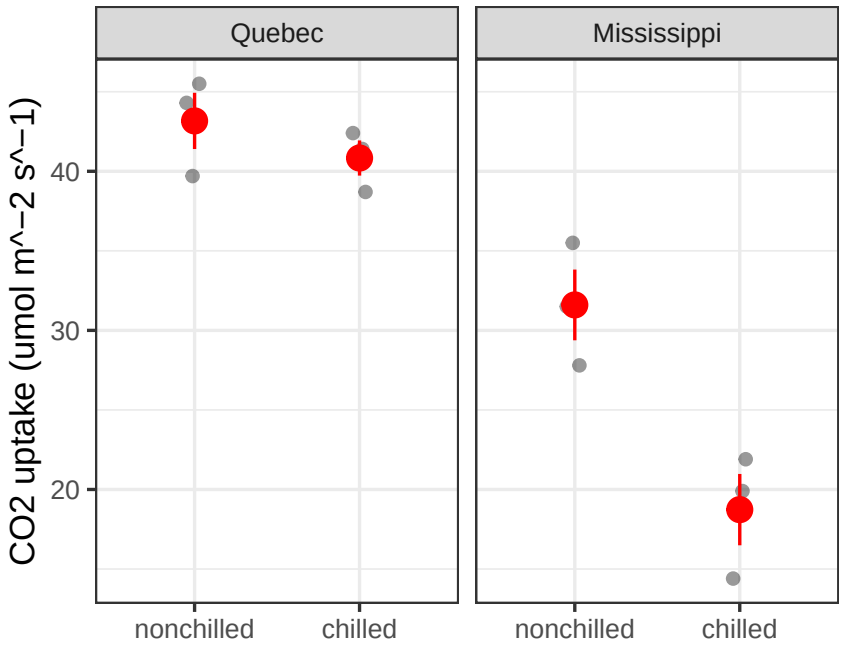


Figure E.5.: Uptake at the highest CO2 concentration (1000 mL/L). Raw data (jittered points, one per plant, n = 3 per group) with the group mean +/- 1 SE drawn on top.

- `geom_jitter(width = 0.1, height = 0)` spreads points sideways only a little, so they stay with their group, and never moves them up or down (which would change their y values).
- Swap `stat_summary()` for `geom_boxplot(outlier.shape = NA)` to show medians and quartiles instead. (`outlier.shape = NA` stops outliers being drawn twice, since the jitter already shows them.)
- Say in the caption what the error bars are (SE, SD, or 95% CI). Readers cannot tell otherwise.
- Make sure each point is an independent replicate before computing an SE. Each plant in `CO2` was measured at seven concentrations, so we kept one concentration; pooling all of them would treat 21 repeated measurements on 3 plants as 21 replicates.

E.5.4. Reordering categories

```
ggplot(CO2, aes(x = reorder(Plant, uptake, FUN = median), y =
 ↪  uptake)) +
  geom_boxplot() +
  labs(x = "Plant", y = "CO2 uptake (umol m^-2 s^-1)")
```

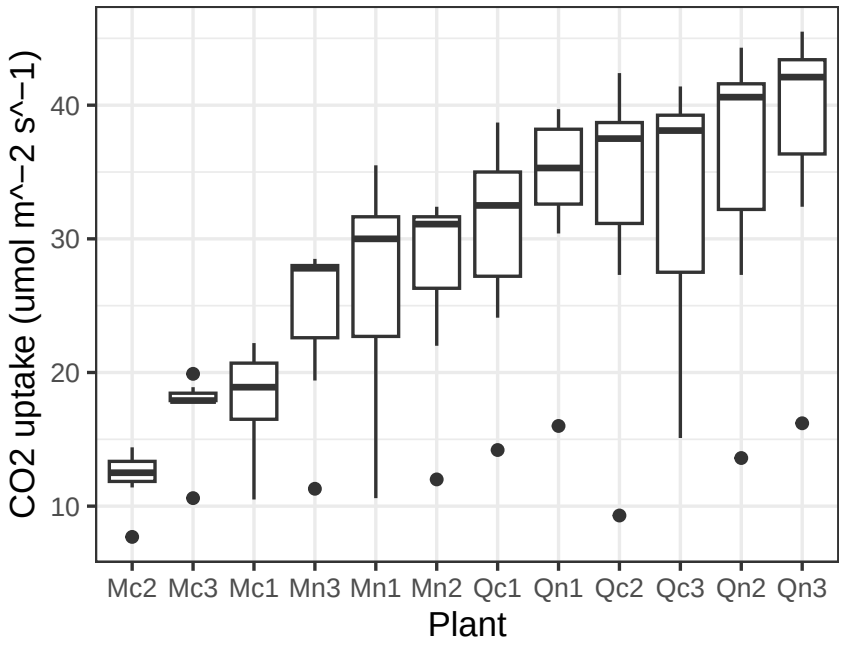


Figure E.6.: Plants ordered by median uptake rather than alphabetically.

To set an order by hand, change the factor before plotting, e.g., `factor(x, levels = c("low", "medium", "high"))`.

E.5.5. One distribution

```
ggplot(CO2, aes(x = uptake)) +
  geom_histogram(binwidth = 2.5, fill = "grey70", color = "black") +
  labs(x = "CO2 uptake (umol m^-2 s^-1)", y = "Number of
   ↪ measurements")
```

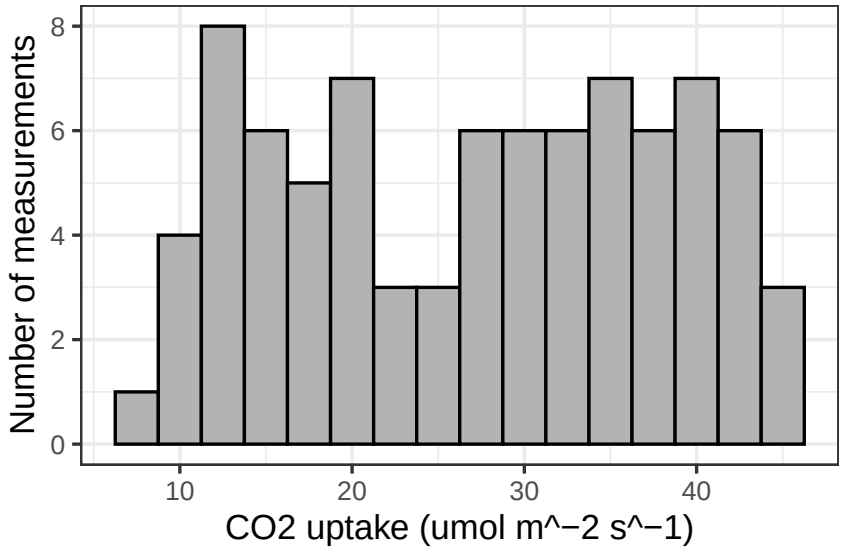


Figure E.7.: A histogram. Try several binwidths; the default is rarely the best.

E.5.6. Log scales

Many ecological variables (body size, abundance, area) span orders of magnitude. Log the *axis*, not the column, so the tick labels stay in original units:

```
ggplot(df, aes(x = body_mass, y = metabolic_rate)) +
  geom_point() +
  scale_x_log10() +
  scale_y_log10()
```

E.6. Polish

- **Titles and labels:** `labs(x = , y = , color = , title = )`. In a report or this book, put the explanation in the figure caption and usually skip `title`.
- **Theme:** set once at the top of a script with `theme_set(theme_bw(base_size = 13))`. `theme_classic()` and `theme_minimal()` are also clean.
- **Legend:** move it with `theme(legend.position = "bottom")`; remove it with `"none"` when the facet labels or a caption already say what the colors mean.
- **Subscripts, superscripts, Greek letters:** use `expression()`, as in the scatterplot above: `CO[2]` is a subscript, `m^-2` a superscript, `mu` gives a Greek mu, `~` is a space, and `*` joins without a space.
- **Plain text in plots:** type `umol`, `~`, and `+/-` rather than pasting special symbols such as degree signs or a micro sign into labels. Pasted symbols can silently disappear when a figure is saved to PDF; `expression()` draws them safely.

E.7. Saving figures

Save with `ggsave()` into your `figs/` folder, and set the size yourself so the text comes out the size you intended:

```
p <- ggplot(CO2, aes(x = conc, y = uptake)) + geom_point()
ggsave("figs/co2_uptake.png", plot = p, width = 6, height = 4, dpi =
 ↪  300)
```

- Width and height are in inches by default. A figure that fills half a page is about 6 x 4.
- Use `.png` for slides and documents, `.pdf` when a journal or your instructor asks for vector graphics.
- If text looks too small in the saved file, increase `base_size` in the theme; do not just make the image bigger.

E.8. Checklist before you hand it in

- ☐ Can someone tell what each axis is, and in what units, without reading your text?
- ☐ Are the raw data visible, or is there a good reason they are not?
- ☐ Does every color, shape, and panel stand for a variable, and is that explained in the legend or caption?
- ☐ Are categories in a sensible order?
- ☐ Does the caption say what the error bars or lines are and what the sample size is?
- ☐ Is the text readable at the size the figure will actually be shown?
- ☐ Would the figure still work in grayscale or for a colorblind reader?

E.9. Where to look next

- *R for Data Science* (2e), chapters on data visualization and communication: https://r4ds.hadley.nz
- The `ggplot2` reference pages, with an example for every function: https://ggplot2.tidyverse.org/reference/
- The Posit `ggplot2` cheat sheet (search "ggplot2 cheat sheet").

F. Your Own Interests: Telling a short story with data

F.1. Goals

For this assignment, you will,

- Share a story with ecological implications, connections, or foundations that interests you.
- Find, cite, and import data from a reliable, credible source to help you tell that story.
- Write R code that helps you understand, analyze, and share data more effectively than you otherwise could or would.
- Distill your story into a headline, a question, and a single, data-rich story.

F.2. Background

Throughout this book, you've been mostly following my script. In this chapter, I want you to follow your own script, Hollywood or otherwise.

Your goal in this chapter is to tell a story, a very short story, using data.

Why a story? Because people remember stories far better than they remember lists of facts. A good data story has the same bones as any other story: a setting (why should anyone care?), a tension or question (what don't we know?), and a resolution (what do the data say?). Knaflic (2015) puts it simply: before you make any graph, you should be able to say, in one sentence, what you want your audience to know and why it matters.

Consider one example. Rosenberg et al. (2019) combined decades of bird surveys and weather radar data and estimated that the breeding bird population of the United States and Canada had declined by nearly 3 billion birds since 1970. That is a story with a setting (birds are everywhere, and we mostly take them for granted), a tension (are there as many as there used to be?), and a resolution delivered by data (no, about 29% fewer). One graph and one sentence carried that story to newspapers around the world.

F.2.1. Start a conversation - don't end it

While Rosenberg et al (2019) showed us an important pattern, they also raised other questions. Is it still happening? Is the situation getting worse, or getting better? why did this happen? They used their data to start a conversation. That is what I hope you will do.

F.3. Your grade depends on how your slides communicate

Share your interests, not your answers. Imagine you are starting a conversation, not ending one.

For this assignment, you will be evaluated on how you use data to share what interests you. You will not be evaluated on answering a deep question. Instead, tell us what interested you, what you learned (show your data), and what questions you still have. If you can, leave us thinking "Huh. I never thought about that before" or "Cool - I'd like to know the answer too" or even "Really? That seems questionable to me. Are you sure?"

F.3.1. What interests you?

Almost *everything* connects to ecological systems. Here are a few:

- the loss of billions of North American birds since 1970;
- the price of gasoline in your local gas station;
- the toilet paper in your bathroom;
- abundances of smallmouth bass in your favorite fishing spot;
- an upcoming U.S. election;
- the sycamore leaves turning yellow in October;
- what you had for breakfast.

All of these can be tied to ecological patterns and processes. How does price influence fossil fuel use and the rate of climate change? What do our politicians respond to, and how important do they think protecting species, clean air, and clean water is to their voters? Was the toilet paper made from boreal forest spruce trees in forests that have never ever been cut before, and if so, what kinds of migratory birds have lost their nesting and breeding habitat as a result? What regulates the abundance of top predators in lake food chains? Are the sycamore leaves are responding to only day length, or is the timing of their autumnal color change shifting as the climate warms?

F.3.2. What topic do you want to spend a little time thinking about?

Don't think about this one too long. Just ask yourself and don't think about the answer.

You might already know what about the workings of the natural world interests you.

If nothing comes immediately to mind, that's OK, give it a second.

If ideas come flooding in because everything is interesting, pick just one – it doesn't really matter which one.

F.4. Preparatory Steps

1. **Pick your topic.** Write it down in one sentence.
2. **Turn it into a question** that data could answer. "Are birds declining?" is a better question than "Birds are cool." "Has the date of first leaf color change in sycamores shifted over the last 30 years?" is better still.
3. **Find a data source.** Try online sources first. An online source will often explain how you should cite it. See the appendix Accessing Public Ecological Databases to get you started. If or when you get stuck, ask a librarian, your instructor, or another knowledgeable person.
4. **Save the data in your `Rwork/data/` folder**, if you download a file, and record where it came from, when you got it, and how to cite it. Data are sacrosanct: never edit the original file by hand.
5. **Find 2-4 reliable, credible sources** that help you explain the background of your story and interpret your results.
6. **Create a new script** at the top level of your `Rwork` folder, and save it with a descriptive name (e.g., `my_story.R`).

F.5. An example: Does a cold night hurt a grass's ability to take up carbon?

Here is an example of the kind of short story I have in mind. It uses a data set that comes with R, so you can run every line of it yourself.

Barnyard grass, *Echinochloa crus-galli*, is a widespread, weedy C_4 grass. C_4 plants tend to be warm-climate specialists, so we might ask: if a plant gets chilled overnight, does it take up less carbon dioxide the next day? And does it matter whether the individual plant comes from a cold place (Quebec) or a warm place (Mississippi)? Potvin et al. (1990) measured CO_2 uptake in plants from both places, half of which had been chilled overnight, across a range of CO_2 concentrations.

F.5.1. The data

```
# CO2 uptake by Echinochloa crus-galli, from Potvin et al. (1990);
# this data set is built into R, see ?CO2
data(CO2)
glimpse(CO2)
```

```
Rows: 84
Columns: 5
$ Plant     <ord> Qn1, Qn1, Qn1, Qn1, Qn1, Qn1, Qn1, Qn2, Qn2, Qn2,
 ↪  Qn2, Qn2, ~
$ Type      <fct> Quebec, Quebec, Quebec, Quebec, Quebec, Quebec,
 ↪  Quebec, Queb~
$ Treatment <fct> nonchilled, nonchilled, nonchilled, nonchilled,
 ↪  nonchilled, ~
$ conc      <dbl> 95, 175, 250, 350, 500, 675, 1000, 95, 175, 250,
 ↪  350, 500, 6~
$ uptake    <dbl> 16.0, 30.4, 34.8, 37.2, 35.3, 39.2, 39.7, 13.6,
 ↪  27.3, 37.1, ~
```

There are six variables:

- `Plant`, a unique identifier for each of 12 plants;
- `Type`, where the plant came from (Quebec or Mississippi);
- `Treatment`, whether the plant was chilled overnight or not;
- `conc`, the ambient CO_2 concentration (mL/L, i.e., ppm);
- `uptake`, the rate of CO_2 uptake (μmol m^{-2} s^{-1}).

F.5.2. Graphing the data

```
p_co2 <- ggplot(CO2, aes(x = conc, y = uptake, colour = Treatment))
 ↪  +
  geom_line(aes(group = Plant)) +
  geom_point() +
  facet_wrap(~ Type) +
  labs(x = "CO2 concentration (mL/L)",
       y = "CO2 uptake (umol / m^2 / s)",
       colour = "Treatment")
p_co2
```

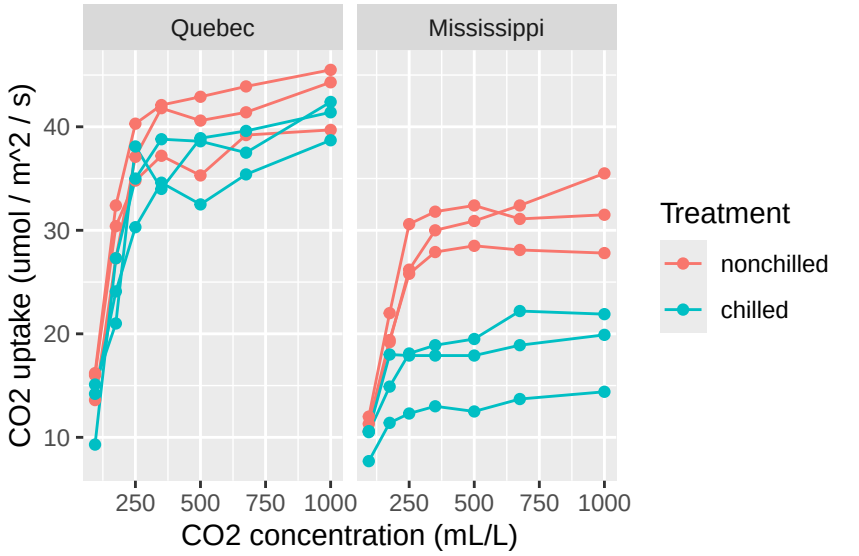


Figure F.1.: Carbon dioxide uptake by barnyard grass increases with ambient CO_2 concentration, but chilling overnight reduces uptake, especially in plants from Mississippi. Each line is one plant (n = 3 plants per group).

```
ggsave("figs/co2_uptake.png", p_co2, width = 6, height = 3.5)
```

F.5.3. A simple analysis

The graph tells most of the story. A simple analysis can help us say how big the effect is. Here, we use only the highest CO_2 concentration, where the differences are largest, and fit a linear model with an interaction between origin and treatment.

```
co2_high <- CO2 |> filter(conc == 1000)

mod <- lm(uptake ~ Type * Treatment, data = co2_high)
tidy(mod)
```

```
# A tibble: 4 x 5
  term                             estimate std.error statistic
   ↪  p.value
  <chr>                               <dbl>     <dbl>     <dbl>
   ↪  <dbl>
1 (Intercept)                          43.2      1.89    22.8
 ↪  0.0000000144
2 TypeMississippi                     -11.6      2.68    -4.32
 ↪  0.00254
3 Treatmentchilled                     -2.33     2.68    -0.872
 ↪  0.409
4 TypeMississippi:Treatmentchilled    -10.5      3.78    -2.78
 ↪  0.0238
```

The interaction term tells us whether chilling has a different effect on plants from Mississippi than on plants from Quebec. With only three plants per group, we should be cautious, but the pattern in the graph is striking.

F.5.4. The two-slide version

- **Slide 1 (background and question).** C_4 grasses are warm-climate specialists. Does a cold night reduce a grass's ability to take up carbon, and does it depend on where the plant comes from?
- **Slide 2 (the data-rich answer).** Headline: *"Southern grasses lose their carbon appetite after a cold night."* Figure F.1, plus one sentence about the size of the effect.

Notice that the headline is a full sentence that states the take-home message. It is not a label like "CO2 uptake results."

F.5.5. Discuss the following questions with your colleagues

1. Is the headline above justified by the data? Is it too strong? Too weak?
2. What would you need to know before claiming that this pattern applies to C_4 grasses in general?
3. Why might plants from Quebec be less affected by chilling than plants from Mississippi?
4. What is the *setting*, the *tension*, and the *resolution* of the story you want to tell?

F.6. Write your own script

Your script should follow the same structure as every other script in this book. Here is a template. Section headers that end in four dashes (`----`) let you navigate your script quickly in RStudio's document outline.

```
# My story: <one-sentence description of your story> ----
# Author: <your name>
# Date: <date>

## Set up ----
library(tidyverse)

## Import data ----
# Source: <who collected the data, where you got it, URL, date
 ↪  accessed>
# Citation: <how the data source asks to be cited>
my_data <- read_csv("data/my_data.csv")

## Explore and clean ----
glimpse(my_data)

## Graph ----
p1 <- ggplot(my_data, aes(x = , y = )) +
  geom_point()
p1
ggsave("figs/my_story.png", p1, width = 6, height = 4)
```

```
## Analysis (optional) ----
### Fit a simple model ----
# mod <- lm(y ~ x, data = my_data)
# summary(mod)
```

F.7. Overthinking: Are your data the data you think they are?

You can skip this on a first read, but come back to it before you write your report.

Before you trust any data set, read its documentation (often called *metadata*). Ask yourself,

- **What exactly was measured, and in what units?** "Abundance" might mean individuals counted, individuals per hectare, or an index relative to some baseline year.
- **Who collected the data, how, and why?** Data collected for one purpose (e.g., counting fish caught by anglers) may give a biased picture of something else (e.g., how many fish are in the lake).
- **Where and when?** Does a data set from one state, one year, or one lab let you say something about the whole world?
- **What's missing?** Gaps in time or space, and how missing values are coded (`NA`, `-999`, blank cells), can change your answer.
- **Is a pattern a cause?** Most stories you can tell with public data are correlational. That is fine, but say so, and think about what else could produce the same pattern.

Your story will be stronger, not weaker, if you are honest about its limitations.

F.8. Written report

Writing is always easiest when you write for a specific audience.

The two main purposes of this report are for you to create your presentation narrative and to provide a record of how credible your data are.

Write this report almost as a verbal, spoken script for your talk. Your report should,

1. introduce and explain your question; tell us why you are interested in this question, and provide enough detail to make sure we fully understand it;
2. tell us where you got your data;

3. shows us your data and explain clearly your chart; imagine we are blind, and explain everything about the chart - what the x-axis is, what the y-axis is, and what the data look like. After that, tell us what you learned from the pattern of data;
4. finally, tell us how you felt about what you learned, and tell us what questions you have now.

After adding a title and a references cited section, that will be your report. It should help prepare you to give the presentation.

F.9. Deliverables

In the end, you will need to create and share the following three items:

1. A slide deck with only **two** slides, to share with the instructor *and* the class,

    a. Slide 1: Background and your question.
    b. Slide 2: The data-rich answer to your question; this will include a graph with a headline, and perhaps the results of a simple analysis.

2. A written report that tells your story, shared with only the instructor. It must,

    - be 400-600 words in length, including sources;
    - use 2-4 reliable, credible sources, included as in-text citations and listed in a References Cited section at the end;
    - cite your data source.

3. Your R script used to create your data-rich answer. This might include,

    - code to access the data from an online source, or code to import a spreadsheet or CSV file from your `data/` folder;
    - code to create the graphical data display, saved to your `figs/` folder;
    - code to analyze the data statistically, *if* you did that.

    Your script MUST include comments that explain what your code does.

F.10. References

Knaflic, C. N. (2015). *Storytelling with Data: A Data Visualization Guide for Business Professionals*. Wiley.

Potvin, C., Lechowicz, M. J., and Tardif, S. (1990). The statistical analysis of ecophysiological response curves obtained from experiments involving repeated measures. *Ecology*, 71, 1389-1400.

Rosenberg, K. V., Dokter, A. M., Blancher, P. J., Sauer, J. R., Smith, A. C., Smith, P. A., Stanton, J. C., Panjabi, A., Helft, L., Parr, M., and Marra, P. P. (2019). Decline of the North American avifauna. *Science*, 366, 120-124.